

CATEGORY THEORY

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1. BASIC CATEGORY THEORY

Definition 1.0.1 (Category). A *category* $\mathcal{C}$ consists of the following data:

- A class of *objects* denoted $\text{Ob}(\mathcal{C})$.
- For each pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, a class

$$\text{Hom}_{\mathcal{C}}(X, Y)$$

of *morphisms* (also called *arrows* or *homs*). If the category $\mathcal{C}$ is clear, then this *hom-class* is also denoted by $\text{Hom}(X, Y)$. It may also be denoted by $\text{hom}_{\mathcal{C}}(X, Y)$ or $\text{hom}(X, Y)$, especially to distinguish from other types of hom's (e.g. internal hom's (Definition 2.0.2))

- For each triple of objects X, Y, Z , a composition law

$$\circ : \text{Hom}_{\mathcal{C}}(Y, Z) \times \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{C}}(X, Z),$$

denoted $(g, f) \mapsto g \circ f$.

- For each object X , an *identity morphism*

$$\text{id}_X \in \text{Hom}_{\mathcal{C}}(X, X).$$

These data satisfy the following axioms:

- (Associativity) For all morphisms $f \in \text{Hom}_{\mathcal{C}}(X, Y)$, $g \in \text{Hom}_{\mathcal{C}}(Y, Z)$, and $h \in \text{Hom}_{\mathcal{C}}(Z, W)$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

- (Identity) For all $f \in \text{Hom}_{\mathcal{C}}(X, Y)$,

$$\text{id}_Y \circ f = f = f \circ \text{id}_X.$$

One often writes $X \in \mathcal{C}$ synonymously with $X \in \text{Ob}(\mathcal{C})$, i.e. to denote that X is an object of $\mathcal{C}$.

We may call a category as above an *ordinary category* to distinguish this notion from the notions of *categories enriched in monoidal categories* (Definition 4.0.1) or higher/ n -categories.

(♠ TODO: TODO: define n -categories)

A category as defined above may be called called a *large category* or a *class category* to emphasize that the hom-classes may be proper classes rather than sets (note, however, that the possibility that hom-classes are sets is not excluded for large categories). Accordingly, a *category* may often refer to a locally small category (Definition 1.0.9), which is a category whose hom-classes are all sets.

Definition 1.0.2. We introduce the following common examples of categories, together with some frequently used notational variants:

- The *category of sets* (Definition 1.0.3), denoted **Set**, also written **Set**, **Sets**, or **(Set)**, has objects all sets and morphisms all set-theoretic functions.
- The *category of groups* (Definition C.0.7), denoted **Grp**, also written **Grp** or **Groups**, has objects all groups (Definition C.0.5) and morphisms all group homomorphisms (Definition C.0.7).
- The *category of abelian groups*, denoted **Ab**, also written **Ab** or **AbGrp**, has objects all abelian groups (Definition C.0.7) and morphisms all group homomorphisms (Definition C.0.7).
- Given a not-necessarily commutative ring with unity R (Definition C.0.1), the *category of (left/right/two-sided) R -modules*, denoted **Mod_R**, also written **R -Mod**, has objects all (left/right/two-sided) R -modules and morphisms the R -module homomorphisms (Definition C.0.10).
- Given a field k , the *category of vector spaces over k* , denoted **Vect_k**, also written **k -Vect** or **Vect(k)**, has objects all k -vector spaces and morphisms all k -linear maps.
- The *category of topological spaces*, denoted **Top**, also written **Top** or **TopSpaces**, has objects all topological spaces and morphisms all continuous maps.

- The *category of rings*, denoted **Ring**, also written **Ring**, has objects all rings with unity and morphisms all unital ring homomorphisms.
- The *category of commutative rings*, denoted **CRing**, also written **CRing** or **CommRing**, has objects all commutative rings with unity and morphisms all unital ring homomorphisms.
- The *category of fields*, denoted **Field**, also written **Field** or **Fields**, has objects all fields and morphisms all field homomorphisms.
- The *category of monoids*, denoted **Mon**, also written **Mon** or **Monoids**, has objects all monoids and morphisms all monoid homomorphisms.
- The *category of semigroups*, denoted **SemiGrp**, also written **SGrp** or **Semigroups**, has objects all semigroups and morphisms all semigroup homomorphisms.
- Given a topological space X , the *category of sheaves of sets on X* , denoted **Sh(X)**, also written **Sh(X)**, has objects all sheaves of sets on X and morphisms all morphisms of sheaves.
- The *category of small categories*, denoted **Cat**, also written **Cat** or **Categories**, has objects all small categories (Definition 1.0.9) and morphisms all functors (Definition 1.1.1) between them.
- Given categories $\mathcal{C}$ and $\mathcal{D}$, the *functor category* (Definition 1.2.1), denoted $[\mathcal{C}, \mathcal{D}]$, also written **Fun($\mathcal{C}, \mathcal{D}$)**, has objects all functors from $\mathcal{C}$ to $\mathcal{D}$ and morphisms all natural transformations between them.

Definition 1.0.3. The category of sets is the (locally small) (Definition 1.0.9) category (Definition 1.0.1)

- whose objects are sets (Definition A.0.1), and
- whose morphisms $X \rightarrow Y$ are set functions (Definition A.0.2) $X \rightarrow Y$.

The category of sets is often denoted by notations such as **Set**, **Set**, **Sets**, **Sets**, **(Set)**, **(Set)**, **(Sets)**, **(Sets)**.

Given a Grothendieck universe (Definition E.0.1) $\mathcal{U}$, the category of (Definition C.0.11) $\mathcal{U}$ -sets (Definition E.0.1) may be denoted by notations such as **Set $_{\mathcal{U}}$** . The category of $\mathcal{U}$ -sets is $\mathcal{U}$ -locally small (Definition 1.0.9) and is $\mathcal{V}$ -small (Definition 1.0.9) for any Grothendieck universe $\mathcal{V}$ containing $\mathcal{U}$.

Definition 1.0.4. 1. The *category of groups* is the locally small (Definition 1.0.9) category (Definition 1.0.1) whose objects are groups (Definition C.0.5) and whose morphisms are group homomorphisms (Definition C.0.7). It is often denoted by notations such as **Grp**.

2. The *category of abelian groups* is the locally small (Definition 1.0.9) category (Definition 1.0.1) whose objects are abelian groups (Definition C.0.5) and whose morphisms are group homomorphisms (Definition C.0.7). It is often denoted by notations such as **Ab**. **Ab** is an abelian category.

Definition 1.0.5. 1. The *category of rings* is the locally small (Definition 1.0.9) category (Definition 1.0.1) whose objects are rings (Definition C.0.1) R and whose morphisms $R \rightarrow S$ are ring homomorphisms (Definition C.0.4). The category of rings over R is often denoted by notations such as **Ring**.

2. The *category of commutative rings* is the full subcategory (Definition 1.3.2) of **Ring** consisting of the commutative rings (Definition C.0.3). It is denoted by notations such as **CommRing** or **CRing**.

Definition 1.0.6. The *category of topological spaces* is the (locally small) (Definition 1.0.9) category (Definition 1.0.1)

- whose objects are topological spaces (Definition B.0.1), and
- whose morphisms are continuous maps (Definition B.0.3).

The category of topological spaces is often denoted by notations such as **Top**, **Top**, etc.

Definition 1.0.7 (Category of opens of a topological space). Let X be a topological space (Definition B.0.1). The *category of opens of X* , sometimes denoted **Open**(X) (or **Open**(X) or **Ouv**(X) (for the French word “ouvert”, meaning open), etc.), is the small (Definition 1.0.9) category (Definition 1.0.1) defined as follows:

- The objects are the open subsets $U \subseteq X$.
- For two open sets $U, V \subseteq X$, the morphism set is

$$\mathrm{Hom}_{\mathbf{Open}(X)}(U, V) = \begin{cases} \{\iota_{U,V}\}, & \text{if } U \subseteq V, \\ \emptyset, & \text{otherwise,} \end{cases}$$

where $\iota_{U,V}$ denotes the inclusion morphism $U \hookrightarrow V$.

- Composition of morphisms is given by composition of set-theoretic inclusions, i.e.

$$\iota_{V,W} \circ \iota_{U,V} = \iota_{U,W} \quad \text{whenever } U \subseteq V \subseteq W.$$

- The identity morphism on an object U is the inclusion $\iota_{U,U} = \mathrm{id}_U$.

See also Definition 5.1.30.

Definition 1.0.8 (Isomorphism in a category). Let $\mathcal{C}$ be a (large) category (Definition 1.0.1), and let $x, y \in \mathrm{Ob}(\mathcal{C})$. A morphism $f \in \mathcal{C}(x, y)$ is called an *isomorphism* if there exists a morphism $g \in \mathcal{C}(y, x)$ such that

$$g \circ f = 1_x \quad \text{and} \quad f \circ g = 1_y.$$

In this case, g is called the *inverse of f* , and x and y are said to be *isomorphic objects* in $\mathcal{C}$. It is standard to write $x \cong y$ if there exists an isomorphism $f : x \rightarrow y$.

In practice, isomorphisms in specific categories may be defined in different, yet equivalent, ways.

Definition 1.0.9 (Locally small category). A (large) category (Definition 1.0.1) $\mathcal{C}$ is called a *locally small category* if for every pair of objects $X, Y \in \mathrm{Ob}(\mathcal{C})$, the collection $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ of morphisms between them is a (small) *set* (as opposed to a proper class). In other words, each hom-class is a set and may even be called a *hom-set*.

In some contexts, a locally small category may simply be called a *category*, especially when genuinely large categories are not considered.

A category $\mathcal{C}$ is called a *small category* if it is a locally small category and the class $\text{Ob}(\mathcal{C})$ of objects is a set.

Given a universe (Definition E.0.1) U , we can define the notion of a *U -locally small category* and of a *U -small category* similarly [GV72, Exposé I, Définition 1.1.]. More explicitly,

1. a U -locally small category is a category such that for every pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, the collection $\text{Hom}_{\mathcal{C}}(X, Y)$ of morphisms between them is a U -set.
2. a U -small category is a category such that $\text{Ob}(\mathcal{C})$ is a U -set and for every pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, the collection $\text{Hom}_{\mathcal{C}}(X, Y)$ of morphisms between them is a U -set; in particular the collection of all objects and morphisms in a U -small category is a U -set.

A *U -category* refers to a U -locally small category.

Remark 1.0.10. Many “concrete” categories considered in “classical mathematics” or outside of more “abstract” category theory tend to be locally small. For example, the categories of sets, groups, R -modules, vector spaces, topological spaces, schemes, manifolds, sheaves on “small enough” sites are all locally small.

Convention 1.0.11. In this document, care should be taken to make it clear whether a category is locally small or large. However, currently such care has not been fully taken, so most statements that just speak of “categories” may not be applicable to large categories.

Definition 1.0.12 (Opposite category). Let $\mathcal{C}$ be a (large) category (Definition 1.0.1). The *opposite category* of $\mathcal{C}$, denoted $\mathcal{C}^{\text{op}}$, is defined as follows:

- The objects of $\mathcal{C}^{\text{op}}$ are the same as those of $\mathcal{C}$.
- For any pair of objects $X, Y \in \mathcal{C}$, the morphisms from X to Y in $\mathcal{C}^{\text{op}}$ are given by the morphisms from Y to X in $\mathcal{C}$:

$$\text{Hom}_{\mathcal{C}^{\text{op}}}(X, Y) := \text{Hom}_{\mathcal{C}}(Y, X).$$

- Composition in $\mathcal{C}^{\text{op}}$ is defined by reversing the order of composition in $\mathcal{C}$. That is, for morphisms $f \in \text{Hom}_{\mathcal{C}^{\text{op}}}(X, Y)$ and $g \in \text{Hom}_{\mathcal{C}^{\text{op}}}(Y, Z)$, their composition is

$$g \circ_{\mathcal{C}^{\text{op}}} f := f \circ_{\mathcal{C}} g.$$

Intuitively, the category $\mathcal{C}^{\text{op}}$ thus “reverses” the direction of all morphisms in $\mathcal{C}$.

1.1. Functors between categories.

Definition 1.1.1. Let $\mathcal{C}$ and $\mathcal{D}$ be (large) categories (Definition 1.0.1).

1. A *functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ (from $\mathcal{C}$ to $\mathcal{D}$) consists of :
 - For each object X in $\mathcal{C}$, an object $F(X)$ in $\mathcal{D}$.
 - For each morphism $f : X \rightarrow Y$ in $\mathcal{C}$, a morphism $F(f) : F(X) \rightarrow F(Y)$ in $\mathcal{D}$, such that:

$$F(\text{id}_X) = \text{id}_{F(X)} \quad \text{for all objects } X \text{ in } \mathcal{C},$$

$$F(g \circ f) = F(g) \circ F(f) \quad \text{for all } X, Y, Z \in \text{Ob}(\mathcal{C}) \text{ and all } f : X \rightarrow Y, g : Y \rightarrow Z \text{ in } \mathcal{C}.$$

Functors as defined above are also referred to as *covariant functors* to distinguish them from contravariant functors

2. A *contravariant functor from $\mathcal{C}$ to $\mathcal{D}$* refers to a covariant functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$. Equivalently, such a functor consists of

- For each object X in $\mathcal{C}$, an object $F(X)$ in $\mathcal{D}$.
- For each morphism $f : X \rightarrow Y$ in $\mathcal{C}$, a morphism $F(f) : F(Y) \rightarrow F(X)$ in $\mathcal{D}$,

such that:

$$F(\text{id}_X) = \text{id}_{F(X)} \quad \text{for all objects } X \text{ in } \mathcal{C},$$

$$F(g \circ f) = F(f) \circ F(g) \quad \text{for all } X, Y, Z \in \text{Ob}(\mathcal{C}) \text{ and all } f : X \rightarrow Y, g : Y \rightarrow Z \text{ in } \mathcal{C}.$$

A synonym for a “contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ ” is a “presheaf on $\mathcal{C}$ with values in $\mathcal{D}$ (Definition 5.0.1)”.

Note that declarations such as “Let $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ be a contravariant functor” can be common; such declarations usually mean “Let F be a contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ ” as opposed to “Let F be a contravariant functor from $\mathcal{C}^{\text{op}}$ to $\mathcal{D}$ ”. further note that a contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ is equivalent to a covariant functor from $\mathcal{C}^{\text{op}}$ to $\mathcal{D}$.

Convention 1.1.2. By a “functor”, without the qualifying adjectives of “covariant” (Definition 1.1.1) or “contravariant” (Definition 1.1.1), we will most usually mean a covariant functor.

1.1.1. Natural transformations between functors.

Definition 1.1.3. Let $\mathcal{C}$ and $\mathcal{D}$ be (large) categories (Definition 1.0.1). Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be functors (Definition 1.1.1).

A *natural transformation η between F and G* is a family of morphisms $\eta_X : F(X) \rightarrow G(X)$ in $\mathcal{D}$, one for each object X in $\mathcal{C}$, such that for every morphism $f : X \rightarrow Y$ in $\mathcal{C}$,

$$G(f) \circ \eta_X = \eta_Y \circ F(f)$$

in $\mathcal{D}$. In other words, the following diagram commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ \eta_X \downarrow & & \downarrow \eta_Y \\ G(X) & \xrightarrow{G(f)} & G(Y) \end{array}$$

We write such a natural transformation by $\eta : F \Rightarrow G$. We might also write $\eta : F \Rightarrow G : \mathcal{C} \rightarrow \mathcal{D}$ to fully specify that η is a natural transformation from F to G , which are in turn functors from $\mathcal{C}$ to $\mathcal{D}$.

If η_X is an isomorphism (Definition 1.0.8) for all objects X of $\mathcal{C}$, then η is said to be a *natural isomorphism*.

Definition 1.1.4. Let $\mathcal{C}$ and $\mathcal{D}$ be (large) categories (Definition 1.0.1). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor (Definition 1.1.1).

1. F is called **full** if for every pair of objects $x, y \in \text{Ob}(\mathcal{C})$, the induced rule/assignment/class function

$$F_{x,y} : \text{Hom}_{\mathcal{C}}(x, y) \rightarrow \text{Hom}_{\mathcal{D}}(F(x), F(y))$$

on Hom-collections is “surjective”, i.e. for all morphisms $g : F(x) \rightarrow F(y)$, there exists some morphism $f : x \rightarrow y$ such that $F(f) = g$.

2. F is called **faithful** if for every pair of objects $x, y \in \text{Ob}(\mathcal{C})$, the induced class function (assignment)

$$F_{x,y} : \text{Hom}_{\mathcal{C}}(x, y) \rightarrow \text{Hom}_{\mathcal{D}}(F(x), F(y))$$

on Hom-collections is “injective”, i.e., for any morphisms $f_1, f_2 \in \text{Hom}_{\mathcal{C}}(x, y)$, if $F(f_1) = F(f_2)$ in $\text{Hom}_{\mathcal{D}}(F(x), F(y))$, then $f_1 = f_2$.

3. F is called **fully faithful** if it is both full and faithful.

Definition 1.1.5 (Image of a functor). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor (Definition 1.1.1) between (large) categories (Definition 1.0.1). The **image of F** is the subcategory **$\text{Im}(F)$** of $\mathcal{D}$ defined by:

- $\text{Ob}(\text{Im}(F)) = \{ F(c) \mid c \in \text{Ob}(\mathcal{C}) \}$, the collection of objects of $\mathcal{D}$ that arise as images of objects of $\mathcal{C}$.
- For $F(x), F(y) \in \text{Ob}(\text{Im}(F))$, the hom-set is

$$\text{Im}(F)(F(x), F(y)) = \{ F(f) \mid f \in \mathcal{C}(x, y) \}.$$

- Composition and identities are inherited from $\mathcal{D}$ and are well-defined by functoriality of F .

Thus $\text{Im}(F)$ is the smallest (not necessarily full (Definition 1.3.2)) subcategory of $\mathcal{D}$ containing all $F(c)$ for $c \in \text{Ob}(\mathcal{C})$ and all $F(f)$ for f a morphism of $\mathcal{C}$.

Definition 1.1.6 (Essential image of a functor). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between (large) categories (Definition 1.0.1). The **essential image of F** is the full subcategory (Definition 1.3.2) of $\mathcal{D}$ whose objects are those $d \in \text{Ob}(\mathcal{D})$ for which there exists an object $c \in \text{Ob}(\mathcal{C})$ such that

$$F(c) \cong d.$$

(Definition 1.0.8) Equivalently, the essential image is given by

$$\text{EssIm}(F) = \{ d \in \text{Ob}(\mathcal{D}) \mid \exists c \in \text{Ob}(\mathcal{C}), F(c) \cong d \},$$

endowed with all morphisms $\mathcal{D}(d, d')$ between such objects.

Equivalently, the essential image of F is the smallest replete (Definition 1.3.3) full subcategory (Definition 1.3.2) of $\mathcal{D}$ containing the image (Definition 1.1.5) of F

Definition 1.1.7. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between (large) categories (Definition 1.0.1). It is said to be essentially surjective if its essential image (Definition 1.1.6) coincides with $\mathcal{D}$.

Definition 1.1.8. An **equivalence of categories** between two (large) categories (Definition 1.0.1) $\mathcal{C}$ and $\mathcal{D}$ consists of a pair of functors (Definition 1.1.1)

$$F : \mathcal{C} \rightarrow \mathcal{D} \quad \text{and} \quad G : \mathcal{D} \rightarrow \mathcal{C}$$

together with natural isomorphisms (Definition 1.1.3)

$$\eta : \text{Id}_{\mathcal{C}} \xrightarrow{\sim} G \circ F \quad \text{and} \quad \epsilon : F \circ G \xrightarrow{\sim} \text{Id}_{\mathcal{D}}.$$

Such functors F and G may be called *(natural) inverses of each other*.

When $\mathcal{C}$ and $\mathcal{D}$ are locally small categories (Definition 1.0.9), F is an equivalence of categories if and only if F is fully faithful (Definition 1.1.4) and essentially surjective (Definition 1.1.7)

Definition 1.1.9. A category $\mathcal{C}$ is called *essentially small* if it is equivalent (Definition 1.1.8) to a small category (Definition 1.0.9), i.e., there exists a small category $\mathcal{D}$ and an equivalence of categories (Definition 1.1.8)

$$F : \mathcal{D} \rightarrow \mathcal{C}.$$

Note that an essentially small category is necessarily locally small (Definition 1.0.9).

1.1.2. Yoneda's lemma.

Theorem 1.1.10 (Yoneda Lemma). Let $\mathcal{C}$ be a locally small category (Definition 1.0.9). Let A be an object of $\mathcal{C}$, and let $F : \mathcal{C} \rightarrow \mathbf{Set}$ be a covariant functor (Definition 1.1.1) to the category of sets (Definition 1.0.3). Let $h^A : \mathcal{C} \rightarrow \mathbf{Set}$ denote the covariant representable functor defined by $h^A(X) = \text{Hom}_{\mathcal{C}}(A, X)$.

There exists a bijection

$$y_{A,F} : \text{Nat}(h^A, F) \xrightarrow{\cong} F(A)$$

between the set of natural transformations (Definition 1.1.3) from h^A to F and the set $F(A)$. This bijection is given by the mapping

$$\alpha \mapsto \alpha_A(\text{id}_A),$$

where $\alpha : h^A \rightarrow F$ is a natural transformation, $\alpha_A : h^A(A) \rightarrow F(A)$ is its component at A , and $\text{id}_A \in h^A(A) = \text{Hom}_{\mathcal{C}}(A, A)$ is the identity morphism.

Furthermore, this isomorphism is natural in both A and F . Explicitly:

1. For any morphism $f : A \rightarrow B$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{array}{ccc} \text{Nat}(h^B, F) & \xrightarrow{y_{B,F}} & F(B) \\ \downarrow - \circ h^f & & \downarrow F(f) \\ \text{Nat}(h^A, F) & \xrightarrow{y_{A,F}} & F(A) \end{array}$$

where $h^f : h^B \rightarrow h^A$ is the natural transformation induced by pre-composition with f .

2. For any natural transformation $\eta : F \rightarrow G$, the following diagram commutes:

$$\begin{array}{ccc} \text{Nat}(h^A, F) & \xrightarrow{y_{A,F}} & F(A) \\ \downarrow \eta \circ - & & \downarrow \eta_A \\ \text{Nat}(h^A, G) & \xrightarrow{y_{A,G}} & G(A) \end{array}$$

Corollary 1.1.11 (Yoneda Embedding). Let $\mathcal{C}$ be a locally small category (Definition 1.0.9). The functor

$$h^\bullet : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}^{\mathcal{C}}$$

(Definition 1.0.12) (Definition 1.2.1) defined on objects by $A \mapsto h^A = \text{Hom}_{\mathcal{C}}(A, -)$ and on morphisms by $f \mapsto h^f = (- \circ f)$ is fully faithful (Definition 1.1.4). That is, for any objects A, B in $\mathcal{C}$, the map

$$\text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Nat}(h^B, h^A)$$

given by sending a morphism $f : A \rightarrow B$ to the natural transformation (Definition 1.1.3) $h^f : h^B \rightarrow h^A$ (pre-composition by f) is a bijection.

Consequently, $\mathcal{C}^{\text{op}}$ embeds as a full subcategory (Definition 1.3.2) of the functor category $\mathbf{Set}^{\mathcal{C}}$.

Theorem 1.1.12 (Contravariant Yoneda Lemma). Let $\mathcal{C}$ be a locally small category (Definition 1.0.9). Let A be an object of $\mathcal{C}$, and let $G : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ be a contravariant functor (Definition 1.1.1) (i.e. a presheaf (Definition 5.0.1)). Let $h_A : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ denote the contravariant representable functor defined by $h_A(X) = \text{Hom}_{\mathcal{C}}(X, A)$.

There exists a bijection natural in A and G :

$$\text{Nat}(h_A, G) \cong G(A)$$

given by $\alpha \mapsto \alpha_A(\text{id}_A)$.

1.1.3. Adjoint functors between categories.

Definition 1.1.13. Let $\mathcal{C}$ and $\mathcal{D}$ be categories (Definition 1.0.1). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ be functors.

An *adjunction between F and G* consists of two natural transformations (Definition 1.1.3): $\eta : \text{Id}_{\mathcal{C}} \Rightarrow GF$ (the *unit*), and $\varepsilon : FG \Rightarrow \text{Id}_{\mathcal{D}}$ (the *counit*)

These must satisfy the triangle identities: For every object $X \in \mathcal{C}$ and $Y \in \mathcal{D}$,

$$\varepsilon_{FX} \circ F(\eta_X) = \text{id}_{FX}$$

$$G(\varepsilon_Y) \circ \eta_{GY} = \text{id}_{GY}.$$

In diagrammatic form, the triangle identities assert that the following are commutative diagrams:

$$\begin{array}{ccc} F(X) & \xrightarrow{F(\eta_X)} & FGF(X) \\ & \searrow \text{id}_{F(X)} & \downarrow \varepsilon_{F(X)} \\ & & F(X) \end{array} \quad \begin{array}{ccc} G(Y) & \xrightarrow{\eta_{G(Y)}} & GFG(Y) \\ & \searrow \text{id}_{G(Y)} & \downarrow G(\varepsilon_Y) \\ & & G(Y) \end{array}$$

We say that F is a *left adjoint to G* and G is a *right adjoint to F* (written $F \dashv G$).

In the case that $\mathcal{C}$ and $\mathcal{D}$ are locally small (Definition 1.0.9) categories (or U -locally small categories if a universe (Definition E.0.1) U is available), we have an adjunction $F \dashv G$ if and

only if for every object X in $\mathcal{C}$ and Y in $\mathcal{D}$ there is a natural isomorphism (Definition 1.1.3)

$$\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \cong \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$$

that is natural in both X and Y . More precisely, there are two functors

$$\mathcal{C}^{\mathrm{op}} \times \mathcal{D} \rightarrow \mathbf{Sets}$$

given by

$$\begin{aligned} (X, Y) &\mapsto \mathrm{Hom}_{\mathcal{D}}(F(X), Y) \\ (X, Y) &\mapsto \mathrm{Hom}_{\mathcal{C}}(X, G(Y)) \end{aligned}$$

respectively, and there is a natural isomorphism between them.

In this case, the *unit of the adjunction* is the natural transformation $\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow GF$ such that,

1. for every $X \in \mathcal{C}$, the morphism $\eta_X : X \rightarrow GF(X)$ (each called a *unit morphism*) in $\mathcal{C}$ is obtained as the image of $\mathrm{id}_{F(X)}$ via the adjoint isomorphism

$$\mathrm{Hom}_{\mathcal{D}}(F(X), F(X)) \cong \mathrm{Hom}_{\mathcal{C}}(X, GF(X)).$$

2. for every $Y \in \mathcal{D}$, the morphism $\epsilon_Y : FG(Y) \rightarrow Y$ (each called a *counit morphism*) in $\mathcal{D}$ is obtained as the image of $\mathrm{id}_{G(Y)}$ via the adjoint isomorphism

$$\mathrm{Hom}_{\mathcal{C}}(G(Y), G(Y)) \cong \mathrm{Hom}_{\mathcal{D}}(FG(Y), Y).$$

Proposition 1.1.14 (Factorization through the counit and the unit). Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and let

$$L : \mathcal{C} \rightarrow \mathcal{D}, \quad R : \mathcal{D} \rightarrow \mathcal{C}$$

be adjoint functors (Definition 1.1.13) with $L \dashv R$. Denote by

$$\eta : \mathrm{Id}_{\mathcal{C}} \Rightarrow R \circ L, \quad \epsilon : L \circ R \Rightarrow \mathrm{Id}_{\mathcal{D}}$$

the unit and counit of the adjunction (Definition 1.1.13).

- (1) For every $X' \in \mathrm{Ob}(\mathcal{C})$ and $X \in \mathrm{Ob}(\mathcal{D})$, a morphism

$$f : L(X') \rightarrow X \quad \text{in } \mathcal{D}$$

factors uniquely as

$$f = \epsilon_X \circ L(g),$$

for some morphism $g : X' \rightarrow R(X)$ in $\mathcal{C}$. Equivalently, the assignment $g \mapsto \epsilon_X \circ L(g)$ gives a bijection

$$\mathrm{Hom}_{\mathcal{C}}(X', R(X)) \cong \mathrm{Hom}_{\mathcal{D}}(L(X'), X),$$

which is precisely the adjunction isomorphism.

- (2) Dually, for every $X \in \mathrm{Ob}(\mathcal{D})$ and $X' \in \mathrm{Ob}(\mathcal{C})$, a morphism

$$h : X \rightarrow R(X') \quad \text{in } \mathcal{C}$$

factors uniquely as

$$h = R(f) \circ \eta_{X'},$$

for some morphism $f : L(X) \rightarrow X'$ in $\mathcal{D}$. Equivalently, the assignment $f \mapsto R(f) \circ \eta_X$ gives a bijection

$$\mathrm{Hom}_{\mathcal{D}}(L(X), X') \cong \mathrm{Hom}_{\mathcal{C}}(X, R(X')),$$

again exhibiting the adjunction isomorphism.

1.2. Diagrams, systems, and limits in categories.

Definition 1.2.1 (Diagram in a category and category of diagrams). Let $\mathcal{C}$ be a (large) category (Definition 1.0.1), and let I be a (large) category (Definition 1.0.1).

1. A *diagram of shape I in $\mathcal{C}$* is a functor (Definition 1.1.1) $D : I \rightarrow \mathcal{C}$. We often denote such a diagram by the family $\{D(i)\}_{i \in \mathrm{Ob}(I)}$ with transition maps given by the functorial image of morphisms in I .

A diagram is also synonymously called a *system*. Moreover, the category I is called the *index category* or the *indexing category of the diagram D* .

2. Given two diagrams $D, E : I \rightarrow \mathcal{C}$, a *morphism of diagrams* is simply a natural transformation (Definition 1.1.3) $D \Rightarrow E$ of the functors D and E .
3. The *category of I -shaped diagrams in $\mathcal{C}$* or simply *diagram category (of I -shaped diagrams in $\mathcal{C}$)*, often denoted $\mathcal{C}^I$, $[I, \mathcal{C}]$, or $\mathrm{Fun}(I, \mathcal{C})$, is the (large) category whose objects are functors $I \rightarrow \mathcal{C}$ (that is, diagrams of shape I in $\mathcal{C}$) and whose morphisms are natural transformations (Definition 1.1.3) between such functors. The category $\mathcal{C}^I$ is also called the *functor category of functors $I \rightarrow \mathcal{C}$* . Equivalently, the functor category $\mathcal{C}^I$ is the category $\mathrm{PreShv}(I^{\mathrm{op}}, \mathcal{C})$ of presheaves (Definition 5.0.1) on I^{op} with values in $\mathcal{C}$ and hence notations for presheaf categories are applicable as notations for functor categories.

If $\mathcal{C}$ is locally small (Definition 1.0.9) and I is small, then $\mathcal{C}^I$ is locally small by Lemma 5.0.5.

Intuitively, a limit of a diagram in a category is a “universal” object with a morphism to (all objects in) the diagram and a colimit, the dual notion, is a “universal” object with a morphism from (all objects in) the diagram.

Definition 1.2.2 (Cones, limits and colimits in a category). Let $\mathcal{C}$ be a (large) category (Definition 1.0.1), let I be a (large) category, and let $D : I \rightarrow \mathcal{C}$ be a diagram (Definition 1.2.1) (Definition 1.2.1).

1. A *cone to the diagram D* is an object $L \in \mathcal{C}$ together with a family of morphisms

$$\{\pi_i : L \rightarrow D(i)\}_{i \in I}$$

such that for every morphism $f : i \rightarrow j$ in I , the diagram

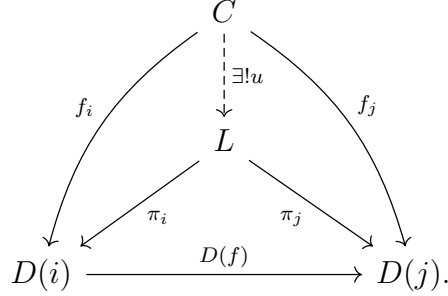
$$\begin{array}{ccc} & L & \\ \pi_i \swarrow & & \searrow \pi_j \\ D(i) & \xrightarrow{D(f)} & D(j) \end{array}$$

commutes, i.e. $D(f) \circ \pi_i = \pi_j$.

2. A cone $(L, \{\pi_i\})$ is called a **limit of D** if it satisfies the following “universal property”: for any cone $(C, \{f_i\})$ over D , there exists a *unique* morphism $u : C \rightarrow L$ such that

$$\pi_i \circ u = f_i \quad \text{for all } i \in I.$$

Visually, the following diagrams commute every morphism $f : i \rightarrow j$ in I :

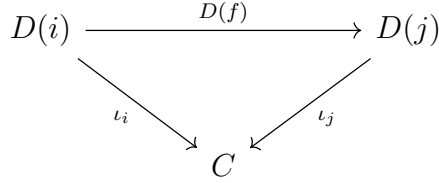


If such a cone exists, then the object L is necessarily unique up to unique isomorphism by the universal property. In this case, L is denoted by $\lim_{i \in I} D$ or $\lim D$.

3. A **cocone from the diagram D** is an object $C \in \mathcal{C}$ together with a family of morphisms

$$\{\iota_i : D(i) \rightarrow C\}_{i \in I}$$

such that for every morphism $f : i \rightarrow j$ in I , the diagram

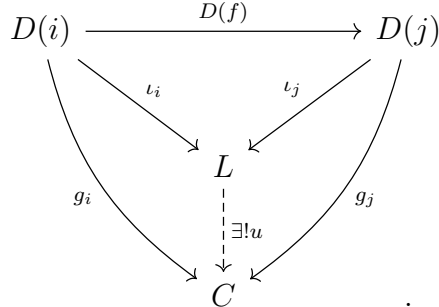


commutes, i.e. $\iota_j \circ D(f) = \iota_i$.

4. A cocone $(L, \{\iota_i\})$ is called a **colimit of D** if it satisfies the following “universal property”: for any cocone $(C, \{g_i\})$ under D , there exists a *unique* morphism $u : L \rightarrow C$ such that

$$u \circ \iota_i = g_i \quad \text{for all } i \in I.$$

Visually, the following diagrams commute every morphism $f : i \rightarrow j$ in I :



If such a cocone exists, then the object L is necessarily unique up to unique isomorphism by the universal property. In this case, L is denoted by $\text{colim}_{i \in I} D$ or $\text{colim } D$.

A limit/colimit is called **finite** (resp. **small**) if the index category I is finite (resp. small).

Some authors use the terms *projective limit* or *inverse limit* to refer to what is defined here as a limit. Similarly, the terms *inductive limit* or *direct limit* are sometimes used to mean a colimit. However, these phrases can have more specific meanings to other authors: a *projective* or *inverse limit* may refer to a limit over a diagram indexed by a codirected poset (Definition 1.2.7). Likewise, an *inductive* or *direct limit* may refer to a colimit over a directed poset (Definition 1.2.7) (see Definition 1.2.12).

Thus, while the terms are sometimes used interchangeably with “limit” and “colimit,” they may also emphasize particular indexing shapes and directions, distinguishing them from general limits and colimits taken over arbitrary small categories.

Theorem 1.2.3. Let $\mathcal{C}$ be a category (Definition 1.0.1), and let $\mathcal{J}$ be a category such that the limits (resp. colimits) (Definition 1.2.2) indexed by $\mathcal{J}$ exist in $\mathcal{C}$. Then the process of taking limits (resp. colimits) defines a functor

$$\lim : \mathcal{C}^{\mathcal{J}} \rightarrow \mathcal{C} \quad (\text{resp. } \text{colim} : \mathcal{C}^{\mathcal{J}} \rightarrow \mathcal{C})$$

from the functor category (Definition 1.2.1) $\mathcal{C}^{\mathcal{J}}$ to $\mathcal{C}$.

More precisely:

1. If $\mathcal{C}$ has all limits indexed by the category $\mathcal{J}$, then the assignment sending each diagram $D : \mathcal{J} \rightarrow \mathcal{C}$ to its limit $\lim D$ extends to a functor

$$\lim : \text{Fun}(\mathcal{J}, \mathcal{C}) \rightarrow \mathcal{C}.$$

This functor sends a natural transformation $\alpha : D \rightarrow D'$ to the unique morphism $\lim D \rightarrow \lim D'$ induced by the universal property of limits.

2. Similarly, if $\mathcal{C}$ has all colimits indexed by $\mathcal{J}$, then the assignment sending each diagram $D : \mathcal{J} \rightarrow \mathcal{C}$ to its colimit $\text{colim } D$ extends to a functor

$$\text{colim} : \text{Fun}(\mathcal{J}, \mathcal{C}) \rightarrow \mathcal{C}.$$

This functor sends a natural transformation $\alpha : D \rightarrow D'$ to the unique morphism $\text{colim } D \rightarrow \text{colim } D'$ induced by the universal property of colimits.

For a diagram indexed by an essentially small category I , we can identify their limits/colimits with those of diagrams indexed by a small category J equivalent to I .

Theorem 1.2.4 (Limits and Colimits indexed by essentially small categories). Let $\mathcal{C}$ be a (large) category (Definition 1.0.1) and let I be an essentially small category (Definition 1.1.9). Suppose $D : I \rightarrow \mathcal{C}$ is a diagram indexed by I (Definition 1.2.1).

If J is a small category (Definition 1.0.9) that is equivalent (Definition 1.1.8) to I via an equivalence of categories

$$F : J \xrightarrow{\sim} I,$$

then the limit and colimit (Definition 1.2.2) of D indexed by I may be identified with the limit and colimit of the composed diagram

$$D \circ F : J \rightarrow \mathcal{C}.$$

In other words, if either $\lim D$ or $\operatorname{colim} D$ exists in $\mathcal{C}$, then so does $\lim(D \circ F)$ and $\operatorname{colim}(D \circ F)$, and these limits and colimits are canonically isomorphic as objects of $\mathcal{C}$.

Definition 1.2.5. Let $\mathcal{C}$ be a (large) category (Definition 1.0.1), let I be a small category (Definition 1.0.9), and let $D : I \rightarrow \mathcal{C}$ be a diagram (Definition 1.2.1).

A limit or colimit (Definition 1.2.2) is called *finite* (resp. *small*) if the indexing category I has finitely many objects and morphisms (resp. if I is a small category (Definition 1.0.9)).

Theorem 1.2.6. Let $\mathcal{C}$ be a category (Definition 1.0.1). Let $F : \mathcal{J} \rightarrow \mathcal{C}$ be a diagram (Definition 1.2.1) where $\mathcal{J}$ is a small category (Definition 1.0.9).

1. The limit (Definition 1.2.2) of F is constructed as the equalizer (Definition 1.2.37) of the pair of morphisms (s, t) : assuming that the products (Definition 1.2.36) and equalizers below exist in $\mathcal{C}$, the limit $\lim F$ exists and

$$\lim F \cong \operatorname{eq} \left(\prod_{j \in \operatorname{Ob}(\mathcal{J})} F(j) \xrightarrow[t]{s} \prod_{\alpha \in \operatorname{Mor}(\mathcal{J})} F(\operatorname{cod}(\alpha)) \right)$$

where $\operatorname{cod}(\alpha)$ stands for the codomain of the morphism α , and the morphisms s and t are induced by the universal property of the product, such that for any morphism $\alpha : i \rightarrow k$ in $\mathcal{J}$, the projection to the factor indexed by α is:

- $\pi_\alpha \circ s = F(\alpha) \circ \pi_i$
- $\pi_\alpha \circ t = \pi_k$

2. The colimit (Definition 1.2.2) of F is constructed as the coequalizer (Definition 1.2.37) of the pair of morphisms (s, t) : assuming that the coproducts (Definition 1.2.36) and coequalizers below exist in $\mathcal{C}$, the colimit $\operatorname{colim} F$ exists and

$$\operatorname{colim} F \cong \operatorname{coeq} \left(\prod_{\alpha \in \operatorname{Mor}(\mathcal{J})} F(\operatorname{dom}(\alpha)) \xrightarrow[t]{s} \prod_{j \in \operatorname{Ob}(\mathcal{J})} F(j) \right)$$

where $\operatorname{dom}(\alpha)$ stands for the domain of the morphism α , and the morphisms s and t are induced by the universal property of the coproduct, such that for any morphism $\alpha : i \rightarrow k$ in $\mathcal{J}$, the injection from the summand indexed by α is:

- $s \circ \iota_\alpha = \iota_k \circ F(\alpha)$
- $t \circ \iota_\alpha = \iota_i$

In particular,

1. If $\mathcal{C}$ has all nonempty finite (resp. small) products and equalizers, then $\mathcal{C}$ has all nonempty finite (resp. small) limits.
2. If $\mathcal{C}$ has all nonempty finite (resp. small) coproducts and coequalizers, then $\mathcal{C}$ has all nonempty finite (resp. small) colimits.
3. If $\mathcal{C}$ has all finite (resp. small) products and equalizers, then $\mathcal{C}$ has all finite (resp. small) limits.
4. If $\mathcal{C}$ has all finite (resp. small) coproducts and coequalizers, then $\mathcal{C}$ has all finite (resp. small) colimits.

Definition 1.2.7 (Partially ordered set). 1. A *partially ordered set* (or *poset*), or *ordered set* is a pair $(P, \leq)$ where P is a set and

$$\leq: P \times P \rightarrow \{\text{true}, \text{false}\}$$

is a binary relation on P satisfying the following axioms for all $a, b, c \in P$:

- *Reflexivity*: $a \leq a$,
- *Antisymmetry*: if $a \leq b$ and $b \leq a$, then $a = b$,
- *Transitivity*: if $a \leq b$ and $b \leq c$, then $a \leq c$.

The relation $\leq$ is called an *order* or a *partial order*

2. A partially ordered set $(P, \leq)$ is called a *directed partially ordered set* if for every pair $a, b \in P$, there exists $c \in P$ such that

$$a \leq c \quad \text{and} \quad b \leq c.$$

3. A partially ordered set $(P, \leq)$ is called a *codirected partially ordered set* (or *downward directed poset*) if for every pair $a, b \in P$, there exists $d \in P$ such that

$$d \leq a \quad \text{and} \quad d \leq b.$$

Definition 1.2.8 (Filtered category). 1. A *filtered category* is a (nonempty, large) category $\mathcal{I}$ satisfying the following conditions:

- For every finite collection of objects $i_1, i_2, \dots, i_n$ in $\mathcal{I}$, there exists an object j and morphisms

$$\phi_k : i_k \rightarrow j, \quad \text{for each } k = 1, \dots, n.$$

- For every pair of morphisms $f, g : i \rightarrow j$ in $\mathcal{I}$, there exists an object k and a morphism

$$h : j \rightarrow k$$

that satisfies

$$h \circ f = h \circ g.$$

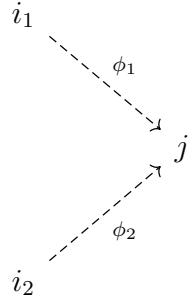


FIGURE 1. *

Condition 1: Upper Bound

$$i \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} j \xrightarrow{\quad h \quad} k$$

FIGURE 2. *

Condition 2: Coequalizing map

In other words, $\mathcal{I}$ is nonempty, any finite diagram of objects admits a cocone (Definition 1.2.2), and any pair of parallel morphisms become equal after post-composition with an appropriate morphism.

2. Dually, a *Cofiltered category* is a category whose opposite category (Definition 1.0.12) is filtered. More explicitly, A cofiltered category is a (nonempty, large) category $\mathcal{I}$ satisfying the following conditions:

- For every finite collection of objects $i_1, i_2, \dots, i_n$ in $\mathcal{I}$, there exists an object j and morphisms

$$\phi_k : j \rightarrow i_k, \quad \text{for each } k = 1, \dots, n.$$

- For every pair of morphisms $f, g : j \rightarrow i$ in $\mathcal{I}$, there exists an object k and a morphism

$$h : k \rightarrow j$$

that satisfies

$$f \circ h = g \circ h.$$

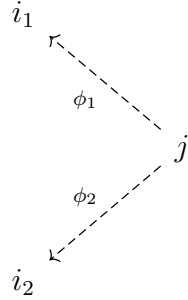


FIGURE 3. *

Condition 1: Lower Bound

$$k \xrightarrow{\quad h \quad} j \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} i$$

FIGURE 4. *

Condition 2: Equalizing map

In other words, $\mathcal{I}$ is nonempty, any finite diagram of objects admits a cone, and any pair of parallel morphisms become equal after pre-composition with an appropriate morphism.

Definition 1.2.9 (Thin category). A *thin category* is a (large) category $\mathcal{C}$ such that for every pair of objects $X, Y \in \mathcal{C}$, the hom-collection

$$\text{Hom}_{\mathcal{C}}(X, Y)$$

has at most one element.

Lemma 1.2.10. Let $(P, \leq)$ be a nonempty poset (Definition 1.2.7).

1. Regarding P as a category whose objects are the elements of P and such that there is a unique arrow $a \rightarrow b$ if and only if $a \leq b$, the category is filtered.
2. Every nonempty small (Definition 1.0.9) thin (Definition 1.2.9) filtered category (Definition 1.2.8) corresponds to a poset in this way.
3. Moreover, the poset P is directed (Definition 1.2.7) if and only if the category is filtered. The poset P is codirected (Definition 1.2.7) if and only if the category is cofiltered.

Definition 1.2.11 (Systems in a category). Let $\mathcal{C}$ be a (large) category. Let I be a (large) category.

1. A diagram/system (Definition 1.2.1) $I \rightarrow \mathcal{C}$ is called *filtered* (resp. *cofiltered*) if I is a filtered (Definition 1.2.8) (resp. cofiltered (Definition 1.2.8)) category.

2. A diagram/system $I \rightarrow C$ is called *directed* (resp. *codirected*) if I is small and thing, i.e. is regardable/comes from (Lemma 1.2.10) a directed (resp. codirected) partially ordered set (Definition 1.2.7). A *direct system* or *inductive system* is synonymous for a directed system and a *inverse system* or *projective system* is synonymous for a codirected system.

One might also speak of a *filtered direct/inductive system* synonymously for a filtered system to emphasize that the indexing category is a general filtered category, rather than a directed poset.

Definition 1.2.12 (Special cases of limits). Let $\mathcal{C}$ be a (large) category. Let I be a (large) category. Let $I \rightarrow \mathcal{C}$ be a diagram (Definition 1.2.1)/system.

- Suppose that the system is a cofiltered system (Definition 1.2.11), i.e. I is a cofiltered category. A limit (Definition 1.2.2) of this diagram is often denoted by

$$\varprojlim_{i \in I} D(i)$$

and may be called a *cofiltered (inverse/projective) limit*. In case that the system is more specifically an inverse/projective system (Definition 1.2.11), i.e. I is a cofiltered poset (Definition 1.2.7), the preferred term for such a limit is *inverse/projective limit*.

- Suppose that the system is a filtered system, i.e. I is a filtered category. A colimit of this diagram is often denoted by

$$\varinjlim_{i \in I} D(i)$$

and may be called a *filtered colimit* or a *direct/inductive/injective limit*. In case that the system is more specifically a direct/inductive system, i.e. I is a filtered poset (Definition 1.2.7), the preferred term for such a limit is *direct/inductive limit*.

Definition 1.2.13 (Cofinal and final sub-posets). Let $(I, \leq)$ be a partially ordered set (Definition 1.2.7), and let $J \subseteq I$ be a sub-poset. Let $(K, \leq)$ be a partially ordered set (Definition 1.2.7), and let $L \subseteq K$ be a sub-poset.

1. J is *cofinal in I* if for every $i \in I$, there exists $j \in J$ such that $i \leq j$.
2. L is *final in K* if for every $k \in K$, there exists $\ell \in L$ such that $\ell \leq k$.

Definition 1.2.14 (Cofinal and final subsystems of direct/inverse systems). (♠ TODO: AI generated, verify) Let $\mathcal{C}$ be a category (Definition 1.0.1).

1. Let $(I, \leq)$ be a directed partially ordered set (Definition 1.2.7), and let $(A_i)_{i \in I}$ be a direct system (Definition 1.2.11) in $\mathcal{C}$ indexed by I . Let $J \subseteq I$ be a sub-poset that is cofinal in I (Definition 1.2.13). The restriction of the direct system to J , denoted $(A_j)_{j \in J}$, is called a *cofinal subsystem* of $(A_i)_{i \in I}$.
2. Let $(K, \leq)$ be a codirected partially ordered set (Definition 1.2.7), and let $(B_k)_{k \in K}$ be an inverse system (Definition 1.2.11) in $\mathcal{C}$ indexed by K . Let $L \subseteq K$ be a sub-poset that is final in K (Definition 1.2.13). The restriction of the inverse system to L , denoted $(B_\ell)_{\ell \in L}$, is called a *final subsystem* of $(B_k)_{k \in K}$.

Theorem 1.2.15 (Cofinal and final subsystems suffice for limits). Let $\mathcal{C}$ be a category (Definition 1.0.1).

1. Let $(I, \leq)$ be a directed partially ordered set (Definition 1.2.7), and let $J \subseteq I$ be a sub-poset. Let $(A_i)_{i \in I}$ be a direct system (Definition 1.2.11) in $\mathcal{C}$ indexed by I . Suppose that J is cofinal in I (Definition 1.2.13). Let $(A_j)_{j \in J}$ denote the restriction of the direct system to J . Suppose that $\varinjlim_{i \in I} A_i$ exists in $\mathcal{C}$. Then $\varinjlim_{j \in J} A_j$ also exists in $\mathcal{C}$ and the canonical morphism

$$(A) \quad \varinjlim_{j \in J} A_j \rightarrow \varinjlim_{i \in I} A_i$$

(Definition 1.2.12) induced by the inclusions $J \hookrightarrow I$ and the transition morphisms of the direct system is an isomorphism.

2. Let $(K, \leq)$ be a codirected partially ordered set (Definition 1.2.7), and let $L \subseteq K$ be a sub-poset. Let $(B_k)_{k \in K}$ be an inverse system (Definition 1.2.11) in $\mathcal{C}$ indexed by K . Suppose that L is final in K (Definition 1.2.13). Let $(B_\ell)_{\ell \in L}$ denote the restriction of the inverse system to L . Suppose that $\varprojlim_{k \in K} B_k$ exists in $\mathcal{C}$. Then $\varprojlim_{\ell \in L} B_\ell$ also exists in $\mathcal{C}$ and the canonical morphism

$$(B) \quad \varprojlim_{k \in K} B_k \rightarrow \varprojlim_{\ell \in L} B_\ell$$

(Definition 1.2.12) induced by the inclusions $L \hookrightarrow K$ and the transition morphisms of the inverse system is an isomorphism.

Proof. 1. Suppose that $\varinjlim_{i \in I} A_i$ exists in $\mathcal{C}$. We show that it satisfies the universal property of $\varinjlim_{j \in J} A_j$, i.e. that $\varinjlim_{i \in I} A_i$ is a cocone (Definition 1.2.2) of $(A_j)_{j \in J}$ and that for any cocone D of $(A_j)_{j \in J}$ there is a unique morphism $\varinjlim_{j \in J} A_j \rightarrow D$ compatible with the structures of these objects as cocones of $(A_j)_{j \in J}$. For every $j \in J$, there is a canonical map A_j to $\varinjlim_{i \in I} A_i$. Moreover, these canonical maps over $j \in J$ admit $\varinjlim_{i \in I} A_i$ as a cocone (Definition 1.2.2) of $(A_j)_{j \in J}$ because $\varinjlim_{i \in I} A_i$ is a cocone of $(A_i)_{i \in I}$.

Suppose that $D \in \mathcal{C}$ is a cocone of $(A_j)_{j \in J}$, so equivalently it is equipped with morphisms $\pi_j : A_j \rightarrow D$ for all $j \in J$ such that for every morphism $f : j_1 \rightarrow j_2$ in J , we have that

$$\begin{array}{ccc} A_{j_1} & & \\ \downarrow A_f & \searrow \pi_{j_1} & \\ & D & \\ \uparrow A_g & \swarrow \pi_{j_2} & \\ A_{j_2} & & \end{array}$$

commutes. To first show the existence of the desired map $\varinjlim_{i \in I} A_i \rightarrow D$, we extend this cocone D of $(A_j)_{j \in J}$ to a cocone of $(A_i)_{i \in I}$. Given $i \in I$, define $\pi_i : A_i \rightarrow D$ as follows: by the cofinality of J in I , there exists some $j \in J$ such that $i \leq j$. Writing

$A_{i \rightarrow j}$ for morphism in $\mathcal{C}$ assigned to $i \rightarrow j$ by the diagram A , let

$$(C) \quad \pi_i = \pi_j \circ A_{i \rightarrow j}.$$

We show that this is well defined, i.e. is independent of the choice of $j \in J$. To this end, suppose that j_1, j_2 are objects of J such that $i \leq j_1, j_2$. Since I is a directed poset, there exists some $j_3 \in J$ such that $j_1, j_2 \leq j_3$. The following diagram commutes:

$$\begin{array}{ccccc} & & A_{j_1} & & \\ & \nearrow & \searrow & \searrow^{\pi_{j_1}} & \\ A_i & & & & D \\ & \searrow & \nearrow & \xrightarrow{\pi_{j_3}} & \\ & & A_{j_3} & & \\ & \nearrow & \searrow_{\pi_{j_2}} & & \\ & & A_{j_2} & & \end{array}$$

Therefore, $\pi_j \circ A_{i \rightarrow j_1}$ and $\pi_j \circ A_{i \rightarrow j_2}$ both equal any of the maps from A_i to D above, so π_i is indeed well defined.

We show that D equipped with the maps $\pi_i : A_i \rightarrow D$ is indeed a cocone of $(A_i)_{i \in I}$. Suppose that $i_1 \rightarrow i_2$ is an arrow in I . We show that

$$\begin{array}{ccc} A_{i_1} & & \\ \downarrow A_{i_1 \rightarrow i_2} & \searrow^{\pi_{i_1}} & \\ & & D \\ & \nearrow_{\pi_{i_2}} & \\ A_{i_2} & & \end{array}$$

commutes. Let $j_1, j_2 \in J$ satisfy $i_1 \leq j_1$ and $i_2 \leq j_2$. By the directedness of I , choose $i \in I$ such that $j_1, j_2 \leq i$ and by the cofinality of J in I , choose $j \in I$ such that $i \leq j$. In particular, we have that $i_1, i_2 \leq j$. By construct, $\pi_{i_1} = \pi_j \circ A_{i_1 \rightarrow j}$ and $\pi_{i_2} = \pi_j \circ A_{i_2 \rightarrow j}$. Therefore,

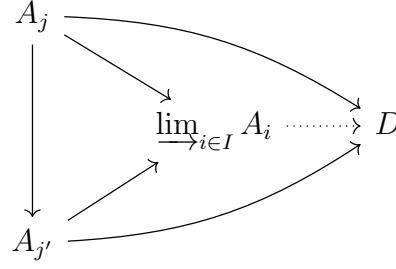
$$\pi_{i_2} \circ A_{i_1 \rightarrow i_2} = \pi_j \circ A_{i_2 \rightarrow j} \circ A_{i_1 \rightarrow i_2} = \pi_j \circ A_{i_1 \rightarrow j} = \pi_{i_1}$$

as desired.

As D equipped with the π_i is a cocone of $(A_i)_{i \in I}$, the universal property of $\varinjlim_{i \in I} A_i$ means that there is a canonical morphism

$$\varinjlim_{i \in I} A_i \rightarrow D.$$

We have so far thus shown that $\varinjlim_{i \in I} A_i$ is a cocone of $(A_j)_{j \in J}$ and that for every cocone D of $(A_j)_{j \in J}$, there is a canonical morphism $\varinjlim_{i \in I} A_i \rightarrow D$ such that, for every morphism $j \rightarrow j'$ in J ,



commutes.

We show that the morphism is unique, i.e. that the desired map $\varinjlim_{i \in I} A_i \rightarrow D$ is unique. To this end, suppose that $\varinjlim_{i \in I} A_i \rightarrow D$ is a morphism making the above diagram commute for all $j \rightarrow j'$ in J . To give a morphism $\varinjlim_{i \in I} A_i \rightarrow D$ is equivalent to giving D as a cocone of A_i (♠ TODO: write the notion that a morphism from an injective limit to an object is equivalent to giving the object as a cocone of the system, assuming that the injective limit exists). Write $\pi_i : A_i \rightarrow D$ for the morphisms of this cocone. For any $j_0 \in J$ such that $i \leq j_0$ in I , we must have that $\pi_i = \pi_{j_0} \circ A_{i \rightarrow j_0}$ where $\pi_{j_0} : A_{j_0} \rightarrow D$ here is written as a morphism for D as a cocone of $(A_i)_{i \in I}$. On the other hand, π_{j_0} as a morphism for D as a cocone of $(A_i)_{i \in I}$ coincides with π_{j_0} as a morphism for D as a cocone of $(A_j)_{j \in J}$ because the above diagram has to commute and because the map $A_{j_0} \rightarrow A_{j_0}$ where the first copy is from $(A_j)_{j \in J}$ and the second copy is from $(A_i)_{i \in I}$ is the identity map. Therefore, π_i must be constructed as in (C), so the morphism is indeed unique as desired. We have thus shown that the desired morphism both exists and is unique as desired.

2. This is dual to the previous part.

□

Definition 1.2.16 (cofinal and final functors). Let $\mathcal{I}$ and $\mathcal{J}$ be categories (Definition 1.0.1), and let $u : \mathcal{I} \rightarrow \mathcal{J}$ be a functor.

1. A **cofinal functor** is a functor $u : \mathcal{I} \rightarrow \mathcal{J}$ such that, for every object $j \in \mathcal{J}$, the comma category (Definition 1.3.12) $j \downarrow u$ is non-empty and a connected category.
2. A **final functor** is a functor $u : \mathcal{I} \rightarrow \mathcal{J}$ such that, for every object $j \in \mathcal{J}$, the comma category (Definition 1.3.12) $u \downarrow j$ is non-empty and a connected category.

(♠ TODO: AI generated, verify)

Theorem 1.2.17 (Cofinal functors reduce colimits, final functors reduce limits). (♠ TODO: AI generated, verify) Let $\mathcal{C}$ be a category (Definition 1.0.1).

1. (Cofinal functors and colimits) Let $u : \mathcal{I} \rightarrow \mathcal{J}$ be a cofinal functor (Definition 1.2.16). Let $F : \mathcal{J} \rightarrow \mathcal{C}$ be a diagram.
 - (a) If $\varinjlim_{j \in \mathcal{J}} F(j)$ exists in $\mathcal{C}$, then $\varinjlim_{i \in \mathcal{I}} F(u(i))$ also exists in $\mathcal{C}$ and the canonical morphism

$$(D) \quad \varinjlim_{i \in \mathcal{I}} F(u(i)) \rightarrow \varinjlim_{j \in \mathcal{J}} F(j)$$

induced by u and the colimit cocone of F is an isomorphism.

- (b) Conversely, if $\varinjlim_{i \in \mathcal{I}} F(u(i))$ exists in $\mathcal{C}$, then $\varinjlim_{j \in \mathcal{J}} F(j)$ also exists in $\mathcal{C}$ and the canonical morphism (D) is an isomorphism.
2. (Final functors and limits) Let $u : \mathcal{I} \rightarrow \mathcal{J}$ be a final functor (Definition 1.2.16). Let $G : \mathcal{J} \rightarrow \mathcal{C}$ be a diagram.
- (a) If $\varinjlim_{j \in \mathcal{J}} G(j)$ exists in $\mathcal{C}$, then $\varinjlim_{i \in \mathcal{I}} G(u(i))$ also exists in $\mathcal{C}$ and the canonical morphism

$$(E) \quad \varinjlim_{j \in \mathcal{J}} G(j) \rightarrow \varinjlim_{i \in \mathcal{I}} G(u(i))$$

induced by u and the limit cone of G is an isomorphism.

- (b) Conversely, if $\varinjlim_{i \in \mathcal{I}} G(u(i))$ exists in $\mathcal{C}$, then $\varinjlim_{j \in \mathcal{J}} G(j)$ also exists in $\mathcal{C}$ and the canonical morphism (E) is an isomorphism.

Proof. We prove part (1); part (2) is dual.

(1a) Assume $\varinjlim_{j \in \mathcal{J}} F(j)$ exists. Write $L = \varinjlim_{j \in \mathcal{J}} F(j)$ and let $\lambda_j : F(j) \rightarrow L$ be the colimit cocone. We show that $(L, \{\lambda_{u(i)}\}_{i \in \mathcal{I}})$ is a colimit cocone of $F \circ u$.

Clearly, for every morphism $\alpha : i \rightarrow i'$ in $\mathcal{I}$, the diagram

$$F(u(i)) \xrightarrow{F(u(\alpha))} F(u(i')) \xrightarrow{\lambda_{u(i')}} L$$

equals $F(u(i)) \xrightarrow{\lambda_{u(i)}} L$ because $(L, \{\lambda_j\})$ is a cocone of F . Thus $(L, \{\lambda_{u(i)}\})$ is a cocone of $F \circ u$.

Let $(D, \{\mu_i : F(u(i)) \rightarrow D\}_{i \in \mathcal{I}})$ be another cocone of $F \circ u$. We must produce a unique morphism $\varphi : L \rightarrow D$ such that $\varphi \circ \lambda_{u(i)} = \mu_i$ for all $i \in \mathcal{I}$.

Existence. For each object $j \in \mathcal{J}$, consider the comma category $j \downarrow u$. Since u is cofinal, $j \downarrow u$ is non-empty and connected. The objects of $j \downarrow u$ are pairs $(i, \beta : j \rightarrow u(i))$ with $i \in \mathcal{I}$ and β a morphism in $\mathcal{J}$. For each such object, define

$$\nu_{(i, \beta)} = \mu_i \circ F(u(\beta))^{\text{op}}$$

where $F(u(\beta)) : F(j) \rightarrow F(u(i))$ and we compose $F(j) \xrightarrow{F(u(\beta))} F(u(i)) \xrightarrow{\mu_i} D$. We claim this assignment is independent of the choice of $(i, \beta) \in \text{Ob}(j \downarrow u)$. Because $j \downarrow u$ is connected, it suffices to show that for every morphism $\gamma : (i, \beta) \rightarrow (i', \beta')$ in $j \downarrow u$ (i.e., $\gamma : i \rightarrow i'$ in $\mathcal{I}$ with $u(\gamma) \circ \beta = \beta'$), we have $\nu_{(i, \beta)} = \nu_{(i', \beta')}$. Indeed,

$$\nu_{(i', \beta')} = \mu_{i'} \circ F(u(\gamma)) \circ F(u(\beta)) = \mu_i \circ F(u(\beta)) = \nu_{(i, \beta)},$$

where we used that $(D, \{\mu_i\})$ is a cocone of $F \circ u$, so $\mu_{i'} \circ F(u(\gamma)) = \mu_i$.

Thus we obtain a well-defined morphism $\mu_j : F(j) \rightarrow D$ for each $j \in \mathcal{J}$. These morphisms form a cocone of F : if $f : j \rightarrow j'$ is a morphism in $\mathcal{J}$, pick any $(i, \beta : j' \rightarrow u(i)) \in \text{Ob}(j' \downarrow u)$; then $(i, \beta \circ f : j \rightarrow u(i)) \in \text{Ob}(j \downarrow u)$, and

$$\mu_{j'} \circ F(f) = \mu_i \circ F(u(\beta)) \circ F(f) = \mu_i \circ F(u(\beta \circ f)) = \mu_j.$$

By the universal property of $L = \varinjlim F$, there exists a unique morphism $\varphi : L \rightarrow D$ such that $\varphi \circ \lambda_j = \mu_j$ for all $j \in \mathcal{J}$. In particular, $\varphi \circ \lambda_{u(i)} = \mu_i$ for all $i \in \mathcal{I}$.

Uniqueness. Suppose $\psi : L \rightarrow D$ is another morphism satisfying $\psi \circ \lambda_{u(i)} = \mu_i$ for all $i \in \mathcal{I}$. For any $j \in \mathcal{J}$, pick $(i, \beta : j \rightarrow u(i)) \in \text{Ob}(j \downarrow u)$. Then

$$\psi \circ \lambda_j = \psi \circ \lambda_{u(i)} \circ F(u(\beta)) = \mu_i \circ F(u(\beta)) = \mu_j = \varphi \circ \lambda_j.$$

Since the morphisms λ_j are jointly epimorphic (as a colimit cocone), $\psi = \varphi$.

We have shown that $(L, \{\lambda_{u(i)}\})$ satisfies the universal property of $\varinjlim_{i \in \mathcal{I}} F(u(i))$. The canonical morphism (D) is the identity on L , hence an isomorphism.

(1b) Assume $\varinjlim_{i \in \mathcal{I}} F(u(i))$ exists. Write $L' = \varinjlim_{i \in \mathcal{I}} F(u(i))$ with cocone $\rho_i : F(u(i)) \rightarrow L'$. We show that L' satisfies the universal property of $\varinjlim_{j \in \mathcal{J}} F(j)$.

The argument is identical to (1a), replacing the role of the given colimit. For each $j \in \mathcal{J}$, the comma category $j \downarrow u$ is non-empty and connected, so we define $\lambda'_j : F(j) \rightarrow L'$ by choosing $(i, \beta : j \rightarrow u(i)) \in \text{Ob}(j \downarrow u)$ and setting $\lambda'_j = \rho_i \circ F(u(\beta))$. Connectedness of $j \downarrow u$ ensures this is well-defined, and the same computation as above shows $(L', \{\lambda'_j\})$ is a cocone of F . The universal property follows by the same extension-and-uniqueness argument. (♠
TODO: AI generated, verify) $\square$

Definition 1.2.18. Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

1. Assume $\mathcal{C}$ admits all finite limits (Definition 1.2.2). A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called *left exact* if it preserves all finite limits (Definition 1.2.5). Explicitly, for every finite diagram $D : \mathcal{J} \rightarrow \mathcal{C}$, the limit

$$\varprojlim(F \circ D)$$

exists and the canonical map

$$F(\varprojlim D) \xrightarrow{\cong} \varprojlim(F \circ D)$$

is an isomorphism.

2. Assume $\mathcal{C}$ admits all finite colimits (Definition 1.2.2). A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called *right exact* if it preserves all finite colimits (Definition 1.2.5). Explicitly, for every finite diagram $D : \mathcal{J} \rightarrow \mathcal{C}$, the colimit

$$\varinjlim(F \circ D)$$

exists and the canonical map

$$\varinjlim(F \circ D) \xrightarrow{\cong} F(\varinjlim D)$$

is an isomorphism.

3. If $\mathcal{C}$ admits both finite limits and finite colimits, a functor F is called *exact* if it is both left exact and right exact.

Lemma 1.2.19. Let $\mathcal{E}$ and $\mathcal{F}$ be topoi (Definition 5.1.55).

1. A functor $F : \mathcal{E} \rightarrow \mathcal{F}$ is left exact (Definition 1.2.18) if and only if it preserves terminal objects (Definition 2.0.6) and pullbacks (Definition 1.2.38).
2. A functor $F : \mathcal{E} \rightarrow \mathcal{F}$ is right exact (Definition 1.2.18) if and only if it preserves initial objects (Lemma 2.0.7) and pushouts (Definition 1.2.25).

Definition 1.2.20. Let $\mathcal{C}$ be a category (Definition 1.0.1), let Z be an object, and let X, Y be objects of $\mathcal{C}$ over (Definition 1.3.4) Z , i.e. morphisms $X \rightarrow Z$ and $Y \rightarrow Z$ are fixed. A *cartesian product of X and Y over Z in $\mathcal{C}$* (or *fiber product* or *pullback diagram*) is an object, often denoted by $X \times_Z Y$, with *projection morphisms* $X \times_Z Y \rightarrow X$ and $X \times_Z Y \rightarrow Y$ that are universal. More precisely, for any object T of $\mathcal{C}$ and morphisms $f_X : T \rightarrow X$, $f_Y : T \rightarrow Y$ such that the compositions $T \xrightarrow{f_X} X \rightarrow Z$ and $T \xrightarrow{f_Y} Y \rightarrow Z$ agree, there exists a unique morphism $u : T \rightarrow X \times_Z Y$ such that the following diagram commutes:

$$\begin{array}{ccccc}
 T & & & & \\
 & \searrow^{f_X} & & \searrow & \\
 & & X \times_Z Y & \longrightarrow & X \\
 & \searrow_{f_Y} & \downarrow & & \downarrow \\
 & & Y & \longrightarrow & Z
 \end{array}$$

(Note: A dotted arrow labeled u points from T to $X \times_Z Y$.)

Equivalently, $X \times_Z Y$ is the limit (Definition 1.2.2) of the diagram (Definition 1.2.1)

$$\begin{array}{ccc}
 & X & \\
 & \downarrow & \\
 Y & \longrightarrow & Z
 \end{array}$$

in $\mathcal{C}$.

The commutative diagram

$$\begin{array}{ccc}
 X \times_Z Y & \longrightarrow & X \\
 \downarrow & & \downarrow \\
 Y & \longrightarrow & Z
 \end{array}$$

may be referred to as a *cartesian square*.

Definition 1.2.21. Let $\mathcal{C}$ be a category (Definition 1.0.1) which admits finite products (Definition 1.2.36). Let Y, Z be objects of $\mathcal{C}$.

1. An *exponential object from Y to Z* is an object Z^Y (also denoted $[Y, Z]$ or $\underline{\text{hom}}(Y, Z)$) equipped with a morphism

$$\text{ev} : Z^Y \times Y \rightarrow Z$$

called the *evaluation morphism*, satisfying the following universal property:

2. For any object X in $\mathcal{C}$ and any morphism $f : X \times Y \rightarrow Z$, there exists a unique morphism $\tilde{f} : X \rightarrow Z^Y$ (often called the *transpose* or *currying of f*) such that the following diagram commutes:

$$\begin{array}{ccc}
X \times Y & \xrightarrow{\tilde{f} \times \text{id}_Y} & Z^Y \times Y \\
& \searrow f & \downarrow \text{ev} \\
& & Z
\end{array}$$

Equivalently, the functor $- \times Y : \mathcal{C} \rightarrow \mathcal{C}$ has a right adjoint (Definition 1.1.13), denoted $(-)^Y : \mathcal{C} \rightarrow \mathcal{C}$.

Definition 1.2.22. Let $\mathcal{C}$ be a category (Definition 1.0.1). $\mathcal{C}$ is called **cartesian closed** if it satisfies the following three conditions:

1. It has a terminal object (Definition 2.0.6) (empty product), often denoted **1**.
2. It has all binary products (Definition 1.2.36) (and thus all finite products). That is, for any objects $A, B \in \mathcal{C}$, the product **$A \times B$** exists.
3. It has exponential objects for all object $Y \rightarrow Z$.

Definition 1.2.23. Let $\mathcal{C}$ be a category with a terminal object (Lemma 2.0.7) **1**. A **subobject classifier** is an object **Ω** together with a morphism **$\text{true} : 1 \rightarrow \Omega$** satisfying the following universal property:

For any monomorphism (Definition 1.3.7) **$m : U \hookrightarrow X$** in $\mathcal{C}$, there exists a unique morphism **$\chi_m : X \rightarrow \Omega$** (called the **characteristic morphism of m**) such that the following diagram is a pullback square (Definition 1.2.38):

$$\begin{array}{ccc}
U & \xrightarrow{!} & 1 \\
\downarrow m & & \downarrow \text{true} \\
X & \xrightarrow{\chi_m} & \Omega
\end{array}$$

where $U \rightarrow 1$ is the unique morphism to the terminal object. Intuitively, Ω acts as a "truth value object" where χ_m assigns "true" to the "points" of X that lie in the subobject U .

Remark 1.2.24. In the category of sets (**Set**), the subobject classifier is the set $\Omega = \{0, 1\}$ (often denoted **2**), and "true" is the function picking out 1. The characteristic morphism χ_m corresponds to the characteristic function of the subset $U \subseteq X$.

Definition 1.2.25. Let $\mathcal{C}$ be a category (Definition 1.0.1), let Z be an object, and let X, Y be objects of $\mathcal{C}$ under (Definition 1.3.4) Z , i.e. morphisms $Z \rightarrow X$ and $Z \rightarrow Y$ are fixed. A **cccartesian sum of X and Y under Z in $\mathcal{C}$** (or **fiber coproduct** or **pushout diagram**) is an object, often denoted by **$X \amalg_Z Y$** , with **inclusion morphisms** $X \rightarrow X \amalg_Z Y$ and $Y \rightarrow X \amalg_Z Y$ that are universal. More precisely, for any object T of $\mathcal{C}$ and morphisms $j_X : X \rightarrow T$, $j_Y : Y \rightarrow T$, there exists a unique morphism $v : X \amalg_Z Y \rightarrow T$ such that the following diagram commutes:

$$\begin{array}{ccccc}
Z & \longrightarrow & X & & \\
\downarrow & & \downarrow & \searrow j_X & \\
Y & \longrightarrow & X \amalg_Z Y & & \\
& \searrow j_Y & \swarrow v & \searrow & \\
& & & & T
\end{array}$$

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Equivalently, $X \amalg_Z Y$ is the colimit (Definition 1.2.2) of the diagram (Definition 1.2.1)

$$X \longleftarrow Z \longrightarrow Y$$

in $\mathcal{C}$.

The commutative diagram

$$\begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y & \longrightarrow & X \amalg_Z Y \end{array}$$

may be referred to as a *cocartesian square*.

Theorem 1.2.29 states the right adjoint functors commute with limits and left adjoint functors commute with colimits.

Definition 1.2.26. Let $\mathcal{C}$ and $\mathcal{D}$ be (large) categories (Definition 1.0.1), $F : \mathcal{C} \rightarrow \mathcal{D}$ a functor (Definition 1.1.1), and $D : \mathcal{J} \rightarrow \mathcal{C}$ a diagram (Definition 1.2.1) indexed by a small category $\mathcal{J}$.

1. Assume that the colimit (Definition 1.2.2) of D exists in $\mathcal{C}$, denoted by $\operatorname{colim}_{\mathcal{J}} D$, with universal cocone (Definition 1.2.2) $\eta : D \Rightarrow \Delta_{\operatorname{colim} D}$. The functor F maps this to a cocone $F(\eta) : F \circ D \Rightarrow \Delta_{F(\operatorname{colim} D)}$ in $\mathcal{D}$.

If the colimit of the composite diagram $F \circ D$ exists in $\mathcal{D}$, denoted by $\operatorname{colim}_{\mathcal{J}} (F \circ D)$, then by the universal property of the colimit, there exists a unique morphism

$$\phi : \operatorname{colim}_{j \in \mathcal{J}} F(D(j)) \longrightarrow F(\operatorname{colim}_{j \in \mathcal{J}} D(j))$$

mediating between the cocone of the composite colimit and the image cocone $F(\eta)$. This unique morphism is called the *canonical colimit comparison morphism*.

2. Dually, assume that the limit of D exists in $\mathcal{C}$, denoted by $\lim_{\mathcal{J}} D$, with universal cone $\varepsilon : \Delta_{\lim D} \Rightarrow D$. The functor F maps this to a cone $F(\varepsilon) : \Delta_{F(\lim D)} \Rightarrow F \circ D$ in $\mathcal{D}$.

If the limit of the composite diagram $F \circ D$ exists in $\mathcal{D}$, denoted by $\lim_{\mathcal{J}} (F \circ D)$, then by the universal property of the limit, there exists a unique morphism

$$\psi : F\left(\lim_{j \in \mathcal{J}} D(j)\right) \longrightarrow \lim_{j \in \mathcal{J}} F(D(j))$$

mediating between the image cone $F(\varepsilon)$ and the cone of the composite limit. This unique morphism is called the *canonical limit comparison morphism*.

Definition 1.2.27. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor (Definition 1.1.1) between (large) categories (Definition 1.0.1) $\mathcal{C}$ and $\mathcal{D}$.

Let $D : \mathcal{J} \rightarrow \mathcal{C}$ be a diagram in $\mathcal{C}$ such that the limit $\lim_{\mathcal{J}} D$ exists in $\mathcal{C}$, with limiting cone $\lambda : \Delta_{\lim D} \Rightarrow D$. The functor F *preserves the limit of D* if the image cone

$$F(\lambda) : \Delta_{F(\lim D)} \cong F \circ \Delta_{\lim D} \Rightarrow F \circ D$$

exhibits $F(\lim_{\mathcal{J}} D)$ as a limit of the composite diagram $F \circ D$ in $\mathcal{D}$. Explicitly, the canonical comparison morphism (Definition 1.2.26)

$$F(\lim_{\mathcal{J}} D) \xrightarrow{\cong} \lim_{\mathcal{J}} (F \circ D)$$

must be an isomorphism.

Dually, let $D : \mathcal{J} \rightarrow \mathcal{C}$ be a diagram such that the colimit $\text{colim}_{\mathcal{J}} D$ exists in $\mathcal{C}$. The functor F **preserves the colimit of D** if the canonical comparison morphism (Definition 1.2.26)

$$\text{colim}_{\mathcal{J}} (F \circ D) \xrightarrow{\cong} F(\text{colim}_{\mathcal{J}} D)$$

is an isomorphism.

Definition 1.2.28. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor (Definition 1.1.1) between (large) categories (Definition 1.0.1) $\mathcal{C}$ and $\mathcal{D}$.

- The functor F is called **continuous** if it preserves all small limits that exist in $\mathcal{C}$. That is, for every small category $\mathcal{J}$ and every diagram $D : \mathcal{J} \rightarrow \mathcal{C}$ having a limit in $\mathcal{C}$, F preserves the limit (Definition 1.2.27) of D .
- The functor F is called **cocontinuous** if it preserves all small colimits that exist in $\mathcal{C}$. That is, for every small category $\mathcal{J}$ and every diagram $D : \mathcal{J} \rightarrow \mathcal{C}$ having a colimit in $\mathcal{C}$, F preserves the colimit (Definition 1.2.27) of D .

Theorem 1.2.29 (Left adjoints preserve colimits; right adjoints preserve limits). (♠ TODO: AI generated, verify) Let $\mathcal{C}$ and $\mathcal{D}$ be categories (Definition 1.0.1). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ form an adjunction (Definition 1.1.13) $F \dashv G$, with unit (Definition 1.1.13) $\eta : \text{Id}_{\mathcal{C}} \Rightarrow GF$ and counit (Definition 1.1.13) $\varepsilon : FG \Rightarrow \text{Id}_{\mathcal{D}}$. Let $\mathcal{J}$ be a category.

1. Let $D : \mathcal{J} \rightarrow \mathcal{C}$ be a diagram (Definition 1.2.1) such that the colimit (Definition 1.2.2) $\text{colim}_{\mathcal{J}} D$ exists in $\mathcal{C}$, with universal cocone $(\lambda_j : D(j) \rightarrow \text{colim}_{\mathcal{J}} D)_{j \in \text{Ob}(\mathcal{J})}$. Then the family

$$(F(\lambda_j) : F(D(j)) \rightarrow F(\text{colim}_{\mathcal{J}} D))_{j \in \text{Ob}(\mathcal{J})}$$

is a colimit cocone (Definition 1.2.2) for the diagram $F \circ D : \mathcal{J} \rightarrow \mathcal{D}$. In other words, for every object $Y \in \mathcal{D}$ and every family of morphisms $(\nu_j : F(D(j)) \rightarrow Y)_{j \in \text{Ob}(\mathcal{J})}$ satisfying $\nu_k \circ F(D(\alpha)) = \nu_j$ for all morphisms $\alpha : j \rightarrow k$ in $\mathcal{J}$, there exists a unique morphism $f : F(\text{colim}_{\mathcal{J}} D) \rightarrow Y$ such that $f \circ F(\lambda_j) = \nu_j$ for all $j \in \text{Ob}(\mathcal{J})$.

2. Let $E : \mathcal{J} \rightarrow \mathcal{D}$ be a diagram such that the limit (Definition 1.2.2) $\lim_{\mathcal{J}} E$ exists in $\mathcal{D}$, with universal cone $(\rho_j : \lim_{\mathcal{J}} E \rightarrow E(j))_{j \in \text{Ob}(\mathcal{J})}$. Then the family

$$(G(\rho_j) : G(\lim_{\mathcal{J}} E) \rightarrow G(E(j)))_{j \in \text{Ob}(\mathcal{J})}$$

is a limit cone (Definition 1.2.2) for the diagram $G \circ E : \mathcal{J} \rightarrow \mathcal{C}$. In other words, for every object $X \in \mathcal{C}$ and every family of morphisms $(\sigma_j : X \rightarrow G(E(j)))_{j \in \text{Ob}(\mathcal{J})}$ satisfying $G(E(\alpha)) \circ \sigma_j = \sigma_k$ for all morphisms $\alpha : j \rightarrow k$ in $\mathcal{J}$, there exists a unique morphism $g : X \rightarrow G(\lim_{\mathcal{J}} E)$ such that $G(\rho_j) \circ g = \sigma_j$ for all $j \in \text{Ob}(\mathcal{J})$.

Proof. (♠ TODO: AI generated, verify)

We prove part 1; part 2 follows by duality in the opposite categories $\mathcal{C}^{\text{op}}$ and $\mathcal{D}^{\text{op}}$, where F and G are swapped and all arrows reversed.

Fix a diagram $D : \mathcal{J} \rightarrow \mathcal{C}$, its colimit $\text{colim}_{\mathcal{J}} D$, and the universal cocone $(\lambda_j)_{j \in \text{Ob}(\mathcal{J})}$. Let $Y \in \mathcal{D}$ and let $(\nu_j : F(D(j)) \rightarrow Y)_{j \in \text{Ob}(\mathcal{J})}$ be a cocone under $F \circ D$. We must construct a unique morphism $f : F(\text{colim}_{\mathcal{J}} D) \rightarrow Y$ with $f \circ F(\lambda_j) = \nu_j$ for all j . The setup is the following diagram, in which we seek the dotted arrow:

$$\begin{array}{ccc}
 F(D(j)) & & \\
 \searrow^{F(\lambda_j)} & & \\
 & F(\text{colim}_{\mathcal{J}} D) & \\
 \searrow_{\nu_j} & \downarrow \exists! f & \\
 & Y &
 \end{array}$$

We first pull the cocone (ν_j) from $\mathcal{D}$ back to $\mathcal{C}$ using G and the unit η . For each $j \in \text{Ob}(\mathcal{J})$, define

$$\mu'_j := G(\nu_j) \circ \eta_{D(j)} : D(j) \longrightarrow G(Y).$$

We verify that $(\mu'_j)_j$ is a cocone from D to $G(Y)$. Let $\alpha : j \rightarrow k$ be any morphism in $\mathcal{J}$. Since $(\nu_j)_j$ is a cocone under $F \circ D$, we have $\nu_k \circ F(D(\alpha)) = \nu_j$. We compute:

$$\mu'_k \circ D(\alpha) = G(\nu_k) \circ \eta_{D(k)} \circ D(\alpha).$$

By the naturality of η applied to the morphism $D(\alpha) : D(j) \rightarrow D(k)$, we have $\eta_{D(k)} \circ D(\alpha) = GF(D(\alpha)) \circ \eta_{D(j)}$. Substituting:

$$G(\nu_k) \circ GF(D(\alpha)) \circ \eta_{D(j)} = G(\nu_k \circ F(D(\alpha))) \circ \eta_{D(j)}.$$

Using the cocone condition $\nu_k \circ F(D(\alpha)) = \nu_j$, this becomes $G(\nu_j) \circ \eta_{D(j)} = \mu'_j$. Thus $(\mu'_j)_j$ is a cocone.

Next we invoke the universal property of $\text{colim}_{\mathcal{J}} D$. Since $(\lambda_j)_j$ is a colimit cocone for D and $(\mu'_j)_j$ is a cocone from D to $G(Y)$, there exists a unique morphism

$$u : \text{colim}_{\mathcal{J}} D \longrightarrow G(Y)$$

such that $u \circ \lambda_j = \mu'_j$ for all $j \in \text{Ob}(\mathcal{J})$.

We then push the morphism u forward to $\mathcal{D}$ using the counit ε . Define

$$f := \varepsilon_Y \circ F(u) : F(\text{colim}_{\mathcal{J}} D) \longrightarrow Y,$$

where $\varepsilon_Y : FG(Y) \rightarrow Y$ is the component of the counit at Y .

We verify that f mediates the cocone (ν_j) . For each $j \in \text{Ob}(\mathcal{J})$, we compute:

$$f \circ F(\lambda_j) = \varepsilon_Y \circ F(u) \circ F(\lambda_j) = \varepsilon_Y \circ F(u \circ \lambda_j).$$

Substituting $u \circ \lambda_j = \mu'_j = G(\nu_j) \circ \eta_{D(j)}$:

$$\varepsilon_Y \circ F(G(\nu_j) \circ \eta_{D(j)}) = \varepsilon_Y \circ FG(\nu_j) \circ F(\eta_{D(j)}).$$

By the naturality of ε applied to the morphism $\nu_j : F(D(j)) \rightarrow Y$, we have the commutative square

$$\begin{array}{ccc} FG(F(D(j))) & \xrightarrow{FG(\nu_j)} & FG(Y) \\ \varepsilon_{F(D(j))} \downarrow & & \downarrow \varepsilon_Y \\ F(D(j)) & \xrightarrow{\nu_j} & Y \end{array}$$

That is, $\varepsilon_Y \circ FG(\nu_j) = \nu_j \circ \varepsilon_{F(D(j))}$. Substituting:

$$f \circ F(\lambda_j) = \nu_j \circ \varepsilon_{F(D(j))} \circ F(\eta_{D(j)}).$$

By the first triangle identity for the adjunction $F \dashv G$, applied to the object $D(j) \in \mathcal{C}$, we have $\varepsilon_{F(D(j))} \circ F(\eta_{D(j)}) = \text{id}_{F(D(j))}$. Therefore:

$$f \circ F(\lambda_j) = \nu_j \circ \text{id}_{F(D(j))} = \nu_j.$$

Finally we establish uniqueness. Suppose $f' : F(\text{colim}_{\mathcal{J}} D) \rightarrow Y$ is another morphism satisfying $f' \circ F(\lambda_j) = \nu_j$ for all j . Define

$$u' := G(f') \circ \eta_{\text{colim}_{\mathcal{J}} D} : \text{colim}_{\mathcal{J}} D \rightarrow G(Y).$$

For each j , we compute:

$$u' \circ \lambda_j = G(f') \circ \eta_{\text{colim}_{\mathcal{J}} D} \circ \lambda_j.$$

By the naturality of η applied to $\lambda_j : D(j) \rightarrow \text{colim}_{\mathcal{J}} D$, we have $\eta_{\text{colim}_{\mathcal{J}} D} \circ \lambda_j = GF(\lambda_j) \circ \eta_{D(j)}$. Hence:

$$u' \circ \lambda_j = G(f') \circ GF(\lambda_j) \circ \eta_{D(j)} = G(f' \circ F(\lambda_j)) \circ \eta_{D(j)}.$$

Since $f' \circ F(\lambda_j) = \nu_j$, this equals $G(\nu_j) \circ \eta_{D(j)} = \mu'_j$. Thus u' is also a mediating morphism for the cocone $(\mu'_j)_j$ through $(\lambda_j)_j$. By the uniqueness of u , we have $u' = u$.

It remains to show that $f' = f$. We compute:

$$\varepsilon_Y \circ F(u') = \varepsilon_Y \circ F(G(f') \circ \eta_{\text{colim}_{\mathcal{J}} D}) = \varepsilon_Y \circ FG(f') \circ F(\eta_{\text{colim}_{\mathcal{J}} D}).$$

By the naturality of ε applied to $f' : F(\text{colim}_{\mathcal{J}} D) \rightarrow Y$, we have $\varepsilon_Y \circ FG(f') = f' \circ \varepsilon_{F(\text{colim}_{\mathcal{J}} D)}$. Hence:

$$\varepsilon_Y \circ F(u') = f' \circ \varepsilon_{F(\text{colim}_{\mathcal{J}} D)} \circ F(\eta_{\text{colim}_{\mathcal{J}} D}).$$

By the first triangle identity applied to $\text{colim}_{\mathcal{J}} D$, we have $\varepsilon_{F(\text{colim}_{\mathcal{J}} D)} \circ F(\eta_{\text{colim}_{\mathcal{J}} D}) = \text{id}_{F(\text{colim}_{\mathcal{J}} D)}$. Therefore:

$$\varepsilon_Y \circ F(u') = f' \circ \text{id}_{F(\text{colim}_{\mathcal{J}} D)} = f'.$$

Since $u' = u$, we obtain $f' = \varepsilon_Y \circ F(u) = f$. This establishes uniqueness.

The cocone $(F(\lambda_j))_j$ therefore satisfies the universal property of a colimit cocone for $F \circ D$, completing the proof of part 1. $\square$

Theorem 1.2.30 states that colimits commutes with colimits and limits commute with limits.

Theorem 1.2.30 (Fubini Theorem for Colimits and Limits). Let $\mathcal{C}$ be a category and let I and J be small categories. Let $D : I \times J \rightarrow \mathcal{C}$ be a bifunctor.

1. Assume the following colimits (Definition 1.2.2) exist in $\mathcal{C}$:

- for every object $i \in I$, $\varinjlim_{j \in J} D(i, j)$
- $\varinjlim_{(i,j) \in I \times J} D(i, j)$
- $\varinjlim_{i \in I} (\varinjlim_{j \in J} D(i, j))$

There is a canonical isomorphism:

$$\varinjlim_{i \in I} \left(\varinjlim_{j \in J} D(i, j) \right) \cong \varinjlim_{(i,j) \in I \times J} D(i, j).$$

2. Assume the following limits (Definition 1.2.2) exist in $\mathcal{C}$:

- for every object $i \in I$, $\varprojlim_{j \in J} D(i, j)$
- $\varprojlim_{(i,j) \in I \times J} D(i, j)$
- $\varprojlim_{i \in I} (\varprojlim_{j \in J} D(i, j))$

There is a canonical isomorphism:

$$\varprojlim_{i \in I} \left(\varprojlim_{j \in J} D(i, j) \right) \cong \varprojlim_{(i,j) \in I \times J} D(i, j).$$

Remark 1.2.31. These results assert that the colimit functor (Theorem 1.2.3) $\text{colim} : \mathcal{C}^{I \times J} \rightarrow \mathcal{C}$ (assuming $\mathcal{C}$ is cocomplete) is isomorphic to the composition of partial colimit functors $\text{colim}_I \circ \text{colim}_J$ and $\text{colim}_J \circ \text{colim}_I$.

Proof. (♠ TODO: AI generated, verify) We prove the colimit version. The limit version follows dually.

Let $L = \varprojlim_{(i,j) \in I \times J} D(i, j)$ denote the limit over the product category. We want to show that $L' = \varprojlim_{i \in I} (\varprojlim_{j \in J} D(i, j))$ is isomorphic to L .

The limit L comes with canonical projections $\pi_{i,j} : L \rightarrow D(i, j)$ for every pair (i, j) .

1. **Constructing a map $\phi : L \rightarrow L'$:** For a fixed $i \in I$, the collection of projections $\{\pi_{i,j} : L \rightarrow D(i, j)\}_{j \in J}$ forms a cone over the diagram $D(i, -)$. By the universal property of $L_i = \varprojlim_{j \in J} D(i, j)$, there exists a unique morphism $\alpha_i : L \rightarrow L_i$ such that for all $j \in J$, the composition $L \xrightarrow{\alpha_i} L_i \rightarrow D(i, j)$ is $\pi_{i,j}$.
The collection $\{\alpha_i : L \rightarrow L_i\}_{i \in I}$ now forms a cone over the diagram $i \mapsto L_i$. By the universal property of $L' = \varprojlim_{i \in I} L_i$, there exists a unique morphism $\phi : L \rightarrow L'$ such that for all $i \in I$, the composition $L \xrightarrow{\phi} L' \rightarrow L_i$ is α_i .
2. **Constructing a map $\psi : L' \rightarrow L$:** Let $p_i : L' \rightarrow L_i$ be the projection from the limit L' , and $q_{i,j} : L_i \rightarrow D(i, j)$ be the projection from the limit L_i . The composition $q_{i,j} \circ p_i : L' \rightarrow D(i, j)$ constitutes a cone over the diagram D indexed by $I \times J$ (i.e., for any morphism $(u, v) : (i, j) \rightarrow (i', j')$, the corresponding diagrams commute). By the universal property of $L = \varprojlim_{(i,j) \in I \times J} D(i, j)$, there exists a unique morphism $\psi : L' \rightarrow L$ such that $\pi_{i,j} \circ \psi = q_{i,j} \circ p_i$.
3. **Verification:** One verifies that $\psi \circ \phi = \text{id}_L$ and $\phi \circ \psi = \text{id}_{L'}$ by checking that these compositions induce the identity on the projections $\pi_{i,j}$ and p_i respectively, invoking the uniqueness part of the universal properties.

□

At times, we may want to consider the limit/colimit of a system of functors. Such limits/colimits can be computed “pointwise”, whenever the pointwise limits/colimits exist.

Proposition 1.2.32 (Pointwise Computation of Limits and Colimits in Functor Categories). Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Let $\text{Fun}(\mathcal{C}, \mathcal{D})$ denote the category of functors from $\mathcal{C}$ to $\mathcal{D}$ (also denoted $\mathcal{D}^{\mathcal{C}}$). Let $\mathcal{J}$ be a small category and let $F : \mathcal{J} \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$ be a diagram of functors, denoted by $j \mapsto F_j$.

1. **Limits are computed pointwise:** Suppose that for every object $C \in \mathcal{C}$, the limit of the diagram $j \mapsto F_j(C)$ exists in $\mathcal{D}$. Then the limit of the diagram F exists in $\text{Fun}(\mathcal{C}, \mathcal{D})$ and is computed pointwise. That is, there is an isomorphism in $\text{Fun}(\mathcal{C}, \mathcal{D})$:

$$\left(\lim_{j \in \mathcal{J}} F_j \right) (C) \cong \lim_{j \in \mathcal{J}} (F_j(C)).$$

The action of this limit functor on a morphism $f : C \rightarrow C'$ in $\mathcal{C}$ is the unique morphism induced by the family $\{F_j(f)\}_{j \in \mathcal{J}}$ via the universal property of limits in $\mathcal{D}$.

2. **Colimits are computed pointwise:** Suppose that for every object $C \in \mathcal{C}$, the colimit of the diagram $j \mapsto F_j(C)$ exists in $\mathcal{D}$. Then the colimit of the diagram F exists in $\text{Fun}(\mathcal{C}, \mathcal{D})$ and is computed pointwise. That is, there is an isomorphism in $\text{Fun}(\mathcal{C}, \mathcal{D})$:

$$(\text{colim}_{j \in \mathcal{J}} F_j) (C) \cong \text{colim}_{j \in \mathcal{J}} (F_j(C)).$$

The action of this colimit functor on a morphism $f : C \rightarrow C'$ in $\mathcal{C}$ is the unique morphism induced by the family $\{F_j(f)\}_{j \in \mathcal{J}}$ via the universal property of colimits in $\mathcal{D}$.

Lemma 1.2.33. (♠ TODO: AI generated, verify) Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be (large) categories (Definition 1.0.1) and let $\mathcal{B}$ be a (large) category (Definition 1.0.1). If $\mathcal{A}_1$ and $\mathcal{A}_2$ are equivalent (Definition 1.1.8) as categories, then the functor categories (Definition 1.2.1) $[\mathcal{A}_1, \mathcal{B}]$ and $[\mathcal{A}_2, \mathcal{B}]$ are equivalent.

More precisely, let $F : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ and $G : \mathcal{A}_2 \rightarrow \mathcal{A}_1$ be functors (Definition 1.1.1) together with natural isomorphisms (Definition 1.1.3)

$$\eta : \text{Id}_{\mathcal{A}_1} \xrightarrow{\sim} G \circ F \quad \text{and} \quad \epsilon : F \circ G \xrightarrow{\sim} \text{Id}_{\mathcal{A}_2}$$

witnessing an equivalence of categories (Definition 1.1.8). Then precomposition (Definition 1.1.1) with F and G induces an equivalence of categories

$$F^* : [\mathcal{A}_2, \mathcal{B}] \rightarrow [\mathcal{A}_1, \mathcal{B}], \quad F^*(H) = H \circ F$$

and

$$G^* : [\mathcal{A}_1, \mathcal{B}] \rightarrow [\mathcal{A}_2, \mathcal{B}], \quad G^*(K) = K \circ G,$$

with natural isomorphisms

$$\eta^* : \text{Id}_{[\mathcal{A}_1, \mathcal{B}]} \xrightarrow{\sim} G^* \circ F^* \quad \text{and} \quad \epsilon^* : F^* \circ G^* \xrightarrow{\sim} \text{Id}_{[\mathcal{A}_2, \mathcal{B}]}$$

induced by η and ϵ respectively.

Definition 1.2.34. Let $\mathcal{C}$ and $\mathcal{J}$ be categories (Definition 1.0.1). The *diagonal functor*

$$\Delta : \mathcal{C} \rightarrow \mathcal{C}^{\mathcal{J}}$$

(Definition 1.2.1) (♠ TODO: constant functor) is the functor sending an object $X \in \mathcal{C}$ to the constant functor $\Delta(X) : \mathcal{J} \rightarrow \mathcal{C}$, which maps every object in $\mathcal{J}$ to X and every morphism in $\mathcal{J}$ to the identity morphism 1_X . For a morphism $f : X \rightarrow Y$ in $\mathcal{C}$, $\Delta(f)$ is the natural transformation (Definition 1.1.3) whose component at every $j \in \mathcal{J}$ is f .

Theorem 1.2.35. Let $\mathcal{C}$ be a locally small (Definition 1.0.9) category and $\mathcal{J}$ be a small index category (Definition 1.0.9).

1. If $\mathcal{C}$ admits all colimits (Definition 1.2.2) of shape (Definition 1.2.1) $\mathcal{J}$, then the colimit functor (Theorem 1.2.3)

$$\text{colim} : \mathcal{C}^{\mathcal{J}} \rightarrow \mathcal{C}$$

is left adjoint (Definition 1.1.13) to the diagonal functor Δ . That is, for any functor $F : \mathcal{J} \rightarrow \mathcal{C}$ and any object $X \in \mathcal{C}$, there is a natural bijection:

$$\text{Hom}_{\mathcal{C}}(\text{colim } F, X) \cong \text{Hom}_{\mathcal{C}^{\mathcal{J}}}(F, \Delta(X)).$$

2. If $\mathcal{C}$ admits all limits of shape $\mathcal{J}$, then the limit functor (Theorem 1.2.3)

$$\lim : \mathcal{C}^{\mathcal{J}} \rightarrow \mathcal{C}$$

is right adjoint (Definition 1.1.13) to the diagonal functor (Definition 1.2.34) Δ . That is, for any object $X \in \mathcal{C}$ and any functor $F : \mathcal{J} \rightarrow \mathcal{C}$, there is a natural bijection:

$$\text{Hom}_{\mathcal{C}}(X, \lim F) \cong \text{Hom}_{\mathcal{C}^{\mathcal{J}}}(\Delta(X), F).$$

These adjunctions characterize limits and colimits via their universal properties.

1.2.1. Some specific kinds of limits.

Definition 1.2.36 (Product in a category). Let $\mathcal{C}$ be a category and let $\{X_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$ indexed by a class I .

1. A *product of the family $\{X_i\}$* is an object P of $\mathcal{C}$ together with a “universal” family of morphisms

$$\pi_i : P \rightarrow X_i, \quad \text{for each } i \in I.$$

More precisely, for any object Y and any family of morphisms $\{f_i : Y \rightarrow X_i\}_{i \in I}$, there exists a unique morphism

$$f : Y \rightarrow P$$

making the following diagram commute for all $i \in I$, i.e. $\pi_i \circ f = f_i$:

$$\begin{array}{ccc} Y & & \\ \downarrow \exists! f & \searrow f_i & \\ \prod X_i & \xrightarrow{\pi_i} & X_i \end{array}$$

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Such a product is often denoted by $\prod_{i \in I} X_i$. If $\prod_{i \in I} X_i$ exists in $\mathcal{C}$, then it is unique up to unique isomorphism by the universal property described above.

Equivalently, the product $\prod_{i \in I} X_i$ is the limit (Definition 1.2.2) of the diagram (Definition 1.2.1) $I \rightarrow \mathcal{C}, i \mapsto X_i$, where I is made into a category whose objects are the members of I and whose morphisms are just the identity morphisms.

2. A **coproduct** (or synonymously **direct sum**) of the family $\{X_i\}$ is an object C of $\mathcal{C}$ together with a “universal” family of morphisms

$$\iota_i : X_i \rightarrow C, \quad \text{for each } i \in I.$$

More precisely, for any object Y and any family of morphisms $\{g_i : X_i \rightarrow Y\}_{i \in I}$, there exists a unique morphism

$$g : C \rightarrow Y$$

making the following diagram commute for all $i \in I$, i.e. $g \circ \iota_i = g_i$:

$$\begin{array}{ccc} X_i & \xrightarrow{\iota_i} & \prod X_i \\ & \searrow g_i & \downarrow \exists! g \\ & & Y \end{array}$$

Such a coproduct is often denoted by $\prod_{i \in I} X_i$ or $\oplus_{i \in I} X_i$. If $\prod_{i \in I} X_i$ exists in $\mathcal{C}$, then it is unique up to unique isomorphism by the universal property described above.

Equivalently, the coproduct $\prod_{i \in I} X_i$ is the colimit (Definition 1.2.2) of the diagram (Definition 1.2.1) $I \rightarrow \mathcal{C}, i \mapsto X_i$, where I is made into a category whose objects are the members of I and whose morphisms are just the identity morphisms.

Definition 1.2.37 (Equalizer in a category). Let $\mathcal{C}$ be a (large) category (Definition 1.0.1) and let $f, g : X \rightarrow Y$ be morphisms in $\mathcal{C}$.

1. An **equalizer of f and g** is an object E together with a morphism

$$e : E \rightarrow X$$

such that

$$f \circ e = g \circ e$$

and for any object Z with morphism $z : Z \rightarrow X$ satisfying

$$f \circ z = g \circ z,$$

there exists a unique morphism $u : Z \rightarrow E$ making the diagram commute:

$$e \circ u = z.$$

$$\begin{array}{ccccc} Z & & & & \\ \downarrow \exists! u & \searrow z & & & \\ E & \xrightarrow{e} & X & \xrightleftharpoons[f]{g} & Y \end{array}$$

If such an equalizer of f and g exists, then we say that the following **equalizer diagram is exact**:

$$E \xrightarrow{e} X \rightrightarrows^f_g Y$$

2. A **coequalizer of f and g** is an object Q together with a morphism

$$q : Y \rightarrow Q$$

such that

$$q \circ f = q \circ g$$

and for any object Z with morphism $w : Y \rightarrow Z$ satisfying

$$w \circ f = w \circ g,$$

there exists a unique morphism $v : Q \rightarrow Z$ making the diagram commute:

$$\begin{array}{ccc} X \rightrightarrows^f_g Y & \xrightarrow{q} & Q \\ & \searrow w & \downarrow \exists! v \\ & & Z \end{array}$$

If such a coequalizer of f and g exists, then we say that the following **coequalizer diagram is exact**:

$$X \rightrightarrows^f_g Y \xrightarrow{q} Q$$

Definition 1.2.38. Let $\mathcal{C}$ be a category (Definition 1.0.1), let Z be an object, and let X, Y be objects of $\mathcal{C}$ over (Definition 1.3.4) Z , i.e. morphisms $X \rightarrow Z$ and $Y \rightarrow Z$ are fixed. A **cartesian product of X and Y over Z in $\mathcal{C}$** (or **fiber product** or **pullback diagram**) is an object, often denoted by $X \times_Z Y$, with **projection morphisms** $X \times_Z Y \rightarrow X$ and $X \times_Z Y \rightarrow Y$ that are universal. More precisely, for any object T of $\mathcal{C}$ and morphisms $f_X : T \rightarrow X$, $f_Y : T \rightarrow Y$ such that the compositions $T \xrightarrow{f_X} X \rightarrow Z$ and $T \xrightarrow{f_Y} Y \rightarrow Z$ agree, there exists a unique morphism $u : T \rightarrow X \times_Z Y$ such that the following diagram commutes:

$$\begin{array}{ccccc} T & & \xrightarrow{f_X} & & X \\ & \searrow u & & \searrow & \downarrow \\ & & X \times_Z Y & \longrightarrow & X \\ & \searrow f_Y & \downarrow & & \downarrow \\ & & Y & \longrightarrow & Z \end{array}$$

Equivalently, $X \times_Z Y$ is the limit (Definition 1.2.2) of the diagram (Definition 1.2.1)

$$\begin{array}{ccc} & X & \\ & \downarrow & \\ Y & \longrightarrow & Z \end{array}$$

in $\mathcal{C}$.

The commutative diagram

$$\begin{array}{ccc} X \times_Z Y & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Z \end{array}$$

may be referred to as a **cartesian square**.

Definition 1.2.39. Let $\mathcal{C}$ be a category

1. Let X be an object in $\mathcal{C}$. If the product $X \times X$ exists, then the **diagonal morphism of X** is the morphism

$$\Delta_X : X \rightarrow X \times X$$

induced by the universal property of $X \times X$ for the identity morphisms $\text{id}_X : X \rightarrow X$ from X to each of the first and second factors of X in $X \times X$. In other words, Δ_f is the unique morphism such that the following equalities hold:

$$p_1 \circ \Delta_X = \text{id}_X, \quad p_2 \circ \Delta_X = \text{id}_X$$

where $p_1, p_2 : X \times X \rightarrow X$ are the two canonical projections.

2. Let $f : X \rightarrow Z$ be a morphism in $\mathcal{C}$. If the cartesian product (Definition 1.2.38) $X \times_Z X$ exists, then the **diagonal morphism of f** is the morphism

$$\Delta_f = \Delta_X : X \rightarrow X \times_Z X$$

induced by the universal property of $X \times_Z X$ for the identity morphisms $\text{id}_X : X \rightarrow X$ from X to each of the first and second factors of X in $X \times_Z X$. In other words, Δ_f is the unique morphism such that the following equalities hold:

$$p_1 \circ \Delta_f = \text{id}_X, \quad p_2 \circ \Delta_f = \text{id}_X$$

where $p_1, p_2 : X \times X \rightarrow X$ are the two canonical projections.

Definition 1.2.40. Let $\mathcal{C}$ be a category (Definition 1.0.1), let Z be an object, and let X, Y be objects of $\mathcal{C}$ under (🔥 TODO: under category) Z , i.e. morphisms $f : Z \rightarrow X$ and $g : Z \rightarrow Y$ are fixed. A **pushout of X and Y under Z in $\mathcal{C}$** is an object, often denoted by $X \amalg_Z Y$, with **inclusion morphisms** $i_X : X \rightarrow X \amalg_Z Y$ and $i_Y : Y \rightarrow X \amalg_Z Y$ that are universal. More precisely, the morphisms satisfy $i_X \circ f = i_Y \circ g$, and for any object Q of $\mathcal{C}$ with morphisms $u_X : X \rightarrow Q$, $u_Y : Y \rightarrow Q$ satisfying $u_X \circ f = u_Y \circ g$, there exists a unique morphism $w : X \amalg_Z Y \rightarrow Q$ such that the following diagram commutes:

$$\begin{array}{ccc} Z & \xrightarrow{f} & X \\ g \downarrow & & \downarrow i_X \\ Y & \xrightarrow{i_Y} & X \amalg_Z Y \end{array} \quad \begin{array}{c} \xrightarrow{u_X} \\ \searrow w \\ \xrightarrow{u_Y} \end{array} \quad \begin{array}{c} \\ \\ Q \end{array}$$

Equivalently, $X \amalg_Z Y$ is the colimit of the diagram (Definition 1.2.1)

$$\begin{array}{ccc} & & X \\ & \nearrow f & \\ Z & & \\ & \searrow g & \\ & & Y \end{array}$$

in $\mathcal{C}$.

The commutative diagram

$$\begin{array}{ccc} Z & \xrightarrow{f} & X \\ g \downarrow & & \downarrow i_X \\ Y & \xrightarrow{i_Y} & X \amalg_Z Y \end{array}$$

may be referred to as a *cocartesian square*.

Note that the notion of cocartesian squares is dual of the notion of cartesian squares (Definition 1.2.38).

1.3. Miscellaneous categorical constructions and definitions.

1.3.1. Subcategory of a category.

Definition 1.3.1 (Subcategory). Let $\mathcal{C}$ be a (large) category (Definition 1.0.1). A *subcategory* $\mathcal{D}$ of $\mathcal{C}$ consists of:

- a subclass of objects $\text{Ob}(\mathcal{D}) \subseteq \text{Ob}(\mathcal{C})$,
- for each pair of objects $X, Y \in \text{Ob}(\mathcal{D})$, a subclass of morphisms

$$\text{Hom}_{\mathcal{D}}(X, Y) \subseteq \text{Hom}_{\mathcal{C}}(X, Y),$$

such that

- for every object $X \in \text{Ob}(\mathcal{D})$, the identity morphism id_X of X in $\mathcal{C}$ lies in $\text{Hom}_{\mathcal{D}}(X, X)$,
- the composition of morphisms in $\mathcal{D}$ is inherited from $\mathcal{C}$ and is closed in $\mathcal{D}$: for morphisms $f \in \text{Hom}_{\mathcal{D}}(X, Y)$ and $g \in \text{Hom}_{\mathcal{D}}(Y, Z)$, their composition $g \circ f$, which is a priori in $\text{Hom}_{\mathcal{C}}(X, Z)$, is in $\text{Hom}_{\mathcal{D}}(X, Z)$.

Definition 1.3.2 (Full subcategory). Let $\mathcal{C}$ be a (large) category (Definition 1.0.1). A *full subcategory* $\mathcal{D}$ of $\mathcal{C}$ is a subcategory (Definition 1.3.1) such that for every pair of objects $X, Y \in \text{Ob}(\mathcal{D})$, the morphism classes coincide:

$$\text{Hom}_{\mathcal{D}}(X, Y) = \text{Hom}_{\mathcal{C}}(X, Y).$$

In other words, a full subcategory includes all morphisms between its objects that exist in the ambient category $\mathcal{C}$.

Definition 1.3.3 (Replete subcategory). Let $\mathcal{C}$ be a (large) category (Definition 1.0.1). A full subcategory (Definition 1.3.2) $\mathcal{D} \subseteq \mathcal{C}$ is called **replete** if it satisfies the following property:

- For all $x \in \text{Ob}(\mathcal{C})$ and $y \in \text{Ob}(\mathcal{D})$, if $x \cong y$ in $\mathcal{C}$, then $x \in \text{Ob}(\mathcal{D})$ as well.

Equivalently, a replete subcategory of $\mathcal{C}$ is a full subcategory closed under isomorphisms in $\mathcal{C}$.

Definition 1.3.4 (Category of objects over a fixed object). Let $\mathcal{C}$ be a category (Definition 1.0.1) and let $X \in \text{Ob}(\mathcal{C})$ be a fixed object.

1. The **category of objects over X** (or synonymously the **slice category of X in $\mathcal{C}$** or the **over category of X in $\mathcal{C}$**), commonly denoted $\mathcal{C}/X$, $\mathcal{C}_{/X}$, or $(\mathcal{C} \downarrow X)$ is the category defined as follows:

- An object of $\mathcal{C}/X$ is a morphism $f: A \rightarrow X$ in $\mathcal{C}$, where $A \in \text{Ob}(\mathcal{C})$.
- A morphism from $f: A \rightarrow X$ to $g: B \rightarrow X$ in $\mathcal{C}/X$ is a morphism $h: A \rightarrow B$ in $\mathcal{C}$ such that the following diagram commutes:

$$\begin{array}{ccc} A & \xrightarrow{h} & B \\ & \searrow f & \downarrow g \\ & & X \end{array}$$

i.e. such that $g \circ h = f$.

- The identity morphisms and composition in $\mathcal{C}/X$ are inherited from $\mathcal{C}$.
2. The **category of objects under X** (or synonymously the **coslice category of X in $\mathcal{C}$** or the **under category of X in $\mathcal{C}$**), commonly denoted $X/\mathcal{C}$, $X \backslash \mathcal{C}$, $\mathcal{C}_{X/}$, or $(X \downarrow \mathcal{C})$, is the category defined as follows:

- An object of $X/\mathcal{C}$ is a morphism $f: X \rightarrow A$ in $\mathcal{C}$, where $A \in \text{Ob}(\mathcal{C})$.
- A morphism from $f: X \rightarrow A$ to $g: X \rightarrow B$ in $X/\mathcal{C}$ is a morphism $h: A \rightarrow B$ in $\mathcal{C}$ such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{f} & A \\ & \searrow g & \downarrow h \\ & & B \end{array}$$

i.e. such that $h \circ f = g$.

- The identity morphisms and composition in $X/\mathcal{C}$ are inherited from $\mathcal{C}$.

Both notions are special cases of comma categories (Definition 1.3.12).

Lemma 1.3.5. Let $\mathcal{C}$ be a category (Definition 1.0.1) and let $X \in \text{Ob}(\mathcal{C})$ be a fixed object. The slice category $\mathcal{C}/X$ (Definition 1.3.4) has X as its final object (Definition 2.0.6).

Proof. This is clear. □

Definition 1.3.6 (Reflective subcategory). Let $\mathcal{C}$ be a (large) category (Definition 1.0.1) and let $\mathcal{D} \subseteq \mathcal{C}$ be a full subcategory (Definition 1.3.2). The subcategory $\mathcal{D}$ is called **reflective**

if the inclusion functor

$$I : \mathcal{D} \hookrightarrow \mathcal{C}$$

admits a left adjoint (Definition 1.1.13)

$$L : \mathcal{C} \rightarrow \mathcal{D}.$$

In this case, L is called the **reflector**, and for each $c \in \text{Ob}(\mathcal{C})$, the unit morphism of the adjunction

$$\eta_c : c \rightarrow I(L(c))$$

is called the **reflection arrow of c into $\mathcal{D}$** .

(♠ TODO: describe what monomorphisms and epimorphisms are like in additive or abelian categories)

1.3.2. Types of morphisms in a category.

Definition 1.3.7 (Monomorphism and Epimorphism in Categories). Let $\mathcal{C}$ be a category (Definition 1.0.1). For objects $A, B \in \mathcal{C}$, let $f : A \rightarrow B$ be a morphism in $\mathcal{C}$.

- The morphism f is called a **monomorphism** (or a **monic morphism**) if for every object X and every pair of morphisms $g_1, g_2 : X \rightarrow A$, the equality $f \circ g_1 = f \circ g_2$ implies $g_1 = g_2$.
- The morphism f is called an **epimorphism** (or an **epic morphism**) if for every object Y and every pair of morphisms $h_1, h_2 : B \rightarrow Y$, the equality $h_1 \circ f = h_2 \circ f$ implies $h_1 = h_2$.

Definition 1.3.8. Let $\mathcal{C}$ be a (large) (Definition 1.0.1) pointed category, i.e. a category with a zero object (Definition 2.0.6) 0 . Let $X, Y \in \text{Ob}(\mathcal{C})$ be an object and let $f : X \rightarrow Y$ be a morphism.

1. A morphism $i : K \rightarrow X$ is called the **kernel of f** if:
 - (a) $f \circ i = 0$, where 0 is the zero morphism $K \rightarrow Y$,
 - (b) for any morphism $g : Z \rightarrow X$ such that $f \circ g = 0$, there exists a unique morphism $u : Z \rightarrow K$ such that $g = i \circ u$.

The kernel, if it exists, is unique up to unique isomorphism (Definition 1.0.8). **ker(f)** denotes the object K determined (up to isomorphism) by a kernel of f .

Equivalently, **ker(f)** is the equalizer (Definition 1.2.37) of f and the 0 morphism $X \rightarrow Y$, see ??.

2. a morphism $p : Y \rightarrow Q$ is called the **cokernel of f** if:
 - (a) $p \circ f = 0$, where 0 is the zero morphism (Definition 2.0.6) $X \rightarrow Q$,
 - (b) for any morphism $g : Y \rightarrow Z$ such that $g \circ f = 0$, there exists a unique morphism $v : Q \rightarrow Z$ such that $g = v \circ p$.

The cokernel, if it exists, is unique up to unique isomorphism. **coker(f)** denotes the object Q determined (up to isomorphism) by a cokernel of f .

Equivalently, **coker(f)** is the coequalizer (Definition 1.2.37) of f and the 0 morphism $X \rightarrow Y$, see ??.

1.3.3. Essential image of a functor between categories.

1.3.4. Types of categories.

Definition 1.3.9 (Complete and Cocomplete Category). Let $\mathcal{C}$ be a category (Definition 1.0.1).

- The category $\mathcal{C}$ is called **complete** (resp. **finitely complete**) if all small limits (Definition 1.2.5) (resp. finite limits) exist in $\mathcal{C}$; that is, for every small diagram $D : J \rightarrow \mathcal{C}$ (with J a small category), the limit $\lim D$ exists and is an object of $\mathcal{C}$.
- The category $\mathcal{C}$ is called **cocomplete** (resp. **finitely cocomplete**) if all small colimits (Definition 1.2.5) (resp. finite colimits) exist in $\mathcal{C}$; that is, for every small diagram $D : J \rightarrow \mathcal{C}$, the colimit $\operatorname{colim} D$ exists and is an object of $\mathcal{C}$.

1.3.5. Categories constructed from other categories.

Definition 1.3.10 (Ind-category). Let $\mathcal{C}$ be a locally small category (Definition 1.0.9).

1. The **Ind-category of $\mathcal{C}$** , denoted $\operatorname{Ind}(\mathcal{C})$, is defined as follows:
 - Objects of $\operatorname{Ind}(\mathcal{C})$ are formal filtered colimits (Definition 1.2.12) of objects in $\mathcal{C}$. More precisely, an object is given by a filtered (Definition 1.2.8) small category I and a functor

$$X : I \rightarrow \mathcal{C}.$$

- Morphisms between objects $X : I \rightarrow \mathcal{C}$ and $Y : J \rightarrow \mathcal{C}$ are defined by

$$\operatorname{Hom}_{\operatorname{Ind}(\mathcal{C})}(X, Y) := \varinjlim_{i \in I} \varprojlim_{j \in J} \operatorname{Hom}_{\mathcal{C}}(X_i, Y_j),$$

(Definition 1.2.12) where X_i and Y_j denote the images of $i \in I$ and $j \in J$ under X and Y , respectively.

The composition of morphisms is induced naturally from composition in $\mathcal{C}$. Hence, $\operatorname{Ind}(\mathcal{C})$ is the completion of $\mathcal{C}$ under filtered colimits. Objects of $\operatorname{Ind}(\mathcal{C})$ are called **Ind-objects of $\mathcal{C}$** .

2. The **Pro-category of $\mathcal{C}$** , denoted $\operatorname{Pro}(\mathcal{C})$, is defined as follows:
 - Objects of $\operatorname{Pro}(\mathcal{C})$ are formal cofiltered limits (Definition 1.2.12) of objects in $\mathcal{C}$. More precisely, an object is given by a cofiltered small category I and a functor

$$X : I \rightarrow \mathcal{C}.$$

- Morphisms between objects $X : I \rightarrow \mathcal{C}$ and $Y : J \rightarrow \mathcal{C}$ are defined by

$$\operatorname{Hom}_{\operatorname{Pro}(\mathcal{C})}(X, Y) := \varprojlim_{j \in J} \varinjlim_{i \in I} \operatorname{Hom}_{\mathcal{C}}(X_i, Y_j),$$

where X_i and Y_j denote the images of $i \in I$ and $j \in J$ under X and Y , respectively.

The composition of morphisms is induced naturally from composition in $\mathcal{C}$.

Hence, $\operatorname{Pro}(\mathcal{C})$ is the completion of $\mathcal{C}$ under cofiltered limits. Objects of $\operatorname{Pro}(\mathcal{C})$ are called **Pro-objects of $\mathcal{C}$** .

Since **Sets** has all limits and colimits (♠ **TODO:**) and hence has all projective and inductive limits and since $\mathcal{C}$ is locally small, $\operatorname{Ind}(\mathcal{C})$ and $\operatorname{Pro}(\mathcal{C})$ are locally small.

The *canonical embedding of a locally small category $\mathcal{C}$ into its Ind-category or Pro-category* is defined by treating each object as a "constant" diagram. Let $\mathbf{1}$ be the category with a single object $*$ and its identity morphism.

1. The Embedding $\iota_{\text{Ind}} : \mathcal{C} \rightarrow \text{Ind}(\mathcal{C})$:

- Objects: An object $C \in \mathcal{C}$ is mapped to the functor $X : \mathbf{1} \rightarrow \mathcal{C}$ sending $*$ to C . This represents C as a formal filtered colimit over the trivial index category.
- Morphisms: Given objects X (representing C) and Y (representing D), the Hom-set in the Ind-category simplifies:

$$\text{Hom}_{\text{Ind}(\mathcal{C})}(X, Y) \cong \varinjlim_{i \in \mathbf{1}} \varprojlim_{j \in \mathbf{1}} \text{Hom}_{\mathcal{C}}(X_i, Y_j) \cong \text{Hom}_{\mathcal{C}}(C, D).$$

2. The Embedding $\iota_{\text{Pro}} : \mathcal{C} \rightarrow \text{Pro}(\mathcal{C})$:

- Objects: An object $C \in \mathcal{C}$ is mapped to the functor $X : \mathbf{1} \rightarrow \mathcal{C}$ sending $*$ to C , representing C as a formal cofiltered limit.
- Morphisms: Similarly, the Pro-morphism definition reduces to:

$$\text{Hom}_{\text{Pro}(\mathcal{C})}(X, Y) \cong \varprojlim_{j \in \mathbf{1}} \varinjlim_{i \in \mathbf{1}} \text{Hom}_{\mathcal{C}}(X_i, Y_j) \cong \text{Hom}_{\mathcal{C}}(C, D).$$

In both cases, the functor ι is fully faithful, allowing us to identify $\mathcal{C}$ with a full subcategory of its respective completion. Given an object C of $\mathcal{C}$, the pro/ind-object $\iota(C)$ may be called the *constant pro/ind-object of C* .

Definition 1.3.11 (Product Category of a Family of Categories). Let $\{\mathcal{C}_i\}_{i \in I}$ be a family of (large) categories (Definition 1.0.1) indexed by a class I . The *product category of the family*, denoted

$$\prod_{i \in I} \mathcal{C}_i,$$

is the very large category (♠ TODO: define very large categories) defined as follows:

- The class of objects is

$$\text{Ob}\left(\prod_{i \in I} \mathcal{C}_i\right) = \prod_{i \in I} \text{Ob}(\mathcal{C}_i),$$

i.e., an object is a family $(A_i)_{i \in I}$ with $A_i \in \text{Ob}(\mathcal{C}_i)$.

- For two objects $(A_i)_i$ and $(B_i)_i$, the morphism class is

$$\text{Hom}_{\prod_{i \in I} \mathcal{C}_i}((A_i)_i, (B_i)_i) = \prod_{i \in I} \text{Hom}_{\mathcal{C}_i}(A_i, B_i).$$

In other words, a morphism $(f_i)_i : (A_i)_i \rightarrow (B_i)_i$ consists of morphisms $f_i : A_i \rightarrow B_i$ in each $\mathcal{C}_i$.

- For morphisms $(f_i)_i : (A_i)_i \rightarrow (B_i)_i$ and $(g_i)_i : (B_i)_i \rightarrow (C_i)_i$, composition is defined componentwise:

$$(g_i)_i \circ (f_i)_i = (g_i \circ_i f_i)_i.$$

- For each object $(A_i)_i$, the identity morphism is given by the family

$$(\text{id}_{A_i})_i.$$

If I is a set, then $\prod_{i \in I} \mathcal{C}_i$ is a large category. If I is a set and if each $\mathcal{C}_i$ is locally small (Definition 1.0.9), then $\prod_{i \in I} \mathcal{C}_i$ is locally small.

In case that I is finite, the notation of $\times$ may be used for product categories, e.g. $\mathcal{C}_i \times \mathcal{C}_j$ denotes the product of two categories $\mathcal{C}_i \times \mathcal{C}_j$.

(♠ **TODO: ordinal, U_α**) If α is an ordinal such that $\mathcal{C}_i$ and I are U_α -large (i.e. they live in $U_{\alpha+1}$), then $\prod_{i \in I} \mathcal{C}_i$ is $U_{\alpha+1}$ -large.

Definition 1.3.12 (Comma Category). Let $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ be categories, and let $S : \mathcal{A} \rightarrow \mathcal{C}$ and $T : \mathcal{B} \rightarrow \mathcal{C}$ be functors.

The *comma category* $(S \downarrow T)$ (also denoted (S, T) or S/T) is defined as follows:

- **Objects:** Triples (A, B, h) , where A is an object of $\mathcal{A}$, B is an object of $\mathcal{B}$, and $h : S(A) \rightarrow T(B)$ is a morphism in $\mathcal{C}$.
- **Morphisms:** A morphism from (A, B, h) to (A', B', h') consists of a pair (f, g) , where $f : A \rightarrow A'$ is a morphism in $\mathcal{A}$ and $g : B \rightarrow B'$ is a morphism in $\mathcal{B}$, such that the following diagram in $\mathcal{C}$ commutes:

$$\begin{array}{ccc} S(A) & \xrightarrow{h} & T(B) \\ S(f) \downarrow & & \downarrow T(g) \\ S(A') & \xrightarrow{h'} & T(B') \end{array}$$

That is, $T(g) \circ h = h' \circ S(f)$.

- **Composition:** Composition is defined component-wise: $(f', g') \circ (f, g) = (f' \circ f, g' \circ g)$.

The general definition of the comma category subsumes several common categorical constructions:

1. **Slice Category (Over-category):** If $\mathcal{A} = \mathcal{C}$, $S = \text{Id}_{\mathcal{C}}$, $\mathcal{B} = \mathbf{1}$ (the terminal category with one object $*$), and $T : \mathbf{1} \rightarrow \mathcal{C}$ is the functor selecting an object $X \in \mathcal{C}$ (i.e., $T(*) = X$), then $(S \downarrow T)$ is the slice category $\mathcal{C}/X$ (Definition 1.3.4).
2. **Coslice Category (Under-category):** If $\mathcal{A} = \mathbf{1}$, $S(*) = X$, $\mathcal{B} = \mathcal{C}$, and $T = \text{Id}_{\mathcal{C}}$, then $(S \downarrow T)$ is the coslice category $X/\mathcal{C}$ (Definition 1.3.4).
3. **Relative Slice Category:** If $S = u : \mathcal{D} \rightarrow \mathcal{C}$ is a functor and $T : \mathbf{1} \rightarrow \mathcal{C}$ picks out an object $U \in \mathcal{C}$, then $(S \downarrow T)$ is the relative slice category (Definition 5.3.1).
4. **Arrow Category:** If $\mathcal{A} = \mathcal{B} = \mathcal{C}$ and $S = T = \text{Id}_{\mathcal{C}}$, then $(S \downarrow T)$ is the arrow category $\mathcal{C}^{\rightarrow}$ (Definition 1.3.13) of $\mathcal{C}$. Its objects are all morphisms in $\mathcal{C}$.

Definition 1.3.13 (Arrow Category). Let $\mathcal{C}$ be a category (Definition 1.0.1). The *arrow category of $\mathcal{C}$* , denoted $\mathcal{C}^{\rightarrow}$ (or sometimes $\text{Arr}(\mathcal{C})$ or $\mathcal{C}^2$), is defined as follows:

- **Objects:** Morphisms $f : A \rightarrow B$ in $\mathcal{C}$. (We may view such an object as a triple (A, B, f)).
- **Morphisms:** A morphism from an object $f : A \rightarrow B$ to an object $f' : A' \rightarrow B'$ is a pair of morphisms (h_0, h_1) in $\mathcal{C}$, where $h_0 : A \rightarrow A'$ and $h_1 : B \rightarrow B'$, such that the

following square commutes:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ h_0 \downarrow & & \downarrow h_1 \\ A' & \xrightarrow{f'} & B' \end{array}$$

That is, $h_1 \circ f = f' \circ h_0$.

- **Composition:** Composition is defined component-wise: $(k_0, k_1) \circ (h_0, h_1) = (k_0 \circ h_0, k_1 \circ h_1)$.

Arrow categories are special cases of comma categories (Definition 1.3.12).

1.3.6. Types of functors.

Definition 1.3.14 (Reflecting a type of morphism). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between (large) categories (Definition 1.1.1), and let $\mathcal{P}$ be a property of morphisms (or more generally a property of sequences or families of morphisms) that is stable under isomorphism (Definition 1.0.8) (e.g. monomorphism, epimorphism (Definition 1.3.7), isomorphism, etc.). We say that F *reflects $\mathcal{P}$ -morphisms* if for every morphism $f : x \rightarrow y$ in $\mathcal{C}$, whenever $F(f)$ has property $\mathcal{P}$ in $\mathcal{D}$, it follows that f has property $\mathcal{P}$ in $\mathcal{C}$.

One can speak of more general notions of reflection (e.g. reflection of limits (Definition 1.2.2), of colimits (Definition 1.2.2), of “exact sequences”, say for abelian categories, etc.)

Proposition 1.3.15 (Reflection of morphism properties by full or faithful functors). Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor (Definition 1.1.1) between locally small categories (Definition 1.0.9).

(1) If F is faithful (Definition 1.1.4), then F reflects (Definition 1.3.14) monomorphisms and epimorphisms (Definition 1.3.7). That is, if $f : x \rightarrow y$ in $\mathcal{C}$ is such that $F(f)$ is a monomorphism (resp. epimorphism) in $\mathcal{D}$, then f is a monomorphism (resp. epimorphism) in $\mathcal{C}$.

(♠ TODO: define split monos and epis) (2) If F is fully faithful, then F reflects isomorphisms, split monomorphisms, and split epimorphisms. That is, if $f : x \rightarrow y$ in $\mathcal{C}$ is such that $F(f)$ is an isomorphism in $\mathcal{D}$, then f is an isomorphism in $\mathcal{C}$.

Definition 1.3.16 (n-ary (Multivariable) Functor). Let I be a finite set with $|I| = n$, and let $\{\mathcal{C}_i\}_{i \in I}$ be (large) categories (Definition 1.0.1), together with another category $\mathcal{D}$. An *n-ary functor* (also called a *multivariable functor*, a *multivariate functor*, or a *multifunctor*) from the categories $\{\mathcal{C}_i\}_{i \in I}$ to $\mathcal{D}$ is a functor (Definition 1.1.1)

$$F : \prod_{i \in I} \mathcal{C}_i \rightarrow \mathcal{D}.$$

(Definition 1.3.11) That is, F assigns:

- to each object $((A_i)_{i \in I})$ in $\prod_{i \in I} \mathcal{C}_i$, an object $F((A_i)_{i \in I})$ in $\mathcal{D}$,
- to each morphism $((f_i)_{i \in I}) : (A_i)_i \rightarrow (B_i)_i$, a morphism $F((f_i)_i) : F((A_i)_i) \rightarrow F((B_i)_i)$ in $\mathcal{D}$,

so that F preserves identities and composition componentwise. For instance, a **bifunctor** is an n -ary functor when $n = 2$, a **ternary functor/trifunctor** is an n -ary functor when $n = 3$, etc.

1.3.7. Algebraic objects in a category.

Definition 1.3.17. Let $\mathcal{C}$ be a (large) category (Definition 1.0.1). A **semigroup object in $\mathcal{C}$** is an object $A \in \mathcal{C}$ such that the product (Definition 1.2.36) $A \times A$ exists in $\mathcal{C}$ together with a morphism

$$\mu : A \times A \rightarrow A,$$

called the **multiplication morphism** such that the associativity diagram

$$\begin{array}{ccc} A \times A \times A & \xrightarrow{\mu \times \text{id}_A} & A \times A \\ \text{id}_A \times \mu \downarrow & & \downarrow \mu \\ A \times A & \xrightarrow{\mu} & A \end{array}$$

commutes.

The semigroup object structure (A, μ, η, ι) is said to be **abelian** or **commutative** if the morphisms $\mu : A \times A \rightarrow A$ and $\mu \circ \tau_{A,A} : A \times A \rightarrow A$ coincide, where $\tau_{A,A} : A \times A \rightarrow A \times A$ is the symmetry morphism swapping the two factors.

Definition 1.3.18. Let $\mathcal{C}$ be a (large) category (Definition 1.0.1) with a final object (Definition 2.0.6). A **monoid object in $\mathcal{C}$** is a semigroup object (Definition 1.3.17) (A, μ) together with a **unit morphism**

$$\eta : 1 \rightarrow A$$

such that the products (Definition 1.2.36) $1 \times A$ and $A \times 1$ exist and the unit diagrams

$$\begin{array}{ccc} 1 \times A & \xrightarrow{\eta \times \text{id}_A} & A \times A \\ \text{pr}_2 \searrow & & \swarrow \mu \\ & A & \end{array}$$

$$\begin{array}{ccc} A \times 1 & \xrightarrow{\text{id}_A \times \eta} & A \times A \\ \text{pr}_1 \searrow & & \swarrow \mu \\ & A & \end{array}$$

commute.

Definition 1.3.19. Let $\mathcal{C}$ be a (large) category (Definition 1.0.1) with a final object (Definition 2.0.6).

A **group object in $\mathcal{C}$** is a monoid object (Definition 1.3.18) (A, μ, η) together with a **inverse morphism**

$$\iota : A \rightarrow A$$

such that the diagrams

$$\begin{array}{ccc} A & \xrightarrow{\Delta} & A \times A \\ & \searrow \eta \circ !_A & \downarrow \mu \circ (\text{id}_A \times \iota) \\ & & A \end{array}$$

$$\begin{array}{ccc} A & \xrightarrow{\Delta} & A \times A \\ & \searrow \eta \circ !_A & \downarrow \mu \circ (\iota \times \text{id}_A) \\ & & A \end{array}$$

commute, where $\Delta : A \rightarrow A \times A$ is the diagonal and $!_A : A \rightarrow 1$ is the unique morphism.

An **abelian group object in $\mathcal{C}$** is a group object as above that also satisfies... (♠ TODO:)

A **morphism of group objects in $\mathcal{C}$** is ...

The category of group objects may be denoted by notations such as **Grp($\mathcal{C}$)**. The category of abelian group objects may be denoted by notations such as **Ab($\mathcal{C}$)**.

Definition 1.3.20. Let $\mathcal{C}$ be a (large) category (Definition 1.0.1) with a terminal object (Definition 2.0.6) 1 .

A **ring object in $\mathcal{C}$** is an object $R \in \mathcal{C}$ equipped with:

- an abelian (Definition 1.3.17) group object (Definition 1.3.19) structure $(R, +, 0, -)$ (written additively), i.e. morphisms

$$+ : R \times R \rightarrow R, \quad 0 : 1 \rightarrow R, \quad - : R \rightarrow R,$$

making $(R, +, 0, -)$ a group object;

- a monoid object (Definition 1.3.18) structure $(R, \cdot, 1_R)$ (written multiplicatively), i.e. morphisms

$$\cdot : R \times R \rightarrow R, \quad 1_R : 1 \rightarrow R,$$

making $(R, \cdot, 1_R)$ a monoid object;

such that the usual distributivity and absorption axioms hold, expressed by the commutativity of the diagrams corresponding to

$$a \cdot (b + c) = a \cdot b + a \cdot c, \quad (a + b) \cdot c = a \cdot c + b \cdot c,$$

and $0 \cdot a = a \cdot 0 = 0$, for all a, b, c (interpreted as morphisms in $\mathcal{C}$).

Definition 1.3.21 (Module object over a ring object in a category). Let $\mathcal{C}$ be a (large) category (Definition 1.0.1) with a terminal object (Definition 2.0.6) 1 . Let R be a ring object (Definition 1.3.20) in $\mathcal{C}$. An **R -module object in $\mathcal{C}$** is an object $M \in \mathcal{C}$ together with:

- an abelian (Definition 1.3.17) group object (Definition 1.3.19) structure $(M, +_M, 0_M, -_M)$ on M ;
- an action morphism

$$\alpha : R \times M \rightarrow M,$$

such that the usual module axioms hold, expressed by commutative diagrams corresponding to

$$\begin{aligned} r \cdot (m_1 +_M m_2) &= r \cdot m_1 +_M r \cdot m_2, & (r_1 + r_2) \cdot m &= r_1 \cdot m +_M r_2 \cdot m, \\ (r_1 \cdot r_2) \cdot m &= r_1 \cdot (r_2 \cdot m), & 1_R \cdot m &= m, & 0_R \cdot m &= 0_M = r \cdot 0_M, \end{aligned}$$

for all $r, r_1, r_2 \in R$ and $m, m_1, m_2 \in M$, interpreted as morphisms and equalities in $\mathcal{C}$.

Definition 1.3.22. Let $\mathcal{C}$ be a (large) category (Definition 1.0.1).

A *groupoid object in $\mathcal{C}$* consists of two objects X_0 (the "object of objects") and X_1 (the "object of morphisms"), together with five structure morphisms:

- *Source* and *Target*: $s, t : X_1 \rightarrow X_0$, such that the fiber product (Definition 1.2.38) $X_1 \times_{s, X_0, t} X_1$ of the morphisms s and t exists in $\mathcal{C}$,
- *Identity*: $e : X_0 \rightarrow X_1$,
- *Composition*: $m : X_1 \times_{s, X_0, t} X_1 \rightarrow X_1$,
- *Inverse*: $i : X_1 \rightarrow X_1$,

such that the following conditions hold (expressing the axioms of a category where every morphism is invertible):

1. **Source/Target identities:**

$$\begin{aligned} s \circ e &= \text{id}_{X_0}, & t \circ e &= \text{id}_{X_0} \\ s \circ m &= s \circ \pi_2, & t \circ m &= t \circ \pi_1 \end{aligned}$$

2. **Associativity:** The following diagram of composition commutes:

$$m \circ (m \times \text{id}_{X_1}) = m \circ (\text{id}_{X_1} \times m)$$

3. **Unitality:**

$$m \circ (e \circ s, \text{id}_{X_1}) = \text{id}_{X_1}, \quad m \circ (\text{id}_{X_1}, e \circ t) = \text{id}_{X_1}$$

4. **Invertibility:**

$$m \circ (i, \text{id}_{X_1}) = e \circ s, \quad m \circ (\text{id}_{X_1}, i) = e \circ t$$

2. MONOIDAL CATEGORIES

(♠ TODO: Carefully check what statements need to assume that the categories are locally small and which do not)

Definition 2.0.1. A *monoidal category* is a (large) category (Definition 1.0.1) $\mathcal{C}$ equipped with:

- a bifunctor (Definition 1.3.16) $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ (Definition 1.3.11) (called the *tensor product*);
- an object $\mathbb{I} \in \text{Ob}(\mathcal{C})$ (often called the *unit object*); common notations for the unit object include $\mathbb{I}$ and $\mathbb{1}$.
- natural isomorphisms (*associator*) $\alpha_{X,Y,Z} : (X \otimes Y) \otimes Z \rightarrow X \otimes (Y \otimes Z)$ for all $X, Y, Z \in \mathcal{C}$;

- natural isomorphisms (*left and right unitors*) $\lambda_X : \mathbb{I} \otimes X \rightarrow X$, $\rho_X : X \otimes \mathbb{I} \rightarrow X$ for all $X \in \mathcal{C}$;

such that the following coherence diagrams commute:

Pentagon coherence: For all $W, X, Y, Z \in \mathcal{C}$, the diagram

$$\begin{array}{ccc}
 ((W \otimes X) \otimes Y) \otimes Z & \xrightarrow{\alpha_{W \otimes X, Y, Z}} & (W \otimes X) \otimes (Y \otimes Z) \xrightarrow{\alpha_{W, X, Y \otimes Z}} W \otimes (X \otimes (Y \otimes Z)) \\
 \alpha_{W, X, Y} \otimes \text{id}_Z \downarrow & & \uparrow \text{id}_W \otimes \alpha_{X, Y, Z} \\
 (W \otimes (X \otimes Y)) \otimes Z & \xrightarrow{\alpha_{W, X \otimes Y, Z}} & W \otimes ((X \otimes Y) \otimes Z).
 \end{array}$$

Triangle coherence: For all $X, Y \in \mathcal{C}$, the diagram

$$\begin{array}{ccc}
 (X \otimes \mathbb{I}) \otimes Y & \xrightarrow{\alpha_{X, \mathbb{I}, Y}} & X \otimes (\mathbb{I} \otimes Y) \\
 \searrow \rho_X \otimes \text{id}_Y & & \downarrow \text{id}_X \otimes \lambda_Y \\
 & & X \otimes Y.
 \end{array}$$

Definition 2.0.2. Let $(\mathcal{C}, \otimes, \mathbb{I})$ be a monoidal category (Definition 2.0.1). Given objects $X, Y \in \text{Ob}(\mathcal{C})$, an *internal hom object from X to Y* is an object $\underline{\text{Hom}}(X, Y) \in \text{Ob}(\mathcal{C})$ together with a morphism

$$\text{ev}_{X, Y} : \underline{\text{Hom}}(X, Y) \otimes X \rightarrow Y$$

such that for every object $Z \in \mathcal{C}$, the assignment

$$\text{hom}_{\mathcal{C}}(Z, \underline{\text{Hom}}(X, Y)) \rightarrow \text{hom}_{\mathcal{C}}(Z \otimes X, Y), \quad f \mapsto \text{ev}_{X, Y} \circ (f \otimes \text{id}_X)$$

is a natural isomorphism of sets.

Definition 2.0.3. A *closed category* is a category $\mathcal{C}$ equipped with the following data:

- A functor

$$\underline{\text{Hom}} : \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathcal{C}$$

called the *internal hom-functor*.

- A *left evaluation morphism*

$$j_X : I \rightarrow \underline{\text{Hom}}(X, X)$$

for each object $X \in \mathcal{C}$, where I is a fixed object serving as a "unit" of the internal hom-structure.

- A family of natural *left composition morphisms*

$$L_{Y, Z}^X : \underline{\text{Hom}}(Y, Z) \otimes \underline{\text{Hom}}(X, Y) \rightarrow \underline{\text{Hom}}(X, Z)$$

(♠ **TODO: TODO: the axioms**) satisfying appropriate associativity and unit axioms that generalize composition.

Definition 2.0.4. A *closed monoidal category* is a monoidal category (Definition 2.0.1) that is a closed category (Definition 2.0.3) where the internal hom (Definition 2.0.2) is right adjoint to the monoidal tensor.

In other words, for all objects $Y, Z \in \text{Ob}(\mathcal{C})$ there is a natural isomorphism

$$\text{hom}_{\mathcal{C}}(Y \otimes X, Z) \cong \text{hom}_{\mathcal{C}}(Y, \underline{\text{Hom}}(X, Z))$$

of Hom-sets natural in Y and Z .

Definition 2.0.5. A *symmetric monoidal category* is a monoidal category (Definition 2.0.1) $(\mathcal{C}, \otimes, \mathbb{I})$ together with a natural isomorphism (symmetry)

$$\gamma_{X,Y} : X \otimes Y \xrightarrow{\cong} Y \otimes X$$

for all $X, Y \in \mathcal{C}$, such that for all $X, Y, Z \in \mathcal{C}$ the following holds:

- $\gamma_{Y,X} \circ \gamma_{X,Y} = \text{id}_{X \otimes Y}$ (involutivity);
- the **hexagon coherence diagrams** commute:

$$\begin{array}{ccc} (X \otimes Y) \otimes Z & \xrightarrow{\alpha_{X,Y,Z}} & X \otimes (Y \otimes Z) \xrightarrow{\gamma_{X,Y \otimes Z}} (Y \otimes Z) \otimes X \\ \gamma_{X,Y} \otimes \text{id}_Z \downarrow & & \uparrow \text{id}_Y \otimes \gamma_{X,Z} \\ (Y \otimes X) \otimes Z & \xrightarrow{\alpha_{Y,X,Z}} & Y \otimes (X \otimes Z) \end{array}$$

and the analogous hexagon with inverse braiding:

$$\begin{array}{ccc} X \otimes (Y \otimes Z) & \xrightarrow{\alpha_{X,Y,Z}^{-1}} & (X \otimes Y) \otimes Z \xrightarrow{\gamma_{X \otimes Y, Z}} Z \otimes (X \otimes Y) \\ \text{id}_X \otimes \gamma_{Y,Z} \downarrow & & \uparrow \gamma_{X,Z} \otimes \text{id}_Y \\ X \otimes (Z \otimes Y) & \xrightarrow{\alpha_{X,Z,Y}^{-1}} & (X \otimes Z) \otimes Y \end{array}$$

- the **symmetry coherence diagram** commutes:

$$\begin{array}{ccc} X \otimes Y & \xrightarrow{\gamma_{X,Y}} & Y \otimes X \\ & \searrow \text{id}_{X \otimes Y} & \downarrow \gamma_{Y,X} \\ & & X \otimes Y \end{array}$$

A *closed symmetric monoidal category* usually refers to a symmetric monoidal category that is closed as a monoidal category (Definition 2.0.4).

2.0.1. Types of objects in a category.

Definition 2.0.6. Let $\mathcal{C}$ be a (large) category (Definition 1.0.1).

1. An object $I \in \mathcal{C}$ is called an *initial object* if for every object $X \in \mathcal{C}$ there exists a unique morphism

$$I \rightarrow X.$$

Equivalently, an initial object is a limit (Definition 1.2.2) of the empty diagram (Definition 1.2.1), if such a limit exists.

2. An object $F \in \mathcal{C}$ is called a **final object** (or **terminal object**) if for every object $X \in \mathcal{C}$ there exists a unique morphism

$$X \rightarrow F.$$

Equivalently, a final object is a colimit (Definition 1.2.2) of the empty diagram (Definition 1.2.1), if such a colimit exists.

3. An object $Z \in \mathcal{C}$ is called a **zero object** if Z is both initial and final in $\mathcal{C}$. In particular, for every object $X \in \mathcal{C}$ there exist unique morphisms

$$Z \rightarrow X \quad \text{and} \quad X \rightarrow Z.$$

In particular, if initial/final/zero objects exist in a category, then they are unique up to unique isomorphism.

Lemma 2.0.7. Let $\mathcal{C}$ be a full subcategory (Definition 1.3.2) of a (large) category (Definition 1.0.1) $\mathcal{D}$. Suppose that $\mathcal{D}$ has an initial object (Definition 2.0.6) I (resp. a final object (Definition 2.0.6) F) and that this object also belongs to $\mathcal{C}$. The object is initial (resp. final) in $\mathcal{C}$.

Definition 2.0.8. Let C and D be categories (Definition 1.0.1), let $S : C^{op} \times C \rightarrow D$ be a functor, and let d be an object of D .

A **wedge from d to S** is a family of morphisms $\omega = \{\omega_c\}_{c \in \text{Ob}(C)}$ in D , denoted

$$\omega : d \rightarrow S,$$

where each component is a morphism $\omega_c : d \rightarrow S(c, c)$, satisfying the wedge condition: for every morphism $f : c \rightarrow c'$ in C , the diagram

$$S(1_c, f) \circ \omega_c = S(f, 1_{c'}) \circ \omega_{c'}$$

commutes in D . (Explicitly, both composites are morphisms $d \rightarrow S(c, c')$).

Dually, a **cowedge from S to d** is a family of morphisms $\beta = \{\beta_c\}_{c \in \text{Ob}(C)}$ in D , denoted

$$\beta : S \rightarrow d,$$

where each component is a morphism $\beta_c : S(c, c) \rightarrow d$, satisfying the **cowedge condition**: for every morphism $f : c \rightarrow c'$ in C , the equality

$$\beta_c \circ S(f, 1_c) = \beta_{c'} \circ S(1_{c'}, f)$$

holds in D (as morphisms from $S(c', c)$ to d).

Definition 2.0.9. Let C and D be categories (Definition 1.0.1) and let $S : C^{op} \times C \rightarrow D$ be a functor.

An **end of S** is a pair (E, ω) , where E is an object of D and $\omega : E \rightarrow S$ is a wedge (Definition 2.0.8), which is universal among all wedges to S . That is, for any object $d \in D$ and any wedge $\nu : d \rightarrow S$, there exists a unique morphism $h : d \rightarrow E$ in D such that $\nu_c = \omega_c \circ h$

for all $c \in \text{Ob}(C)$. The object E is uniquely determined up to unique isomorphism and is denoted by

$$\int_{c \in C} S(c, c) \quad \text{or} \quad \int_C S.$$

Dually, a **coend of S** is a pair (K, β) , where K is an object of D and $\beta : S \rightarrow K$ is a cowedge, which is universal among all cowedges from S . That is, for any object $d \in D$ and any cowedge $\gamma : S \rightarrow d$, there exists a unique morphism $g : K \rightarrow d$ in D such that $\gamma_c = g \circ \beta_c$ for all $c \in \text{Ob}(C)$. The object K is denoted by

$$\int^{c \in C} S(c, c) \quad \text{or} \quad \int^C S.$$

3. ADDITIVE AND ABELIAN CATEGORIES

(♠ TODO: Carefully check what statements need to assume that the categories are locally small and which do not)

Definition 3.0.1 (Additive category). Let $\mathcal{A}$ be a locally small category (Definition 1.0.9).

1. $\mathcal{A}$ is said to be a **preadditive category** if the following hold:
 - For any two objects A, B in $\mathcal{A}$, the set $\text{Hom}_{\mathcal{A}}(A, B)$ is an abelian group (Definition C.0.5), and composition of morphisms is bilinear.
 - There is a zero object (Definition 2.0.6) 0 in $\mathcal{A}$.

Equivalently, a preadditive category $\mathcal{A}$ is a (necessarily locally small) category enriched in (Definition 4.0.1) the monoidal category (Definition 2.0.1) Ab that also possesses a zero object.

2. If $\mathcal{A}$ is preadditive, then it is called **additive** if it additionally satisfies the following:
 - For any two objects A, B in $\mathcal{A}$, there exists a product object $A \times B$ (Definition 1.2.36), often written $A \oplus B$, called the **direct sum of A and B** . In fact, $A \oplus B$ is not only a product but also a coproduct of A and B .

Given a finite collection $\{A_i\}_i$ of objects A_i in an additive category $\mathcal{A}$, we may more generally speak of the **direct sum** $\bigoplus_i A_i$; it has canonical injections from and projections to each A_i .

Lemma 3.0.2. (♠ TODO: AI generated, verify) Let $\mathcal{A}$ be a preadditive category (Definition 3.0.1) with a zero object (Definition 2.0.6) 0 . Then for every object $A \in \mathcal{A}$, the Hom-groups (Definition 3.0.1) $\text{Hom}_{\mathcal{A}}(A, 0)$ and $\text{Hom}_{\mathcal{A}}(0, A)$ are the zero group (Definition C.0.6) (i.e., the trivial abelian group consisting of a single element, the zero morphism).

In particular, there is exactly one morphism $A \rightarrow 0$ and exactly one morphism $0 \rightarrow A$, both of which are the zero morphism in the respective Hom-groups.

Definition 3.0.3 (Additive functor). Let $\mathcal{A}$ and $\mathcal{B}$ be pre-additive categories (Definition 3.0.1). A functor

$$F : \mathcal{A} \rightarrow \mathcal{B}$$

is an **additive functor** if for every pair of objects $A, A' \in \mathcal{A}$, the induced map

$$F_{A,A'} : \text{Hom}_{\mathcal{A}}(A, A') \rightarrow \text{Hom}_{\mathcal{B}}(F(A), F(A'))$$

is a group homomorphism of abelian groups, or equivalently if it is enriched over the category Ab of abelian groups (Definition 4.0.1).

In the case that $\mathcal{A}$ and $\mathcal{B}$ are additive categories (Definition 3.0.1), this condition implies that F is compatible with the additive structures on $\mathcal{A}$ and $\mathcal{B}$. More precisely,

- F sends the zero object $0_{\mathcal{A}}$ of $\mathcal{A}$ to the zero object $0_{\mathcal{B}}$ of $\mathcal{B}$, i.e.,

$$F(0_{\mathcal{A}}) = 0_{\mathcal{B}}.$$

- F preserves finite direct sums: For any finite family of objects $\{A_i\}_{i=1}^n$ in $\mathcal{A}$,

$$F\left(\bigoplus_{i=1}^n A_i\right) \cong \bigoplus_{i=1}^n F(A_i)$$

via the canonical isomorphism induced by F applied to the canonical injections and projections.

Definition 3.0.4 (Abelian category). Let $\mathcal{A}$ be a category. The category $\mathcal{A}$ is an **abelian category** if:

- $\mathcal{A}$ is an additive category (Definition 3.0.1).
- Every morphism $f : A \rightarrow B$ has a kernel $\ker(f)$ and a cokernel $\text{coker}(f)$ (Definition 1.3.8).
- For every morphism $f : A \rightarrow B$, the canonical morphism $\text{coim}(f) \rightarrow \text{im}(f)$ is an isomorphism, where

$$\text{coim}(f) = \text{coker}(\ker(f) \rightarrow A), \quad \text{im}(f) = \ker(B \rightarrow \text{coker}(f)).$$

(♠ TODO: I think I need to re-check this definition) (♠ TODO: coimage)

.

It is also worth considering Grothendieck's additional axioms for abelian categories (Definition 3.0.5).

Definition 3.0.5 (Grothendieck's axioms for abelian categories (Ab1–Ab5)). Let $\mathcal{A}$ be an abelian category (Definition 3.0.4).

Grothendieck introduced the following hierarchy of additional axioms to express stronger completeness and exactness properties in $\mathcal{A}$ — we note that Ab1, Ab2, and Ab2* are already satisfied for any abelian category:

- **Ab1**: Every morphism in $\mathcal{A}$ has a kernel and a cokernel (Definition 1.3.8).
- **Ab2**: Every monic (Definition 1.3.7) in $\mathcal{A}$ is the kernel of its cokernel.
- **Ab2***: Every epi in $\mathcal{A}$ is the cokernel of its kernel.
- **AB3**: The category $\mathcal{A}$ is cocomplete (Definition 1.3.9).

- Since $\mathcal{A}$ is abelian (and hence admits equalizers (Definition 1.2.37) as kernels), this is equivalent to requiring that $\mathcal{A}$ has all small coproducts (Definition 1.2.36) (direct sums).
- **AB4**: The category $\mathcal{A}$ satisfies AB3, and coproducts are exact (Definition 3.0.7) (when regarded as functors $\bigoplus_I : \mathcal{A}^I \rightarrow \mathcal{A}$).
 - That is, the coproduct of a family of short exact sequences is a short exact sequence. Explicitly, for any family of short exact sequences $0 \rightarrow A_i \rightarrow B_i \rightarrow C_i \rightarrow 0$ indexed by a set I , the sequence

$$0 \rightarrow \bigoplus_{i \in I} A_i \rightarrow \bigoplus_{i \in I} B_i \rightarrow \bigoplus_{i \in I} C_i \rightarrow 0$$

is exact in $\mathcal{A}$.

- **AB5**: The category $\mathcal{A}$ satisfies AB3, and filtered colimits (Definition 1.2.12) are exact (Definition 3.0.7) (when regarded as functors $\varinjlim_J : \mathcal{A}^J \rightarrow \mathcal{A}$).
 - Equivalently, for any filtered (Definition 1.2.8) index category J and any directed system (Definition 1.2.11) of short exact sequences $0 \rightarrow A_j \rightarrow B_j \rightarrow C_j \rightarrow 0$, the colimit sequence

$$0 \rightarrow \varinjlim A_j \rightarrow \varinjlim B_j \rightarrow \varinjlim C_j \rightarrow 0$$

is exact.

- Note: AB5 implies AB4. An abelian category satisfying AB5 and having a generator (Definition 5.1.60) is called a *Grothendieck category*.
- **AB6**: The category $\mathcal{A}$ satisfies AB3, and for any object X and any family of filtered subobjects $\{F_i\}_{i \in I}$ of X (where each F_i is a filter of subobjects), the intersection commutes with the limit:

$$\bigcap_{i \in I} (\varinjlim_{j \in F_i} U_{i,j}) = \varinjlim_{(j_i) \in \prod F_i} (\bigcap_{i \in I} U_{i,j_i}).$$

(This axiom is less commonly cited but appears in Grothendieck's Tohoku paper).

- **AB5***: The category $\mathcal{A}$ is complete (Definition 1.3.9) (i.e., has all small products).
- **AB4***: The category $\mathcal{A}$ satisfies AB3* and products are exact.
 - Note: This is rarely satisfied for module categories (e.g., it fails for Abelian groups), but it is satisfied for the category of sheaves on a space.
- **AB5***: The category $\mathcal{A}$ satisfies AB3* and filtered limits (inverse limits) are exact.

Notes:

- AB5 implies AB4, and AB4 implies AB3.
- AB5* implies AB4*, and AB4* implies AB3*.

Proposition 3.0.6. The following are examples of abelian categories (Definition 3.0.4):

1. The category of R - S bimodules where R, S are (not necessarily commutative) rings (Definition C.0.1) (??).
2. The category **Ab** of abelian groups and group homomorphisms is abelian.
3. The category $\mathbf{Vect}_k$ of vector spaces over a field k and k -linear maps is abelian.

4. More generally, if R is a noetherian ring, then the category of finitely generated R -modules is abelian.
5. For a ringed space $(X, \mathcal{O}_X)$, the category of $\mathcal{O}_X$ -modules (Definition 5.3.9) is abelian. (♠ TODO: a quasi-coherent sheaf on a locally ringed space)
6. If X is a scheme (or more generally a locally ringed space), the category of quasi-coherent sheaves on X is abelian.
7. For any essentially small category (Definition 1.1.9) $\mathcal{C}$ and any abelian category $\mathcal{A}$, the functor category $[\mathcal{C}, \mathcal{A}]$ (Definition 1.2.1) and the category $\text{PreShv}(\mathcal{C}, \mathcal{A})$ of presheaves (Definition 5.0.1) are abelian. (♠ TODO: apparently, the essentially smallness condition is removable, provided that the sheafification functor exists. However, the essentially small assumption is needed to show that the category of sheaves of $\mathcal{O}$ -modules is a Grothendieck abelian category. Verify all this. Moreover, when working with a big site of a scheme, one typically fixes a universe or work relative to a cardinal cutoff to treat it as essentially small)
8. For any site (Definition 5.1.13) $(\mathcal{C}, J)$ on an essentially small category (Definition 1.1.9) $\mathcal{C}$ and any abelian category $\mathcal{A}$, the category $\text{Shv}(\mathcal{C}, \mathcal{A})$ of sheaves (Definition 5.1.32) is abelian.
9. For any site (Definition 5.1.13) $(\mathcal{C}, J)$ on an essentially small category (Definition 1.1.9) $\mathcal{C}$ and a sheaf of rings (Definition 5.1.32) $\mathcal{O}$ on $\mathcal{C}$, the category $\mathbf{Mod}(\mathcal{O})$ of $\mathcal{O}$ -modules (Definition 5.3.9) is an abelian category.

Definition 3.0.7. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 3.0.3) between abelian categories (Definition 3.0.4).

1. F is called *left exact* if it preserves all finite limits (Definition 1.2.2); since F is an additive functor between abelian categories, this is equivalent to saying that F preserves kernels (Definition 1.3.8)¹

Equivalently, for every left exact sequence in $\mathcal{A}$

$$0 \rightarrow A' \xrightarrow{f} A \xrightarrow{g} A''$$

the sequence

$$0 \rightarrow F(A') \xrightarrow{F(f)} F(A) \xrightarrow{F(g)} F(A'')$$

is exact at $F(A')$ and $F(A)$ (i.e., F preserves monomorphisms (Definition 1.3.7) and exactness at the first two terms).

2. Dually, F is called *right exact* if it preserves all finite colimits (Definition 1.2.2); since F is an additive functor between abelian categories, this is equivalent to saying that F preserves cokernels (Definition 1.3.8).

¹To see this equivalence, note that

- finite products (Definition 1.2.36) coincide with finite direct sums (Definition 3.0.1) in abelian categories,
- additive functors preserve finite direct sums by definition,
- kernels (Definition 1.3.8) in a pointed category are equalizers (Definition 1.2.37), and all equalizers in abelian categories are kernels,
- if all finite products and equalizers exist in a general category, then all finite limits exist in the category (Theorem 1.2.6)

Equivalently, for every right exact sequence in $\mathcal{A}$

$$A' \xrightarrow{f} A \xrightarrow{g} A'' \rightarrow 0,$$

the sequence

$$F(A') \xrightarrow{F(f)} F(A) \xrightarrow{F(g)} F(A'') \rightarrow 0$$

is exact at $F(A)$ and $F(A'')$ (i.e., F preserves epimorphisms (Definition 1.3.7) and exactness at the last two terms).

3. F is called **exact** if it is both left and right exact.

The additive functor F is left/right exact if and only if it is left/right exact (Definition 1.2.18) in the more general sense, i.e. if it preserves all finite (Definition 1.2.5) limits/colimits (Definition 1.2.2)

Lemma 3.0.8. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 3.0.3) between abelian categories (Definition 3.0.4). The functor F is left exact/right exact/exact (Definition 3.0.7) as an additive functor if and only if it is left exact/right exact/exact (Definition 1.2.18) as a functor.

3.1. Simple and semisimple objects in an additive category.

Definition 3.1.1. Let $\mathcal{C}$ be a category (Definition 1.0.1). Let $X \in \text{Ob}(\mathcal{C})$ be an object. A **subobject of X** refers to a monomorphism (Definition 1.3.7) $i : Y \hookrightarrow X$ in $\mathcal{C}$. We regard two subobjects (Y, i) and (Y', i') of X as isomorphic if there exists an isomorphism $f : Y \rightarrow Y'$ such that $i = i' \circ f$. One often leaves the monomorphism i implicit, suppressing it from the notation. One also often write $Y \subseteq X$ to denote that Y is a subobject of X .

Unlike the notion of a subobject, the notion of a quotient object is more appropriate to speak of for an abelian category rather than a more general additive category.

Definition 3.1.2. Let $\mathcal{C}$ be an abelian category (Definition 3.0.4). Let $X \in \text{Ob}(\mathcal{C})$ be an object. Let $i : A \hookrightarrow X$ be a subobject (Definition 3.1.1). The cokernel (Definition 1.3.8) $\pi : X \rightarrow X/A := \text{coker}(i)$ is called the **quotient object of X by A** . The object X/A is determined up to canonical isomorphism.

Definition 3.1.3. Let $\mathcal{C}$ be an abelian category (Definition 3.0.4). Let $X \in \text{Ob}(\mathcal{C})$, and let $A \hookrightarrow B \hookrightarrow X$ be subobjects of X . The quotient object (Definition 3.1.2) $B/A := \text{coker}(A \hookrightarrow B)$ is called a **subquotient object of X** .

Definition 3.1.4. Let $\mathcal{C}$ be an additive category (Definition 3.0.1) and $X \in \text{Ob}(\mathcal{C})$. A subobject (Definition 1.3.1) $i : Y \hookrightarrow X$ is called a **direct summand of X** if there exists a morphism $p : X \rightarrow Y$ such that $p \circ i = \text{id}_Y$. In this case, X is isomorphic to the direct sum (Definition 3.0.1) $Y \oplus Z$ for some object Z , with i the canonical inclusion.

Definition 3.1.5. Let $\mathcal{C}$ be an additive category (Definition 3.0.1). An object $S \in \text{Ob}(\mathcal{C})$ is called **simple** (or **irreducible**) if $S \neq 0$ and the only subobjects (Definition 3.1.1) of S are 0 and S itself (up to isomorphism of subobjects).

Definition 3.1.6. Let $\mathcal{C}$ be an additive category (Definition 3.0.1). An object $X \in \text{Ob}(\mathcal{C})$ is called **semisimple** if it is isomorphic to a finite direct sum (Definition 3.0.1) of simple objects (Definition 3.1.5) in $\mathcal{C}$.

Definition 3.1.7. An additive category (Definition 3.0.1) $\mathcal{C}$ is called a *semisimple category* if every object of $\mathcal{C}$ is semisimple (Definition 3.1.6).

(♠ TODO: TODO: Give examples of monoidal categories)

4. CATEGORIES ENRICHED IN MONOIDAL CATEGORIES

(♠ TODO: Carefully check what statements need to assume that the categories are locally small and which do not)

We might want to study categories whose hom-sets carry additional structure beyond that of a set (e.g. an abelian group, vector space); the theory of categories enriched (Definition 4.0.1) in monoidal categories accommodates such desires. In fact, enriched categories can have hom-objects which are not sets.

Definition 4.0.1 (Category enriched in a monoidal category). Let $(\mathcal{V}, \otimes, \mathbf{1})$ be a monoidal category (Definition 2.0.1). A *category enriched in $\mathcal{V}$* (or a *$\mathcal{V}$ -enriched category* or a *$\mathcal{V}$ -category*) $\mathcal{C}$ consists of the following data:

- A class $\text{Ob}(\mathcal{C})$ of *objects*. As with regular categories (Definition 1.0.1), we may write $X \in \text{Ob}(\mathcal{C})$ or $X \in \mathcal{C}$ to mean that X is an object of $\mathcal{C}$.
- For each pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, an object $\underline{\text{Hom}}_{\mathcal{C}}(X, Y) \in \text{Ob}(\mathcal{V})$ of *morphisms*; it is an object of the monoidal category $\mathcal{V}$. It is also often denoted by notations such as $\mathcal{C}(X, Y)$, $\underline{\text{Hom}}(X, Y) = \underline{\text{Hom}}_{\mathcal{C}}(X, Y)$, or $\text{Mor}(X, Y) = \text{Mor}_{\mathcal{C}}(X, Y)$.
- For each triple $X, Y, Z \in \text{Ob}(\mathcal{C})$, a *composition morphism*

$$\mu_{X,Y,Z} : \underline{\text{Hom}}_{\mathcal{C}}(Y, Z) \otimes \underline{\text{Hom}}_{\mathcal{C}}(X, Y) \rightarrow \underline{\text{Hom}}_{\mathcal{C}}(X, Z).$$

It is a morphism in $\mathcal{V}$.

- For each object X , a *unit morphism* $\eta_X : \mathbf{1} \rightarrow \underline{\text{Hom}}_{\mathcal{C}}(X, X)$ in $\mathcal{V}$.

These data satisfy the following axioms:

- (Associativity) For all $W, X, Y, Z \in \text{Ob}(\mathcal{C})$, the following diagram in $\mathcal{V}$ commutes:

$$\begin{array}{ccc}
 (\underline{\text{Hom}}_{\mathcal{C}}(Z, W) \otimes \underline{\text{Hom}}_{\mathcal{C}}(Y, Z)) \otimes \underline{\text{Hom}}_{\mathcal{C}}(X, Y) & \xrightarrow{\alpha} & \underline{\text{Hom}}_{\mathcal{C}}(Z, W) \otimes (\underline{\text{Hom}}_{\mathcal{C}}(Y, Z) \otimes \underline{\text{Hom}}_{\mathcal{C}}(X, Y)) \\
 \downarrow \mu \otimes \text{id} & & \downarrow \text{id} \otimes \mu \\
 \underline{\text{Hom}}_{\mathcal{C}}(Y, W) \otimes \underline{\text{Hom}}_{\mathcal{C}}(X, Y) & & \underline{\text{Hom}}_{\mathcal{C}}(Z, W) \otimes \underline{\text{Hom}}_{\mathcal{C}}(X, Z) \\
 \downarrow \mu & & \downarrow \mu \\
 \underline{\text{Hom}}_{\mathcal{C}}(X, W) & \xlongequal{\quad\quad\quad} & \underline{\text{Hom}}_{\mathcal{C}}(X, W)
 \end{array}$$

where α is the associativity constraint in $\mathcal{V}$.

- (Unit) For all $X, Y \in \text{Ob}(\mathcal{C})$, the following diagrams commute:

$$\begin{array}{ccc}
\mathbf{1} \otimes \underline{\mathrm{Hom}}_{\mathcal{C}}(X, Y) & \xrightarrow{\eta_Y \otimes \mathrm{id}} & \underline{\mathrm{Hom}}_{\mathcal{C}}(Y, Y) \otimes \underline{\mathrm{Hom}}_{\mathcal{C}}(X, Y) \\
& \searrow \lambda & \downarrow \mu \\
& & \underline{\mathrm{Hom}}_{\mathcal{C}}(X, Y) \\
\underline{\mathrm{Hom}}_{\mathcal{C}}(X, Y) \otimes \mathbf{1} & \xrightarrow{\mathrm{id} \otimes \eta_X} & \underline{\mathrm{Hom}}_{\mathcal{C}}(X, Y) \otimes \underline{\mathrm{Hom}}_{\mathcal{C}}(X, X) \\
& \searrow \rho & \downarrow \mu \\
& & \underline{\mathrm{Hom}}_{\mathcal{C}}(X, Y)
\end{array}$$

where λ and ρ are the left and right unit constraints in $\mathcal{V}$.

- Example 4.0.2.**
1. A category enriched in (Definition 4.0.1) **Sets** is equivalent to a locally small category (Definition 1.0.9).
 2. A category enriched in **Ab** is equivalent to a pre-additive category (Definition 3.0.1).
(♠ TODO: TODO: define an R -linear category and a k -linear category)
 3. Let R be a commutative ring. A category enriched in $R\text{-Mod}$ is equivalent to a R -linear category.
 4. Let k be a field. A category enriched in $k\text{-Vect}$ is equivalent to a k -linear category.

Definition 4.0.3. Let $\mathcal{V}$ be a monoidal category (Definition 2.0.1) and let $\mathcal{C}$ and $\mathcal{D}$ be categories enriched in $\mathcal{V}$ (Definition 4.0.1). A $\mathcal{V}$ -functor between $\mathcal{C}$ and $\mathcal{D}$ or an enriched functor between $\mathcal{C}$ and $\mathcal{D}$, written

$$F : \mathcal{C} \rightarrow \mathcal{D},$$

consists of:

- a function on objects $F : \mathrm{Ob}(\mathcal{C}) \rightarrow \mathrm{Ob}(\mathcal{D})$,
- for all $A, B \in \mathrm{Ob}(\mathcal{C})$, morphisms in $\mathcal{V}$,

$$F_{A,B} : \mathcal{C}(A, B) \rightarrow \mathcal{D}(FA, FB),$$

such that the following diagrams in $\mathcal{V}$ commute:

$$\begin{array}{ccc}
\mathcal{C}(B, C) \otimes \mathcal{C}(A, B) & \xrightarrow{\circ_{\mathcal{C}}} & \mathcal{C}(A, C) \\
\downarrow F_{B,C} \otimes F_{A,B} & & \downarrow F_{A,C} \\
\mathcal{D}(FB, FC) \otimes \mathcal{D}(FA, FB) & \xrightarrow{\circ_{\mathcal{D}}} & \mathcal{D}(FA, FC)
\end{array}$$

and for all $A \in \mathrm{Ob}(\mathcal{C})$, the unit compatibility condition holds:

$$\iota_{FA} = F_{A,A} \circ \iota_A.$$

Definition 4.0.4. Let $\mathcal{V}$ be a monoidal category (Definition 2.0.1), let $\mathcal{A}, \mathcal{B}$ be $\mathcal{V}$ -categories (Definition 4.0.1), and let $F, G : \mathcal{A} \rightarrow \mathcal{B}$ be $\mathcal{V}$ -functors (Definition 4.0.3).

A $\mathcal{V}$ -natural transformation $\alpha : F \Rightarrow G$ consists of a family of morphisms in $\mathcal{V}_0$:

$$\alpha_x : I \rightarrow \mathcal{B}(Fx, Gx)$$

indexed by objects $x \in \mathrm{Ob}(\mathcal{A})$, satisfying the $\mathcal{V}$ -naturality axiom.

For every pair of objects $x, y \in \text{Ob}(\mathcal{A})$, the following hexagonal diagram in $\mathcal{V}_0$ commutes (expressing the enriched analogue of $\alpha_y \circ F(f) = G(f) \circ \alpha_x$):

$$\begin{array}{ccccc}
\mathcal{A}(x, y) & \xrightarrow{F_{xy}} & \mathcal{B}(Fx, Fy) & \xrightarrow[\cong]{\lambda^{-1}} & I \otimes \mathcal{B}(Fx, Fy) \\
\downarrow G_{xy} & & & & \downarrow \alpha_y \otimes 1 \\
\mathcal{B}(Gx, Gy) & & & & \mathcal{B}(Fy, Gy) \otimes \mathcal{B}(Fx, Fy) \quad \text{Here,} \\
\downarrow \rho^{-1} \cong & & & & \downarrow M_{Fx, Fy, Gy}^{\mathcal{B}} \\
\mathcal{B}(Gx, Gy) \otimes I & \xrightarrow{1 \otimes \alpha_x} & \mathcal{B}(Gx, Gy) \otimes \mathcal{B}(Fx, Gx) & \xrightarrow{M_{Fx, Gx, Gy}^{\mathcal{B}}} & \mathcal{B}(Fx, Gy)
\end{array}$$

$\lambda : I \otimes X \cong X$ and $\rho : X \otimes I \cong X$ denote the left and right unit isomorphisms of the monoidal category $\mathcal{V}$, and $M^{\mathcal{B}}$ denotes the composition in $\mathcal{B}$.

Definition 4.0.5. Let $\mathcal{V} = (\mathcal{V}_0, \otimes, I, \alpha, \lambda, \rho)$ be a monoidal category (Definition 2.0.1), and let $\mathcal{A}$ be a $\mathcal{V}$ -category (Definition 4.0.1) with object set $\text{Ob}(\mathcal{A})$ and hom-objects $\mathcal{A}(X, Y) \in \text{Ob}(\mathcal{V}_0)$.

The *underlying ordinary category* of $\mathcal{A}$, denoted $\mathcal{A}_0$, is the category (Definition 1.0.1) defined as follows:

1. The objects of $\mathcal{A}_0$ are the objects of $\mathcal{A}$, i.e., $\text{Ob}(\mathcal{A}_0) = \text{Ob}(\mathcal{A})$.
2. For any two objects $X, Y \in \text{Ob}(\mathcal{A}_0)$, the set of morphisms is defined by

$$\text{Hom}_{\mathcal{A}_0}(X, Y) = \text{Hom}_{\mathcal{V}_0}(I, \mathcal{A}(X, Y)),$$

where $\text{Hom}_{\mathcal{V}_0}$ denotes the hom-set in the ordinary category $\mathcal{V}_0$.

3. The identity morphism $\text{id}_X \in \text{Hom}_{\mathcal{A}_0}(X, X)$ is the element corresponding to the identity morphism $j_X : I \rightarrow \mathcal{A}(X, X)$ in $\mathcal{V}$.
4. The composition of morphisms $f \in \text{Hom}_{\mathcal{A}_0}(X, Y)$ and $g \in \text{Hom}_{\mathcal{A}_0}(Y, Z)$ (represented by morphisms $f : I \rightarrow \mathcal{A}(X, Y)$ and $g : I \rightarrow \mathcal{A}(Y, Z)$ in $\mathcal{V}_0$) is defined to be the composition $g \circ f$ in $\mathcal{V}_0$ given by the following diagram in $\mathcal{V}_0$:

$$I \xrightarrow{\cong} I \otimes I \xrightarrow{g \otimes f} \mathcal{A}(Y, Z) \otimes \mathcal{A}(X, Y) \xrightarrow{M_{XYZ}} \mathcal{A}(X, Z),$$

where M_{XYZ} is the composition morphism of the enriched category $\mathcal{A}$.

Definition 4.0.6. Let C be a category enriched in a monoidal category (Definition 4.0.1) $\mathcal{V}$. Given an object X of C , the *functor of points* h_X is the functor (Definition 1.1.1)/presheaf (Definition 5.0.1) $C^{\text{op}} \rightarrow \mathcal{V}$ given by $T \mapsto \text{Hom}_C(T, X)$. A functor $C^{\text{op}} \rightarrow \mathcal{V}$ (or equivalently, a presheaf on C valued in $\mathcal{V}$) is said to be *representable* if it is naturally isomorphic (Definition 1.1.3) to some functor h_X of points for an object X of C .

Dually, a functor $C \rightarrow \mathcal{V}$ is called *co-representable* if it is naturally isomorphic to a functor $T \mapsto \text{Hom}_C(X, T)$ for an object X in C .

For instance, we may speak of these notions when $\mathcal{V}$ is the monoidal category **Sets**, i.e. C is a locally small category (Definition 1.0.9).

Theorem 4.0.7 ((Enriched) Yoneda lemma). (♠ TODO: functor of enriched categories, natural transform) Let C be a $\mathcal{V}$ -enriched category (Definition 4.0.1) where $\mathcal{V}$ is a monoidal

category (Definition 2.0.1), and let $F : C^{\text{op}} \rightarrow \mathcal{V}$ be a $\mathcal{V}$ -functor. For any object $X \in C$, there is a natural isomorphism:

$$\text{Nat}_{\mathcal{V}}(h_X, F) \cong F(X),$$

where $h_X = \text{Hom}_C(-, X)$ is the representable functor associated to X (Definition 4.0.6), and $\text{Nat}_{\mathcal{V}}(h_X, F)$ is the $\mathcal{V}$ -object of $\mathcal{V}$ -natural transformations.

(♠ **TODO: end**) Alternatively, if $\mathcal{V}$ is closed (Definition 2.0.4) and locally small (Definition 1.0.9), then

$$\text{Nat}_{\mathcal{V}}(h_X, F) \cong \int_{T \in C} [\text{Hom}_C(T, X), F(T)],$$

where $[_, _]$ denotes the internal hom (Definition 2.0.2) in $\mathcal{V}$ and $\int$ is the end.

In particular, when $\mathcal{V} = \mathbf{Sets}$ (so C is simply a locally small category (Definition 1.0.9)), the (classical) Yoneda lemma states the following: let $F : C^{\text{op}} \rightarrow \mathbf{Sets}$ a functor, and let X be an object of C . Then there is a natural isomorphism

$$\text{Nat}(h_X, F) \cong F(X).$$

5. SITES, SHEAVES AND TOPOI

Definition 5.0.1 (Presheaf on a category). Let C and $\mathcal{A}$ be (large) categories (Definition 1.0.1).

1. A *presheaf* $\mathcal{F}$ on C with values in $\mathcal{A}$ is a functor

$$\mathcal{F} : C^{\text{op}} \rightarrow \mathcal{A}.$$

In other words, a presheaf $\mathcal{F}$ on C with values in $\mathcal{A}$ is simply a contravariant functor (Definition 1.1.1) from C to $\mathcal{A}$. Explicitly, for every object U in C , one has an object $\mathcal{F}(U)$ in $\mathcal{A}$ (called the *U -valued sections/sections evaluated at U of $\mathcal{F}$* , cf. Definition 5.3.11), and for every morphism $f : V \rightarrow U$ in C , one has a morphism (called the *restriction map*)

$$\mathcal{F}(f) : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$$

in $\mathcal{A}$, such that for all composable morphisms $W \xrightarrow{g} V \xrightarrow{f} U$ in C , the following diagram in $\mathcal{A}$ commutes:

$$\begin{array}{ccccc} & & \mathcal{F}(f \circ g) & & \\ & \nearrow & & \searrow & \\ \mathcal{F}(U) & \xrightarrow{\mathcal{F}(f)} & \mathcal{F}(V) & \xrightarrow{\mathcal{F}(g)} & \mathcal{F}(W) \end{array}$$

That is,

$$\mathcal{F}(g) \circ \mathcal{F}(f) = \mathcal{F}(f \circ g),$$

and for every object U in C , $\mathcal{F}(\text{id}_U) = \text{id}_{\mathcal{F}(U)}$.

2. Let $\mathcal{F}, \mathcal{G} : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ be two presheaves on $\mathcal{C}$ with values in $\mathcal{A}$. A **morphism of presheaves**

$$\varphi : \mathcal{F} \rightarrow \mathcal{G}$$

is a natural transformation of functors (Definition 1.1.3): for each object U of $\mathcal{C}$, one has a morphism

$$\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

in $\mathcal{A}$, such that for every morphism $f : V \rightarrow U$ in $\mathcal{C}$, the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\mathcal{F}(f)} & \mathcal{F}(V) \\ \varphi_U \downarrow & & \downarrow \varphi_V \\ \mathcal{G}(U) & \xrightarrow{\mathcal{G}(f)} & \mathcal{G}(V) \end{array}$$

commutes, i.e.,

$$\varphi_V \circ \mathcal{F}(f) = \mathcal{G}(f) \circ \varphi_U$$

for all objects and morphisms in $\mathcal{C}$.

3. Given a universe (Definition E.0.1) U , a **U -presheaf on $\mathcal{C}$** typically refers to a presheaf of U -sets on $\mathcal{C}$.
4. The **presheaf category/category of $\mathcal{A}$ -valued presheaves on $\mathcal{C}$** is the (large) category whose objects are the presheaves on $\mathcal{C}$ with values in $\mathcal{A}$ and whose morphisms are the presheaf morphisms. Common notations for the presheaf category include, but are not limited to: $\mathcal{A}^{\mathcal{C}^{\text{op}}}$, $\text{PreShv}(\mathcal{C}, \mathcal{A})$, $[\mathcal{C}^{\text{op}}, \mathcal{A}]$. If the value category $\mathcal{A}$ is clear from context, then notations such as $\text{PreShv}(\mathcal{C})$ are also common. Note that the presheaf category $\text{PreShv}(\mathcal{C}, \mathcal{A})$ is equivalent to the category of functors (Definition 1.2.1) $\mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ and hence notations for the functor categories are applicable as notations for presheaf categories.

Definition 5.0.2. Let $\mathcal{C}$ be a category (Definition 1.0.1), and let $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ be a presheaf (Definition 5.0.1) on $\mathcal{C}$.

A **subfunctor of F** is a presheaf G together with a natural transformation (Definition 1.1.3) $\iota : G \hookrightarrow F$ such that for every object U of $\mathcal{C}$, the component map $\iota_U : G(U) \rightarrow F(U)$ is **injective**. Equivalently, a subfunctor of F is a subobject (Definition 1.3.1) of F as an object of $\text{PSh}(\mathcal{C}, \mathbf{Sets})$ (Proposition 5.0.3).

Equivalently, a subfunctor of F can be described as a collection of subsets $S(U) \subseteq F(U)$ for each object U of $\mathcal{C}$, such that for every morphism $f : V \rightarrow U$ in $\mathcal{C}$, the restriction map $F(f) : F(U) \rightarrow F(V)$ satisfies $F(f)(S(U)) \subseteq S(V)$. The two descriptions are equivalent: given a collection $\{S(U)\}_U$, one defines $G(U) = S(U)$ and ι_U as the inclusion map; conversely, given an injective natural transformation $\iota : G \hookrightarrow F$, one sets $S(U) = \text{im}(\iota_U) \subseteq F(U)$.

Proposition 5.0.3. Let $\mathcal{C}$ be a category (Definition 1.0.1). The category of presheaves on $\mathcal{C}$, denoted by $\text{PSh}(\mathcal{C}) = [\mathcal{C}^{\text{op}}, \mathbf{Set}]$, is a topos (Definition 5.1.55), and limits and colimits in this functor category are computed pointwise.

A morphism $\alpha : F \rightarrow G$ between presheaves on $\mathcal{C}$ is a **monomorphism** in the functor category $\text{PSh}(\mathcal{C})$ if and only if for every object U of $\mathcal{C}$, the component map $\alpha_U : F(U) \rightarrow G(U)$ is an **injective function**.

Equivalently, a subobject of a presheaf F in $\mathbf{PSh}(\mathcal{C})$ is represented by a subfunctor (Definition 5.0.2) of F , i.e., an equivalence class of monomorphisms $\iota : G \hookrightarrow F$ such that each component $\iota_U : G(U) \rightarrow F(U)$ is injective.

Proposition 5.0.4 (Images of Morphisms in Set-Valued Presheaves are Computed Pointwise). Let $\mathcal{C}$ be a category (Definition 1.0.1). Let $F, G : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ be presheaves (Definition 5.0.1) on $\mathcal{C}$, and let $\alpha : F \rightarrow G$ be a natural transformation (Definition 1.1.3).

Define a presheaf $\text{Im}(\alpha)$ by setting, for each object U of $\mathcal{C}$:

$$\text{Im}(\alpha)(U) = \text{im}(\alpha_U : F(U) \rightarrow G(U)) \subseteq G(U)$$

and for each morphism $f : V \rightarrow U$ in $\mathcal{C}$, letting $\text{Im}(\alpha)(f)$ be the restriction of $G(f) : G(U) \rightarrow G(V)$ to the subset $\text{Im}(\alpha)(U) \subseteq G(U)$, which lands in $\text{Im}(\alpha)(V)$ by naturality of α .

Let $\iota : \text{Im}(\alpha) \hookrightarrow G$ be the inclusion natural transformation (componentwise subset inclusion), and let $\pi : F \twoheadrightarrow \text{Im}(\alpha)$ be the factorization of α through the pointwise images, so that $\alpha = \iota \circ \pi$. Then:

1. $\pi : F \twoheadrightarrow \text{Im}(\alpha)$ is an epimorphism (Definition 1.3.7) in the functor category $\mathbf{PSh}(\mathcal{C}) = [\mathcal{C}^{\text{op}}, \mathbf{Set}]$.
2. $\iota : \text{Im}(\alpha) \hookrightarrow G$ is a monomorphism (Definition 1.3.7) in $\mathbf{PSh}(\mathcal{C})$, equivalently a subfunctor (Definition 5.0.2) inclusion Proposition 5.0.3.
3. The factorization $F \xrightarrow{\pi} \text{Im}(\alpha) \xrightarrow{\iota} G$ is the *image of α in $\mathbf{PSh}(\mathcal{C})$* . In other words, the image of a morphism of set-valued presheaves is *computed pointwise*.

Lemma 5.0.5. Let $\mathcal{C}$ be a small category (Definition 1.0.9) (resp. U -small category where U is some universe (Definition E.0.1)) and let $\mathcal{A}$ be a locally small (Definition 1.0.9) category (resp. U -locally small category). The presheaf category $\mathbf{PreShv}(\mathcal{C}, \mathcal{A})$ (Definition 5.0.1) is locally small (resp. U -locally small).

Proof. A morphism $\mathcal{F} \rightarrow \mathcal{G}$ in $\mathbf{PreShv}(\mathcal{C}, \mathcal{A})$ is a natural transformation (Definition 1.1.3) of the functors $\mathcal{F}, \mathcal{G} : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$. Such a natural transformation is encoded by a family $(\eta_C)_C$ of morphisms (satisfying certain conditions) $\eta_C : \mathcal{F}(C) \rightarrow \mathcal{G}(C)$ in $\mathcal{A}$ over objects C of $\mathcal{C}^{\text{op}}$. The product $\prod_{C \in \text{Ob } \mathcal{C}^{\text{op}}} \text{Hom}_{\mathcal{A}}(\mathcal{F}(C), \mathcal{G}(C))$ is a product of (U -small) sets indexed by a (U -small) set, and the collection of natural transformations is a subset of this set. Therefore, $\text{Hom}_{\mathbf{PreShv}(\mathcal{C}, \mathcal{A})}(\mathcal{F}, \mathcal{G})$ is a (U -small) set. $\square$

Lemma 5.0.6. (♠ TODO: AI generated, verify) Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be (large) categories (Definition 1.0.1) and let $\mathcal{B}$ be a (large) category (Definition 1.0.1). If $\mathcal{A}_1$ and $\mathcal{A}_2$ are equivalent (Definition 1.1.8) as categories, then the presheaf categories (Definition 5.0.1) $\mathbf{PreShv}(\mathcal{A}_1, \mathcal{B})$ and $\mathbf{PreShv}(\mathcal{A}_2, \mathcal{B})$ are equivalent.

More precisely, let $F : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ and $G : \mathcal{A}_2 \rightarrow \mathcal{A}_1$ be functors (Definition 1.1.1) together with natural isomorphisms (Definition 1.1.3)

$$\eta : \text{Id}_{\mathcal{A}_1} \xrightarrow{\sim} G \circ F \quad \text{and} \quad \epsilon : F \circ G \xrightarrow{\sim} \text{Id}_{\mathcal{A}_2}$$

witnessing an equivalence of categories (Definition 1.1.8). Then precomposition with $F^{\text{op}} : \mathcal{A}_1^{\text{op}} \rightarrow \mathcal{A}_2^{\text{op}}$ (Definition 1.0.12) and $G^{\text{op}} : \mathcal{A}_2^{\text{op}} \rightarrow \mathcal{A}_1^{\text{op}}$ induces an equivalence of categories

$$(F^{\text{op}})^* : \text{PreShv}(\mathcal{A}_2, \mathcal{B}) \rightarrow \text{PreShv}(\mathcal{A}_1, \mathcal{B}), \quad (F^{\text{op}})^*(\mathcal{H}) = \mathcal{H} \circ F^{\text{op}}$$

and

$$(G^{\text{op}})^* : \text{PreShv}(\mathcal{A}_1, \mathcal{B}) \rightarrow \text{PreShv}(\mathcal{A}_2, \mathcal{B}), \quad (G^{\text{op}})^*(\mathcal{K}) = \mathcal{K} \circ G^{\text{op}},$$

with natural isomorphisms

$$(\eta^{\text{op}})^* : \text{Id}_{\text{PreShv}(\mathcal{A}_1, \mathcal{B})} \xrightarrow{\sim} (G^{\text{op}})^* \circ (F^{\text{op}})^* \quad \text{and} \quad (\epsilon^{\text{op}})^* : (F^{\text{op}})^* \circ (G^{\text{op}})^* \xrightarrow{\sim} \text{Id}_{\text{PreShv}(\mathcal{A}_2, \mathcal{B})}$$

induced by η and ϵ respectively.

Equivalently, this follows from Lemma 1.2.33 applied to $\mathcal{A}_1^{\text{op}}$ and $\mathcal{A}_2^{\text{op}}$, together with the fact that $\mathcal{A}_1^{\text{op}}$ and $\mathcal{A}_2^{\text{op}}$ are equivalent whenever $\mathcal{A}_1$ and $\mathcal{A}_2$ are.

Remark 5.0.7. Even when $\mathcal{C}$ is $(U-)$ locally small and $(\mathcal{A}$ is $(U-)$ locally small), $\text{PreShv}(\mathcal{C}, \mathcal{A})$ may not be locally small.

Remark 5.0.8. In practice, one might treat $\mathcal{C}$ as if it were a small category, even when it is technically not a small category, to treat $\text{PreShv}(\mathcal{C}, \mathcal{A})$ as if it were a locally small category. For example, in algebraic geometry, we might take $\mathcal{C}$ to be the small étale site $X_{\text{ét}}$ of some scheme X ; this is not a small category. Instead, we might technically replace $X_{\text{ét}}$ with a small category, e.g. the set of étale maps $U \rightarrow X$ of the following form: U is obtained by patching the schemes attached to quotients of rings of the form $A[T_1, T_2, \dots]$ where $A = \Gamma(V, \mathcal{O}_X)$ for some open affine $V \subset X$ and $\{T_1, T_2, \dots\}$ is a fixed countable set of symbols. Such technicalities of replacing $\mathcal{C}$ with a small category are often glossed over.

Lemma 5.0.9. Let $\mathcal{C}$ be a (large) category (Definition 1.0.1). The category of set-valued presheaves (Definition 5.0.1) on $\mathcal{C}$ has a final object (Definition 2.0.6), namely the presheaf $\mathcal{C}^{\text{op}} \rightarrow \mathbf{Sets}$ which sends any object T of $\mathcal{C}$ to a singleton set $\{*\}$.

5.1. Sieves, Grothendieck topologies, and sites.

5.1.1. Sieves.

Definition 5.1.1 ([GV72, Exposé I Définition 4.1]). Let $\mathcal{C}$ be a (large) category (Definition 1.0.1).

1. A **sieve S on the category $\mathcal{C}$** is a full subcategory (Definition 1.3.2) $\mathcal{D}$ that is downward closed (Definition 5.1.2). In other words, for any morphism $f : V \rightarrow U$ in $\mathcal{C}$ where U is an object of $\mathcal{D}$, the object V must also be an object of $\mathcal{D}$ (and hence f is in $\mathcal{D}$ as well as $\mathcal{D}$ is a full subcategory of $\mathcal{C}$).
2. A **sieve S on an object $U \in \text{Ob}(\mathcal{C})$** is a sieve of the category $\mathcal{C}/U$ (Definition 1.3.4). Equivalently, a sieve S on U is a collection of morphisms in $\mathcal{C}$ with codomain U that is closed under precomposition by any compatible morphism in $\mathcal{C}$. In other words, S is a sieve if for every $f : V \rightarrow U$ in S and morphism $g : W \rightarrow V$ in $\mathcal{C}$, the composition $f \circ g : W \rightarrow U$ is also in S .

Given a morphism $f : V \rightarrow U$ in a sieve S , we also say that **f factors through U** .

If $\mathcal{C}$ is a locally small category (Definition 1.0.9), then sieves on objects X of $\mathcal{C}$ bijectively correspond to subfunctors (Definition 5.0.2) of h_X in the category $\mathbf{PSh}(\mathcal{C}) = [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ (Definition 5.0.1)

Definition 5.1.2 (Downward/upward closed subcategory). Let $\mathcal{C}$ be a (large) category (Definition 1.0.1) and $\mathcal{D} \hookrightarrow \mathcal{C}$ a full subcategory (Definition 1.3.2).

1. We say that $\mathcal{D}$ is **downward closed** if for every object $d \in \text{Ob}(\mathcal{D})$ and every morphism $f : c \rightarrow d$ in $\mathcal{C}$ (with $c \in \text{Ob}(\mathcal{C})$), the domain object c belongs to $\text{Ob}(\mathcal{D})$ (and hence $f \in \mathcal{D}$ by fullness of the embedding).
2. We say that $\mathcal{D}$ is **upward closed** if for every object $d \in \text{Ob}(\mathcal{D})$ and every morphism $f : d \rightarrow c$ in $\mathcal{C}$ (with $c \in \text{Ob}(\mathcal{C})$), the codomain object c belongs to $\text{Ob}(\mathcal{D})$ (and hence $f \in \mathcal{D}$ by fullness of the embedding).

Proposition 5.1.3. Let $\mathcal{C}$ be a category (Definition 1.0.1). Let X be an object of $\mathcal{C}$. The sieves (Definition 5.1.1) on X are in bijection with the downward closed (Definition 5.1.2) full subcategories of the slice category $\mathcal{C}/X$ (Definition 1.3.4).

Definition 5.1.4. Let $\mathcal{C}$ be a category (Definition 1.0.1) and let U be an object of $\mathcal{C}$. The **maximal sieve on U** is the collection of all morphisms with target U ; clearly, it is a sieve (Definition 5.1.1) on U .

Lemma 5.1.5 (Sieve containing the identity is maximal). Let $\mathcal{C}$ be a category (Definition 1.0.1), let U be an object of $\mathcal{C}$, and let R be a sieve on U (Definition 5.1.1). If the identity morphism id_U belongs to R , then R is the maximal sieve (Definition 5.1.4) on U .

Proof. Let $f : V \rightarrow U$ be any morphism in $\mathcal{C}$ with codomain U . Since R is a sieve, it is closed under precomposition. As $\text{id}_U \in R$ and $f = f \circ \text{id}_U$, we have $f \in R(V)$. Thus, R contains every morphism with codomain U , so R is the maximal sieve on U . $\square$

Definition 5.1.6. Let $\mathcal{C}$ be a category (Definition 1.0.1) and $U \in \mathcal{C}$ an object. Let $\mathcal{F} = \{f_i : U_i \rightarrow U\}_{i \in I}$ be a family of morphisms with codomain U .

The **sieve generated by $\mathcal{F}$** , denoted $[\mathcal{F}]$, $\langle \mathcal{F} \rangle$, or $S(\mathcal{F})$, is the smallest sieve on U (Definition 5.1.1) containing all the morphisms in $\mathcal{F}$.

Explicitly, a morphism $h : V \rightarrow U$ belongs to the generated sieve if and only if h factors through some morphism in $\mathcal{F}$. That is, there exists an index $i \in I$ and a morphism $g : V \rightarrow U_i$ such that

$$h = f_i \circ g.$$

Definition 5.1.7. Let $\mathcal{C}$ be a category, let $U \in \text{Ob}(\mathcal{C})$, and let S be a sieve on U (Definition 5.1.1). For a morphism $f : V \rightarrow U$ in $\mathcal{C}$, the **pullback sieve f^*S** (or **basechange sieve $S \times_U V$**) on V is defined by

$$f^*S = \{g : W \rightarrow V \mid f \circ g \in S\}.$$

In other words, f^*S consists of all morphisms into V whose composite with f belongs to the sieve S on U . It is indeed a sieve (Lemma 5.1.8).

Lemma 5.1.8. Let $\mathcal{C}$ be a category, let $U \in \text{Ob}(\mathcal{C})$, and let S be a sieve on U (Definition 5.1.1). Let $f : V \rightarrow U$ be a morphism in $\mathcal{C}$. The pullback sieve (Definition 5.1.7) f^*S is indeed a sieve (Definition 5.1.1) on V .

Proof. Let $g : W \rightarrow V$ be a morphism in f^*S , and let $h : X \rightarrow W$ be a morphism in $\mathcal{C}$. The composition $g \circ h : X \rightarrow V$ is also in f^*S because $f \circ (g \circ h) = (f \circ g) \circ h$, the map $(f \circ g)$ is in S , and S is a sieve. $\square$

Proposition 5.1.9 (see [GV72, Exposé I, Proposition 4.2.]). Let $\mathcal{C}$ be a locally small category (Definition 1.0.9) (or a $\mathcal{U}$ -locally small category (Definition 1.0.9) if a Grothendieck universe (Definition E.0.1) $\mathcal{U}$ is available), and let X be an object of $\mathcal{C}$. Denote by $h_X = \text{Hom}_{\mathcal{C}}(-, X) : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ the representable functor associated to X .

There exists a bijective correspondence between:

1. sieves (Definition 5.1.1) on X , and
2. subfunctors (Definition 5.0.2) of h_X (i.e., subobjects of h_X in the functor category (Definition 1.2.1) $\text{PSh}(\mathcal{C}) = [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ (Definition 5.0.1)).

Explicitly, given a sieve S on X , one defines a subfunctor $F_S \subseteq h_X$ by setting

$$F_S(U) = \{f \in \text{Hom}_{\mathcal{C}}(U, X) \mid f \in S\}$$

for each object U of $\mathcal{C}$, with the restriction maps inherited from h_X . Conversely, given a subfunctor $F \subseteq h_X$, one recovers a sieve by taking

$$S_F = \bigcup_{U \in \text{Ob}(\mathcal{C})} F(U) \subseteq \bigcup_{U \in \text{Ob}(\mathcal{C})} \text{Hom}_{\mathcal{C}}(U, X),$$

which is a sieve because the naturality condition on F ensures closure under precomposition.

These two constructions are mutually inverse, establishing a bijection. Moreover, this correspondence is natural in X : if $g : Y \rightarrow X$ is a morphism in $\mathcal{C}$, then the pullback map $h_Y \rightarrow h_X$ induced by composition with g sends a sieve S on X to the sieve $g^*(S)$ on Y generated by precomposing elements of S with g .

5.1.2. Grothendieck pretopologies.

Definition 5.1.10. Let $\mathcal{C}$ be a category (Definition 1.0.1). A *basis for a Grothendieck topology* (also called a *Grothendieck pretopology* or simply a *pretopology*) on $\mathcal{C}$ is a collection of families $K(U)$ of morphisms for each object $U \in \mathcal{C}$, called *coverings* or *covering families*, satisfying the following axioms:

1. **(Identity)** For every isomorphism $U' \rightarrow U$, the singleton family $\{U' \rightarrow U\}$ is in $K(U)$.
2. **(Base Change)** If $\{U_i \rightarrow U\}_{i \in I}$ is a covering family in $K(U)$ and $V \rightarrow U$ is any morphism in $\mathcal{C}$, then the fiber products (Definition 1.2.38) $U_i \times_U V$ exist, and the family of projections $\{U_i \times_U V \rightarrow V\}_{i \in I}$ is in $K(V)$.
3. **(Composition)** If $\{U_i \rightarrow U\}_{i \in I}$ is in $K(U)$ and for each $i \in I$, $\{V_{ij} \rightarrow U_i\}_{j \in J_i}$ is in $K(U_i)$, then the composite family $\{V_{ij} \rightarrow U_i \rightarrow U\}_{i \in I, j \in J_i}$ is in $K(U)$.

Given a family $\{U_i \rightarrow U\}_{i \in I}$ in $K(U)$, the sieve (Definition 5.1.1) that it generates (Definition 5.1.6) is called a **covering sieve of U (generated by $\{U_i \rightarrow U\}_{i \in I}$)**.

Definition 5.1.11 (Čech nerve of a family of morphisms to an object). (♠ TODO: AI generated, verify)

Let $\mathcal{C}$ be a category (Definition 1.0.1).

For every object U of $\mathcal{C}$ and every family of morphisms $\{U_i \rightarrow U\}_{i \in I}$ in $\mathcal{C}$ such that all fiber products (Definition 1.2.38) $U_{i_0} \times_U U_{i_1} \times_U \cdots \times_U U_{i_n}$ exist for all $n \geq 0$ and all $(i_0, \dots, i_n) \in I^{n+1}$, one defines the **Čech nerve of $\{U_i \rightarrow U\}_{i \in I}$** , denoted by $\check{C}(\{U_i \rightarrow U\}_{i \in I})$, as a simplicial object (Definition 10.0.1) in the slice category (Definition 1.3.4) $\mathcal{C}/U$ (Definition 1.3.4).

Concretely, for each $[n] \in \Delta$ (Definition 8.0.3), the value of the Čech nerve at $[n]$ is the coproduct (Definition 1.2.36) in $\mathcal{C}/U$

$$\check{C}(\{U_i \rightarrow U\})_n := \coprod_{(i_0, \dots, i_n) \in I^{n+1}} (U_{i_0} \times_U U_{i_1} \times_U \cdots \times_U U_{i_n} \longrightarrow U),$$

where the morphism to U is the canonical projection, assuming that this coproduct and the relevant fiber products exist.

For each order-preserving map $\theta : [m] \rightarrow [n]$ in Δ , the induced morphism

$$\check{C}(\{U_i \rightarrow U\})_\theta : \check{C}(\{U_i \rightarrow U\})_n \longrightarrow \check{C}(\{U_i \rightarrow U\})_m$$

in $\mathcal{C}/U$ is the coproduct of the canonical projections between fiber products induced by θ : for each $(i_0, \dots, i_n) \in I^{n+1}$, the component at index $(i_{\theta(0)}, \dots, i_{\theta(m)}) \in I^{m+1}$ is the morphism

$$U_{i_0} \times_U \cdots \times_U U_{i_n} \longrightarrow U_{i_{\theta(0)}} \times_U \cdots \times_U U_{i_{\theta(m)}}$$

induced by the projections among the factors.

The face maps (Definition 8.0.11) $d^k : \check{C}(\{U_i \rightarrow U\})_1 \rightrightarrows \check{C}(\{U_i \rightarrow U\})_0$ correspond to the two projections from each pairwise fiber product $U_{i_0} \times_U U_{i_1}$, and the degeneracy maps $s^k : \check{C}(\{U_i \rightarrow U\})_n \rightarrow \check{C}(\{U_i \rightarrow U\})_{n+1}$ correspond to the diagonal morphisms.

When $\mathcal{A}$ is a category (Definition 1.0.1) and $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ is a functor, applying F levelwise yields a cosimplicial object in $\mathcal{A}$, denoted by $F(\check{C}(\{U_i \rightarrow U\})_\bullet)$, whose value at $[n]$ is

$$F(\check{C}(\{U_i \rightarrow U\})_n) = \coprod_{(i_0, \dots, i_n) \in I^{n+1}} F(U_{i_0} \times_U U_{i_1} \times_U \cdots \times_U U_{i_n}).$$

Definition 5.1.12 (Sheaf on a Grothendieck pretopology). (♠ TODO: AI generated, verify)

Let $\mathcal{C}$ be a category (Definition 1.0.1) equipped with a Grothendieck pretopology (Definition 5.1.10) K . Let $\mathcal{A}$ be a category (Definition 1.0.1).

For every object U of $\mathcal{C}$ and every covering family (Definition 5.1.10) $\{U_i \rightarrow U\}_{i \in I} \in K(U)$, one forms the Čech nerve (Definition 5.1.11) of the covering family: this is a simplicial object in the slice category $\mathcal{C}/U$ whose value at $[n] \in \Delta$ is the disjoint union (or equivalently, the

family indexed by I^{n+1} of $(n+1)$ -fold fiber products

$$\check{C}(\{U_i \rightarrow U\})_n = \coprod_{(i_0, \dots, i_n) \in I^{n+1}} U_{i_0} \times_U U_{i_1} \times_U \cdots \times_U U_{i_n},$$

with face maps induced by the projections between fiber products of different lengths and degeneracy maps induced by the diagonal morphisms.

Applying a presheaf (Definition 5.0.1) $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ (Definition 1.0.12) to the Čech nerve yields a cosimplicial object in $\mathcal{A}$, hence a diagram indexed by Δ . The *matching object of F with respect to the covering family $\{U_i \rightarrow U\}_{i \in I}$* is defined as the limit (Definition 1.2.2)

$$M(\{U_i \rightarrow U\}_{i \in I}, F) := \varprojlim_{[n] \in \Delta} F(\check{C}(\{U_i \rightarrow U\})_n),$$

provided this limit exists in $\mathcal{A}$.

A presheaf $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ is called a *sheaf on $(\mathcal{C}, K)$ valued in $\mathcal{A}$* if, for every object U of $\mathcal{C}$ and every covering family $\{U_i \rightarrow U\}_{i \in I} \in K(U)$, the limit defining $M(\{U_i \rightarrow U\}_{i \in I}, F)$ exists and the canonical morphism

$$F(U) \longrightarrow M(\{U_i \rightarrow U\}_{i \in I}, F)$$

is an isomorphism in $\mathcal{A}$.

When $\mathcal{A}$ is the category of sets (Definition 1.0.3), we simply say that F is a *sheaf on $(\mathcal{C}, K)$* .

5.1.3. Coverages.

5.1.4. Grothendieck topologies.

Definition 5.1.13 (Grothendieck topology). Let $\mathcal{U}$ be a universe (Definition E.0.1).

1. (See [GV72, Exposé II, Définition 1.1]) Let $\mathcal{C}$ be a category (Definition 1.0.1). A *Grothendieck topology on $\mathcal{C}$* assigns to each object U of $\mathcal{C}$ a collection $J(U)$ of sieves (Definition 5.1.1) $\{U_i \rightarrow U\}_{i \in I}$, each called a *covering sieve of U* , satisfying:
 - (a) (Stability under “base change”): If $S \in J(U)$ is a covering sieve of an object U , and $f : V \rightarrow U$ is any morphism in $\mathcal{C}$, then the pullback sieve (Definition 5.1.7) f^*S is a covering sieve of V .
 - (b) (Local character condition) If S is a sieve on U , and if there exists a covering sieve $R \in J(U)$ such that for all $f : V \rightarrow U$ in R the pullback sieve (Definition 5.1.7) f^*S is in $J(V)$, then $S \in J(U)$.
 - (c) The maximal sieve (Definition 5.1.4) is a covering sieve.

Some will refer to a Grothendieck topology as simply a *topology*, not to be confused with the related, but less general, notion of a topology on a set (Definition B.0.1).

2. (See [GV72, Exposé II, 1.1.5]) A *site* is a category $\mathcal{C}$ equipped with a Grothendieck topology.

When we are working with a Grothendieck pretopology (Definition 5.1.10) K on a category $\mathcal{C}$, we may regard $\mathcal{C}$ as a site by equipping it with the Grothendieck topology generated by (Definition 5.1.16) K . In this case, we may call $\mathcal{C}$ equipped with this Grothendieck topology a *site with a pretopology*. Note that the Stacks project notion of

“site” is that of a small (Definition 1.0.9) site (i.e. a site whose underlying category is small) with a pretopology, cf. [Sta25, Tag 00V0] for a discussion for this choice, [Sta25, Tag 00VH] for the definition and [Sta25, Tags 0005 and 0015] for the Stacks project’s convention of “categories” (what we call small categories (Definition 1.0.9)) and “big categories” (what we call locally small categories (Definition 1.0.9)). (♠ TODO: talk more about how, at least when the axiom of universes is assumed, topoi coming from general Grothendieck topology are equivalent to those on pretopologies).

3. (See [GV72, Exposé II, Définition 1.2]) Let $(\mathcal{C}, J)$ be a site. A family of morphisms $(U_i \rightarrow U)_{i \in I}$ is called a *covering family of U (with respect to the site/topology)* or a *cover of U (with respect to the site/topology)* if the sieve generated by (Definition 5.1.6) the family is a covering sieve of U .
4. (See [GV72, Exposé II, Définition 3.0.1]) Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13), where J is a Grothendieck topology on $\mathcal{C}$.

A family G of objects $\mathcal{C}$ is called a *topologically generating family of the site/topology* or a *generating family/collection of the site/topology* if for every object $X \in \mathcal{C}$, there is a covering family $\{X_\alpha \rightarrow X\}_{\alpha \in A}$ of X such that every X_α is a member of G .

Equivalently, the Grothendieck topology J is the smallest Grothendieck topology containing all covers of the U_i . Also equivalently, for any $S \in J(X)$, the sieve S contains a covering family $\{V_i \rightarrow X\}$ such that each morphism $V_i \rightarrow X$ factors through some member of G . (♠ TODO: Verify that these claimed equivalences are indeed equivalences)

Proposition 5.1.14 (Sieve containing a covering sieve is covering). Let $\mathcal{C}$ be a category (Definition 1.0.1), and let J be a Grothendieck topology (Definition 5.1.13) on $\mathcal{C}$. Let U be an object of $\mathcal{C}$, let S be a covering sieve (Definition 5.1.13) of U , and let R be a sieve on U (Definition 5.1.1) such that $S \subseteq R$.

Then R is also a covering sieve of U .

Proof. By the local character condition of Definition 5.1.13, it suffices to show that for all $g : V \rightarrow U$ in S that $g^*R \in J(V)$. Since $g \in S$, note that $g \in R$. By the definition of a pullback sieve (Definition 5.1.7), $g^*R = \{h : W \rightarrow V : g \circ h \in R\}$. In particular, $\text{id}_V : V \rightarrow V$ is in g^*R because $g \circ \text{id}_V = g \in R$. Any sieve on V that contains the identity morphism is the maximal sieve on V (Lemma 5.1.5). Therefore, g^*R is the maximal sieve, which is a covering sieve by the axioms of a Grothendieck topology. Therefore, $g^*R \in J(V)$ as desired. $\square$

Definition 5.1.15. 1. An *small site* is a site (Definition 5.1.13) whose underlying category is small (Definition 1.0.9).

2. A *locally small site* is a site whose underlying category is locally small (Definition 1.0.9).
3. Given a Grothendieck universe (Definition E.0.1) $\mathcal{U}$, a *$\mathcal{U}$ -site* is a site (Definition 5.1.13) whose underlying category (Definition 1.0.1) is a $\mathcal{U}$ -(locally small) category (Definition 1.0.9) and that has a $\mathcal{U}$ -small topologically generating family (Definition 5.1.13); a Grothendieck topology making the site a $\mathcal{U}$ -site is called a *$\mathcal{U}$ -topology*.

A $\mathcal{U}$ -site is called *$\mathcal{U}$ -small* if its underlying category is $\mathcal{U}$ -small (Definition 1.0.9).

Definition 5.1.16. Let $\mathcal{C}$ be a category equipped with a Grothendieck pretopology (Definition 5.1.10) K . The *Grothendieck topology generated by K* , denoted J_K , is the Grothendieck topology constructed as follows: a sieve (Definition 5.1.1) S on an object U belongs to

$J_K(U)$ if and only if there exists a covering family (Definition 5.1.10) $\{U_i \rightarrow U\}_{i \in I} \in K(U)$ such that for every $i \in I$, the morphism $U_i \rightarrow U$ belongs to S , i.e. the sieve generated by (Definition 5.1.6) $\{U_i \rightarrow U\}_{i \in I}$ is a sub-sieve of S .

It is the coarsest Grothendieck topology (Definition 5.1.13) on $\mathcal{C}$ such that every member in $K(U)$ is a covering family (Definition 5.1.13) for J_K , i.e. generates (Definition 5.1.6) a covering sieve (Definition 5.1.13). See Theorem 5.1.17.

Theorem 5.1.17. Let $\mathcal{C}$ be a category equipped with a Grothendieck pretopology (Definition 5.1.10) K .

1. The Grothendieck topology (Definition 5.1.13) J_K generated by (Definition 5.1.16) K is indeed a Grothendieck topology.
2. J_K is the coarsest Grothendieck topology such that every member of $K(U)$ is a covering family (Definition 5.1.13) for J_K , i.e. generates (Definition 5.1.6) a covering sieve (Definition 5.1.13).

Proof. 1. We first show that J is stable under “base change”. Let $S \in J_K(U)$. By the definition of J_K , there exists some covering family $\mathcal{F} = \{\alpha_i : U_i \rightarrow U\}_{i \in I} \in K(U)$ such that the sieve $S(\mathcal{F})$ generated by $\mathcal{F}$ is a sub-sieve of S , i.e. that S contains

$$S(\mathcal{F}) = \{h : W \rightarrow U \text{ in } \mathcal{C} : \exists i \in I \exists \alpha : W \rightarrow U_i \text{ such that } h = \alpha_i \circ \alpha\},$$

and let $f : V \rightarrow U$ be a morphism in $\mathcal{C}$. We need to show that the pullback sieve (Definition 5.1.7)

$$f^*S = \{g : X \rightarrow V \text{ in } \mathcal{C} : f \circ g \in S\}$$

contains

$$S((f^*\alpha_i)_{i \in I}) = \{g : X \rightarrow V \text{ in } \mathcal{C} : \exists i \in I \exists \tilde{\alpha} : X \rightarrow U_i \times_U V \text{ such that } g = f^*\alpha_i \circ \tilde{\alpha}\}.$$

Since $S(\mathcal{F}) \subseteq S$, we have that $f^*(S(\mathcal{F})) \subseteq f^*S$.

Say that $g : X \rightarrow V$ is a member of $S((f^*\alpha_i)_{i \in I})$, so there exists $i \in I$ and $\tilde{\alpha} : X \rightarrow U_i \times_U V$ such that $g = f^*\alpha_i \circ \tilde{\alpha}$. In other words, we have a diagram

$$(F) \quad \begin{array}{ccccc} X & & \xrightarrow{g} & & V \\ & \searrow \tilde{\alpha} & & \nearrow f^*\alpha_i & \\ & U_i \times_U V & & & \\ & \downarrow & & \downarrow f & \\ & U_i & \xrightarrow{\alpha_i} & & V \end{array}$$

The composition $f \circ g$ then coincides with the composition $\alpha_i \circ \alpha$, which is a member of S . Thus, g is a member of f^*S , so the desired inclusion $S((f^*\alpha_i)_{i \in I}) \subseteq f^*S$ holds. Therefore, J is indeed stable under base change.

We show that J satisfies the “local character condition”. Let U be an object of $\mathcal{C}$, let S be a sieve (Definition 5.1.1) on U , let R be a sieve that is a member of $J(U)$ such that for all $f : V \rightarrow U$ in R the pullback sieve (Definition 5.1.7) f^*S is a member of $J(V)$. We show that S is a member of $J(U)$. Say that R is generated by the member

$\{\alpha_i : U_i \rightarrow U\}_{i \in I}$ of $K(U)$. By hypothesis, the pullback sieve $\alpha_i^* S$ is a member of $J(U_i)$. Thus, there exists a family $\mathcal{F}_i = \{\beta_{ij} : V_{ij} \rightarrow U_i\}_{j \in J_i} \in K(U_i)$ that generates (Definition 5.1.6) $\alpha_i^* S$. The composition

$$\mathcal{F}' = \{\alpha_i \circ \beta_{ij} : V_{ij} \rightarrow U\}_{i \in I, j \in J_i}$$

of the families $\{\alpha_i\}$ and $\mathcal{F}_i$ is a family in $K(U)$ by the composition axiom of Grothendieck pretopologies. To show that $S \in J(U)$, it now suffices to show that $S(\mathcal{F}') \subseteq S$ (Definition 5.1.6). Let $g : W \rightarrow U$ be any morphism in the sieve $S(\mathcal{F}')$. By the definition of $S(\mathcal{F}')$, there exists some $i \in I$, $j \in J_i$, and a morphism $h : W \rightarrow V_{ij}$ such that

$$g = (\alpha_i \circ \beta_{ij}) \circ h = \alpha_i \circ (\beta_{ij} \circ h)$$

Since $\mathcal{F}_i$ generates a sub-sieve of $\alpha_i^* S$, and since $\beta_{ij} \in \mathcal{F}_i$, we have $\beta_{ij} \in \alpha_i^* S$. In other words, $\alpha_i \circ \beta_{ij} \in S$. Since S is a sieve and hence is closed under precomposition, we have that $g \in S$. We have thus shown that $S(\mathcal{F}') \subseteq S$ as desired.

We show that the maximal sieve (Definition 5.1.4) is a covering sieve of J_K . By the axioms of a Grothendieck pretopology, the singleton family $\mathcal{F} = \{\text{id}_U : U \rightarrow U\}$ is a member of $K(U)$. The sieve $S(\mathcal{F})$ generated by (Definition 5.1.6) this family consists of all the morphisms $g : V \rightarrow U$ in $\mathcal{C}$ such that there exists a morphism $h : V \rightarrow U$ such that $g = \text{id}_U \circ h$, i.e. $g = h$. Thus, the sieve $S(\mathcal{F})$ is in fact the maximal sieve. Thus, the maximal sieve is a covering sieve of J_K .

2. (♠ TODO:)

□

Definition 5.1.18. Let $(\mathcal{C}, J)$ and $(\mathcal{C}, J')$ be two sites (Definition 5.1.13) on the same underlying category $\mathcal{C}$. We say that the topology J is *finer than* J' (or equivalently, that J' is *coarser than* J) if for every object U in $\mathcal{C}$, there is an inclusion

$$J'(U) \subseteq J(U)$$

That is, every sieve (Definition 5.1.1) on U which is a covering sieve for J' is also a covering sieve for J .

Definition 5.1.19 (cf. [GV72, Exposé III, 3.1]). Let $\mathcal{C}, \mathcal{D}$ be categories (Definition 1.0.1), let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a functor (Definition 1.1.1), and let $(\mathcal{D}, K)$ be a site (Definition 5.1.13). The *induced topology* $u^* K$ (also called the *pullback topology*) on $\mathcal{C}$ is the Grothendieck topology given as follows: for every object $V \in \mathcal{C}$, a sieve (Definition 5.1.1) $S \in u^* K(V)$ is a covering sieve if and only if its image $u(S)$ under u is a covering sieve in $K(u(V))$. Equivalently, it is the finest (Definition 5.1.18) Grothendieck topology (Definition 5.1.13) making u a continuous functor (Definition 5.2.1). (♠ TODO: this equivalence)

Proposition 5.1.20. Let $\mathcal{C}, \mathcal{D}$ be categories (Definition 1.0.1), let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a functor (Definition 1.1.1), and let $(\mathcal{D}, K)$ be a site (Definition 5.1.13).

The functor $u : (\mathcal{C}, u^* K) \rightarrow (\mathcal{D}, K)$ (Definition 5.1.19) is continuous (Definition 5.2.1).

Proof. Let U be an object of $\mathcal{C}$. Let $S \in u^* K(U)$ be a covering sieve of U , i.e. $u(S)$ is a covering sieve of $u(U)$ in K . In particular, the sieve generated by $u(S)$ is a covering sieve in K and hence u is continuous. □

Theorem 5.1.21 (Comparison Lemma, cf. [GV72, Exposé III, Théorème 4.1]). Let $\mathcal{C}$ be a small category (Definition 1.0.9), let $(\mathcal{D}, K)$ be a site (Definition 5.1.13), let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a fully faithful functor (Definition 1.1.4), and equip $\mathcal{C}$ with the Grothendieck topology (Definition 5.1.13) J induced by (Definition 5.1.19) u . Let $\mathcal{A}$ be a locally small (Definition 1.0.9) complete category (Definition 1.3.9), i.e. a category which admits all small limits (Definition 1.2.5).

If the objects of $\mathcal{C}$ form a topologically generating family (Definition 5.1.13) (♠ TODO: I need to check that the condition of covering in SGA really matches the notion of topologically generating family) of $\mathcal{D}$, then the direct image (Definition 5.3.2)/restriction functor

$$u^s : \mathrm{Sh}(\mathcal{D}, K; \mathcal{A}) \rightarrow \mathrm{Sh}(\mathcal{C}, J; \mathcal{A})$$

is an equivalence of categories (Definition 1.1.8). In particular, $\mathrm{Sh}(\mathcal{D}, K; \mathcal{A})$ is a locally small category (Definition 1.0.9).

Proof. Note that the direct image indeed is a functor between the categories of sheaves by Proposition 5.2.2 because $u : \mathcal{C} \rightarrow \mathcal{D}$ is a continuous functor (Definition 5.2.1) between the sites by Proposition 5.1.20. (♠ TODO: talk about the right Kan extension — in particular, how does it relate to the left Kan extension and to the direct image/inverse image of a functor?) (♠ TODO: continue)

We explicitly construct the inverse functor and prove the equivalence without assuming *a priori* that $\mathrm{Sh}(\mathcal{D}, K)$ is locally small.

Step 1: Construction of the Inverse Functor u_*

We define the functor $u_* : \mathrm{Sh}(\mathcal{C}, J) \rightarrow \mathrm{Sh}(\mathcal{D}, K)$ as the **Right Kan Extension** of a sheaf F along u .

For a sheaf $F \in \mathrm{Sh}(\mathcal{C}, J)$ and an object $d \in \mathcal{D}$, define:

$$u_*F(d) = \lim_{(c,f) \in (u \downarrow d)^{op}} F(c)$$

Here, the limit is taken over the opposite of the comma category $(u \downarrow d)$, whose objects are pairs $(c, f : u(c) \rightarrow d)$.

Verification of Well-Definedness (Size): Since $\mathcal{C}$ is a small category and $\mathcal{D}$ is locally small, the collection of objects in $(u \downarrow d)$ is a set (indexed by objects of $\mathcal{C}$ and hom-sets of $\mathcal{D}$). Therefore, the limit defining $u_*F(d)$ is a **small limit of sets**, which exists in **Set**. Thus, u_*F takes values in **Set** rather than proper classes.

*Verification that u_*F is a Sheaf:* Since limits commute with limits, and the sheaf condition is a limit condition, the Right Kan Extension of a sheaf along a continuous functor is a sheaf. The density condition ensures that covers in $\mathcal{D}$ are "seen" by $\mathcal{C}$, ensuring the sheaf condition is preserved.

Step 2: The Unit of Adjunction ($u^*u_* \cong \mathrm{Id}$)

We examine u^*u_*F for $F \in \text{Sh}(\mathcal{C}, J)$. Evaluating at an object $c_0 \in \mathcal{C}$:

$$(u^*u_*F)(c_0) = u_*F(u(c_0)) = \lim_{(c,f) \in (u \downarrow u(c_0))^{op}} F(c)$$

Since u is **fully faithful**, the comma category $(u \downarrow u(c_0))$ has a terminal object: $(c_0, \text{id}_{u(c_0)})$. The limit over a category with a terminal object is isomorphic to the value at that object. Thus, $(u^*u_*F)(c_0) \cong F(c_0)$, concluding $u^*u_* \cong \text{Id}_{\text{Sh}(\mathcal{C})}$.

Step 3: The Coint of Adjunction ($H \cong u_*u^*H$)

This is the critical step that uses the **Density Condition** to control the size of sheaves on $\mathcal{D}$. Let $H \in \text{Sh}(\mathcal{D}, K)$. We construct a map $\eta_d : H(d) \rightarrow u_*(u^*H)(d)$. By definition:

$$u_*(u^*H)(d) = \lim_{u(c) \rightarrow d} H(u(c))$$

There is a canonical map η_d induced by the morphisms $H(f) : H(d) \rightarrow H(u(c))$ for each $f : u(c) \rightarrow d$.

To see that η_d is an isomorphism:

1. **Density as a Cover:** By hypothesis, the family of morphisms $\mathcal{S} = \{f : u(c) \rightarrow d \mid c \in \mathcal{C}\}$ generates a covering sieve S on d .
2. **Sheaf Property:** Since H is a sheaf on $(\mathcal{D}, K)$, $H(d) \cong \text{Match}(S, H)$.
3. **Matching Families:** The limit over the comma category $(u \downarrow d)$ is exactly the set of compatible families indexed by the generators of the sieve. Because $\mathcal{S}$ generates S , the data of a matching family on $\mathcal{S}$ extends uniquely to the whole sieve S .

The canonical map $H(d) \rightarrow \lim_{u(c) \rightarrow d} H(u(c))$ is therefore an isomorphism, so $H \cong u_*(u^*H)$.

Step 4: Conclusion of Equivalence and Local Smallness

We have established natural isomorphisms $u^*u_* \cong \text{Id}$ and $\text{Id} \cong u_*u^*$, establishing an equivalence. Since $\mathcal{C}$ is small, $\text{Sh}(\mathcal{C}, J)$ is a locally small category. Since $\text{Sh}(\mathcal{D}, K)$ is equivalent to it, $\text{Sh}(\mathcal{D}, K)$ is itself locally small. Specifically:

$$\text{Hom}_{\mathcal{D}}(H, G) \cong \text{Hom}_{\mathcal{C}}(u^*H, u^*G)$$

where the latter is a set.

Since $\mathcal{C}$ is small and $\mathcal{A}$ is locally small, the category of presheaves on $\mathcal{C}$ valued in $\mathcal{A}$ is locally small by Lemma 5.0.5. Therefore, the category of sheaves is locally small. $\square$

Proposition 5.1.22 ([GV72, Exposé II, Proposition 6.3.1]). Let $\mathcal{C}$ be a site (Definition 5.1.13), and let γ be a “kind of algebraic structure defined by finite projective limits” (🔥 **TODO:** Probably the correct terminology for this is “essentially algebraic categories” or “Lawvere theories”) (e.g. γ can be the notion of group objects or abelian group objects (Definition 1.3.19), ring objects (Definition 1.3.20), or module objects (Definition 1.3.21)). The following categories are equivalent (Definition 1.1.8):

1. The category of sheaves (Definition 5.1.32) on $\mathcal{C}$ with values in the category of γ -sets,
2. The category of γ -objects of the category of sheaves of sets on $\mathcal{C}$, and

3. The category of γ -objects of the category of presheaves (Definition 5.0.1) of sets on $\mathcal{C}$ whose underlying presheaf is a sheaf of sets.

Definition 5.1.23. Let $\mathcal{C}$ be a locally small (Definition 1.0.9) category (Definition 1.0.1). An object X in $\mathcal{C}$ is called *quasi-compact* (or *compact*, or *finitely presentable* depending on context) if the functor $h^X = \text{hom}_{\mathcal{C}}(X, -)$ represented by (Definition 4.0.6) X , preserves filtered colimits (Definition 1.2.12).

Explicitly, this means that for any filtered diagram (Definition 1.2.1) $D : \mathcal{I} \rightarrow \mathcal{C}$ such that the colimit (Definition 1.2.2) $\text{colim}_{i \in \mathcal{I}} D_i$ exists in $\mathcal{C}$, the canonical map

$$\text{colim}_{i \in \mathcal{I}} \text{hom}_{\mathcal{C}}(X, D_i) \xrightarrow{\cong} \text{hom}_{\mathcal{C}}(X, \text{colim}_{i \in \mathcal{I}} D_i)$$

is a bijection.

Definition 5.1.24. Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13). Assume that $\mathcal{C}$ is locally small (Definition 1.0.9).

An object U in $\mathcal{C}$ is called *quasi-compact (relative to the topology J)* if every covering sieve (Definition 5.1.13) $S \in J(U)$ contains a finite set of morphisms $\{f_i : V_i \rightarrow U\}_{i=1}^n$ which generates (Definition 5.1.6) a covering sieve (Definition 5.1.13).

Equivalently, if coverings are defined via a basis (or pretopology) (Definition 5.1.10), an object U is quasi-compact if every covering family $\{U_i \rightarrow U\}_{i \in I}$ admits a finite subfamily $\{U_i \rightarrow U\}_{i \in I_0}$ (with $I_0 \subseteq I$ finite) which is also a covering family.

Definition 5.1.25. (♠ TODO: does this need local smallness) Let $(\mathcal{C}, J)$ be a locally small (Definition 1.0.9) site (Definition 5.1.13). A morphism $f : X \rightarrow Y$ in $\mathcal{C}$ is called *quasi-compact* if for every morphism $Z \rightarrow Y$ where the source Z is a quasi-compact object with respect to J (Definition 5.1.24), the fiber product (Definition 1.2.38) $X \times_Y Z$ exists and is also a quasi-compact object with respect to J .

Definition 5.1.26. Let $(\mathcal{C}, J)$ be a locally small (Definition 1.0.9) site (Definition 5.1.13).

An object X in $\mathcal{C}$ is called *quasi-separated with respect to J* if the product (Definition 1.2.36) $X \times X$ exists and the diagonal morphism (Definition 1.2.39) $\Delta : X \rightarrow X \times X$ is a quasi-compact morphism with respect to J .

Explicitly, this means that for any two morphisms $U \rightarrow X$ and $V \rightarrow X$ from quasi-compact objects U and V , the fiber product $U \times_X V$ is a quasi-compact object, assuming that the relevant fiber products exist.

Definition 5.1.27. Let $(\mathcal{C}, J)$ be a site where $\mathcal{C}$ is locally small (Definition 1.0.9). An object $X \in \mathcal{C}$ is called a *coherent object* if it satisfies two conditions:

1. X is quasi-compact (Definition 5.1.24) relative to the topology J .
2. X is quasi-separated (Definition 5.1.26) relative to the topology J .

Definition 5.1.28. A site (Definition 5.1.13) $(\mathcal{C}, J)$ is called a *coherent site* if it satisfies the following conditions:

1. The category $\mathcal{C}$ is locally small with a small topologically generating family (Definition 5.1.13).

2. The category $\mathcal{C}$ admits all finite limits (Definition 1.2.2).
3. Every object in $\mathcal{C}$ is quasi-compact relative to (Definition 5.1.24) the topology J .
4. The collection of quasi-compact objects relative to J is closed under fiber products (Definition 1.2.38). Explicitly, if $X \rightarrow Z$ and $Y \rightarrow Z$ are morphisms where X, Y, Z are quasi-compact, then $X \times_Z Y$ is also quasi-compact.

Definition 5.1.29 (Slice site). Let $(\mathcal{C}, \tau)$ be a site (Definition 5.1.13), where τ is a Grothendieck topology on the (locally small or U -locally small (Definition 1.0.9), if a universe (Definition E.0.1) U is available) category $\mathcal{C}$. For a fixed object X in $\mathcal{C}$, the *slice site* (or the *over site*, the *site on the slice category $\mathcal{C}_{/X}$* , the *site induced on the over category $\mathcal{C}_{/X}$* , the *localization of the site $\mathcal{C}$ at the object X* , etc.) $(\mathcal{C}_{/X}, \tau_{/X})$ is the site whose underlying category is the slice category $\mathcal{C}_{/X}$ (Definition 1.3.4), and whose Grothendieck topology $\tau_{/X}$ (also denoted by notations such as $\tau|_X$ or τ/X) is defined by declaring a family of morphisms $\{f_i : Y_i \rightarrow Y\}$ in $\mathcal{C}_{/X}$ to be a covering if and only if the family $\{f_i : Y_i \rightarrow Y\}$ is a covering in $(\mathcal{C}, \tau)$.

Definition 5.1.30. Let (X, τ_X) be a topological space. The *small site associated to X* or *the site of open covers of X* or *the canonical site on $\text{Open } X$* is the category $\text{Open}(X)$ of open subsets (Definition 1.0.7) of X with inclusion morphisms, equipped with the canonical Grothendieck topology (Definition 5.1.13) generated by (Definition 5.1.16) the Grothendieck pretopology (Definition 5.1.10) whose covering families $\{U_i \rightarrow U\}_{i \in I}$, for $U \in \text{Open}(X)$ are families of morphisms in $\text{Open}(X)$ such that $\bigcup_{i \in I} U_i = U$. In other words, $\{U_i \rightarrow U\}_{i \in I}$ is a covering for the pretopology if it is an open coverings (Definition B.0.2).

Proposition 5.1.31. (♠ TODO: many definitions) The following are common examples of sites:

1. Let X be a scheme. The following categories equipped with their respective Grothendieck topologies form *small sites associated to X* :
 - The *small Zariski site* $\text{Zar}(X)$: the category of open immersions $U \rightarrow X$ with the Zariski topology given by open covers.
 - The *small étale site* $\text{Ét}(X)$: the category of étale morphisms (Definition D.0.1) $U \rightarrow X$ equipped with the étale topology defined by étale coverings.
 - The *small Nisnevich site* $\text{Nis}(X)$: the category of étale morphisms $U \rightarrow X$ with the Nisnevich topology given by Nisnevich covers.
 - The *small fppf site* $\text{fppf}(X)$: the category of flat morphisms locally of finite presentation $U \rightarrow X$ equipped with the fppf topology.
 - The *small fpqc site* $\text{fpqc}(X)$: the category of flat morphisms $U \rightarrow X$ (not necessarily locally of finite presentation), with the fpqc topology.
 - The *small crystalline site* $\text{Cris}(X)$: the category of PD-thickenings over X equipped with the crystalline topology.

5.1.5. Sheaf on a site.

Definition 5.1.32 (Sheaf on a site). Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13). Let $\mathcal{A}$ be a (large) category (Definition 1.0.1).

1. A presheaf (Definition 5.0.1) $\mathcal{F} : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ (Definition 1.0.12) is called a *sheaf on the site $(\mathcal{C}, J)$ valued in $\mathcal{A}$* if, for every object U of $\mathcal{C}$ and every covering sieve (Definition 5.1.13)

$S \in J(U)$, the limit (Definition 1.2.2)

$$\varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} \mathcal{F}|_{\mathcal{D}_S}(V),$$

exists and the canonical natural morphism

$$\mathcal{F}(U) \rightarrow \varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} \mathcal{F}|_{\mathcal{D}_S}(V)$$

is an isomorphism. Here, $\mathcal{D}_S \hookrightarrow \mathcal{C}/U$ (Definition 1.3.4) is the full downward-closed subcategory (Definition 5.1.2) such that $\text{Ob}(\mathcal{D}_S) = \{(f : V \rightarrow U) : f \in S(V)\}$,

In particular, when we are working with a Grothendieck pretopology (Definition 5.1.10) K on a category $\mathcal{C}$, we may speak of sheaves on the site whose Grothendieck topology is the one generated by (Definition 5.1.16) K .

2. Given sheaves $\mathcal{F}, \mathcal{G} : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ on the site $(\mathcal{C}, J)$, a *morphism between the sheaves* is a morphism (Definition 5.0.1) between $\mathcal{F}$ and $\mathcal{G}$ as presheaves.
3. Let U be a universe (Definition E.0.1). A *U -sheaf* typically refers to a U -presheaf that is a sheaf for a U -site. In other words, a U -sheaf is a sheaf on a site whose underlying category is U -locally small (Definition 1.0.9) and which has a U -small topologically generating family such that the sheaf is valued in U -sets.
4. The *sheaf category/category of $\mathcal{A}$ -valued sheaves on $\mathcal{C}$* is the (large) category defined as the full subcategory of $\text{PreShv}(\mathcal{C}, \mathcal{A})$ whose objects are the sheaves on $\mathcal{C}$ with values in $\mathcal{A}$. Common notations for the sheaf category include $\text{Shv}(\mathcal{C}, \mathcal{A})$, $\text{Shv}(\mathcal{C}, J, \mathcal{A})$, $\text{Sh}(\mathcal{C}, \mathcal{A})$, $\text{Sh}(\mathcal{C}, J, \mathcal{A})$. If the value category $\mathcal{A}$ is clear from context, then notations such as $\text{Shv}(\mathcal{C})$, $\text{Shv}(\mathcal{C}, J)$, $\text{Sh}(\mathcal{C})$, $\text{Sh}(\mathcal{C}, J)$ are also common.

Theorem 5.1.33 (Limits of presheaf diagrams are computed sectionwise). Let $\mathcal{C}$ and $\mathcal{A}$ be categories (Definition 1.0.1). Let I be a category. Let $D : I \rightarrow \text{PreShv}(\mathcal{C}, \mathcal{A})$ (Definition 5.0.1) be a diagram (Definition 1.2.1) of presheaves on $\mathcal{C}$ valued in $\mathcal{A}$, written $i \mapsto D(i)$.

Suppose that for every object U of $\mathcal{C}$, the limit (Definition 1.2.2) $\lim_{i \in I} D(i)(U)$ exists in $\mathcal{A}$. Then the limit of the diagram D exists in $\text{PreShv}(\mathcal{C}, \mathcal{A})$, and it is computed sectionwise: for every object U of $\mathcal{C}$, there is a canonical isomorphism in $\mathcal{A}$:

$$\left(\lim_{i \in I} D(i) \right) (U) \cong \lim_{i \in I} (D(i)(U)).$$

Proof. For each object U of $\mathcal{C}$, let

$$P(U) := \lim_{i \in I} D(i)(U),$$

with limit projections $\pi_{i,U} : P(U) \rightarrow D(i)(U)$ for each $i \in I$.

We first show that the assignment on objects

$$P : \text{Ob}(\mathcal{C}) \longrightarrow \text{Ob}(\mathcal{A}), \quad U \longmapsto P(U),$$

extends to a functor $\mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$. For each morphism $f : V \rightarrow U$ in $\mathcal{C}$, each presheaf $D(i)$ provides a restriction morphism $D(i)(f) : D(i)(U) \rightarrow D(i)(V)$. The family of compositions

$$D(i)(f) \circ \pi_{i,U} : P(U) \rightarrow D(i)(V), \quad i \in I,$$

forms a cone (Definition 1.2.2) from $P(U)$ to the diagram $(j \mapsto D(j)(V))$ in $\mathcal{A}$. By the universal property of the limit $P(V) = \lim_{j \in I} D(j)(V)$, there exists a unique morphism $P(f) : P(U) \rightarrow P(V)$ such that for all $i \in I$:

$$\pi_{i,V} \circ P(f) = D(i)(f) \circ \pi_{i,U}.$$

In other words, the following diagram commutes for every $i \in I$:

$$\begin{array}{ccc} P(U) & \xrightarrow{P(f)} & P(V) \\ \pi_{i,U} \downarrow & & \downarrow \pi_{i,V} \\ D(i)(U) & \xrightarrow{D(i)(f)} & D(i)(V) \end{array}$$

Next we verify functoriality. For any object U , the identity morphism $\text{id}_{P(U)}$ satisfies $\pi_{i,U} \circ \text{id}_{P(U)} = D(i)(\text{id}_U) \circ \pi_{i,U}$ for all i , since each $D(i)$ is a functor. By uniqueness of $P(\text{id}_U)$, we have $P(\text{id}_U) = \text{id}_{P(U)}$. For composable morphisms $W \xrightarrow{g} V \xrightarrow{f} U$, both $P(f \circ g)$ and $P(g) \circ P(f)$ satisfy the same universal property: for all i :

$$\pi_{i,W} \circ P(f \circ g) = D(i)(f \circ g) \circ \pi_{i,U} = D(i)(g) \circ D(i)(f) \circ \pi_{i,U} = D(i)(g) \circ \pi_{i,V} \circ P(f) = \pi_{i,W} \circ P(g) \circ P(f).$$

By uniqueness, $P(f \circ g) = P(g) \circ P(f)$; thus $P : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ is a presheaf.

Finally we establish the universal property in the presheaf category. Let Q be a presheaf on $\mathcal{C}$ valued in $\mathcal{A}$, and let $(\alpha_i : Q \rightarrow D(i))_{i \in I}$ be a cone from Q to the diagram D in $\text{PreShv}(\mathcal{C}, \mathcal{A})$. For each object U of $\mathcal{C}$, the family $(\alpha_{i,U} : Q(U) \rightarrow D(i)(U))_{i \in I}$ forms a cone from $Q(U)$ to the diagram $(i \mapsto D(i)(U))$ in $\mathcal{A}$. By the universal property of $P(U) = \lim_{i \in I} D(i)(U)$, there exists a unique morphism $\beta_U : Q(U) \rightarrow P(U)$ such that for all $i \in I$: (♠ TODO: this part could use a diagram)

$$\pi_{i,U} \circ \beta_U = \alpha_{i,U}.$$

We verify that $(\beta_U)_{U \in \text{Ob}(\mathcal{C})}$ is a morphism of presheaves $\beta : Q \rightarrow P$. For any morphism $f : V \rightarrow U$ in $\mathcal{C}$ and any $i \in I$: (♠ TODO: this part could use a diagram)

$$\pi_{i,V} \circ \beta_V \circ Q(f) = \alpha_{i,V} \circ Q(f) = D(i)(f) \circ \alpha_{i,U} = D(i)(f) \circ \pi_{i,U} \circ \beta_U = \pi_{i,V} \circ P(f) \circ \beta_U.$$

Since the limit projections $(\pi_{i,V})_{i \in I}$ are jointly monic, we obtain $\beta_V \circ Q(f) = P(f) \circ \beta_U$. Hence β is a natural transformation, i.e., a morphism of presheaves. If $\beta', \beta'' : Q \rightarrow P$ are two morphisms of presheaves satisfying $\pi_{i,U} \circ \beta'_U = \alpha_{i,U} = \pi_{i,U} \circ \beta''_U$ for all i and U , then $\beta'_U = \beta''_U$ by the uniqueness clause in the universal property of each limit $P(U)$; hence $\beta' = \beta''$.

Therefore, P is the limit of D in $\text{PreShv}(\mathcal{C}, \mathcal{A})$, computed sectionwise. □

Theorem 5.1.34 (Limits of sheaf diagrams are computed sectionwise). (♠ TODO: AI generated, verify) Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13). Let $\mathcal{A}$ be a (large) category (Definition 1.0.1). Let I be a category. Let $D : I \rightarrow \text{Shv}(\mathcal{C}, J, \mathcal{A})$ (Definition 5.1.32) be a diagram (Definition 1.2.1) of sheaves on $(\mathcal{C}, J)$ valued in $\mathcal{A}$, written $i \mapsto D(i)$.

Suppose that for every object U of $\mathcal{C}$, the limit (Definition 1.2.2) $\lim_{i \in I} D(i)(U)$ exists in $\mathcal{A}$. Then the limit of the diagram D exists in $\text{Shv}(\mathcal{C}, J, \mathcal{A})$, and it is computed sectionwise: for

every object U of $\mathcal{C}$, there is a canonical isomorphism in $\mathcal{A}$:

$$\left(\lim_{i \in I} D(i) \right) (U) \cong \lim_{i \in I} (D(i)(U)).$$

No smallness assumptions are required on I , $\mathcal{C}$, or $\mathcal{A}$. In particular, the index category I need not be small, the site $(\mathcal{C}, J)$ need not have a small topologically generating family, and the value category $\mathcal{A}$ need not be locally small or possess all limits of any size.

Proof. For each object U of $\mathcal{C}$, let

$$P(U) := \lim_{i \in I} D(i)(U),$$

with limit projections $\pi_{i,U} : P(U) \rightarrow D(i)(U)$ for each $i \in I$. The cone from $P(U)$ to the diagram $(j \mapsto D(j)(U))$ is illustrated by:

$$\begin{array}{ccccc} & & P(U) & & \\ & \swarrow \pi_{j,U} & \downarrow & \searrow \pi_{i,U} & \\ D(j)(U) & \xrightarrow{\pi_{k,U}} & D(i)(U) & & \\ & \downarrow D(\alpha)_U & \swarrow D(\beta)_U & & \\ & D(k)(U) & & & \end{array}$$

By Theorem 5.1.33, the assignment $U \mapsto P(U)$ defines a presheaf on $\mathcal{C}$ valued in $\mathcal{A}$, and it is the limit of the diagram D viewed as a diagram in $\text{PreShv}(\mathcal{C}, \mathcal{A})$.

Next we show that P is a sheaf. Let U be an object of $\mathcal{C}$ and let $S \in J(U)$ be a covering sieve. The sheaf condition requires that $P(U) \cong \varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} P(V)$, where $\mathcal{D}_S \hookrightarrow \mathcal{C}/U$ is the full subcategory such that $\text{Ob}(\mathcal{D}_S) = \{(f : V \rightarrow U) : f \in S(V)\}$. We have not yet established that this limit exists; its existence will follow from the joint limit argument below.

Since each $D(i)$ is a sheaf on $(\mathcal{C}, J)$ valued in $\mathcal{A}$ (Definition 5.1.32), we have canonical isomorphisms for all $i \in I$:

$$D(i)(U) \xrightarrow{\cong} \varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} D(i)(V).$$

Consider the bifunctor $F : I \times (\mathcal{D}_S)^{\text{op}} \rightarrow \mathcal{A}$ given by $F(i, (V \rightarrow U)) = D(i)(V)$. There are two ways to form iterated limits over this bifunctor. Taking the limit first over $(\mathcal{D}_S)^{\text{op}}$ and then over I yields $\lim_{i \in I} D(i)(U) = P(U)$, since each $D(i)$ is a sheaf. Taking the limit first over I gives $P(V)$ for each $(V \rightarrow U)$, and then taking the limit over $(\mathcal{D}_S)^{\text{op}}$ yields $\varprojlim_{(V \rightarrow U)} P(V)$. The relationship between these two iterated limits is:

We first verify that $P(U)$ satisfies the universal property of the joint limit $\lim_{(i, (V \rightarrow U)) \in I \times (\mathcal{D}_S)^{\text{op}}} F(i, (V \rightarrow U))$.

For each pair $(i, (V \rightarrow U))$, define the projection

$$\rho_{i,(V \rightarrow U)} : P(U) \xrightarrow{\pi_{i,U}} D(i)(U) \xrightarrow{\cong} \varprojlim_{(W \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} D(i)(W) \longrightarrow D(i)(V),$$

where the last map is the limit projection. These morphisms form a cone from $P(U)$ to the diagram F over $I \times (\mathcal{D}_S)^{\text{op}}$.

To verify the universal property, let N be any object of $\mathcal{A}$ equipped with a cone $(\sigma_{i,(V \rightarrow U)} : N \rightarrow D(i)(V))_{i \in I, (V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}}$. For each fixed $i \in I$, the family $(\sigma_{i,(V \rightarrow U)})_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}}$ forms a cone from N to the diagram $((W \rightarrow U) \mapsto D(i)(W))$ over $(\mathcal{D}_S)^{\text{op}}$. By the universal property of the limit $\varprojlim_{(W \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} D(i)(W) \cong D(i)(U)$, there exists a unique morphism $\tau_i : N \rightarrow D(i)(U)$ such that for all $(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}$:

$$\rho_{i,(V \rightarrow U)}^{\text{lim}} \circ \tau_i = \sigma_{i,(V \rightarrow U)},$$

where $\rho_{i,(V \rightarrow U)}^{\text{lim}}$ denotes the limit projection $D(i)(U) \cong \varprojlim_W D(i)(W) \rightarrow D(i)(V)$.

The family $(\tau_i : N \rightarrow D(i)(U))_{i \in I}$ forms a cone from N to the diagram $(j \mapsto D(j)(U))$ over I , because for any morphism $\alpha : i \rightarrow j$ in I and any $(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}$, naturality of the cone gives:

$$D(\alpha)_V \circ \sigma_{i,(V \rightarrow U)} = \sigma_{j,(V \rightarrow U)}.$$

Since the limit projections $(\rho_{j,(V \rightarrow U)}^{\text{lim}})_{(V \rightarrow U)}$ are jointly monic, we obtain $D(\alpha)_U \circ \tau_i = \tau_j$. By the universal property of $P(U) = \lim_{i \in I} D(i)(U)$, there exists a unique morphism $\theta : N \rightarrow P(U)$ such that for all $i \in I$:

$$\pi_{i,U} \circ \theta = \tau_i.$$

By construction, for all $(i, (V \rightarrow U))$:

$$\rho_{i,(V \rightarrow U)} \circ \theta = \sigma_{i,(V \rightarrow U)},$$

and θ is unique with this property because the morphisms $(\rho_{i,(V \rightarrow U)})$ are jointly monic (they factor through the limit projections of $P(U)$, which are jointly monic).

This establishes that $P(U)$ is the joint limit over $I \times (\mathcal{D}_S)^{\text{op}}$. By ??, since the joint limit exists and for each $(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}$ the partial limit $\lim_{i \in I} D(i)(V) = P(V)$ exists (by hypothesis), the iterated limit in the other order also exists and is canonically isomorphic to the joint limit:

$$P(U) \xrightarrow{\cong} \varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} \left(\lim_{i \in I} D(i)(V) \right) = \varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} P(V).$$

This is exactly the sheaf condition for P at U with respect to the covering sieve S . Since U and S were arbitrary, P is a sheaf on $(\mathcal{C}, J)$ valued in $\mathcal{A}$.

Finally we establish that P is the sheaf limit. We have shown that P is both a sheaf and the presheaf limit of D . It remains to verify that P satisfies the universal property of the limit within the sheaf category $\text{Shv}(\mathcal{C}, J, \mathcal{A})$. Let Q be a sheaf on $(\mathcal{C}, J)$ valued in $\mathcal{A}$, and let $(\alpha_i : Q \rightarrow D(i))_{i \in I}$ be a cone from Q to D in $\text{Shv}(\mathcal{C}, J, \mathcal{A})$. Viewing this as a cone in $\text{PreShv}(\mathcal{C}, \mathcal{A})$, the universal property of P as the presheaf limit yields a unique morphism of

presheaves $\beta : Q \rightarrow P$ such that for all $i \in I$, $\pi_i \circ \beta = \alpha_i$. Since Q and P are both sheaves, β is automatically a morphism in $\text{Shv}(\mathcal{C}, J, \mathcal{A})$. Uniqueness follows from the uniqueness of β as a presheaf morphism.

Therefore P is the limit of D in $\text{Shv}(\mathcal{C}, J, \mathcal{A})$, computed sectionwise. $\square$

Proposition 5.1.35 (Sheaves on a pretopology vs. sheaves on the generated topology). (♠
 TODO: AI generated, verify)

Let $\mathcal{C}$ be a category (Definition 1.0.1) equipped with a Grothendieck pretopology (Definition 5.1.10) K . Let J_K be the Grothendieck topology generated by K (Definition 5.1.16). Let $\mathcal{A}$ be a category (Definition 1.0.1).

A presheaf (Definition 5.0.1) $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ is a sheaf on $(\mathcal{C}, K)$ valued in $\mathcal{A}$ (Definition 5.1.12) if and only if F is a sheaf on the site $(\mathcal{C}, J_K)$ valued in $\mathcal{A}$ (Definition 5.1.32).

In other words, the full subcategory of $\text{PreShv}(\mathcal{C}, \mathcal{A})$ consisting of sheaves with respect to K coincides with the full subcategory consisting of sheaves with respect to J_K .

Proof. (♠ TODO: AI generated, verify)

($\Rightarrow$) Suppose F is a sheaf on $(\mathcal{C}, K)$. Let U be an object of $\mathcal{C}$ and let $S \in J_K(U)$ be a covering sieve. By the definition of J_K (Definition 5.1.16), there exists a covering family $\{U_i \rightarrow U\}_{i \in I} \in K(U)$ whose generated sieve is contained in S .

The sheaf condition for the pretopology asserts that the canonical morphism

$$F(U) \longrightarrow \varprojlim_{[n] \in \Delta} F(\check{C}(\{U_i \rightarrow U\})_n)$$

is an isomorphism, where $\check{C}(\{U_i \rightarrow U\})$ is the Čech nerve of the covering family.

The sieve S contains the sieve generated by $\{U_i \rightarrow U\}_{i \in I}$, so there is a canonical functor from the index category of the limit over S to the index category of the Čech nerve (via the inclusion of the sub-diagram). This functor is cofinal: every object $(V \rightarrow U)$ in the sieve S admits a map from some object in the Čech nerve, since by definition of J_K , any morphism in S factors through some U_i . By the cofinality of limits (a limit over a diagram indexed by Δ can be computed as a limit over any cofinal sub-diagram), it follows that

$$\varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} F(V) \cong \varprojlim_{[n] \in \Delta} F(\check{C}(\{U_i \rightarrow U\})_n),$$

so $F(U) \cong \varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} F(V)$, meaning F satisfies the sheaf condition for S .

($\Leftarrow$) Suppose F is a sheaf on $(\mathcal{C}, J_K)$. Let $\{U_i \rightarrow U\}_{i \in I}$ be a covering family in $K(U)$. The sieve S generated by this family belongs to $J_K(U)$, so the sheaf condition for F with respect to S gives an isomorphism

$$F(U) \cong \varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} F(V).$$

The Čech nerve of $\{U_i \rightarrow U\}_{i \in I}$ embeds as a cofinal sub-diagram into the diagram indexed by $(\mathcal{D}_S)^{\text{op}}$. (Every morphism $V \rightarrow U$ in S factors through some U_i , and the simplicial structure of the Čech nerve captures all higher-order matching conditions.) Therefore,

$$F(U) \cong \varprojlim_{[n] \in \Delta} F(\check{C}(\{U_i \rightarrow U\})_n),$$

which is precisely the sheaf condition for the pretopology K .

Therefore, the two notions of sheaf coincide. $\square$

Definition 5.1.36 (Sheaf on a topological space). Let X be a topological space (Definition B.0.1), let $\mathcal{D}$ be a category (Definition 1.0.1); a posteriori, it is necessary for $\mathcal{D}$ to have a terminal object (Lemma 2.0.7) for the following definition to be non-void. Let $\mathcal{F}$ be a presheaf valued in $\mathcal{D}$ on X (Definition 5.1.38). Then $\mathcal{F}$ is a *sheaf* if it satisfies the following additional condition (known as the *sheaf axioms*):

For every open set $U \subseteq X$ and every open cover (Definition B.0.2) $\{U_i\}_{i \in I}$ of U , let $\mathcal{J}$ be the diagram (Definition 1.2.1) in the category of opens (Definition 1.0.7) of U consisting of the inclusions $U_i \cap U_j \hookrightarrow U_i$ for all $i, j \in I$. Then $\mathcal{F}$ is a sheaf if the limit (Definition 1.2.2) of the diagram $\mathcal{F} \circ \mathcal{J}$ exists in $\mathcal{D}$ and the natural morphism

$$\mathcal{F}(U) \rightarrow \lim_{j \in \mathcal{J}} \mathcal{F}(j)$$

is an isomorphism. More precisely, $\mathcal{J} : J \rightarrow \text{Open}(U)$ should be the functor whose index category J consists of

1. An object i for every $i \in I$ and an object (i, j) for every pair $i, j \in I$,
2. Morphisms $p_1 : (i, j) \rightarrow i$ and $p_2 : (i, j) \rightarrow j$ for every pair $i, j \in I$

and which sends the objects and morphisms as follows:

1. $\mathcal{J}(i) = U_i$
2. $\mathcal{J}(i, j) = U_i \cap U_j$
3. $\mathcal{J}(p_1) : U_i \cap U_j \hookrightarrow U_i$
4. $\mathcal{J}(p_2) : U_i \cap U_j \hookrightarrow U_j$.

In particular, taking $U = \emptyset$ and taking the empty open cover of the empty set, $\mathcal{F}(\emptyset)$ must be the terminal object (Lemma 2.0.7) of $\mathcal{D}$

In the case that $\mathcal{D}$ admits all small limits (Definition 1.2.5), the sheaf condition is equivalent (♠ TODO: this equivalence) to the following: For every open set $U \subset X$ and every open cover $\{U_i\}_{i \in I}$ of U , the following equalizer diagram is exact (Definition 1.2.37):

$$\mathcal{F}(U) \rightarrow \prod_{i \in I} \mathcal{F}(U_i) \rightrightarrows \prod_{i, j \in I} \mathcal{F}(U_i \cap U_j).$$

Here, the morphism $\mathcal{F}(U) \rightarrow \prod_{i \in I} \mathcal{F}(U_i)$ and the two morphisms $\prod_{i \in I} \mathcal{F}(U_i) \rightrightarrows \prod_{i, j \in I} \mathcal{F}(U_i \cap U_j)$ are induced by the restriction maps (Definition 5.1.38) $\mathcal{F}(U) \rightarrow \mathcal{F}(U_i)$ and $\mathcal{F}(U_i) \rightarrow \mathcal{F}(U_i \cap U_j)$.

In the case that $\mathcal{D}$ is some subcategory (Definition 1.3.1) of the category of sets (Definition 1.0.3), the sheaf condition is equivalent to the following: For every open set $U \subseteq X$ and every open cover (Definition B.0.2) $\{U_i\}_{i \in I}$ of U ,

- (Locality) If $s, t \in \mathcal{F}(U)$ are such that $s|_{U_i} = t|_{U_i}$ for all i , then $s = t$.
- (Gluing) If for each i there is $s_i \in \mathcal{F}(U_i)$ such that for all i, j one has $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$, then there exists a unique $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i .

Equivalently, a sheaf on a topological space X may be defined as a sheaf on (Definition 5.1.32) the site (Definition 5.1.13) of opens on X (Definition 5.1.30).

Definition 5.1.37 (Morphism of presheaves on a topological space). Let X be a topological space (Definition B.0.1), let $\mathcal{D}$ be a category (Definition 1.0.1) and let $\mathcal{F}$, and $\mathcal{G}$ be presheaves valued in $\mathcal{D}$ on X (Definition 5.1.38).

A *morphism of presheaves* $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a collection of maps

$$\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

in $\mathcal{D}$ defined for every open set $U \subseteq X$, such that the maps are compatible with restriction: for every inclusion $V \subseteq U$ of open sets, the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi(U)} & \mathcal{G}(U) \\ \downarrow \rho_V^U & & \downarrow \rho_V^U \\ \mathcal{F}(V) & \xrightarrow{\varphi(V)} & \mathcal{G}(V) \end{array}$$

commutes.

Equivalently, φ is a natural transformation (Definition 1.1.3) of $\mathcal{F}, \mathcal{G}$, regarded as (Definition 5.1.38) functors

$$\mathcal{F}, \mathcal{G} : \mathbf{Open}(X)^{\text{op}} \rightarrow \mathcal{D}.$$

If $\mathcal{F}$ and $\mathcal{G}$ are sheaves, then a *morphism of sheaves* is a morphism between $\mathcal{F}$ and $\mathcal{G}$ as presheaves.

Definition 5.1.38 (Presheaf on a topological space). Let X be a topological space (Definition B.0.1). Let $\mathcal{D}$ be a category.

A *presheaf (of objects of $\mathcal{D}$ /valued in $\mathcal{D}$) on X* is a rule $\mathcal{F}$ that assigns:

- to each open set $U \subseteq X$, an object $\mathcal{F}(U) \in \text{Ob } \mathcal{D}$, called the *sections of $\mathcal{F}$ over U* ,
- to each inclusion of open sets $V \subseteq U$, a morphism

$$\rho_V^U : \mathcal{F}(U) \rightarrow \mathcal{F}(V), \quad s \mapsto s|_V,$$

in the category $\mathcal{D}$ called the *restriction map* such that the following conditions hold:

- (Identity) For every open set $U \subseteq X$, the restriction map ρ_U^U is the identity on $\mathcal{F}(U)$.
- (Transitivity) For inclusions $W \subseteq V \subseteq U$ of open sets, one has

$$\rho_W^U = \rho_W^V \circ \rho_V^U.$$

For instance, we may speak of a *presheaf of sets/groups/rings/etc. on the topological space X* .

Equivalently, a presheaf on X (of objects in a category $\mathcal{D}$) is a functor (Definition 1.1.1)

$$\mathbf{Open}(X)^{\mathrm{op}} \rightarrow \mathcal{D}$$

from the opposite of the category $\mathbf{Open}(X)$ (Definition 1.0.7) of open subsets of X (see also Definition 5.0.1) .

Equivalently, a presheaf on X is a presheaf on the category $\mathbf{Open}(X)$ in the sense of Definition 5.0.1 .

The sections object $\mathcal{F}(U)$ is also denoted by $\Gamma(U, \mathcal{F})$ (see Definition 5.3.11) . Moreover, the object $\mathcal{F}(X) = \Gamma(X, \mathcal{F})$ is called the *global sections object of $\mathcal{F}$* . This agrees with the notion of global sections as discussed in Definition 5.3.11.

Definition 5.1.39 ($\mathcal{P}$ -torsion sheaf of abelian groups). (♠ TODO: must the site have a small topological generating family?) Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13) with a small topological generating family (Definition 5.1.13) (or a U -small topologically generating family if a universe (Definition E.0.1) U is available), and let $\mathcal{P}$ be a collection of prime numbers. A sheaf (Definition 5.1.32) of abelian groups $\mathcal{F}$ on $(\mathcal{C}, J)$ is called a *$\mathcal{P}$ -torsion sheaf* if for every object U of $\mathcal{C}$, every section $s \in \mathcal{F}(U)$, and every $p \in \mathcal{P}$, there exists some $n \geq 1$ such that $p^n s = 0$ in $\mathcal{F}(U)$.

Definition 5.1.40 (Torsion sheaf of abelian groups). (♠ TODO: must the site have a small topological generating family?) Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13) with a small topological generating family (Definition 5.1.13) (or a U -small topologically generating family if a universe (Definition E.0.1) U is available). A sheaf (Definition 5.4.13) of abelian groups $\mathcal{F}$ on $(\mathcal{C}, J)$ is called a *torsion sheaf of abelian groups* if for every object U in $\mathcal{C}$ and every section $s \in \mathcal{F}(U)$, there exists a nonzero integer n such that $ns = 0$ in $\mathcal{F}(U)$.

5.1.6. Sheafy functors.

Definition 5.1.41. Let $\mathcal{A}, \mathcal{B}$ be categories (Definition 1.0.1), let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a functor (Definition 1.1.1), and let $(\mathcal{D}, K)$ be a site (Definition 5.1.13).

Assume at least one of the following:

1. A sheafification functor (Definition 5.2.20)

$$a : \mathrm{PreSh}(\mathcal{D}, K; \mathcal{B}) \rightarrow \mathrm{Sh}(\mathcal{D}, K; \mathcal{B})$$

exists.

2. F is a continuous functor (Definition 1.2.28) and $\mathcal{B}$ admits limits (Theorem 1.2.35) indexed by covering sieves (Definition 5.1.13) of K .

Define the *sheafy functor* $\mathcal{F}$ of F under each of the above assumptions as follows:

1. As the functor

$$\mathcal{F} : \mathrm{PreSh}(\mathcal{D}, \mathcal{A}) \rightarrow \mathrm{Sh}(\mathcal{D}, K; \mathcal{B})$$

given by $S \mapsto a(F \circ S)$. By restriction, there is an induced functor

$$\mathrm{Sh}(\mathcal{D}, K; \mathcal{A}) \rightarrow \mathrm{Sh}(\mathcal{D}, K; \mathcal{B})$$

on the category of sheaves on $\mathcal{D}$ valued in $\mathcal{A}$; we usually denote this restricted functor by $\mathcal{F}$ as well.

2. As the functor

$$\mathcal{F} : \mathrm{Sh}(\mathcal{D}, K; \mathcal{A}) \rightarrow \mathrm{Sh}(\mathcal{D}, K; \mathcal{B})$$

given by $S \mapsto F \circ S$. Due to the assumptions on F and $\mathcal{B}$, $F \circ S$ is a sheaf.

Note that if both assumptions hold, then the two functors

$$\mathrm{Sh}(\mathcal{D}, K; \mathcal{A}) \rightarrow \mathrm{Sh}(\mathcal{D}, K; \mathcal{B})$$

are naturally isomorphic.

Sheafy functors are often notated with scripted letters.

Proposition 5.1.42. Let $(\mathcal{D}, K)$ be a site (Definition 5.1.13). Let $\mathcal{A}, \mathcal{B}$ be categories (Definition 1.0.1). There is an equivalence (Definition 1.1.8)

$$\mathrm{Sh}(\mathcal{D}, K; \mathcal{A} \times \mathcal{B}) \cong \mathrm{Sh}(\mathcal{D}, K; \mathcal{A}) \times \mathrm{Sh}(\mathcal{D}, K; \mathcal{B})$$

(Definition 5.1.32) (Definition 1.3.11).

Proof. (♠ TODO:) □

Theorem 5.1.43. Let $(\mathcal{D}, K)$ be a site (Definition 5.1.13). Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be categories (Definition 1.0.1). Let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ and $G : \mathcal{A}^{\mathrm{op}} \times \mathcal{C} \rightarrow \mathcal{B}$ be functors such that for any fixed $A \in \mathcal{A}$, there is an adjunction (Definition 1.1.13) $F(A, -) \dashv G(A, -)$. Assume the following:

- there is a sheafification functor (Definition 5.2.20)

$$a : \mathrm{PreSh}(\mathcal{D}, \mathcal{C}) \rightarrow \mathrm{Sh}(\mathcal{D}, K; \mathcal{C}),$$

and

- $\mathcal{B}$ admits limits (Definition 1.2.2) indexed by the covering sieves in the Grothendieck topology (Definition 5.1.13) K .

Let $\mathcal{F}$ be the composition

$$\mathrm{Sh}(\mathcal{D}, K; \mathcal{A}) \times \mathrm{Sh}(\mathcal{D}, K; \mathcal{B}) \cong \mathrm{Sh}(\mathcal{D}, K; \mathcal{A} \times \mathcal{B}) \rightarrow \mathrm{Sh}(\mathcal{D}, K; \mathcal{C})$$

of the equivalence of Proposition 5.1.42 with the sheafy functor (Definition 5.1.41) of F .

For a fixed sheaf $\mathcal{R} \in \mathrm{Sh}(\mathcal{D}, K; \mathcal{A})$, let $\mathcal{G}_{\mathcal{R}} : \mathrm{Sh}(\mathcal{D}, K; \mathcal{C}) \rightarrow \mathrm{Sh}(\mathcal{D}, K; \mathcal{B})$ be the functor defined section-wise by

$$\mathcal{T} \mapsto (U \mapsto G(\mathcal{R}(U), \mathcal{T}(U))).$$

Note that since $G(\mathcal{R}(U), -)$ is a right adjoint (Definition 1.1.13), it preserves (Theorem 1.2.29) the limits (Definition 1.2.2) indexed by the covering sieves of K ; thus, $\mathcal{G}_{\mathcal{R}}(\mathcal{T})$ is a sheaf whenever $\mathcal{T}$ is a sheaf.

Furthermore, the assignment $(\mathcal{R}, \mathcal{T}) \mapsto \mathcal{G}_{\mathcal{R}}(\mathcal{T})$ defines a functor

$$\mathcal{G} : \mathrm{Sh}(\mathcal{D}, K; \mathcal{A})^{\mathrm{op}} \times \mathrm{Sh}(\mathcal{D}, K; \mathcal{C}) \rightarrow \mathrm{Sh}(\mathcal{D}, K; \mathcal{B}).$$

Then for all $\mathcal{R} \in \text{Sh}(\mathcal{D}, K; \mathcal{A})$, there is an adjunction

$$\mathcal{F}(\mathcal{R}, -) \dashv \mathcal{G}_{\mathcal{R}}(-)$$

induced by the pointwise adjunctions $F(\mathcal{R}(U), -) \dashv G(\mathcal{R}(U), -)$ and the universal property of sheafification.

Proof. (♠ TODO:) □

Definition 5.1.44. 1. Let $\mathcal{C}$ be a site, and let $\mathcal{A}$ and $\mathcal{B}$ be sheaves (Definition 5.1.32) of (not necessarily commutative) rings (Definition C.0.1) on $\mathcal{C}$.

- (a) An $(\mathcal{A}, \mathcal{B})$ -bimodule (or a bimodule over $(\mathcal{A}, \mathcal{B})$) is a sheaf (Definition 5.1.32) $\mathcal{M}$ of abelian groups on $\mathcal{C}$ equipped with a left $\mathcal{A}$ -module structure given by a morphism of sheaves (Definition 5.1.32) of sets

$$\lambda : \mathcal{A} \times \mathcal{M} \longrightarrow \mathcal{M},$$

and a right $\mathcal{B}$ -module structure given by a morphism of sheaves of sets

$$\rho : \mathcal{M} \times \mathcal{B} \longrightarrow \mathcal{M},$$

such that the actions are compatible. Specifically, for every object U in $\mathcal{C}$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the equality

$$\lambda_U(a, \rho_U(m, b)) = \rho_U(\lambda_U(a, m), b)$$

holds in $\mathcal{M}(U)$. In standard multiplicative notation where $\lambda(a, m)$ is denoted $a \cdot m$ and $\rho(m, b)$ is denoted $m \cdot b$, this condition is the associativity axiom

$$(a \cdot m) \cdot b = a \cdot (m \cdot b).$$

In particular, for every object $U \in \mathcal{C}$, the abelian group $\mathcal{M}(U)$ has the structure of an $\mathcal{A}(U) - \mathcal{B}(U)$ -bimodule (Definition C.0.8).

- (b) Assume that a sheafification functor (Definition 5.2.20)

$$\text{PreShv}(\mathcal{C}, \mathbf{Rings}) \rightarrow \text{Shv}(\mathcal{C}, \mathbf{Rings})$$

exists. For rings A and B , a A - B -bimodule or bimodule over (A, B) on $\mathcal{C}$ is a bimodule over $(\underline{A}, \underline{B})$ where $\underline{A}$ and $\underline{B}$ are the constant sheaves (Definition 5.2.29) on $\mathcal{C}$ with values A and B respectively.

- (c) Let $\mathcal{M}$ and $\mathcal{N}$ be $(\mathcal{A}, \mathcal{B})$ -bimodules. A homomorphism of $(\mathcal{A}, \mathcal{B})$ -bimodules (or an $(\mathcal{A}, \mathcal{B})$ -linear morphism) is a morphism of sheaves of abelian groups $f : \mathcal{M} \rightarrow \mathcal{N}$ such that for every object U of $\mathcal{C}$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the following compatibility conditions hold:

$$f_U(a \cdot m) = a \cdot f_U(m) \quad \text{and} \quad f_U(m \cdot b) = f_U(m) \cdot b.$$

We denote the category of $(\mathcal{A}, \mathcal{B})$ -bimodules, with morphisms being morphisms of sheaves of abelian groups compatible with both the left $\mathcal{A}$ -action and the right $\mathcal{B}$ -action, by $\mathcal{A}\text{-}\mathcal{B}\text{-Mod}$ or sometimes by ${}_{\mathcal{A}}\text{Mod}_{\mathcal{B}}$ (♠ TODO: talk about how bimodules can be identifies with left/right modules)

2. Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13). Let $\mathcal{O}$ be a sheaf of (not necessarily commutative) rings on $(\mathcal{C}, J)$ (Definition 5.1.32), i.e. $((\mathcal{C}, J), \mathcal{O})$ is a ringed site (Definition 5.4.9).

- (a) An (left/right/two-sided) $\mathcal{O}$ -module consists of the following data:

- A sheaf $\mathcal{F}$ of abelian groups on $(\mathcal{C}, J)$,
- for every object $U \in \mathcal{C}$, the structure of an (left/right/two-sided) $\mathcal{O}(U)$ -module on $\mathcal{F}(U)$,

such that for every morphism $f : V \rightarrow U$ in $\mathcal{C}$, the restriction map

$$\rho_{U,V} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$$

is $\mathcal{O}(U)$ -linear when the $\mathcal{O}(U)$ -action on $\mathcal{F}(V)$ is defined via the natural ring homomorphism

$$\mathcal{O}(U) \rightarrow \mathcal{O}(V)$$

induced by f .

(b) Let $\mathcal{F}$ and $\mathcal{G}$ be $\mathcal{O}$ -modules (Definition 5.3.9).

A *morphism of $\mathcal{O}$ -modules* $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves (Definition 5.1.32) of abelian groups such that, for every object $U \in \mathcal{C}$, the component map

$$\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

is $\mathcal{O}(U)$ -linear, i.e. it satisfies

$$\varphi_U(r \cdot s) = r \cdot \varphi_U(s) \quad \text{for all } r \in \mathcal{O}(U), s \in \mathcal{F}(U).$$

The collection of all $\mathcal{O}$ -modules together with their morphisms of $\mathcal{O}$ -modules forms the *category of $\mathcal{O}$ -modules*, denoted $\mathbf{Mod}(\mathcal{O})$. The Hom functors $\mathrm{Hom}_{\mathbf{Mod}(\mathcal{O})}$ of this category may be denoted by $\mathrm{Hom}_{\mathcal{O}}$.

In case that a sheafification functor (Definition 5.2.20)

$$\mathrm{PreShv}(\mathcal{C}, \mathbf{Rings}) \rightarrow \mathrm{Shv}(\mathcal{C}, \mathbf{Rings})$$

exists, a left, right, two-sided $\mathcal{O}$ -module (and morphisms thereof) is equivalent to a $(\mathcal{O}, \mathbb{Z})$ -bimodule, $(\mathbb{Z}, \mathcal{O})$ -bimodule, and $(\mathcal{O}, \mathcal{O})$ -bimodule (and morphisms thereof) respectively, where $\mathbb{Z}$ is the constant sheaf (Definition 5.2.29) of the integer ring $\mathbb{Z}$. In particular, the category of $(\mathbb{Z}, \mathbb{Z})$ -bimodules is equivalent to the category of sheaves of abelian groups on $\mathcal{C}$.

Definition 5.1.45 (Tensor product of bimodules). Let R, S, T be (not necessarily commutative) rings (Definition C.0.1), let M be an R - S bimodule (Definition C.0.8), and let N be an S - T bimodule. In the free abelian group $\mathbb{Z}[M \times N]$ generated by the Cartesian product $M \times N$, let U be the subgroup generated by elements of the form (♠ TODO: subgroup generated)

$$\begin{aligned} (m + m', n) - (m, n) - (m', n), \\ (m, n + n') - (m, n) - (m, n'), \\ (m \cdot s, n) - (m, s \cdot n), \end{aligned}$$

for all $m, m' \in M$, $n, n' \in N$, and $s \in S$. The *tensor product of M and N over S* is the quotient abelian group

$$M \otimes_S N := \mathbb{Z}[M \times N]/U.$$

The image of an element of the form $(m, n) \in M \times N$ in $M \otimes_S N$ is denoted $m \otimes n$ and called a *pure tensor*. In general, the elements of $M \otimes_S N$ are finite sums

$$\sum_{i=1}^n m_i \otimes n_i \quad m_i \in M, n_i \in N$$

of pure tensors. Thus, the pure tensors satisfy the following relations:

$$\begin{aligned}(m + m') \otimes n &= m \otimes n + m' \otimes n \\ m \otimes (n + n') &= m \otimes n + m \otimes n' \\ (m \cdot s) \otimes n &= m \otimes (s \cdot n)\end{aligned}$$

This tensor product becomes naturally an R - T bimodule with left action and right action defined by

$$\begin{aligned}r \cdot (m \otimes n) &= (r \cdot m) \otimes n, \\ (m \otimes n) \cdot t &= m \otimes (n \cdot t),\end{aligned}$$

for all $r \in R$, $t \in T$, $m \in M$, and $n \in N$.

Inductively, given rings $R_0, \dots, R_k$ and $R_{i-1} - R_i$ -bimodules M_i for $i = 1, \dots, k$, we may speak of the tensor product

$$M_0 \otimes_{R_1} M_1 \otimes_{R_2} \cdots \otimes_{R_{k-1}} M_k;$$

tensor products are associative(♠ TODO:), so parentheses are not strictly needed to notate them. Its *pure tensors* are elements of the form $m_0 \otimes m_1 \otimes \cdots \otimes m_k$ for $m_i \in M_i$, and its general elements are finite sums

$$\sum_{j=1}^n m_{0j} \otimes m_{1j} \otimes \cdots \otimes m_{kj} \quad m_{ij} \in M_i.$$

of pure tensors. It also has a natural $R_0 - R_k$ -bimodule structure.

Given a ring R and a two-sided R -module M , we may also speak of the *n -fold tensor product* $M^{\otimes n} = M^{\otimes_{R^n}}$

Definition 5.1.46. Let $\mathcal{Ring}$ be the category of (associative, with identity) rings (Definition 1.0.5). Let (R, S) and (R', S') be objects in $\mathcal{Ring}^2$ (Definition 1.3.11). Let M be an (R, S) -bimodule (Definition C.0.8) and M' be an (R', S') -bimodule. A *bimodule morphism over a morphism* $(f, g) \in \text{Hom}_{\mathcal{Ring}^2}((R, S), (R', S'))$ is a group homomorphism $\phi : M \rightarrow M'$ such that for all $r \in R, s \in S, m \in M$:

$$\phi(r \cdot m \cdot s) = f(r) \cdot \phi(m) \cdot g(s)$$

Definition 5.1.47. Let $\mathcal{Ring}$ be the category of (associative, with identity) rings (Definition 1.0.5). Define the *category of all bimodules over all rings* as the category, denoted by notations such as $\mathcal{Bimod}$, as follows:

- Objects: Triples (R, S, M) where (R, S) is an object of $\mathcal{Ring}^2$ (Definition 1.3.11) and M is an (R, S) -bimodule (Definition C.0.8).
- Morphisms: A morphism $(f, g, \phi) : (R, S, M) \rightarrow (R', S', M')$ where (f, g) is a morphism in $\mathcal{Ring}^2$ and $\phi : M \rightarrow M'$ is a bimodule morphism over (f, g) .
- Composition: $(f', g', \phi') \circ (f, g, \phi) = (f' \circ f, g' \circ g, \phi' \circ \phi)$.

Note that there is a canonical projection functor $\mathcal{Bimod} \rightarrow \mathcal{Ring}^2$ defined by $(R, S, M) \mapsto (R, S)$ and $(f, g, \phi) \mapsto (f, g)$. This functor exhibits $\mathcal{Bimod}$ as a fibered category (♠ TODO:

define fibered category) over $\mathcal{R}ing^2$. For any morphism $(f, g) : (R, S) \rightarrow (R', S')$, the cartesian lift of an object (R', S', M') is given by the (R, S) -bimodule ${}_f M'_g$ where actions are defined via restriction of scalars: $r \cdot m \cdot s = f(r) \cdot m \cdot g(s)$.

Definition 5.1.48. The *category of (tensor) composable bimodules*, denotable by notations such as $\mathcal{B}imod \times_{\mathcal{R}ing} \mathcal{B}imod$, is the (♠ TODO: fiber product in what sense) fiber product of the category $\mathcal{B}imod$ (Definition 5.1.47) of all bimodules over all rings with itself over the category $\mathcal{R}ing$ (Definition 1.0.5) of all (associative, unital) rings with respect to the functors

$$\begin{aligned} \mathcal{B}imod &\rightarrow \mathcal{R}ing^2 \xrightarrow{\text{pr}_1} \mathcal{R}ing \\ \mathcal{B}imod &\rightarrow \mathcal{R}ing^2 \xrightarrow{\text{pr}_2} \mathcal{R}ing \end{aligned}$$

where pr_1 and pr_2 are the projection functors.

Specifically, it is the category whose objects are pairs of bimodules $((R, S, M), (S, T, N))$ sharing a common middle ring S , and whose morphisms are pairs of morphisms $((f, g, \phi), (g, h, \psi))$ such that the second ring component of the first morphism and the first ring component of the second morphism coincide as $g : S \rightarrow S'$.

Definition 5.1.49. Let $\mathcal{B}imod \times_{\mathcal{R}ing} \mathcal{B}imod$ be the category of tensor composable bimodules (Definition 5.1.48). The *tensor product functor* $\otimes : \mathcal{B}imod \times_{\mathcal{R}ing} \mathcal{B}imod \rightarrow \mathcal{B}imod$ is defined as follows:

- On objects: $\otimes((R, S, M), (S, T, N)) = (R, T, M \otimes_S N)$ (Definition 5.1.45).
- On morphisms: $\otimes((f, g, \phi), (g, h, \psi)) = (f, h, \phi \otimes \psi)$, where $\phi \otimes \psi$ is the unique group homomorphism satisfying $(\phi \otimes \psi)(m \otimes n) = \phi(m) \otimes \psi(n)$.

Proposition 5.1.50. The tensor product functor is associative up to a natural isomorphism. For any three composable bimodules ${}_R M_S$, ${}_S N_T$, and ${}_T P_U$, there exists a natural isomorphism in $\mathcal{B}imod$:

$$\alpha_{M,N,P} : (M \otimes_S N) \otimes_T P \xrightarrow{\cong} M \otimes_S (N \otimes_T P)$$

defined by the mapping $((m \otimes n) \otimes p) \mapsto (m \otimes (n \otimes p))$. This data, along with the unit bimodules ${}_R R_R$, endows $\mathcal{B}imod$ with the structure of a bicategory (Definition 11.1.2).

Definition 5.1.51. The *category of Hom composable bimodules*, (♠ TODO: there needs to be a discussion like Definition 5.1.48 but for Hom's)

Definition 5.1.52 (Sheafified Tensor Product). (♠ TODO: Consolidate with) Let $(\mathcal{D}, K)$ be a site (Definition 5.1.13) that admits a small topologically generating family (Definition 5.1.13) (e.g. by being essentially small) and let $\mathcal{R}, \mathcal{S}, \mathcal{T}$ be sheaves of rings (Definition 5.1.32) on $(\mathcal{D}, K)$. Let $\mathcal{M}$ be a sheaf (Definition 5.1.32) of $\mathcal{R}$ - $\mathcal{S}$ bimodules (Definition 5.3.9) and $\mathcal{N}$ be a sheaf of $\mathcal{S}$ - $\mathcal{T}$ bimodules. The *sheafified tensor product*, denoted $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}$, is the sheaf of $\mathcal{R}$ - $\mathcal{T}$ bimodules defined as the sheafification (Definition 5.2.20) of the section-wise tensor product presheaf:

$$U \mapsto \mathcal{M}(U) \otimes_{\mathcal{S}(U)} \mathcal{N}(U).$$

(Definition 5.1.45) Explicitly, $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N} = a(U \mapsto \mathcal{M}(U) \otimes_{\mathcal{S}(U)} \mathcal{N}(U))$, where a is the sheafification functor, which exists by Theorem 5.2.27. (♠ TODO: to apply the theorem, it seems necessary to show that the category of sheaves of $\mathcal{R}$ - $\mathcal{T}$ bimodules is complete, cocomplete

(probably need sheafification of pointwise construction), has exact filtered colimits, and the IPC property)

In particular, we have a bi-additive functor

$$\otimes : {}_{\mathcal{R}}\mathbf{Mod}_{\mathcal{S}} \times {}_{\mathcal{S}}\mathbf{Mod}_{\mathcal{T}} \rightarrow {}_{\mathcal{R}}\mathbf{Mod}_{\mathcal{T}}$$

In the framework of Definition 5.1.41, the sheafified tensor product is the evaluation of the sheafy functor $\mathcal{F}$ induced by the tensor product functor (Definition 5.1.49) $F = \otimes : \mathbf{Bimod} \times_{\mathcal{R}ing} \mathbf{Bimod} \rightarrow \mathbf{Bimod}$ (Definition 5.1.47) (Definition 5.1.48); (♠ TODO: must show that the category $\mathbf{Bimod}$ is complete, cocomplete, has exact essentially small filtered limits, and satisfies the IPC property to argue that there is a sheafification functor by Theorem 5.2.27) (♠ TODO: show that the “composable” subcategory of sheaves of bimodules over all sheaves of rings on the site is equivalence to the category of sheaves valued in the caetgory of all composable bimodules) Specifically, if $\mathbb{M} := (\mathcal{R}, \mathcal{S}, \mathcal{M}) \in \mathbf{Sh}(\mathcal{D}, K; \mathbf{Bimod})$ and $\mathbb{N} := (\mathcal{S}, \mathcal{T}, \mathcal{N}) \in \mathbf{Sh}(\mathcal{D}, K; \mathbf{Bimod})$, then the sheafified tensor product is the module component of the sheaf $\mathcal{F}(\mathbb{M}, \mathbb{N}) \in \mathbf{Sh}(\mathcal{D}, K; \mathbf{Bimod})$, which is the triple $(\mathcal{R}, \mathcal{T}, \mathcal{M} \otimes_{\mathcal{S}} \mathcal{N})$.

Definition 5.1.53 (Sheafified Hom). Let $(\mathcal{D}, K)$ be a site (Definition 5.1.13) that admits a small topologically generating family (Definition 5.1.13) (e.g. by being essentially small) and let $\mathcal{R}, \mathcal{S}, \mathcal{T}$ be sheaves of rings on $(\mathcal{D}, K)$. Let $\mathcal{M}$ be a sheaf (Definition 5.1.32) of $\mathcal{R}$ - $\mathcal{S}$ bimodules (Definition 5.3.9) and $\mathcal{N}$ be a sheaf of $\mathcal{R}$ - $\mathcal{T}$ bimodules.

The *sheafified Hom*, denoted by notations such as $\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{N})$, $\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{N})$, or $\underline{\mathcal{H}om}_{\mathcal{R}}(\mathcal{M}, \mathcal{N})$, is the sheaf of $\mathcal{S}$ - $\mathcal{T}$ bimodules defined section-wise by:

$$\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{N})(U) = \mathrm{Hom}_{\mathbf{Sh}(\mathcal{D}/U, K|_U; \mathbf{Mod}(\mathcal{R}|_U))}(\mathcal{M}|_U, \mathcal{N}|_U).$$

(Definition 1.3.4) (Definition 5.1.29) (Definition 5.2.30)

Equivalently, it is the unique sheaf which satisfies the right adjoint property relative to the sheafified tensor product (Theorem 5.1.54).

In the framework of Definition 5.1.41, the sheafified Hom corresponds to the right adjoint $\mathcal{G}$ to the sheafified tensor product $\mathcal{F}$. While the general framework defines sheafy functors via section-wise application, the internal Hom for sheaves of modules requires the restriction-based definition (using $\mathcal{D}/U$) to correctly represent the right adjoint in the category of sheaves, ensuring compatibility with the varying ring structure $\mathcal{R}$.

(♠ TODO: the following would actually need to be modified to have a notion of “Hom-composable” category of functors) Also in the framework of Definition 5.1.41, the sheafified Hom is the evaluation of the sheafy functor $\mathcal{G}$ induced by the algebraic Hom functor $G : \mathcal{A}^{op} \times \mathcal{C} \rightarrow \mathcal{B}$ under assumption (2). Given input sheaves $\mathbb{M} := (\mathcal{R}, \mathcal{S}, \mathcal{M})$ and $\mathbb{N} := (\mathcal{R}, \mathcal{T}, \mathcal{N})$, the result is the sheaf $\mathcal{G}(\mathbb{M}, \mathbb{N}) = (\mathcal{S}, \mathcal{T}, \mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{N}))$. Because G is a right adjoint in the base categories, it is a continuous functor (i.e., it preserves limits); thus, $\mathcal{G}$ can be defined section-wise as in Case (2) of the framework. This section-wise construction $U \mapsto \mathrm{Hom}_{\mathcal{R}(U)}(\mathcal{M}(U), \mathcal{N}(U))$ is naturally isomorphic to the restriction-based internal Hom defined above, as both satisfy the same local-to-global universal property.

Theorem 5.1.54 (Tensor-Hom Adjunction for Sheaves of Bimodules). Let $(\mathcal{D}, K)$ be a site (Definition 5.1.13) that admits a small topologically generating family (Definition 5.1.13)

(e.g. by being essentially small). Let $\mathcal{R}, \mathcal{S}, \mathcal{T}$ be sheaves (Definition 5.1.32) of rings on $(\mathcal{D}, K)$. Let $\mathcal{M}$ be a sheaf of $\mathcal{R}$ - $\mathcal{S}$ bimodules (Definition 5.3.9), $\mathcal{N}$ be a sheaf of $\mathcal{S}$ - $\mathcal{T}$ bimodules, and $\mathcal{P}$ be a sheaf of $\mathcal{R}$ - $\mathcal{T}$ bimodules.

Then there is a natural isomorphism:

$$\mathrm{Hom}_{\mathrm{Sh}(\mathcal{R}\text{-}\mathcal{T})}(\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}, \mathcal{P}) \cong \mathrm{Hom}_{\mathrm{Sh}(\mathcal{S}\text{-}\mathcal{T})}(\mathcal{N}, \mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{P}))$$

where $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}$ is the sheafified tensor product and $\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{P})$ is the sheaf of local $\mathcal{R}$ -linear homomorphisms.

Proof. (♠ TODO: Verify this proof We apply Theorem 5.1.43 by specifying the categories $\mathcal{A}, \mathcal{B}, \mathcal{C}$ and the functors F, G as follows:

1. **Categories:** Let $\mathcal{A}$ be the category whose objects are triples (R, S, M) where M is an R - S bimodule. Define $\mathcal{B}$ and $\mathcal{C}$ similarly for S - T and R - T bimodules.
2. **Base Functors:** Define $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ by $(M, N) \mapsto M \otimes_S N$ and $G : \mathcal{A}^{\mathrm{op}} \times \mathcal{C} \rightarrow \mathcal{B}$ by $(M, P) \mapsto \mathrm{Hom}_R(M, P)$. At each component, $F(A, -) \dashv G(A, -)$ is the standard algebraic tensor-hom adjunction for bimodules.
3. **Sheaf Interpretation:** The sheaves $\mathcal{M}, \mathcal{N}, \mathcal{P}$ are viewed as objects in $\mathrm{Sh}(\mathcal{D}, K; \mathcal{A})$, $\mathrm{Sh}(\mathcal{D}, K; \mathcal{B})$, and $\mathrm{Sh}(\mathcal{D}, K; \mathcal{C})$ respectively, where the ring components of these sheaves are fixed to be the sheaves $\mathcal{R}, \mathcal{S}, \mathcal{T}$.

By Theorem 5.1.43, the sheafy functor $\mathcal{F}$ associated to the tensor product is:

$$\mathcal{F}(\mathcal{M}, \mathcal{N}) = a(U \mapsto \mathcal{M}(U) \otimes_{\mathcal{S}(U)} \mathcal{N}(U))$$

which is precisely the definition of the sheaf of $\mathcal{R}$ - $\mathcal{T}$ bimodules $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}$.

The right adjoint functor $\mathcal{G}_{\mathcal{M}}$ is defined section-wise by:

$$\mathcal{G}_{\mathcal{M}}(\mathcal{P}) : U \mapsto \mathrm{Hom}_{\mathcal{R}(U)}(\mathcal{M}(U), \mathcal{P}(U))$$

Per the theorem, since G is a right adjoint, $\mathcal{G}_{\mathcal{M}}(\mathcal{P})$ is a sheaf. In the category of modules, this section-wise Hom sheaf is naturally isomorphic to the internal sheaf $\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{P})$ because any morphism of sheaves $\mathcal{M}|_U \rightarrow \mathcal{P}|_U$ is uniquely determined by its action on sections when the sheafification of the section-wise Hom agrees with the internal Hom.

Thus, the general adjunction $\mathcal{F}(\mathcal{M}, -) \dashv \mathcal{G}_{\mathcal{M}}(-)$ specializes to the required isomorphism of Hom-sets in the category of sheaves of bimodules. $\square$

5.1.7. Topos.

Definition 5.1.55 (Topos). There are multiple notions of a topos depending on the context (geometric vs. logical).

1. A **Grothendieck topos** (or **sheaf topos**) is a category (Definition 1.0.1) equivalent (Definition 1.1.8) to the category of sheaves (Definition 5.1.32) of sets on a small site. That is, there exists a small site $(\mathcal{C}, J)$ such that the category is equivalent to $\mathrm{Sh}(\mathcal{C}, J, \mathbf{Sets})$ (Definition 1.0.3).

If a Grothendieck universe (Definition E.0.1) $\mathcal{U}$ is available, then a $\mathcal{U}$ -(Grothendieck/sheaf)topos is a category equivalent to (Definition 1.1.8) the category of sheaves of $\mathcal{U}$ -sets (Definition E.0.1) on a $\mathcal{U}$ -small site (Definition 5.1.15) $(\mathcal{C}, J)$ [GV72, Exposé IV Définition 1.1].

2. An **elementary topos** is a category which has all finite limits (Definition 1.2.2), is cartesian closed (Definition 1.2.22), and has a subobject classifier (Definition 1.2.23).

Unless otherwise specified, a **topos** refers to a Grothendieck/sheaf topos. The plural for topos is **topoi**. *Remark:* Every Grothendieck topos is an elementary topos, but the converse is not true (e.g., the category of finite sets is an elementary topos but not a Grothendieck topos).

Definition 5.1.56. [GV72, Exposé IV Définition 3.1], see [nLa25a, Definition 2.1] Let E and E' be topoi (Definition 5.1.55) (or $\mathcal{U}$ -topoi (Definition 5.1.55) if a Grothendieck universe (Definition E.0.1) is available). A **(continuous) morphism $f : E \rightarrow E'$ of topoi** is a triple $f = (f_*, f^*, \varphi)$ consisting of

1. A functor $f_* : E \rightarrow E'$, called the **direct image functor of f** ,
2. A functor $f^* : E' \rightarrow E$, called the **inverse image functor of f** ,
3. An adjunction (Definition 1.1.13) $\varphi : f^* \dashv f_*$

such that f^* is left exact (Definition 1.2.18), i.e. commutes with finite limits. Note that the adjunction implies that f^* is right exact as well.

This notion of morphism of topoi is also referred to as **geometric morphism of topoi**.

Definition 5.1.57. Let $\mathcal{E}$ and $\mathcal{F}$ be topoi (Definition 5.1.55). A **geometric morphism from $\mathcal{F}$ to $\mathcal{E}$** , denoted $f : \mathcal{F} \rightarrow \mathcal{E}$, is a pair of adjoint functors (f^*, f_*)

$$f^* : \mathcal{E} \rightleftarrows \mathcal{F} : f_*$$

where:

1. The functor f^* (called the **inverse image functor**) is left adjoint to the functor f_* (called the **direct image functor**).
2. The inverse image functor f^* is left exact, meaning it preserves finite limits (Definition 1.2.2).

A **morphism of topoi** usually refers to a geometric morphism between topoi.

Lemma 5.1.58 (see e.g. [GV72, Exposé IV 5.1]). Let E be a topos (Definition 5.1.55) and let X be an object of E . The slice category $E_{/X}$ (Definition 1.3.4) (Definition 1.3.4) is itself a topos.

Definition 5.1.59. Let $\mathcal{C}$ be a (large) category (Definition 1.0.1). In view of Lemma 5.1.58, the slice category $E_{/X} = E/X$ (Definition 1.3.4) (Definition 1.3.4) is often called the **slice topos** or **over topos**, etc.

Definition 5.1.60 (Generator of a category). Let $\mathcal{C}$ be a category (Definition 1.0.1).

1. An object $G \in \mathcal{C}$ is called a **generator** (or **separator**) if for every pair of distinct morphisms $f, g : X \rightarrow Y$ in $\mathcal{C}$, there exists a morphism $h : G \rightarrow X$ such that

$$f \circ h \neq g \circ h.$$

In case that $\mathcal{C}$ is locally small (Definition 1.0.9), this is equivalent to the condition that the representable functor (Definition 4.0.6)

$$\mathrm{Hom}_{\mathcal{C}}(G, -) : \mathcal{C} \rightarrow \mathbf{Set}$$

is faithful (Definition 1.1.4), which in turn is equivalent to the condition that for every object $X \in \mathcal{C}$, there exists an epimorphism

$$\bigoplus_{i \in I} G \twoheadrightarrow X$$

for some indexing set I , where $\bigoplus$ denotes the coproduct (Definition 1.2.36) in $\mathcal{C}$.

2. A family $\{G_i\}_{i \in I}$ is called a **generating family** if for every pair of distinct morphisms $f, g : X \rightarrow Y$ in $\mathcal{C}$, there exists some index $i \in I$ and a morphism $h : G_i \rightarrow X$ such that

$$f \circ h \neq g \circ h.$$

In case $\mathcal{C}$ is locally small, this is equivalent to the condition that the collection of representable functors

$$\{\mathrm{Hom}_{\mathcal{C}}(G_i, -) : \mathcal{C} \rightarrow \mathbf{Set}\}_{i \in I}$$

is jointly faithful, which in turn is equivalent to the condition that for every object $X \in \mathcal{C}$, there exists a family of objects $\{G_i\}_{i \in J}$ from the generating set indexed by some set J , and an epimorphism

$$\bigoplus_{i \in J} G_i \twoheadrightarrow X.$$

Proposition 5.1.61. (♠ TODO: find in SGA) Let $\mathcal{E}$ be a topos (Definition 5.1.55) that has all small coproducts (Definition 1.2.36) (which is true for any Grothendieck topos). A family of objects $\{G_i\}_{i \in I}$ is a generating family (Definition 5.1.60) if and only if for every object $X \in \mathcal{E}$, there exists a set J and an epimorphism

$$\coprod_{j \in J} G_{i_j} \twoheadrightarrow X$$

where each $i_j \in I$.

Theorem 5.1.62. (♠ TODO: find in SGA) Let $(\mathcal{C}, J)$ be a small (Definition 1.0.9) site (Definition 5.1.13), and let $\mathcal{E} = \mathrm{Sh}(\mathcal{C}, J)$ be the topos (Definition 5.1.55) of sheaves (Definition 5.1.32) of sets on it. Let $a \circ y : \mathcal{C} \rightarrow \mathcal{E}$ be the composite of the Yoneda embedding (Corollary 1.1.11) followed by the sheafification functor (Definition 5.2.20).

The family of sheaves $\{(a \circ y)(U) \mid U \in \mathrm{Ob}(\mathcal{C})\}$ is a generating family (Definition 5.1.60) for the topos $\mathcal{E}$.

(♠ TODO:)

Definition 5.1.63. A Grothendieck topos (Definition 5.1.55) $\mathcal{E}$ is called a *coherent topos* if there exists a coherent site (Definition 5.1.28) $(\mathcal{C}, J)$ such that $\mathcal{E}$ is equivalent to the category of sheaves (Definition 5.1.32) $\mathcal{E} \simeq \text{Sh}(\mathcal{C}, J)$ on that site.

Alternatively, an intrinsic characterization [GSDV72, Exposé VI] is that $\mathcal{E}$ is coherent if there exists a generating family (Definition 5.1.60) $\mathcal{G}$ of objects in $\mathcal{E}$ such that:

1. Every object in $\mathcal{G}$ is coherent (Definition 5.1.27) (quasi-compact (Definition 5.1.24) and quasi-separated (Definition 5.1.26)).
2. The full subcategory generated by $\mathcal{G}$ is closed under fiber products (Definition 1.2.38).

It is a theorem that the full subcategory of *all* coherent objects in a coherent topos is closed under finite limits and finite colimits (forming a so-called *pretopos*).

Definition 5.1.64 ([nLa25b, Definition 1.1]). Let E be a topos (Definition 5.1.55). A *point of E* is a geometric morphism (Definition 5.1.57) $x : \mathbf{Sets} \rightarrow E$ (Definition 1.0.3).

Definition 5.1.65. Let $\mathcal{E}$ be a topos (Definition 5.1.55), and let $p : \mathbf{Set} \rightarrow \mathcal{E}$ be a point of the topos (Definition 5.1.64). By definition, p is a geometric morphism, so it consists of an adjoint pair of functors (p^*, p_*) where the inverse image functor $p^* : \mathcal{E} \rightarrow \mathbf{Set}$ is left exact.

1. The inverse image functor $p^* : \mathcal{E} \rightarrow \mathbf{Set}$ is called the *stalk functor associated with the point p* .
2. For any object (sheaf) $F \in \mathcal{E}$, the set $p^*(F)$ is called the *stalk of F at the point p* .

Remark 5.1.66. If $\mathcal{E}$ is the topos of sheaves on a topological space X , a classical point $x \in X$ induces a point of the topos $p_x : \mathbf{Set} \rightarrow \text{Sh}(X)$. In this case, the stalk functor p_x^* is equivalent to the classical stalk functor at x , which is defined as the colimit of sections over open neighborhoods of x .

$$p_x^*(F) \cong \varinjlim_{x \in U} F(U)$$

Definition 5.1.67. A topos (Definition 5.1.55) $\mathcal{E}$ is said to *have enough points* if the collection of all points of $\mathcal{E}$ is jointly faithful.

Explicitly, this means that for any two distinct morphisms $f, g : X \rightarrow Y$ in $\mathcal{E}$, there exists a point $p : \mathbf{Set} \rightarrow \mathcal{E}$ such that the induced maps on stalks are distinct:

$$p^*(f) \neq p^*(g).$$

Equivalently, a morphism f in $\mathcal{E}$ is an isomorphism if and only if for every point p , the map $p^*(f)$ is an isomorphism of sets.

Theorem 5.1.68. Let $\mathcal{E}$ be a coherent topos (Definition 5.1.63). Then $\mathcal{E}$ has enough points (Definition 5.1.67).

Theorem 5.1.69 (Deligne's Completeness Theorem for Coherent Topoi). Let $\mathcal{E}$ be a coherent topos (Definition 5.1.63). Then there exists a compact (Definition E.0.17) Hausdorff (Definition E.0.18) topological space X and an exact functor $\mathcal{E} \rightarrow \text{Sh}(X)$ that reflects isomorphisms.

5.2. Morphism of sites and topoi.

5.2.1. Continuous functors between sites and Morphisms between sites.

Definition 5.2.1. Let $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ be sites (Definition 5.1.13) (or $\mathcal{U}$ -sites (Definition 5.1.15) if a Grothendieck universe (Definition E.0.1) $\mathcal{U}$ is available).

A functor $u : \mathcal{C} \rightarrow \mathcal{D}$ is said to be a **continuous functor of sites** if for every object U of $\mathcal{C}$ and covering sieve (Definition 5.1.13) $S \in J(U)$, the sieve (Definition 5.1.1) in $\mathcal{D}$ generated by (Definition 5.1.16) the image $\{u(\phi) | \phi \in S\}$ is a covering sieve in $K(u(U))$.

Note that this differs from the definition of continuous morphisms of sites given in [GV72, Exposé III, Définition 1.1]. See also Proposition 5.2.2.

Proposition 5.2.2. Let $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ be sites (Definition 5.1.13) (or $\mathcal{U}$ -sites (Definition 5.1.15) if a Grothendieck universe (Definition E.0.1) $\mathcal{U}$ is available). Let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a functor. Consider the following conditions:

1. u is continuous (Definition 5.2.1), i.e. for every object U of $\mathcal{C}$ and covering sieve (Definition 5.1.13) $S \in J(U)$, the sieve (Definition 5.1.1) in $\mathcal{D}$ generated by (Definition 5.1.16) the image $\{u(\phi) | \phi \in S\}$ is a covering sieve in $K(u(U))$.
2. For any (large) category $\mathcal{A}$ and every sheaf (Definition 5.1.32) F on $\mathcal{D}$ valued in $\mathcal{A}$, the presheaf (Definition 5.0.1) $u^s \mathcal{F} : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}, X \mapsto F(u(X))$ is a sheaf on $\mathcal{D}$.
3. For every sheaf (Definition 5.1.32) of sets F on $\mathcal{D}$, the pushforward/direct image (Definition 5.3.2) $u^s \mathcal{F} : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Sets}, X \mapsto F(u(X))$, which is a priori a presheaf (Definition 5.0.1), is in fact a sheaf on $\mathcal{D}$. Note that this is the definition of continuous functor of sites given in [GV72, Exposé III, Définition 1.1].

Condition (1) implies Condition (2), which in turn implies Condition (3). If $\mathcal{C}$ and $\mathcal{D}$ are locally small (Lemma 5.0.5) and a sheafification functor (Definition 5.2.21)

$$(G) \quad a : \text{PreShv}(\mathcal{D}, \mathbf{Sets}) \rightarrow \text{Shv}(\mathcal{D}, \mathbf{Sets})$$

exists (e.g. $\mathcal{D}$ has a topologically generating family (Definition 5.1.13) or is essentially small (Definition 1.1.9), see Theorem 5.2.27), then Condition (3) implies (1) (♠ TODO: might need to impose these conditions for other things that might use this statement)

Proof. Clearly, (2) implies (3). We show that (1) implies (2). Suppose that U is an object of $\mathcal{C}$ and that $S \in J(U)$ is a covering sieve (Definition 5.1.13). Since u is continuous, the sieve $u(S)^+$ on $u(U)$ generated by (Definition 5.1.6) the image $u(S)$ of S under u is a covering sieve (Definition 5.1.13) of $u(U)$ in K . Write $\mathcal{D}_S \hookrightarrow \mathcal{C}/U$ (Definition 1.3.4) for the full downward-closed subcategory (Definition 5.1.2) such that $\text{Ob}(\mathcal{D}_S) = \{(f : V \rightarrow U) : f \in S(V)\}$. Write $\mathcal{D}_{u(S)^+} \hookrightarrow \mathcal{D}/u(U)$ for the analogous subcategory. The functor u induces a functor $u_U : \mathcal{D}_S \rightarrow \mathcal{D}_{u(S)^+}, (V \rightarrow U) \mapsto (u(V) \rightarrow u(U))$. Due to the continuity of u , this functor is cofinal (Definition 1.2.16) (♠ TODO: prove that u_U is cofinal: for every $(g : W \rightarrow u(U)) \in \mathcal{D}_{u(S)^+}$, the comma category $(g \downarrow u_U)$ is non-empty and connected — follows because g factors through some $u(f)$ with $f \in S$, so non-emptiness is clear; connectedness uses that S is a sieve)

We want to show that the limit

$$(H) \quad \varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} (u^s \mathcal{F})|_{\mathcal{D}_S}(V)$$

exists and the canonical natural morphism

$$u^s \mathcal{F}(U) \rightarrow \varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} (u^s \mathcal{F})|_{\mathcal{D}_S}(V)$$

is an isomorphism. (H) equals

$$\varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} \mathcal{F}|_{\mathcal{D}_S}(u(V)).$$

Moreover, $u^s \mathcal{F}(U) = \mathcal{F}(u(U))$. Since $\mathcal{F}$ is a sheaf on $(\mathcal{D}, K)$, there is a canonical isomorphism

$$\mathcal{F}(u(U)) \cong \varprojlim_{W \rightarrow u(U) \in (\mathcal{D}_{u(S)+})^{\text{op}}} \mathcal{F}(W).$$

Because u_U is cofinal, the limit above, which is assumed to exist is isomorphic to

$$\varprojlim_{(V \rightarrow U) \in \mathcal{D}_S^{\text{op}}} \mathcal{F}(u(V))$$

Theorem 1.2.17 as needed.

Now assume that $\mathcal{C}$ is locally small (Definition 1.0.9) and that a sheafification functor (Definition 5.2.21) (G) exists, we show that (3) implies (1). Let U be an object of $\mathcal{C}$ and let $S \in J(U)$ be any covering sieve. Let $\tilde{h}_{u(U)}$ be the sheafification of the representable functor $h_{u(U)} : \mathcal{D} \rightarrow \mathbf{Sets}$. Condition (3) implies that $u^s \tilde{h}_{u(U)} = \tilde{h}_{u(U)} \circ u$ is a sheaf on $\mathcal{C}$. In particular, we have an isomorphism

$$(u^s \tilde{h}_{u(U)})(U) \cong \varprojlim_{(f: V \rightarrow U) \in S^{\text{op}}} (u^s \tilde{h}_{u(U)})(V).$$

The left-hand side is $\tilde{h}_{u(U)}(u(U))$. Note that there is a canonical map $h_{u(U)} \rightarrow \tilde{h}_{u(U)}$ of presheaves and a canonical section $\text{id}_{u(U)} \in h_{u(U)}(u(U))$. Let us still call by $\text{id}_{u(U)}$ the image of this canonical section under the canonical function $h_{u(U)}(u(U)) \rightarrow \tilde{h}_{u(U)}(u(U))$. The isomorphism above then identifies this section $\text{id}_{u(U)} \in \tilde{h}_{u(U)}(u(U))$ as an element of $\varprojlim_{(f: V \rightarrow U) \in S^{\text{op}}} \tilde{h}_{u(U)}(u(V))$ — this element is given by the compatible family $\mathcal{G} := \{\eta_{u(V)}(u(f))\}_{f \in S}$ of elements of $\tilde{h}_{u(U)}(u(V))$ where $\eta : h_{u(U)} \rightarrow \tilde{h}_{u(U)}$ is the canonical morphism. To show that the sieve $R := S(u(S))$ generated by $u(S)$ is a covering sieve of $u(S)$, it is equivalent to show (♠ TODO: somehow, the following should be true: a sieve $S \in K(X)$ is a covering if and only if the image of the identity id_X under the canonical map $h_X(X) \rightarrow \tilde{h}_X(X)$ is in the image of the map $\varinjlim_{f: V \rightarrow X \in S} \tilde{h}_X(V) \rightarrow \tilde{h}_X(X)$) that the section $\text{id}_{u(U)} \in \tilde{h}_{u(U)}(u(U))$ lies in the image of the canonical map

$$(I) \quad \varprojlim_{(g: W \rightarrow u(U)) \in R^{\text{op}}} \tilde{h}_{u(U)}(W) \rightarrow \tilde{h}_{u(U)}(u(U)).$$

Since $u(S)$ is by construction a subcollection of R , we have a restriction map

$$\rho : \varprojlim_{(g:W \rightarrow u(U)) \in R^{\text{op}}} \tilde{h}_{u(U)}(W) \rightarrow \varprojlim_{(g:V \rightarrow U) \in S^{\text{op}}} \tilde{h}_{u(U)}(u(V)).$$

Given any member $g : W \rightarrow u(U)$ of the sieve R , we show that the family $\{\eta_W(g)\}$ is well-defined in the inverse limit in (I) over R , i.e. that these $\eta_W(g)$ are compatible in the following sense: for any $g : W \rightarrow u(U)$ in R and any morphism $k : W' \rightarrow W$ in $\mathcal{D}$, the restriction of $\eta(g)$ to W' along k equals $\eta(g \circ k)$. Since $g \in R$, we have that $g = u(f) \circ h$ for some $h : W \rightarrow u(V)$. Thus, $g \circ k = u(f) \circ (h \circ k)$. The section $\eta(g)$ is the image of g via $\eta_W : h_{u(U)}(W) \rightarrow \tilde{h}_{u(U)}(W)$. Restricting the image of g to W' along k is exactly the image of $g \circ k$ under $\eta_{W'} : h_{u(U)}(W') \rightarrow \tilde{h}_{u(U)}(W')$. Thus, the desired compatibility holds. The canonical map (I) is the composition

$$\varprojlim_{(g:W \rightarrow u(U)) \in R^{\text{op}}} \tilde{h}_{u(U)}(W) \xrightarrow{\rho} \varprojlim_{(g:V \rightarrow U) \in S^{\text{op}}} \tilde{h}_{u(U)}(u(V)) \cong \tilde{h}_{u(U)}(u(U))$$

and hence the image of this map contains the section $\text{id}_{u(U)} \in \tilde{h}_{u(U)}(u(U))$ from earlier. $\square$

Definition 5.2.3. Let (X, τ_X) and (Y, τ_Y) be topological spaces (Definition B.0.1), and let $f : X \rightarrow Y$ be a continuous map (Definition B.0.3). Let $\text{Open}(X)$ and $\text{Open}(Y)$ be their respective categories of open sets (Definition 1.0.7) with inclusion morphisms, equipped with the canonical (Definition 5.1.30) Grothendieck topologies (Definition 5.1.13) given by open coverings.

Define the functor

$$f^{-1} : \text{Open}(Y) \rightarrow \text{Open}(X), \quad U \mapsto f^{-1}(U).$$

It is a continuous functor of sites (Definition 5.2.1) from $\text{Open}(Y)$ to $\text{Open}(X)$ which induces a site morphism (Definition 5.2.5)

$$f : (\text{Open}(X), \text{can}) \rightarrow (\text{Open}(Y), \text{can})$$

.

Definition 5.2.4. Let $f : X \rightarrow Y$ be a morphism of schemes, and consider one of the common Grothendieck topologies on schemes such as the Zariski, étale, Nisnevich, fppf, fpqc, or crystalline topology. Denote by $\mathbf{C}(X)$ and $\mathbf{C}(Y)$ the corresponding small sites (Definition 5.1.13) of X and Y (i.e., categories of morphisms to X and Y respectively equipped with one of these topologies).

Then the base change functor

$$f^{-1} : \mathbf{C}(Y) \rightarrow \mathbf{C}(X), \quad (V \rightarrow Y) \mapsto (V \times_Y X \rightarrow X)$$

(Definition 1.2.38) is a continuous functor. It in fact induces a morphism of sites (Definition 5.2.5)

$$f : (\mathbf{C}(X), \tau_{\mathbf{C}}) \rightarrow (\mathbf{C}(Y), \tau_{\mathbf{C}})$$

where $\tau_{\mathbf{C}}$ denotes the chosen topology (Zariski, étale, Nisnevich, fppf, fpqc, crystalline).

Definition 5.2.5. Let $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ be sites (Definition 5.1.13). A *morphism of sites* $f : (\mathcal{D}, K) \rightarrow (\mathcal{C}, J)$ consists of a continuous functor (Definition 5.2.1) $u : \mathcal{C} \rightarrow \mathcal{D}$ such

that the inverse image functor (Definition 5.3.3) $u^* : \text{Sh}(\mathcal{D}, K; \mathbf{Sets}) \rightarrow \text{Sh}(\mathcal{C}, J; \mathbf{Sets})$ (Definition 5.1.32) is left exact (Definition 1.2.18), i.e. commutes with finite projective limits (Definition 1.2.12).

When the sites are small (Definition 5.1.15), a morphism of sites is exactly a continuous and flat functor (Definition 5.2.17).

Definition 5.2.6. Let $(\mathcal{C}_1, J_1)$ and $(\mathcal{C}_2, J_2)$ be sites (Definition 5.1.13). An *equivalence of sites* from $(\mathcal{C}_1, J_1)$ to $(\mathcal{C}_2, J_2)$ consists of an equivalence of categories (Definition 1.1.8)

$$F : \mathcal{C}_1 \rightarrow \mathcal{C}_2, \quad G : \mathcal{C}_2 \rightarrow \mathcal{C}_1$$

with natural isomorphisms (Definition 1.1.3)

$$\eta : \text{Id}_{\mathcal{C}_1} \xrightarrow{\sim} G \circ F, \quad \epsilon : F \circ G \xrightarrow{\sim} \text{Id}_{\mathcal{C}_2}$$

such that both F and G are continuous functors of sites (Definition 5.2.1). In this situation, we say that $(\mathcal{C}_1, J_1)$ and $(\mathcal{C}_2, J_2)$ are *equivalent as sites* (or simply *equivalent sites*).

(♠ TODO: these are probably the idncued/pullback topology Definition 5.1.19)

Remark 5.2.7. The continuity of both F and G ensures that the Grothendieck topologies J_1 and J_2 correspond under the equivalence of the underlying categories. In particular, a sieve S on an object U of $\mathcal{C}_1$ is J_1 -covering if and only if the sieve generated by $\{F(\phi) \mid \phi \in S\}$ is J_2 -covering on $F(U)$, and similarly for G .

Lemma 5.2.8. (♠ TODO: AI generated, verify) Let $(\mathcal{C}_1, J_1)$ and $(\mathcal{C}_2, J_2)$ be sites (Definition 5.1.13) that are equivalent as sites (Definition 5.2.6), and let $\mathcal{B}$ be a (large) category (Definition 1.0.1). Then the sheaf categories (Definition 5.1.32) $\text{Shv}(\mathcal{C}_1, J_1, \mathcal{B})$ and $\text{Shv}(\mathcal{C}_2, J_2, \mathcal{B})$ are equivalent (Definition 1.1.8).

More precisely, let $F : \mathcal{C}_1 \rightarrow \mathcal{C}_2$ and $G : \mathcal{C}_2 \rightarrow \mathcal{C}_1$ be the functors witnessing the equivalence of sites, with natural isomorphisms

$$\eta : \text{Id}_{\mathcal{C}_1} \xrightarrow{\sim} G \circ F, \quad \epsilon : F \circ G \xrightarrow{\sim} \text{Id}_{\mathcal{C}_2}.$$

Precomposition with F and G induces functors

$$F^* : \text{Shv}(\mathcal{C}_2, J_2, \mathcal{B}) \rightarrow \text{Shv}(\mathcal{C}_1, J_1, \mathcal{B}), \quad F^*(\mathcal{H}) = \mathcal{H} \circ F$$

and

$$G^* : \text{Shv}(\mathcal{C}_1, J_1, \mathcal{B}) \rightarrow \text{Shv}(\mathcal{C}_2, J_2, \mathcal{B}), \quad G^*(\mathcal{K}) = \mathcal{K} \circ G,$$

which are well-defined because F and G are continuous (Definition 5.2.1) (so precomposition sends sheaves to sheaves). The natural isomorphisms η and ϵ induce natural isomorphisms

$$\eta^* : \text{Id}_{\text{Shv}(\mathcal{C}_1, J_1, \mathcal{B})} \xrightarrow{\sim} G^* \circ F^* \quad \text{and} \quad \epsilon^* : F^* \circ G^* \xrightarrow{\sim} \text{Id}_{\text{Shv}(\mathcal{C}_2, J_2, \mathcal{B})},$$

thereby exhibiting an equivalence of categories.

Proposition 5.2.9. (♠ TODO: AI generated, verify) Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be an equivalence of categories (Definition 1.1.8) with quasi-inverse (Definition 1.1.8) $G : \mathcal{D} \rightarrow \mathcal{C}$. Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13). Then there exists a unique Grothendieck topology (Definition 5.1.13) K on $\mathcal{D}$ such that $F : (\mathcal{C}, J) \rightarrow (\mathcal{D}, K)$ and $G : (\mathcal{D}, K) \rightarrow (\mathcal{C}, J)$ are both continuous functors of sites (Definition 5.2.1) (i.e., F and G form an equivalence of sites (Definition 5.2.6)).

Explicitly, K is the topology on $\mathcal{D}$ defined by: for $U \in \mathcal{D}$, a sieve (Definition 5.1.1) S on U is K -covering if and only if the sieve on $G(U)$ generated by $\{G(\varphi) \mid \varphi \in S\}$ is J -covering. Equivalently, K is the induced topology (Definition 5.1.19) G^*J along G , or equivalently the topology for which F is cocontinuous (Definition 5.2.11) and G is continuous.

Proof. Existence: Define K as above. Since G is a functor, K is a Grothendieck topology on $\mathcal{D}$ (it is the pullback of J along G). By construction, $G : (\mathcal{D}, K) \rightarrow (\mathcal{C}, J)$ is continuous. To see that $F : (\mathcal{C}, J) \rightarrow (\mathcal{D}, K)$ is continuous, let S be a J -covering sieve on $V \in \mathcal{C}$. The sieve generated by $\{F(\psi) \mid \psi \in S\}$ on $F(V)$ is K -covering because its pullback along G is the sieve generated by $\{GF(\psi) \mid \psi \in S\}$, which contains the J -covering sieve S (since $\eta : \text{Id}_{\mathcal{C}} \rightarrow GF$ is a natural isomorphism, the map $S \rightarrow GF(S)$ is an isomorphism of sieves). Hence F is continuous.

Uniqueness: Suppose K' is another topology on $\mathcal{D}$ making F and G continuous. Then $\text{id}_{\mathcal{D}} : (\mathcal{D}, K) \rightarrow (\mathcal{D}, K')$ is continuous (since G is continuous for both, and F is a quasi-inverse), and similarly $\text{id}_{\mathcal{D}} : (\mathcal{D}, K') \rightarrow (\mathcal{D}, K)$ is continuous. Hence $K = K'$. $\square$

Proposition 5.2.10. Let $(\mathcal{C}, J)$ and $(\mathcal{C}, J')$ be two sites (Definition 5.1.13) on the same underlying category $\mathcal{C}$. If the topology J is finer (Definition 5.1.18) than J' on the category $\mathcal{C}$, then the identity functor

$$\text{id}_{\mathcal{C}} : (\mathcal{C}, J) \rightarrow (\mathcal{C}, J')$$

is a continuous functor (Definition 5.2.1) between sites and in fact induces a site morphism (Definition 5.2.5).

$$(\mathcal{C}, J') \rightarrow (\mathcal{C}, J).$$

5.2.2. cocontinuous functors between sites.

Definition 5.2.11 ([GV72, Exposé III, Définition 2.1.]). Let $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ be sites (Definition 5.1.13) (or $\mathcal{U}$ -sites (Definition 5.1.15) if a Grothendieck universe (Definition E.0.1) $\mathcal{U}$ is available).

A functor $u : \mathcal{C} \rightarrow \mathcal{D}$ is said to be a *cocontinuous functor* if for every object $X \in \text{Ob}(\mathcal{C})$ and every covering sieve (Definition 5.1.13) $R \in K(u(X))$, the sieve S on X generated by (Definition 5.1.6) the set of morphisms $\{f : Y \rightarrow X \mid u(f) \in R\}$ is a covering sieve in $J(X)$.

Proposition 5.2.12 ([GV72, Exposé III, Proposition 2.2.]). Let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between sites $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$. The functor u is cocontinuous if and only if the precomposition/inverse image functor (Definition 5.2.14) $u^* : \widehat{\mathcal{D}} \rightarrow \widehat{\mathcal{C}}$, defined by $P \mapsto P \circ u^{\text{op}}$ for any presheaf P on $\mathcal{D}$, restricts to a functor $u^* : \text{Sh}(\mathcal{D}, K; \mathbf{Sets}) \rightarrow \text{Sh}(\mathcal{C}, J; \mathbf{Sets})$ between the categories of sheaves.

Definition 5.2.13. Let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a functor between categories. Let $\mathcal{A}$ be a (large) category (Definition 1.0.1).

For any presheaf (Definition 5.0.1) $\mathcal{F} \in \text{PreSh}(\mathcal{C}; \mathcal{A})$, the *left Kan extension of $\mathcal{F}$ along u* is the presheaf $\text{Lan}_u \mathcal{F} \in \text{PreSh}(\mathcal{D}; \mathcal{A})$ defined (Definition 7.5.5), assuming that all colimits (Definition 1.2.2) below exist, by:

$$(\text{Lan}_u \mathcal{F})(V) = \varinjlim_{(V \downarrow u)} \mathcal{F}(U),$$

where the colimit (Definition 1.2.2) is taken over the comma category $(V \downarrow u)$ (Definition 1.3.12) of pairs $(U, V \rightarrow u(U))$ with $U \in \mathcal{C}$.

Now assuming that Let $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ are sites (Definition 5.1.13), that $u : \mathcal{C} \rightarrow \mathcal{D}$ is a cocontinuous functor of sites, and that the presheaf category $\text{PreSh}(\mathcal{D}, K; \mathcal{A})$ has sheafification (Definition 5.2.20), this construction induces (Theorem 7.1.7) a functor at the level of sheaves — for any sheaf (Definition 5.1.32) $\mathcal{G} \in \text{Sh}(\mathcal{C}, J; \mathcal{A})$, we define the *sheafified left Kan extension* as:

$$u_! \mathcal{G} = a(\text{Lan}_u \mathcal{G}),$$

where a is the sheafification functor (Definition 5.2.20) on $\text{PreSh}(\mathcal{D}, K; \mathcal{A})$. In particular, $u_!$ yields a functor

$$u_! : \text{Sh}(\mathcal{C}, J; \mathcal{A}) \rightarrow \text{Sh}(\mathcal{D}, K; \mathcal{A}).$$

Definition 5.2.14. Let $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ be sites (Definition 5.1.13) and let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a cocontinuous functor of sites (Definition 5.2.11). Let $\mathcal{A}$ be a (large) category (Definition 1.0.1).

The *pullback/inverse image functor under u*

$$u^* : \text{Sh}(\mathcal{D}, K; \mathcal{A}) \rightarrow \text{Sh}(\mathcal{C}, J; \mathcal{A})$$

is defined as the restriction of the precomposition functor $u_* : \text{PreSh}(\mathcal{D}; \mathcal{A}) \rightarrow \text{PreSh}(\mathcal{C}; \mathcal{A})$ given by $P \mapsto P \circ u$.

Specifically, for any sheaf $\mathcal{H} \in \text{Sh}(\mathcal{D}, K; \mathcal{A})$, the *pullback sheaf* is:

$$(u^* \mathcal{H})(U) = \mathcal{H}(u(U)).$$

The inverse image u^* of a cocontinuous functor of sites should not be confused with the notion of the inverse image of a continuous functor of sites (Definition 5.3.3), as these two notions are different constructions, even when u is both continuous and cocontinuous. On the other hand, note that the “inverse image of a cocontinuous functor of sites” and the “direct image of a continuous functor of sites (Definition 5.3.2)” share the same constructions, i.e. by precomposition. More precisely, if $u : \mathcal{C} \rightarrow \mathcal{D}$ is a continuous (Definition 5.2.1) and cocontinuous functor of sites, then the direct image u_* of u as a continuous functor (Definition 5.3.2) and the inverse image u^* of u as a cocontinuous functor are exactly the same.

By the properties of cocontinuous functors, if $\mathcal{H}$ is a sheaf on $(\mathcal{D}, K)$, then $u^* \mathcal{H}$ is automatically a sheaf on $(\mathcal{C}, J)$ without further sheafification. This functor u^* serves as the right adjoint to the sheafified left Kan extension (Definition 5.2.13) $u_!$. (♠ TODO: prove the adjunction)

5.2.3. Morphisms of sites and of topoi. There are multiple types of morphisms of topoi induced by functors between sites, depending on whether said functors are continuous or cocontinuous (with additional hypotheses)

Definition 5.2.15. Let $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ be small sites (Definition 5.1.15). let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a morphism of sites (Definition 5.2.5).

The *morphism of topoi induced by u* is the topos morphism (Definition 5.1.56) $f : \mathrm{Sh}(\mathcal{D}, K; \mathbf{Sets}) \rightarrow \mathrm{Sh}(\mathcal{C}, J; \mathbf{Sets})$ given by $f_* = u_*$ (Definition 5.3.2), which is defined by precomposition, and $f^* = a \circ \mathrm{Lan}_u$ where Lan_u is left Kan extension (Definition 5.2.13) and a is sheafification (Definition 5.2.20).

It turns out that there is indeed an adjunction $f^* \dashv f_*$ (♠ TODO: why; apparently, there is a presheaf level adjunction between Lan and u^* .) and that f^* is indeed left exact (♠ TODO: why).

Therefore, whenever we talk about a property or a notion concerning a topos morphism, that property or notion may be sensibly extended to concern sites.

The notion of a topos morphism induced by a morphism of small sites should not be mistaken for the notion of a topos morphism induced by (Definition 5.2.18) a cocontinuous (Definition 5.2.11) flat functor (Definition 5.2.17) of small sites.

See also Definition 5.2.19.

Remark 5.2.16. In fact, for every morphism $f : \mathcal{E} \rightarrow \mathcal{F}$ of topoi, there are some sites $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ (which may even be taken to be small) whose categories of sheaves of sets are $\mathcal{E}$ and $\mathcal{F}$ and there is some morphism $u : (\mathcal{D}, K) \rightarrow (\mathcal{C}, J)$ whose induced topos morphism is f . However, this assignment is not injective.

Definition 5.2.17. A functor (Definition 1.1.1) $u : \mathcal{C} \rightarrow \mathcal{D}$ of categories (Definition 1.0.1) is said to be *flat* if the Left Kan extension (Definition 5.2.13) functor

$$\mathrm{Lan}_u : \mathrm{PreShv}(\mathcal{C}, \mathbf{Sets}) \rightarrow \mathrm{PreShv}(\mathcal{D}, \mathbf{Sets})$$

is left exact (Definition 1.2.18).

Definition 5.2.18. Let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a cocontinuous functor of small sites (Definition 5.2.11) that is also flat (Definition 5.2.17). The *topos morphism induced by u* is the morphism of topoi (Definition 5.1.56)

$$f : \mathrm{Sh}(\mathcal{C}, J) \rightarrow \mathrm{Sh}(\mathcal{D}, K)$$

defined by the following pair of adjoint functors (Definition 1.1.13):

- The **inverse image** $f^* : \mathrm{Sh}(\mathcal{D}, K) \rightarrow \mathrm{Sh}(\mathcal{C}, J)$ is the inverse image (Definition 5.2.14)/precomposition functor u^* , given by $f^*(\mathcal{H}) = \mathcal{H} \circ u$. Because u is cocontinuous, $\mathcal{H} \circ u$ is automatically a sheaf (Definition 5.1.32); because u is flat (Definition 5.2.17), f^* is left exact (Definition 1.2.18).
- The **direct image** $f_* : \mathrm{Sh}(\mathcal{C}, J) \rightarrow \mathrm{Sh}(\mathcal{D}, K)$ is the right adjoint (Definition 1.1.13) to the precomposition functor. (♠ TODO: why does this exist? Apparently, this right adjoint is the right kan extension) (♠ TODO: Also apparently, there is an adjoint functor theorem that states that a colimit-preserving functor between Grothendieck topoi always has a right adjoint)

The notion of a topos morphism induced by a cocontinuous flat functor of small sites should not be mistaken for the notion of a topos morphism induced by a morphism of small sites (Definition 5.2.15) (which is in particular a continuous flat functor).

See also Definition 5.2.19.

Definition 5.2.19. Let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a functor (Definition 1.1.1) of small sites (Definition 5.1.15) that is continuous (Definition B.0.3), cocontinuous (Definition 5.2.11), and flat (Definition 5.2.17). The *(essential) morphism of topoi induced by u* usually refers to the morphism of topoi induced by u as a (Definition 5.2.18) cocontinuous (Definition 5.2.11) and flat functor (Definition 5.2.17), and not the morphism of topoi induced by u as a (Definition 5.2.15) morphism of sites. (♠ TODO: is morphism of sites = continuous + flat functor?)

More explicitly, this morphism of topoi is the morphism

$$f : \mathrm{Sh}(\mathcal{C}, J) \rightarrow \mathrm{Sh}(\mathcal{D}, K)$$

whose

- Inverse image f^* is precomposition/inverse image (Definition 5.2.14) u^* of u as a cocontinuous functor, not as a continuous functor
- Direct image f_* is right adjoint to f^* .

Furthermore, since u is continuous, f^* has a left adjoint denoted by $f_!$ and called by names such as the *lower shriek of f* , the *exceptional direct image of f* , or the *extension by zero of f* , and which is given by (Definition 5.2.15) $f_! = a \circ \mathrm{Lan}_u$.

To summarize, there are adjunctions $f_! \dashv f^* \dashv f_*$.

Nevertheless, one can talk about the morphism of topoi in the other direct — regarding u as a continuous and flat functor, one has a topos morphism

$$g : \mathrm{Sh}(\mathcal{D}, K) \rightarrow \mathrm{Sh}(\mathcal{C}, J)$$

in the other direction whose

1. Inverse image g^* is given by $a \circ \mathrm{Lan}_u$
2. Direct image g_* is given by precomposition/direct image (Definition 5.3.2) u_* of u as a continuous functor, not as a cocontinuous functor.

Furthermore, since u is cocontinuous, g_* has a right adjoint, denoted by $g^!$ and called by names such as the *upper shriek of g* or the *exceptional inverse image of g* .

To summarize, there are adjunctions $g^* \dashv g_* \dashv g^!$.

Moreover, in the above notations, $f_! = g^*$, $f^* = g_*$, and $f_* = g^!$.

5.2.4. Sheafification.

Definition 5.2.20. Let $\mathcal{C}$ be a site (Definition 5.1.13) and let $\mathcal{A}$ be a (large) category (Definition 1.0.1).

Assuming that the presheaf (Definition 5.0.1) category $\mathrm{PreShv}(\mathcal{C}, \mathcal{A})$ (and hence the sheaf (Definition 5.1.32) category $\mathrm{Shv}(\mathcal{C}, \mathcal{A})$) is locally small (Definition 1.0.9) (or U -locally small if a Grothendieck universe (Definition E.0.1) U is available), a *sheafification functor* refers to a functor

$$a : \mathrm{PreShv}(\mathcal{C}, \mathcal{A}) \rightarrow \mathrm{Shv}(\mathcal{C}, \mathcal{A})$$

that is left adjoint (Definition 1.1.13) to the inclusion functor

$$i : \text{Shv}(\mathcal{C}, \mathcal{A}) \hookrightarrow \text{PreShv}(\mathcal{C}, \mathcal{A}).$$

In other words, a is a reflector of i . If such a sheafification functor exists, then it is unique up to unique natural isomorphism. Given a presheaf P , the sheafification $a(P)$ is also sometimes called the *sheaf associated to P* . See Theorem 5.2.27 for common conditions under which sheafification exists.

Definition 5.2.21. Let $\mathcal{A}$ be a category (Definition 1.0.1) with a terminal object (Lemma 2.0.7). Let X be a topological space (Definition B.0.1).

1. Let $\mathcal{F}$ be a presheaf (Definition 5.1.38) on X valued in $\mathcal{A}$.

A *sheafification of $\mathcal{F}$* is a pair $(\mathcal{F}^+, \theta)$, where $\mathcal{F}^+$ is a sheaf on X and $\theta : \mathcal{F} \rightarrow \mathcal{F}^+$ is a morphism of presheaves, satisfying the following universal property:

For any sheaf $\mathcal{G}$ on X and any morphism of presheaves $\phi : \mathcal{F} \rightarrow \mathcal{G}$, there exists a unique morphism of sheaves $\psi : \mathcal{F}^+ \rightarrow \mathcal{G}$ such that $\psi \circ \theta = \phi$.

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\theta} & \mathcal{F}^+ \\ & \searrow \phi & \downarrow \psi \\ & & \mathcal{G} \end{array}$$

2. A *sheafification functor* refers to a functor

$$a : \text{PreShv}(X, \mathcal{A}) \rightarrow \text{Shv}(X, \mathcal{A})$$

that is left adjoint (Definition 1.1.13) to the inclusion functor

$$i : \text{Shv}(X, \mathcal{A}) \hookrightarrow \text{PreShv}(X, \mathcal{A}).$$

In other words, a is a reflector of i . If such a sheafification functor exists, then it is unique up to unique natural isomorphism. Given a presheaf P , the sheafification $a(P)$ is also sometimes called the *sheaf associated to P* . The sheafification functor on $\text{PreShv}(X, \mathcal{A})$ is a special case of the general definition of sheafification functor (Definition 5.2.20), applied to the site (Definition 5.1.13) whose underlying category is $\text{Open}(X)$ (Definition 1.0.7) and whose Grothendieck topology is the canonical one (Definition 5.1.30).

If $\mathcal{A}$ is a category admitting direct colimits (Definition 1.2.12) (e.g. the category of sets, groups, abelian groups, modules over rings, or vector spaces over fields), then the sheaf $\mathcal{P}^+$ associated to a presheaf $\mathcal{P}$ valued in $\mathcal{A}$ may be constructed by assigning to each open set $U \subseteq X$ the set (or object in $\mathcal{D}$)

$$\mathcal{P}^+(U) := \left\{ s = (s_x)_{x \in U} \in \prod_{x \in U} \mathcal{P}_x \left| \begin{array}{l} \forall x \in U, \\ \exists \text{ an open } V \subseteq U \text{ with } x \in V, \\ \exists t \in \mathcal{P}(V) \text{ such that} \\ \forall y \in V, s_y = t_y \end{array} \right. \right\}.$$

where $\mathcal{P}_x$ is the stalk of $\mathcal{P}$ at x , and t_y is the germ of t at y . In particular, $\mathcal{P}^+$ exists, and $\mathcal{P} \mapsto \mathcal{P}^+$ is a sheafification functor.

Equivalently, the assignment $\mathcal{P} \mapsto \mathcal{P}^+$ is the sheafification functor as defined in Definition 5.2.20 in the case that the site is the site of opens (Definition 5.1.30) on the topological space X . See also Theorem 5.2.27.

Theorem 5.2.22. Let X be a topological space (Definition B.0.1).

1. Let $\mathcal{F}$ be a presheaf (Definition 5.1.38) of sets (or sets with structure, e.g. groups, rings, abelian groups, R -modules, etc.). The sheafification (Theorem 5.2.27) $(\mathcal{F}^+, \theta)$ exists and is unique up to unique isomorphism. The sheaf $\mathcal{F}^+$ can be constructed as the sheaf of sections of the espace étalé (space of stalks) associated to $\mathcal{F}$. Specifically, for any open $U \subseteq X$, $\mathcal{F}^+(U)$ is the set of functions $s : U \rightarrow \prod_{x \in U} \mathcal{F}_x$ such that:
 - (a) For each $x \in U$, $s(x) \in \mathcal{F}_x$.
 - (b) For each $x \in U$, there exists an open neighborhood $V \subseteq U$ of x and a section $t \in \mathcal{F}(V)$ such that $s(y) = t_y$ for all $y \in V$ (where t_y denotes the germ of t at y).
2. A sheafification functor (Definition 5.2.21)

$$a : \text{PreShv}(X, \mathcal{A}) \rightarrow \text{Shv}(X, \mathcal{A})$$

exists.

This theorem is a special case of Theorem 5.2.27 in the case that the site is the site associated to X (Definition 5.1.30).

Definition 5.2.23 (cf. [KS06, Definition 3.1.10]). Let $\mathcal{A}$ be a category (Definition 1.0.1) that admits small (Definition 1.2.5) products (Definition 1.2.36) and small filtered limits (Definition 1.2.12). For any family $\{I_s\}_{s \in S}$ of small and filtered categories (Definition 1.2.8) and any family $\{\alpha_s : I_s \rightarrow \mathcal{A}\}_{s \in S}$ indexed by a small set S , there is a canonical morphism

$$\text{colim} \left(\prod_s I_s \xrightarrow{k \mapsto \prod_s \alpha_s(\pi_s(k))} A \right) \rightarrow \prod_s \text{colim} \alpha_s$$

We say that $\mathcal{A}$ satisfies the *IPC-property* (*inductive-limit-product commutation property*) if any such canonical morphism is an isomorphism.

Proposition 5.2.24 (cf. [KS06, Proposition 3.1.11(i)]). (♠ TODO: think about whether I can be made large) Let $\{\mathcal{A}_i\}_{i \in I}$ be a family of category (Definition 1.0.1) that admits small (Definition 1.2.5) products (Definition 1.2.36) and small filtered limits (Definition 1.2.12) indexed by a set I . If the categories $\mathcal{A}_i$ satisfy the IPC-property (Definition 5.2.23), then so does the product category (Definition 1.3.11) $\prod_{i \in I} \mathcal{A}_i$.

Proposition 5.2.25 ([KS06, Proposition 3.1.11(ii), Corollary 3.1.12, Corollary 3.1.13.]). 1.

- The category of sets (Definition 1.0.3) satisfies the IPC-property (Definition 5.2.23)
2. Given a small category (Definition 1.0.9) $\mathcal{C}$, the functor category (Definition 1.2.1) $[\mathcal{C}, \mathbf{Sets}]$ satisfies the IPC-property.
3. Let R be a (not necessarily commutative) (Definition 1.0.5) ring. The category of (left) R -modules (Definition C.0.11) satisfies the IPC-property; in particular, the category of abelian groups (Definition 1.0.4) and the category of R - S -bimodules (Definition C.0.8), given rings R and S , satisfy the IPC-property.

Proposition 5.2.26 ([KS06, Proposition 3.1.11(iii)]). (♠ TODO: think about whether local smallness can be removed) Let $\mathcal{A}$ be a locally small (Lemma 5.0.5) category (Definition 1.0.1)

that admits small (Definition 1.2.5) products (Definition 1.2.36) and small filtered limits (Definition 1.2.12).

Assume that there exist a set I and a functor $\lambda : \mathcal{A} \rightarrow \mathbf{Sets}^I$ (Definition 1.0.3) (Definition 1.2.1) such that

- λ commutes with small (Definition 1.2.5) products (Definition 1.2.36)
- λ commutes with small filtered inductive limits (Definition 1.2.12)
- λ is conservative (♠ TODO: conservative)

Then $\mathcal{A}$ satisfies the IPC-property

The following expresses the idea that sheafification functors exist for small enough sites when the value category is nice enough.

Theorem 5.2.27. cf. [GV72, Exposé II, Théorème 3.4]

1. Let U be a universe. Let $\mathcal{C}$ be a U -site (Definition 5.1.13). A sheafification functor (Definition 5.2.21)

$$a : \text{PreShv}(\mathcal{C}, U\text{-}\mathbf{Sets}) \rightarrow \text{Shv}(\mathcal{C}, U\text{-}\mathbf{Sets}).$$

exists.

2. Let $\mathcal{C}$ be a site whose underlying category is locally small (Definition 1.0.9) and which has a topologically generating family (Definition 5.1.13) that is a set (rather than a proper class). A sheafification functor

$$a : \text{PreShv}(\mathcal{C}, \mathbf{Sets}) \rightarrow \text{Shv}(\mathcal{C}, \mathbf{Sets})$$

exists.

(♠ TODO: It may be the case that I need to assume $\mathcal{A}$ to be locally small; if so, recheck all the statements that use this.)

3. (see e.g. [nLa25c, 3], [KS06, Theorem 17.4.7]) Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13) on an essentially small category (Definition 1.1.9) $\mathcal{C}$. Suppose that the category $\mathcal{A}$ is complete, cocomplete (Definition 1.3.9), that small filtered colimits (Definition 1.2.12) in $\mathcal{A}$ are exact (Definition 1.2.18), and that $\mathcal{A}$ satisfies the IPC-property (Definition 5.2.23). A sheafification functor (Definition 5.2.20)

$$a : \text{PreShv}(\mathcal{C}, \mathcal{A}) \rightarrow \text{Shv}(\mathcal{C}, \mathcal{A})$$

exists.

(♠ TODO: state as a fact that the category of groups and k -algebras are complete, cocomplete, with small filtered colimits that are exact) This is true for instance of $\mathcal{A} = \mathbf{Set}$ (Proposition 5.2.25), $\mathbf{Grp}$, $k\text{-}\mathbf{Alg}$ for a field k , or $\mathbf{Mod}_R$ (Proposition 5.2.25) for a (not necessarily commutative unital) ring R (Definition C.0.1).

Remark 5.2.28. If the presheaf is valued in nice “algebraic category”, e.g. groups, abelian groups, rings, modules over a ring, etc., then the sheafification is again valued in that category. (♠ TODO: Make this more precise.)

Definition 5.2.29 (Constant sheaf on a site). Let $\mathcal{C}$ be a category (Definition 1.0.1), let $\mathcal{A}$ be a category, and let A be an object of $\mathcal{A}$.

1. The *constant presheaf on $\mathcal{C}$ with value A* is the presheaf (Definition 5.0.1) P defined by

$$P(U) = A$$

for every object U of $\mathcal{C}$ such that every morphism $f : V \rightarrow U$ in $\mathcal{C}$ induces the identity map $A = P(U) \rightarrow P(V) = A$.

2. Let $\mathcal{C}$ be a site (Definition 5.1.13) and assume that a sheafification functor (Definition 5.2.20)

$$a : \text{Shv}(\mathcal{C}, \mathcal{A}) \rightarrow \text{PreShv}(\mathcal{C}, \mathcal{A})$$

exists (e.g. see Theorem 5.2.27). The *constant sheaf on $\mathcal{C}$ with value A* , or the *constant sheaf on $\mathcal{C}$ associated to A* commonly denoted $\underline{A}$ or sometimes just A by abuse of notation, is the sheaf associated to (Theorem 5.2.27) the constant presheaf P with value A above.

3. Let $\mathcal{C}$ be a site. Let $\mathcal{O}$ be a sheaf of (not-necessarily commutative) rings on $\mathcal{C}$. Assume that the global sections ring $\Gamma(\mathcal{O})$ (Definition 5.3.11) exists. A *constant $\mathcal{O}$ -module* is an $\mathcal{O}$ -module (Definition 5.3.9) $\mathcal{F}$ which is isomorphic as a sheaf to the constant sheaf on $\mathcal{C}$ with value M where M is a module of the ring $\Gamma(\mathcal{O})$. Note that sheafification functors exist for presheaves/sheaves valued in Ab (Theorem 5.2.27).

In case that $\mathcal{O}$ is the constant sheaf associated to A for some (not-necessarily commutative) ring A , a constant $\mathcal{O}$ -module is simply called a *constant A -module*.

Definition 5.2.30. Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13), let $\mathcal{F}$ be a presheaf (Definition 5.0.1) on $\mathcal{C}$ valued in some category, and let $U \in \text{Ob}(\mathcal{C})$.

The *restriction of $\mathcal{F}$ to U* is the sheaf $\mathcal{F}|_U$ on the induced site $(\mathcal{C}/U, J/U)$ (Definition 5.1.29) defined by

$$\mathcal{F}|_U(f : V \rightarrow U) = \mathcal{F}(V).$$

More precisely,

$$\mathcal{F}|_U = \mathcal{F} \circ j_U^{\text{op}}$$

where $j_U : \mathcal{C}/U \rightarrow \mathcal{C}$ is the forgetful functor, which is a continuous (Definition 5.2.1) and flat functor (Definition 5.2.17), and $j_U^{\text{op}} : (\mathcal{C}/U)^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$ (Definition 1.0.12) is the opposite functor.

If $\mathcal{F}$ is a sheaf on $(\mathcal{C}, J)$, then $\mathcal{F}|_U$ is a sheaf on $(\mathcal{C}/U, J/U)$.

Definition 5.2.31 (Locally constant sheaf on a site). Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13).

1. Let $\mathcal{A}$ be a (large) category, and let A be an object of $\mathcal{A}$. (♠ TODO: If such a sheafification functor exist, does a sheafification functor exist when restricted to an object U ?) Assume that a sheafification functor (Definition 5.2.20)

$$a : \text{PreShv}(\mathcal{C}, J, \mathcal{A}) \rightarrow \text{Shv}(\mathcal{C}, J, \mathcal{A})$$

(Definition 5.1.32) (Definition 5.0.1) exists (e.g. see Theorem 5.2.27). Let U be an object of $\mathcal{C}$.

A sheaf $\mathcal{F}$ on $\mathcal{C}$ with values in $\mathcal{A}$ is said to be a *locally constant sheaf on U with value A* if there exists a covering sieve (Definition 5.1.13) $\{U_i \rightarrow U\}$ of every object U in $\mathcal{C}$ such that for each i , the restriction $\mathcal{F}|_{U_i}$ (Definition 5.2.30) is isomorphic to the constant sheaf (Definition 5.2.29) $\underline{A}$ on the slice site $\mathcal{C}_{/U_i}$ (Definition 5.1.29).

If $\mathcal{C}$ has a final object, then we may say that $\mathcal{F}$ is a *locally constant sheaf with value A* if it is a locally constants sheaf on the final object with value A .

2. Let $\mathcal{O}$ be a sheaf of (not-necessarily commutative) rings on $\mathcal{C}$. A *locally constant $\mathcal{O}$ -module* is a $\mathcal{O}$ -module (Definition 5.3.9) $\mathcal{F}$ such that there exists a covering sieve (Definition 5.1.13) $\{U_i \rightarrow U\}$ for every object U in $\mathcal{C}$ such that for each i , the restriction (Definition 5.2.30) $\mathcal{F}|_{U_i}$ is isomorphic, as an $\mathcal{O}|_{U_i}$ -module, to a constant $\mathcal{O}|_{U_i}$ -module (Definition 5.2.29)².

We additionally say that $\mathcal{F}$ is

- (a) *locally free of rank r over $\mathcal{O}$* if there exists a covering $\{U_i \rightarrow U\}$ such that $\mathcal{F}|_{U_i}$ is isomorphic as a $\mathcal{O}|_{U_i}$ -module to $(\mathcal{O}|_{U_i})^{\oplus r}$ for each i .
- (b) *free of rank r over $\mathcal{O}$* if $\mathcal{F} \cong \mathcal{O}^{\oplus r}$ as $\mathcal{O}$ -modules.
- (c) *of finite type* if there exists a covering $\{U_i \rightarrow U\}$ such that $\mathcal{F}|_{U_i}$ generated by finitely many sections over U_i as an $\mathcal{O}|_{U_i}$ -module. In other words, there is an epimorphism

$$(\mathcal{O}|_{U_i})^{\oplus n_i} \rightarrow \mathcal{F}|_{U_i}$$

of $\mathcal{O}|_{U_i}$ -modules for each i .

In case that $\mathcal{O}$ is the constant sheaf on $\mathcal{C}$ associated to A (Definition 5.2.29) for some (not-necessarily commutative) ring A , a locally constant $\mathcal{O}$ -module is simply called a *locally constant A -module*.

5.3. Inverse image and direct image of sheaves.

Definition 5.3.1 (Relative Slice Category). Let $\mathcal{C}, \mathcal{D}$ be categories (Definition 1.0.1), let $u : \mathcal{D} \rightarrow \mathcal{C}$ be a functor (Definition 1.1.1) and let U be an object of $\mathcal{C}$. The *relative slice category* (also called the *relative over-category* or *category of objects over U relative to u*), denoted $(u \downarrow U)$ or u/U , is defined as follows:

- **Objects:** Pairs (V, φ) , where V is an object of $\mathcal{D}$ and $\varphi : u(V) \rightarrow U$ is a morphism in $\mathcal{C}$.
- **Morphisms:** A morphism from (V, φ) to (V', φ') is a morphism $f : V \rightarrow V'$ in $\mathcal{D}$ such that the following triangle in $\mathcal{C}$ commutes:

$$\begin{array}{ccc} u(V) & \xrightarrow{u(f)} & u(V') \\ & \searrow \varphi & \swarrow \varphi' \\ & U & \end{array}$$

That is, $\varphi' \circ u(f) = \varphi$.

- **Composition:** Inherited from $\mathcal{D}$.

Slice categories are special cases of comma categories (Definition 1.3.12).

²The global sections rings $\Gamma(\mathcal{O}|_{U_i})$ (Definition 5.3.11) of the sheaves $\mathcal{O}|_{U_i}$ on the slice sites $\mathcal{C}/_{U_i}$ (Definition 5.1.29) exist because each $\mathcal{C}/_{U_i}$ has a final object (Definition 2.0.6)(Lemma 1.3.5), so we may speak of constant $\mathcal{O}|_{U_i}$ -modules.

Definition 5.3.2. Let $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ be sites (Definition 5.1.13) with small topological generating families (Definition 5.1.13) (or U -small topologically generating families if a universe (Definition E.0.1) U is available), and let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a continuous functor of sites (Definition 5.2.1).

Let $\mathcal{A}$ be a (large) category which has all small (or U -small) products (Definition 1.2.36).

For any sheaf (Definition 5.1.32)

$$\mathcal{F} \in \text{Sh}(\mathcal{D}, K; \mathcal{A}),$$

Define the *pushforward/direct image sheaf* $u^s \mathcal{F}$ by

$$u^s \mathcal{F} := \mathcal{F} \circ u : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}.$$

The continuity of u implies that $u^s \mathcal{F}$ is indeed a sheaf on $(\mathcal{C}, J)$ (Proposition 5.2.2). The assignment $\mathcal{F} \mapsto u^s \mathcal{F}$ defines the *direct image/pushforward functor*

$$u^s : \text{Sh}(\mathcal{D}, K; \mathcal{A}) \rightarrow \text{Sh}(\mathcal{C}, J; \mathcal{A}).$$

If u is the functor underlying a site morphism (Definition 5.2.5) $f : (\mathcal{D}, K) \rightarrow (\mathcal{C}, J)$, we may alternatively denote $u^s \mathcal{F}$ by $f_* \mathcal{F}$ and call it the *direct image/pushforward of $\mathcal{F}$ under f* ; the assignment $\mathcal{F} \mapsto f_* \mathcal{F}$ is then the *direct image/pushforward functor*.

$$f_* : \text{Sh}(\mathcal{D}, K; \mathcal{A}) \rightarrow \text{Sh}(\mathcal{C}, J; \mathcal{A}).$$

Note that while the continuous functor u and the site morphism f point in opposite directions, the definition $f_* := u^s$ ensures that f_* corresponds to the standard geometric pushforward used in topology and algebraic geometry.

Note that the “inverse image of a cocontinuous functor of sites (Definition 5.2.14)” and the “direct image of a continuous functor of sites” share the same constructions, i.e. by precomposition. More precisely, if $u : \mathcal{C} \rightarrow \mathcal{D}$ is a continuous (Definition 5.2.1) and cocontinuous functor of sites, then the direct image u_* of u as a continuous functor and the inverse image u^* of u as a cocontinuous functor (Definition 5.2.14) are exactly the same.

Definition 5.3.3. Let $(\mathcal{C}, J)$ and $(\mathcal{D}, K)$ be sites (Definition 5.1.13) with small topological generating families (Definition 5.1.13), and let $u : \mathcal{C} \rightarrow \mathcal{D}$ be a continuous functor of sites (Definition 5.2.1). Let $\mathcal{A}$ be a (large) category such that the presheaf category (Definition 5.0.1) $\text{PreSh}(\mathcal{D}, K; \mathcal{A})$ has sheafification (Definition 5.2.20).

For any sheaf (Definition 5.1.32)

$$\mathcal{G} \in \text{Sh}(\mathcal{C}, J; \mathcal{A}),$$

the *inverse image/pullback sheaf of $\mathcal{G}$ under u* is defined, assuming that all colimits below exist, as:

$$u_s \mathcal{G} : \mathcal{D}^{\text{op}} \rightarrow \mathcal{A}, \quad V \mapsto a \left(\varinjlim_{(V \downarrow u)} \mathcal{G}(U) \right),$$

where a is the sheafification (Definition 5.2.20) functor of presheaves and the colimit (Definition 1.2.2) is taken over the comma category $(V \downarrow u)$ (Definition 1.3.12) of pairs $(U, V \rightarrow u(U))$ with $U \in \mathcal{C}$.

The assignment $\mathcal{G} \mapsto u_s \mathcal{G}$ defines the *inverse image/pullback functor*

$$u_s : \mathrm{Sh}(\mathcal{C}, J; \mathcal{A}) \rightarrow \mathrm{Sh}(\mathcal{D}, K; \mathcal{A}).$$

If u is the functor underlying a site morphism (Definition 5.2.5) $f : (\mathcal{D}, K) \rightarrow (\mathcal{C}, J)$, we may alternatively denote $u_s \mathcal{G}$ by $f^* \mathcal{G}$ (or sometimes by $f^{-1} \mathcal{G}$) and call it the *inverse image/pullback of $\mathcal{G}$ under f* .

Note that while the continuous functor u and the site morphism f point in opposite directions, the identification $f^* := u_s$ ensures that f^* corresponds to the standard geometric pullback. In the case of topological spaces, this recovers the usual construction involving colimits over open neighborhoods to obtain stalks followed by sheafification.

The inverse image f^* of a continuous functor of sites should not be confused with the notion of the inverse image of a cocontinuous functor of sites (Definition 5.2.14), as these two notions are different constructions, even when f is both continuous and cocontinuous.

Theorem 5.3.4. (♠ TODO: It is likely that some more restrictions are needed; e.g. must C and D be small and $\mathcal{A}$ locally small to ensure that we can talk about hom's between sheaves?) Let (C, J) and (D, K) be sites (Definition 5.1.13), and let $u : C \rightarrow D$ be a continuous functor (Definition 5.2.1) of sites. Let $\mathcal{A}$ be a (large) category such that the presheaf category (Definition 5.0.1) $\mathrm{PreSh}(\mathcal{D}, K; \mathcal{A})$ has sheafification (Definition 5.2.20).

Then the inverse image functor (Definition 5.3.2)

$$u_s : \mathrm{Sh}(D, K; \mathcal{A}) \rightarrow \mathrm{Sh}(C, J; \mathcal{A})$$

(assuming that the inverse images of all sheaves on (D, K) valued in $\mathcal{A}$ exist by virtue of $\mathcal{A}$ admitting enough colimits (Definition 1.2.12)) and the direct image functor (Definition 5.3.3)

$$u^s : \mathrm{Sh}(C, J; \mathcal{A}) \rightarrow \mathrm{Sh}(D, K; \mathcal{A})$$

form an adjoint pair (Definition 1.1.13), i.e., for any sheaves $\mathcal{F} \in \mathrm{Sh}(D, K; \mathcal{A})$ and $\mathcal{G} \in \mathrm{Sh}(C, J; \mathcal{A})$, there is a natural isomorphism

$$\mathrm{Hom}_{\mathrm{Sh}(C, J)}(u_s \mathcal{F}, \mathcal{G}) \cong \mathrm{Hom}_{\mathrm{Sh}(D, K)}(\mathcal{F}, u^s \mathcal{G}).$$

Definition 5.3.5 (Pushforward (direct image) of a sheaf). Let $f : X \rightarrow Y$ be a continuous map between topological spaces (Definition B.0.1), and let $\mathcal{F}$ be a presheaf on X valued in a category $\mathcal{D}$ (Definition 5.1.38) with a terminal object (Lemma 2.0.7). The *pushforward* or *direct image presheaf* $f_* \mathcal{F}$ on Y is the presheaf valued in $\mathcal{D}$ on Y (Definition 5.0.1) defined as follows: For every open set $V \subseteq Y$, the value of the pushforward is given by

$$f_* \mathcal{F}(V) := \mathcal{F}(f^{-1}(V)).$$

For an inclusion of open sets $V' \subseteq V$ in Y , the restriction morphism

$$\mathrm{res}_{V, V'}^{f_* \mathcal{F}} : f_* \mathcal{F}(V) \rightarrow f_* \mathcal{F}(V')$$

is defined as the restriction morphism of $\mathcal{F}$ associated with the inclusion of preimages $f^{-1}(V') \subseteq f^{-1}(V)$ in X :

$$\text{res}_{f^{-1}(V), f^{-1}(V')}^{\mathcal{F}} : \mathcal{F}(f^{-1}(V)) \rightarrow \mathcal{F}(f^{-1}(V')).$$

If $\mathcal{F}$ is a sheaf (Definition 5.1.36), then so is $f_*\mathcal{F}$. In this case, it is equivalent to define $f_*\mathcal{F}$ as the direct image (Definition 5.3.2) $(f^{-1})^s\mathcal{F}$ of $\mathcal{F}$ under the continuous functor $f^{-1} : \text{Open } Y \rightarrow \text{Open } X$ (Definition 5.2.3) (Definition 5.2.3) (Definition 5.1.30), which is also equivalent to the direct image (Definition 5.3.2) of $\mathcal{F}$ under the site morphism (Definition 5.2.5) $\text{Open } X \rightarrow \text{Open } Y$ whose underlying continuous functor is f^{-1} .

(♠ TODO: comment on what is happening on the stalks)

Definition 5.3.6 (Pullback (inverse image) of a sheaf). Let $f : X \rightarrow Y$ be a continuous map between topological spaces (Definition B.0.1). Let $\mathcal{D}$ be a category with a terminal object (Definition 2.0.6).

1. Let $\mathcal{G}$ be a presheaf on Y valued in a $\mathcal{D}$ (Definition 5.1.38). The *pullback* or *inverse image presheaf* $f^{-1}\mathcal{G}$ on X is defined as the presheaf

$$U \mapsto \varinjlim_{V \supseteq f(U)} \mathcal{G}(V)$$

provided these colimits (Definition 1.2.2) exist for all U , where U ranges over open subsets of X and the colimit is taken over all open subsets $V \subseteq Y$ containing $f(U)$.

2. If $\mathcal{G}$ is a sheaf (Definition 5.1.36) valued in $\mathcal{D}$, then we can define the *pullback* or *inverse image sheaf* $f^*\mathcal{G}$ on X as the sheaf associated to the presheaf (Definition 5.2.20) $f^{-1}\mathcal{G}$, assuming it exists.

Assuming that a sheafification functor (Definition 5.2.20) exists, one may equivalently define $f^*\mathcal{G}$ via Definition 5.3.3 — More concretely, $f^*\mathcal{G}$ is the following equivalent constructions:

- The inverse image (Definition 5.3.3) $(f^{-1})_s\mathcal{G}$ of $\mathcal{G}$ under the continuous functor $f^{-1} : \text{Open } Y \rightarrow \text{Open } X$, $W \mapsto f^{-1}(W)$ (Definition 5.1.30) (Definition 5.2.3).
- The inverse image (Definition 5.3.3) of $\mathcal{G}$ under the site morphism (Definition 5.2.5) $\text{Open } X \rightarrow \text{Open } Y$ whose underlying continuous functor (Definition 5.2.1) is f^{-1}

If $\mathcal{D}$ is the category of sets (Definition 1.0.3), of groups, of abelian groups, of modules over rings, for example, then the relevant colimits (Definition 1.2.2) needed to define the inverse image presheaves and the sheafification needed to define the inverse image sheaves exist.

Theorem 5.3.7. Let $f : X \rightarrow Y$ be a continuous map between topological spaces (Definition B.0.1). Let $\mathcal{A}$ be a locally small category (Definition 1.0.9) such that $\mathcal{A}$ is cocomplete (Definition 1.3.9), i.e. admits small colimits (Definition 1.2.5), and such that $\text{PreShv}(Y, \mathcal{A})$ admits sheafification (Definition 5.2.21); for example, the categories of sets, of groups, of abelian groups, of rings, of modules over a ring, etc. satisfy these hypotheses.

The inverse image functor (Definition 5.3.6)

$$f^* : \text{Sh}(Y, \mathcal{A}) \rightarrow \text{Sh}(X, \mathcal{A})$$

and the direct image functor (Definition 5.3.5)

$$f_* : \mathrm{Sh}(X, \mathcal{A}) \rightarrow \mathrm{Sh}(Y, \mathcal{A})$$

form an adjoint pair (Definition 1.1.13) $f^* \dashv f_*$, i.e., for any sheaves $\mathcal{F} \in \mathrm{Sh}(Y, \mathcal{A})$ and $\mathcal{G} \in \mathrm{Sh}(X, \mathcal{A})$, there is a natural isomorphism

$$\mathrm{Hom}_{\mathrm{Sh}(X, \mathcal{A})}(f^* \mathcal{F}, \mathcal{G}) \cong \mathrm{Hom}_{\mathrm{Sh}(Y, \mathcal{A})}(\mathcal{F}, f_* \mathcal{G}).$$

Definition 5.3.8. (♠ TODO: think about if assumptions on the small site) Let $f : X \rightarrow Y$ be a morphism of schemes, and let $\mathbf{C}(X)$ and $\mathbf{C}(Y)$ be small sites associated to X and Y respectively, equipped with a Grothendieck topologies for which the continuous functor (Definition 5.2.1) $f^{-1} : \mathbf{C}(Y) \rightarrow \mathbf{C}(X)$ (Definition 5.2.4) on sites underlies a site morphism (Definition 5.2.5) $f : \mathbf{C}(X) \rightarrow \mathbf{C}(Y)$. For example, the common Grothendieck topologies such as Zariski, étale, Nisnevich, fppf, fpqc, or crystalline satisfy this hypothesis(♠ TODO: cite these).

1. Let $\mathcal{A}$ be a (large) category. Given a sheaf $\mathcal{F} \in \mathrm{Sh}(\mathbf{C}(X); \mathcal{A})$, the *direct image along f* is the sheaf $f_* \mathcal{F}$ defined as the direct image (Definition 5.3.2) of $\mathcal{F}$ under site morphism $f : \mathbf{C}(X) \rightarrow \mathbf{C}(Y)$. In particular, it is an object of $\mathrm{Sh}(\mathbf{C}(Y); \mathcal{A})$ if it exists. If $f_* \mathcal{F}$ exists for all sheaves $\mathcal{F} \in \mathrm{Sh}(\mathbf{C}(X); \mathcal{A})$, then f_* yields a functor

$$f_* : \mathrm{Sh}(\mathbf{C}(X); \mathcal{A}) \rightarrow \mathrm{Sh}(\mathbf{C}(Y); \mathcal{A}).$$

2. We have (??) a direct image functor

$$f_* : \mathrm{Sh}(\mathbf{C}(X); \mathrm{Ab}) \rightarrow \mathrm{Sh}(\mathbf{C}(Y); \mathrm{Ab}).$$

In fact, if f yields a ringed site morphism $f : (\mathbf{C}(X), \mathcal{O}_X) \rightarrow (\mathbf{C}(Y), \mathcal{O}_Y)$, then this induces (??) a functor

$$f_* : \mathrm{Mod}(\mathcal{O}_X) \rightarrow \mathrm{Mod}(\mathcal{O}_Y).$$

3. Let $\mathcal{A}$ be a (large) category such that the presheaf category (Definition 5.0.1) $\mathrm{PreSh}(\mathcal{D}, K; \mathcal{A})$ has sheafification (Definition 5.2.20). Given a sheaf $\mathcal{G} \in \mathrm{Sh}(\mathbf{C}(Y); \mathcal{A})$, the *inverse image of $\mathcal{G}$ along f* is the sheaf $f^* \mathcal{G}$ defined as the inverse image (Definition 5.3.3) of $\mathcal{G}$ under the continuous functor (Definition 5.2.1) $f^{-1} : \mathbf{C}(Y) \rightarrow \mathbf{C}(X)$ (Definition 5.2.4) on sites. In particular, it is an object of $\mathrm{Sh}(\mathbf{C}(X); \mathcal{A})$ and f^* yields a functor

$$f^* : \mathrm{Sh}(\mathbf{C}(Y); \mathcal{A}) \rightarrow \mathrm{Sh}(\mathbf{C}(X); \mathcal{A}),$$

4. We have (??) an inverse image functor

$$f^* : \mathrm{Sh}(\mathbf{C}(Y); \mathrm{Ab}) \rightarrow \mathrm{Sh}(\mathbf{C}(X); \mathrm{Ab}).$$

In fact, if f yields a ringed site morphism $f : (\mathbf{C}(X), \mathcal{O}_X) \rightarrow (\mathbf{C}(Y), \mathcal{O}_Y)$, then this induces (??) a functor

$$f^* : \mathrm{Mod}(\mathcal{O}_Y) \rightarrow \mathrm{Mod}(\mathcal{O}_X).$$

5.3.1. *Sheaf of rings on a site and modules over a sheaf of rings on a site.*

Definition 5.3.9. 1. Let $\mathcal{C}$ be a site, and let $\mathcal{A}$ and $\mathcal{B}$ be sheaves (Definition 5.1.32) of (not necessarily commutative) rings (Definition C.0.1) on $\mathcal{C}$.

- (a) An $(\mathcal{A}, \mathcal{B})$ -bimodule (or a bimodule over $(\mathcal{A}, \mathcal{B})$) is a sheaf (Definition 5.1.32) $\mathcal{M}$ of abelian groups on $\mathcal{C}$ equipped with a left $\mathcal{A}$ -module structure given by a morphism of sheaves (Definition 5.1.32) of sets

$$\lambda : \mathcal{A} \times \mathcal{M} \longrightarrow \mathcal{M},$$

and a right $\mathcal{B}$ -module structure given by a morphism of sheaves of sets

$$\rho : \mathcal{M} \times \mathcal{B} \longrightarrow \mathcal{M},$$

such that the actions are compatible. Specifically, for every object U in $\mathcal{C}$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the equality

$$\lambda_U(a, \rho_U(m, b)) = \rho_U(\lambda_U(a, m), b)$$

holds in $\mathcal{M}(U)$. In standard multiplicative notation where $\lambda(a, m)$ is denoted $a \cdot m$ and $\rho(m, b)$ is denoted $m \cdot b$, this condition is the associativity axiom

$$(a \cdot m) \cdot b = a \cdot (m \cdot b).$$

In particular, for every object $U \in \mathcal{C}$, the abelian group $\mathcal{M}(U)$ has the structure of an $\mathcal{A}(U) - \mathcal{B}(U)$ -bimodule (Definition C.0.8).

- (b) Assume that a sheafification functor (Definition 5.2.20)

$$\text{PreShv}(\mathcal{C}, \mathbf{Rings}) \rightarrow \text{Shv}(\mathcal{C}, \mathbf{Rings})$$

exists. For rings A and B , a A - B -bimodule or bimodule over (A, B) on $\mathcal{C}$ is a bimodule over $(\underline{A}, \underline{B})$ where $\underline{A}$ and $\underline{B}$ are the constant sheaves (Definition 5.2.29) on $\mathcal{C}$ with values A and B respectively.

- (c) Let $\mathcal{M}$ and $\mathcal{N}$ be $(\mathcal{A}, \mathcal{B})$ -bimodules. A homomorphism of $(\mathcal{A}, \mathcal{B})$ -bimodules (or an $(\mathcal{A}, \mathcal{B})$ -linear morphism) is a morphism of sheaves of abelian groups $f : \mathcal{M} \rightarrow \mathcal{N}$ such that for every object U of $\mathcal{C}$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the following compatibility conditions hold:

$$f_U(a \cdot m) = a \cdot f_U(m) \quad \text{and} \quad f_U(m \cdot b) = f_U(m) \cdot b.$$

We denote the category of $(\mathcal{A}, \mathcal{B})$ -bimodules, with morphisms being morphisms of sheaves of abelian groups compatible with both the left $\mathcal{A}$ -action and the right $\mathcal{B}$ -action, by $\mathcal{A}\text{-}\mathcal{B}\text{-Mod}$ or sometimes by ${}_{\mathcal{A}}\text{Mod}_{\mathcal{B}}$ (♠ TODO: talk about how bimodules can be identifies with left/right modules)

2. Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13). Let $\mathcal{O}$ be a sheaf of (not necessarily commutative) rings on $(\mathcal{C}, J)$ (Definition 5.1.32), i.e. $((\mathcal{C}, J), \mathcal{O})$ is a ringed site (Definition 5.4.9).

- (a) An (left/right/two-sided) $\mathcal{O}$ -module consists of the following data:

- A sheaf $\mathcal{F}$ of abelian groups on $(\mathcal{C}, J)$,
- for every object $U \in \mathcal{C}$, the structure of an (left/right/two-sided) $\mathcal{O}(U)$ -module on $\mathcal{F}(U)$,

such that for every morphism $f : V \rightarrow U$ in $\mathcal{C}$, the restriction map

$$\rho_{U,V} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$$

is $\mathcal{O}(U)$ -linear when the $\mathcal{O}(U)$ -action on $\mathcal{F}(V)$ is defined via the natural ring homomorphism

$$\mathcal{O}(U) \rightarrow \mathcal{O}(V)$$

induced by f .

(b) Let $\mathcal{F}$ and $\mathcal{G}$ be $\mathcal{O}$ -modules (Definition 5.3.9).

A *morphism of $\mathcal{O}$ -modules* $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves (Definition 5.1.32) of abelian groups such that, for every object $U \in \mathcal{C}$, the component map

$$\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

is $\mathcal{O}(U)$ -linear, i.e. it satisfies

$$\varphi_U(r \cdot s) = r \cdot \varphi_U(s) \quad \text{for all } r \in \mathcal{O}(U), s \in \mathcal{F}(U).$$

The collection of all $\mathcal{O}$ -modules together with their morphisms of $\mathcal{O}$ -modules forms the *category of $\mathcal{O}$ -modules*, denoted $\mathbf{Mod}(\mathcal{O})$. The Hom functors $\mathrm{Hom}_{\mathbf{Mod}(\mathcal{O})}$ of this category may be denoted by $\mathbf{Hom}_{\mathcal{O}}$.

In case that a sheafification functor (Definition 5.2.20)

$$\mathrm{PreShv}(\mathcal{C}, \mathbf{Rings}) \rightarrow \mathrm{Shv}(\mathcal{C}, \mathbf{Rings})$$

exists, a left, right, two-sided $\mathcal{O}$ -module (and morphisms thereof) is equivalent to a $(\mathcal{O}, \mathbb{Z})$ -bimodule, $(\mathbb{Z}, \mathcal{O})$ -bimodule, and $(\mathcal{O}, \mathcal{O})$ -bimodule (and morphisms thereof) respectively, where $\mathbb{Z}$ is the constant sheaf (Definition 5.2.29) of the integer ring $\mathbb{Z}$. In particular, the category of $(\mathbb{Z}, \mathbb{Z})$ -bimodules is equivalent to the category of sheaves of abelian groups on $\mathcal{C}$.

Lemma 5.3.10. Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13) on a locally small category $\mathcal{C}$ or be a U -site for some universe U (Definition E.0.1). Let R be a not-necessarily commutative unital ring (Definition C.0.1). Let $\mathcal{O}$ be the constant sheaf (Definition 5.2.29) valued in R , which exists by Theorem 5.2.27; it is also a sheaf of rings on $(\mathcal{C}, J)$ (Definition 5.1.32).

The following categories are equivalent:

- $\mathbf{Mod}(\mathcal{O})$ of $\mathcal{O}$ -modules (Definition 5.3.9) is equivalent (Definition 1.1.8)
- $\mathrm{Shv}(\mathcal{C}, R\text{-}\mathbf{mod})$ of sheaves (Definition 5.1.32) on $\mathcal{C}$ valued in $R\text{-}\mathbf{mod}$ (Definition 1.0.2).

In particular, the categories of modules over the constant ring $\mathbb{Z}$ on $\mathcal{C}$ is equivalent to the category $\mathrm{Shv}(\mathcal{C}, \mathbf{Ab})$ of sheaves on $\mathcal{C}$ valued in $\mathbf{Ab}$.

5.3.2. Sections.

Definition 5.3.11. Let $\mathcal{C}$ be a (large) category (Definition 1.0.1), and let $\mathcal{D}$ be a (large) category (Definition 1.0.1). Let $\mathcal{F} : \mathcal{C}^{op} \rightarrow \mathcal{D}$ be a presheaf valued in $\mathcal{D}$ (Definition 5.0.1).

1. For an object $U \in \mathcal{C}$, the *sections functor evaluated at U* is the functor

$$\Gamma(U, -) : \mathrm{PSh}(\mathcal{C}, \mathcal{D}) \rightarrow \mathcal{D}$$

defined by

$$\Gamma(U, \mathcal{F}) := \mathcal{F}(U),$$

i.e., the value of the presheaf $\mathcal{F}$ at the object U .

2. The *global sections of $\mathcal{F}$* is the object $\Gamma(\mathcal{F})$ of $\mathcal{D}$ defined as the limit (Definition 1.2.2)

$$\Gamma(\mathcal{F}) = \varprojlim_{U \in \mathcal{C}^{\text{op}}} \mathcal{F}(U)$$

assuming that such a limit exists, where the limit is taken over objects $U \in \mathcal{C}$ and the restriction morphisms $\mathcal{F}(V) \rightarrow \mathcal{F}(U)$ in $\mathcal{D}$ for morphisms $U \rightarrow V$ in $\mathcal{C}$.

If a final object (Definition 2.0.6) $*$ $\in \mathcal{C}$ exists, then $\Gamma(\mathcal{F})$ exists and coincides with $\Gamma(*, \mathcal{F}) = \mathcal{F}(*)$. The construction $\Gamma(\mathcal{F})$ is functorial; in particular, if $\Gamma(\mathcal{F})$ exists for all $\mathcal{F}$ in $\text{PSh}(\mathcal{C}, \mathcal{D})$, e.g. if limits of (Definition 1.2.2) diagrams in $\mathcal{D}$ indexed by $\mathcal{C}$ exist, then Γ is a functor

$$\Gamma : \text{PSh}(\mathcal{C}, \mathcal{D}) \rightarrow \mathcal{D}$$

called the *global sections functor on $\text{PSh}(\mathcal{C}, \mathcal{D})$* .

Proposition 5.3.12. Let $\mathcal{C}$ be an essentially small category (Definition 1.1.9) and let $\mathcal{A}$ be an abelian category (Definition 3.0.4).

1. Let $U \in \text{Ob}(\mathcal{C})$ be some fixed object. The sections functor (Definition 5.3.11)

$$\Gamma(U, -) : \text{PSh}(\mathcal{C}, \mathcal{A}) \rightarrow \mathcal{A}$$

is left exact (Definition 3.0.7). (♠ TODO: state that presheaves and sheaves valued in an abelian category form abelian categories)

2. Assume that $\Gamma(\mathcal{F})$ exists for all $\mathcal{F}$ in $\text{PSh}(\mathcal{C}, \mathcal{A})$ so that Γ is a functor

$$\Gamma : \text{PSh}(\mathcal{C}, \mathcal{A}) \rightarrow \mathcal{A}.$$

The functor Γ is left exact.

Proof. Recall that $\text{PSh}(\mathcal{C}, \mathcal{D})$ is an abelian category (Proposition 3.0.6) since $\mathcal{C}$ is essentially small. (♠ TODO: Talk about how limits are left exact and) $\square$

5.4. Sheaf cohomology.

Lemma 5.4.1 (cf. [Wei94, Porism 2.2.7, Comparison Theorem 2.2.6, Comparison Theorem 2.3.7]). Let $\mathcal{A}$ be an abelian category (Definition 3.0.4).

1. Let

$$\cdots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

be a chain complex (Definition 6.0.1) with P_i projective. For every left resolution (Definition 5.4.3) $Q_\bullet \rightarrow N$ of an object N , every map $M \rightarrow N$ lifts to a complex map (Definition 6.0.1) $P_\bullet \rightarrow Q_\bullet$ unique up to chain homotopy.

2. Let

$$0 \rightarrow M \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \cdots$$

be a (co)chain complex (Definition 6.0.1) with I^i injective. For every right resolution (Definition 5.4.3) $N \rightarrow Q^\bullet$ of an object N , every map $N \rightarrow M$ lifts to a complex map (Definition 6.0.1) $Q^\bullet \rightarrow I^\bullet$ unique up to chain homotopy.

Proof. 1. The map $P_0 \rightarrow M \rightarrow N$ lifts to a map $P_0 \rightarrow Q_0$ because P_0 is projective and $Q_0 \rightarrow N$ is an epimorphism. Inductively suppose that there are morphisms $P_i \rightarrow Q_i$ for $0 \leq i \leq n$, where $n \geq 0$ that make

$$\begin{array}{ccccccccc} P_n & \longrightarrow & P_{n-1} & \longrightarrow & \cdots & \longrightarrow & P_0 & \longrightarrow & M & \longrightarrow & 0 \\ \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & \\ Q_n & \longrightarrow & Q_{n-1} & \longrightarrow & \cdots & \longrightarrow & Q_0 & \longrightarrow & N & \longrightarrow & 0 \end{array}$$

into a commuting diagram are established. The morphism $Q_n \rightarrow Q_{n-1}$ (where we let $Q_{-1} = N$ and $P_{-1} = M$ here in case that $n = 0$) acts as 0 when restricted to $\mathfrak{J} := \text{im}(P_{n+1} \rightarrow P_n \rightarrow Q_n)$ because the composition

$$P_{n+1} \rightarrow P_n \rightarrow Q_n \rightarrow Q_{n-1}$$

equals the composition

$$P_{n+1} \rightarrow P_n \rightarrow P_{n-1} \rightarrow Q_{n-1}.$$

In other words, $\mathfrak{J}$ is a subobject (Definition 3.1.1) of $\ker(Q_n \rightarrow Q_{n-1})$ (Definition 1.3.8), which is isomorphic to $\text{im}(Q_{n+1} \rightarrow Q_n)$ by the acyclicity of the sequence of the Q_i 's. Therefore, we have a map $P_{n+1} \twoheadrightarrow \mathfrak{J} \hookrightarrow \text{im}(Q_{n+1} \rightarrow Q_n)$ along with an epimorphism $Q_{n+1} \twoheadrightarrow \text{im}(Q_{n+1} \rightarrow Q_n)$. Since P_{n+1} is projective, the former map lifts to a map $P_{n+1} \rightarrow Q_{n+1}$ in a way that is compatible with the latter, i.e. the following commutes:

$$\begin{array}{ccc} P_{n+1} & & \\ \downarrow \text{dotted} & \searrow & \\ Q_{n+1} & \longrightarrow & \text{im}(Q_{n+1} \rightarrow Q_n). \end{array}$$

By induction, this shows that $M \rightarrow N$ lifts to a morphism $P_\bullet \rightarrow Q_\bullet$ of complexes.

We show that the morphism of complexes is unique up to chain homotopy, i.e. if $f_1, f_2 : P_\bullet \rightarrow Q_\bullet$ are two morphisms of complexes, then $h := f_1 - f_2$ is null homotopic. We construct a chain contraction $\{s_n : P_n \rightarrow Q_{n+1}\}$ of h by induction on n . If $n < 0$, then set $s_n = 0$. If $n = 0$, note that the composition $P_0 \xrightarrow{h_0} Q_0 \rightarrow N$ equals the composition $P_0 \rightarrow M \xrightarrow{0} N$, so $\text{im}(h_0)$ is a subobject of $\ker(Q_0 \rightarrow N) \cong \text{im}(Q_1 \rightarrow Q_0)$. The projectivity of P_0 thus yields a lift $s_0 : P_0 \rightarrow Q_1$ such that h_0 equals the composition $P_0 \xrightarrow{s_0} Q_1 \xrightarrow{d} Q_0$:

$$\begin{array}{ccc} & P_0 & \\ & \downarrow h_0 & \\ Q_1 & \xrightarrow{d} & Q_0 \end{array} \quad \begin{array}{c} \nearrow s_0 \\ \text{dotted} \end{array}$$

Note moreover that $h_0 = ds_0 + s_{-1}d$ because $s_{-1} = 0$. Inductively suppose that we have maps s_i for $i \leq n$ such that $h_n = ds_n + s_{n-1}d$ or equivalently that $ds_n = h_n - s_{n-1}d$. Consider the map $h_{n+1} - s_n d : P_{n+1} \rightarrow Q_{n+1}$. Compute

$$d(h_{n+1} - s_n d) = dh_{n+1} - ds_n d = dh_{n+1} - (h_n - s_{n-1}d)d = (dh_{n+1} - h_n d) + s_{n-1}dd = 0$$

Therefore, $\text{im}(h_{n+1} - s_n d)$ is a subobject of $\ker(Q_{n+1} \rightarrow Q_n) \cong \text{im}(Q_{n+2} \rightarrow Q_{n+1})$, which is in turn a quotient of Q_{n+2} . Since P_{n+1} is projective, there is a morphism $s_{n+1} : P_{n+1} \rightarrow Q_{n+2}$ such that $ds_{n+1} = h_{n+1} - s_n d$.

$$\begin{array}{ccc} & & P_{n+1} \\ & \nearrow s_{n+1} & \downarrow h_n - s_{n-1}d = ds_n \\ Q_{n+2} & \xrightarrow{d} & \text{im}(Q_{n+2} \rightarrow Q_{n+1}) \cong \ker(Q_{n+1} \rightarrow Q_n) \end{array}$$

The s_n thus form a chain contraction as needed.

2. This is simply dual to the previous part.

□

Proposition 5.4.2 (cf.[Wei94, Lemma 2.4.1]). Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 3.0.3) between abelian categories (Definition 3.0.4). Let A be an object of $\mathcal{A}$.

1. Suppose that F is right exact (Definition 3.0.7), and suppose that a projective resolution (Definition 5.4.3)

$$\cdots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow A \rightarrow 0$$

of A exists in $\mathcal{A}$. Let

$$\cdots \rightarrow Q_2 \rightarrow Q_1 \rightarrow Q_0 \rightarrow A \rightarrow 0$$

be any projective resolution of A in $\mathcal{A}$. For all n , there are natural isomorphisms

$$H_n(F(P_\bullet)) \cong H_n(F(Q_\bullet)).$$

In other words, the left derived objects $L_n F(A)$ (Definition 5.4.8) is well defined.

2. Suppose that F is left exact (Definition 3.0.7), and suppose that a injective resolution (Definition 5.4.3)

$$0 \rightarrow A \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \cdots$$

of A exists in $\mathcal{A}$. Let

$$0 \rightarrow A \rightarrow Q^0 \rightarrow Q^1 \rightarrow Q^2 \rightarrow \cdots$$

be any injective resolution of A in $\mathcal{A}$. For all n , there are natural isomorphisms

$$H_n(F(I^\bullet)) \cong H_n(F(Q^\bullet)).$$

In other words, the right derived objects $R_n F(A)$ (Definition 5.4.8) is well defined.

Proof. 1. By Lemma 5.4.1, there is a lift $f : P_\bullet \rightarrow Q_\bullet$ of the identity map $A \rightarrow A$ unique up to chain homotopy. There are then induced natural maps $H_n(F(f)) : H_n(F(P_\bullet)) \rightarrow H_n(F(Q_\bullet))$ (??). There is also a lift $f' : Q_\bullet \rightarrow P_\bullet$ of the identity map $A \rightarrow A$ unique up to chain homotopy, and this also induces natural maps $H_n(F(f')) : H_n(F(Q_\bullet)) \rightarrow H_n(F(P_\bullet))$. The chain maps f and f' are in fact chain homotopy inverses because Lemma 5.4.1 also implies that any lifts $P_\bullet \rightarrow P_\bullet$ and $Q_\bullet \rightarrow Q_\bullet$ of the identity map $A \rightarrow A$ are chain homotopic to the identity chain maps. Therefore, $H_n(F(f))$ and $H_n(F(f'))$ are inverses of each other as morphisms in $\mathcal{B}$ by ??.

2. This is dual to the previous part.

□

Definition 5.4.3. Let $\mathcal{A}$ be an abelian category (Definition 3.0.4) and let $\mathcal{X}$ be a class of objects in $\mathcal{A}$. Let M be an object of $\mathcal{A}$.

1. A **right resolution of M** is a cochain complex (Definition 6.0.1) $I^\bullet$ with $I^i = 0$ for $i < 0$ and a map $M \rightarrow I^0$ such that the augmented complex

$$0 \rightarrow M \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \dots$$

is exact.

2. A **left resolution of M** is a chain complex (Definition 6.0.1) $P_\bullet$ with $P_i = 0$ for $i < 0$ and a map $P_0 \rightarrow M$ such that the augmented complex

$$\dots P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

is exact.

3. An **$\mathcal{X}$ -left resolution** of an object $M \in \mathcal{A}$ a left resolution (Definition 5.4.3) by objects of $\mathcal{X}$, i.e. an exact complex

$$\dots \rightarrow X_2 \rightarrow X_1 \rightarrow X_0 \rightarrow M \rightarrow 0$$

with each $X_i \in \mathcal{X}$.

4. An **$\mathcal{X}$ -right resolution** of an object $M \in \mathcal{A}$ a right resolution (Definition 5.4.3) by objects of $\mathcal{X}$, i.e. an exact complex

$$0 \rightarrow M \rightarrow X^0 \rightarrow X^1 \rightarrow X^2 \rightarrow \dots$$

with each $X_i \in \mathcal{X}$.

5. A **projective resolution of M** is a left resolution $P^\bullet$ for which the objects P^i are all projective.
6. An **injective resolution of M** is a right resolution $I^\bullet$ for which the objects I^i are all injective.

The notion of right/left resolutions above may be regarded as special cases of the notions of right/left resolutions of Definition 7.6.1 where $\mathcal{A}$ is embedded into the category $\mathbf{Ch}(\mathcal{A})$ by identifying an object A of $\mathcal{A}$ with the (co)chain complex concentrated in degree 0 associated to A — the right/left resolutions as defined above are then right/left resolutions of $A^\bullet$ by cochain/chain complexes $K^\bullet$ whose negative degree components are 0.

Definition 5.4.4. Let $\mathcal{A}$ be an abelian category (Definition 3.0.4).

1. $\mathcal{A}$ is said to **have enough injectives** if for every object A in $\mathcal{A}$, there is an monomorphism (Definition 1.3.7) $A \rightarrow I$ with I an injective object of $\mathcal{A}$.
2. $\mathcal{A}$ is said to **have enough projectives** if for every object A in $\mathcal{A}$, there is a epimorphism (Definition 1.3.7) $P \rightarrow A$ with P a projective object of $\mathcal{A}$.

Theorem 5.4.5 (e.g. see [Sta25, Tag 01DU]). For any site (Definition 5.1.13) $(\mathcal{C}, J)$ on an essentially small category (Definition 1.1.9) $\mathcal{C}$ and a sheaf of rings (Definition 5.1.32) $\mathcal{O}$ on $\mathcal{C}$, the category $\mathbf{Mod}(\mathcal{O})$ of $\mathcal{O}$ -modules (Definition 5.3.9) is an abelian category (Definition 3.0.4) that has enough injectives (Definition 5.4.4). In fact, there is a functorial injective embedding (♠ TODO: what does this mean?)

(♠ TODO: Apparently, if an abelian category satisfies AB5 and admits a generator (whatever that means), then it has enough injectives. Find a reference for this. Moreover, apparently, the category of sheaves of abelian groups on a general U -site satisfies these criteria and hence has enough injectives.)

(♠ TODO: According to Weibel exercise 2.3.7, if $\mathcal{A}$ is complete and has enough injectives and I is (essentially) small, then the functor category $\mathcal{A}^I$ has enough injectives)

Theorem 5.4.6. 1. Examples of abelian categories with enough injectives include:

- The category of abelian groups.
- Any category of modules over a ring; equivalently, any category of bimodules over two rings.
- The category of sheaves of abelian groups on a ringed space or more generally on an essentially small site.

2. Examples of abelian categories with enough projectives include:

- Any category of modules over a ring; equivalent, any category of bimodules over two rings.
- The category of finitely generated modules over a semisimple ring.

Lemma 5.4.7 (cf. [Wei94, Lemma 2.2.5, Lemma 2.3.6]). Let $\mathcal{A}$ be an abelian category (Definition 3.0.4) and let $\mathcal{X}$ be a class of objects in $\mathcal{A}$.

1. If $\mathcal{A}$ has enough objects of class $\mathcal{X}$ on the right, then for every object $A \in \mathcal{A}$ there exists an $\mathcal{X}$ -right resolution of A (Definition 5.4.3).
2. If $\mathcal{A}$ has enough objects of class $\mathcal{X}$ on the left, then for every object $A \in \mathcal{A}$ there exists an $\mathcal{X}$ -left resolution of A (Definition 5.4.3).

Note that this is a special case of ?? obtained by letting the complex $M^\bullet$ be the complex such that

$$M^i = \begin{cases} A & \text{if } i = 0 \\ 0 & \text{otherwise} \end{cases}.$$

In particular,

- If $\mathcal{A}$ has enough injective objects (Definition 5.4.4), then for every object $A \in \mathcal{A}$ there exists an injective resolution of A (Definition 5.4.3).
- If $\mathcal{A}$ has enough projective objects (Definition 5.4.4), then for every object $A \in \mathcal{A}$ there exists a projective resolution of A (Definition 5.4.3).
- If $F : \mathcal{A} \rightarrow \mathcal{B}$ is a left (resp. right) exact functor (Definition 3.0.7) between abelian categories and $\mathcal{A}$ has enough F -acyclic objects on the right (resp. left), then for every object $A \in \mathcal{A}$, there exists an right (resp. left) F -acyclic resolution of A .

Proof. 1. Let $A \in \mathcal{A}$ be an object. Since $\mathcal{A}$ has enough objects of class $\mathcal{X}$ of the right, there is an object X_0 of $\mathcal{X}$ and a monomorphism $\varepsilon_0 : A \rightarrow X_0$. Let $A_0 = \text{coker } \varepsilon_0$ (Definition 1.3.8). Inductively, given an object A_{n-1} of $\mathcal{A}$, choose an object X_n of $\mathcal{X}$ and a monomorphism $\varepsilon_n : A_{n-1} \hookrightarrow X_n$. Let $A_n = \text{coker } \varepsilon_n$. In particular, there is a

surjection $X_n \twoheadrightarrow A_n$. Let d_n be the composition

$$X_{n-1} \twoheadrightarrow A_{n-1} \xrightarrow{\varepsilon_n} X_n.$$

The chain complex

$$0 \rightarrow A \xrightarrow{\varepsilon_0} X_0 \xrightarrow{d_0} X_1 \xrightarrow{d_1} \dots$$

is thus an $\mathcal{X}$ -right resolution of A .

2. This is simply the dual statement of the next statement.

□

Definition 5.4.8. (♠ TODO: I think that the definition of derived categories might be doable for more general kinds of resolutions? Perhaps it is that if I have a right exact functor F , then $L^i F$ can be computed with resolutions of F -acyclic objects?) (♠ TODO: Apparently, left/right derived functors may be defined for functors that are additive and preserve finite coproducts, and not necessarily right/left exact; the exactness condition ensures that the zeroth derived functor agrees with F .) Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories (Definition 3.0.4), and let

$$F : \mathcal{A} \rightarrow \mathcal{B}$$

be an additive functor (Definition 3.0.3).

1. Suppose that the functor F is right exact (Definition 3.0.7) and suppose that $A \in \mathcal{A}$ is an object for which a projective resolution (Definition 5.4.3)

$$\dots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow A \rightarrow 0$$

exists in $\mathcal{A}$. We define the *left derived object* $L_n F A \in \mathcal{B}$ by applying F to obtain a complex

$$\dots \rightarrow F(P_2) \rightarrow F(P_1) \rightarrow F(P_0) \rightarrow 0$$

and letting $L_n F(A)$ be the n -th homology object of this complex in $\mathcal{B}$:

$$L_n F(A) := H_n(F(P_\bullet)).$$

The object $L_n F(A)$ is independent of the choice of projective resolution up to natural isomorphism (Proposition 5.4.2).

By convention, set $L_n F = 0$ for $n < 0$.

The *higher left derived objects* refer to the object $L_n F(A)$ for $n > 0$.

2. Suppose that the functor F is right exact (Definition 3.0.7) and that $\mathcal{A}$ has enough projectives (Definition 5.4.4). The *left derived functors* refer to the family of functors

$$L_n F : \mathcal{A} \rightarrow \mathcal{B}, \quad A \mapsto L_n F(A).$$

The *higher left derived functors* refer to the functors $L_n F$ for $n > 0$.

3. Suppose that the functor F is left exact (Definition 3.0.7) and suppose that $A \in \mathcal{A}$ is an object for which a injective resolution (Definition 5.4.3)

$$0 \rightarrow A \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \dots$$

exists in $\mathcal{A}$. We define the *right derived object* $R_n F A \in \mathcal{B}$, also often denoted by $R^n F A$, by applying F to obtain a complex

$$0 \rightarrow F(I^0) \rightarrow F(I^1) \rightarrow F(I^2) \rightarrow \dots$$

and letting $R_n F(A)$ be the n -th cohomology object of this complex in $\mathcal{B}$:

$$R_n F(A) := H^n(F(I_\bullet)).$$

The object $R_n F(A)$ is independent of the choice of injective resolution up to natural isomorphism (Proposition 5.4.2).

By convention, set $R_n F = 0$ for $n < 0$.

The *higher right derived objects* refer to the object $R_n F(A)$ for $n > 0$.

4. Suppose that the functor F is left exact (Definition 3.0.7) and that $\mathcal{A}$ has enough injectives (Definition 5.4.4). The *right derived functors* refer to the family of functors

$$R_n F : \mathcal{A} \rightarrow \mathcal{B}, \quad A \mapsto R_n F(A).$$

The right derived functors are also often denoted by $R^n F$. The *higher right derived functors* refer to the functors $R_n F$ for $n > 0$.

Definition 5.4.9. (♠ TODO: there are places where sites and sheaves of rings on them are used, but it would be better to just have them be ringed sites.)

A *ringed site* is a site (Definition 5.1.13) $(\mathcal{C}, J)$ with a small topological generating family (Definition 5.1.13) equipped with a sheaf (Definition 5.1.32) of (not necessarily commutative) rings $\mathcal{O}$. If the Grothendieck topology J is clear in context, one may even write that $(\mathcal{C}, \mathcal{O})$ is a ringed site.

Definition 5.4.10. Let $((\mathcal{C}, J), \mathcal{O}_{\mathcal{C}})$ and $((\mathcal{D}, K), \mathcal{O}_{\mathcal{D}})$ be ringed sites (Definition 5.4.9).

A *morphism of ringed sites* $f : ((\mathcal{C}, J), \mathcal{O}_{\mathcal{C}}) \rightarrow ((\mathcal{D}, K), \mathcal{O}_{\mathcal{D}})$ is a pair $(f, f^\#)$ where:

1. $f : (\mathcal{C}, J) \rightarrow (\mathcal{D}, K)$ is a morphism of sites (Definition 5.2.5).
2. $f^\# : \mathcal{O}_{\mathcal{D}} \rightarrow f_* \mathcal{O}_{\mathcal{C}}$ (Definition 5.3.2) is a morphism of sheaves of rings (Definition 5.1.32) on the site $(\mathcal{D}, K)$.

Equivalently, the morphism $f^\#$ a morphism can be described by the adjoint morphism (Theorem 5.3.4) $f^\# : f^{-1} \mathcal{O}_{\mathcal{D}} \rightarrow \mathcal{O}_{\mathcal{C}}$ of sheaves of rings on $(\mathcal{C}, J)$, where f^{-1} is the inverse image functor (Definition 5.3.3) of sheaves.

Definition 5.4.11. A *ringed topos* is a pair $(E, \mathcal{O}_E)$ consisting of a topos (Definition 5.1.55) E and a ring object (Definition 1.3.20) $\mathcal{O}_E$ in E (often called the *structure sheaf*).

Definition 5.4.12. Let $(E, \mathcal{O}_E)$ and $(E', \mathcal{O}_{E'})$ be ringed topoi (Definition 5.4.11). A *morphism of ringed topoi* $f : (E, \mathcal{O}_E) \rightarrow (E', \mathcal{O}_{E'})$ is a pair $(f, f^\#)$ where:

1. $f : E \rightarrow E'$ is a geometric morphism of topoi (Definition 5.1.56).
2. $f^\# : \mathcal{O}_{E'} \rightarrow f_* \mathcal{O}_E$ is a morphism of ring objects (Definition 1.3.20) in E' , where f_* is the direct image functor (Definition 5.1.56) of f .

Equivalently, the second datum can be given by the adjoint morphism $f^\# : f^* \mathcal{O}_{E'} \rightarrow \mathcal{O}_E$ of ring objects in E , where f^* is the inverse image functor (Definition 5.1.56) of f .

Definition 5.4.13. (♠ TODO: generalize to more general abelian value categories)

Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13) on a locally small category (Definition 1.0.9) or a U -site for some universe (Definition E.0.1) U . Let $\mathcal{O}$ be a sheaf of rings (Definition 5.1.32)

on $\mathcal{C}$, so that $(\mathcal{C}, J, \mathcal{O})$ is a ringed site (Definition 5.4.9). Recall that the category $\mathbf{Mod}(\mathcal{O})$ (Definition 5.3.9) of $\mathcal{O}$ -modules is abelian and has enough injectives (Definition 5.4.4) (Theorem 5.4.5).

Assume that global sections objects $\Gamma(\mathcal{G})$ (Definition 5.3.11) exist for all objects $\mathcal{G}$ of $\mathbf{Sh}(\mathcal{C}, \mathbf{Ab})$ ³ so that Γ is a functor

$$\mathbf{Sh}(\mathcal{C}, \mathbf{Ab}) \rightarrow \mathbf{Ab},$$

which is a left exact functor (Definition 3.0.7) (Proposition 5.3.12). Note that Γ restricts to a left exact functor

$$\mathbf{Mod}(\mathcal{O}) \rightarrow \mathbf{Ab}.$$

If $\mathcal{C}$ has a final object (Definition 2.0.6) $*$ as well, then recall that $\Gamma(\mathcal{F}) = \mathcal{F}(*)$.

Let $\mathcal{F}$ be an object of $\mathbf{Mod}(\mathcal{O})$.

1. For each integer $n \geq 0$, the *n -th (abelian) (global) sheaf cohomology group of $\mathcal{F}$* is

$$H^n(\mathcal{C}, J; \mathcal{F}) := R^n\Gamma(\mathcal{F}),$$

where $R^n\Gamma$ is the n -th right derived functor (Definition 5.4.8) of the global sections functor Γ (Definition 5.3.11).

In particular, each H^n is a functor

$$H^n : \mathbf{Mod}(\mathcal{O}) \rightarrow \mathbf{Ab}.$$

2. Given an object $U \in \mathcal{C}$ and for each integer $n \geq 0$, the *n -th (abelian) sheaf cohomology group of $\mathcal{F}$ of sections at U* is

$$H^n(U, \mathcal{F}) := (R^n\Gamma(U, -))(\mathcal{F}),$$

where $R^n\Gamma(U, -)$ is the n -th right derived functor (Definition 5.4.8) of the sections functor $\Gamma(U, -)$ evaluated at U (Definition 5.3.11).

In particular, $H^n(U, \mathcal{F})$ can be regarded as the n th global sheaf cohomology group of the restriction $\mathcal{F}|_U$ (Definition 5.2.30) of $\mathcal{F}$ to U .

5.4.1. Čech cohomology of a sheaf of groups.

Definition 5.4.14 (Čech Cohomology of a Sheaf of Groups). Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13), and $\mathcal{G}$ a sheaf (Definition 5.1.32) of groups (not necessarily abelian) on it. (♠
TODO: define cech cohomology of the covering and cech cohomology)

1. For an object $U \in \mathcal{C}$, and a covering family $\mathfrak{U} = \{U_i \rightarrow U\}_{i \in I} \in J(U)$, the *Čech complex $C^\bullet(\mathfrak{U}, \mathcal{G})$* is defined by:

$$C^n(\mathfrak{U}, \mathcal{G}) := \prod_{(i_0, \dots, i_n) \in I^{n+1}} \mathcal{G}(U_{i_0} \times_U \cdots \times_U U_{i_n}),$$

with face maps defined by restriction and group operations giving a cosimplicial object.

³for example, this occurs when $\mathcal{C}$ is essentially small (Definition 1.1.9)

5.4.2. Sheaf cohomology of a sheaf of groups.

Definition 5.4.15 (Torsors under a Sheaf of Groups). (♠ TODO: group object, action by a group object) Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13) and $\mathcal{G}$ a sheaf of groups on it (Definition 5.1.32). A $\mathcal{G}$ -torsor (or synonymously a *principal $\mathcal{G}$ -bundle* or *principal $\mathcal{G}$ -homogeneous space*, etc.) is a sheaf $\mathcal{P}$ of sets on $(\mathcal{C}, J)$ equipped with a (right) action of $\mathcal{G}U$ such that (♠ TODO: This needs some genuine fixing to be definable on a general site, not just one given by a pretopology) (Definition 5.1.29)

1. For each object U of $\mathcal{C}$, the action of the group $G(U)$ on $\mathcal{P}(U)$ is free and transitive whenever $\mathcal{P}(U)$ is nonempty.
2. $\mathcal{P}$ is locally trivial. More explicitly, for each object U of $\mathcal{C}$, there exists some covering $\{V_i \rightarrow U\}_{i \in I}$ in J such that there is an isomorphism

$$\mathcal{P}|_{U_i} \cong \mathcal{G}|_{U_i}$$

(Definition 5.2.30) as a sheaves with $\mathcal{G}|_{U_i}$ actions for each i .

A $\mathcal{G}$ -torsor is called *trivial on U* if it is isomorphic (as a sheaf with a right action of $\mathcal{G}$) to $\mathcal{G}|_U$ acting on itself by right multiplication, or equivalently, if $\mathcal{G}(U) \neq \emptyset$. In particular, any $\mathcal{G}$ -torsor is *locally* trivial.

Definition 5.4.16 (Nonabelian Sheaf Cohomology: H^0 and H^1). Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13) and $\mathcal{G}$ a sheaf (Definition 5.1.32) of groups on $(\mathcal{C}, J)$. For $U \in \mathcal{C}$, we define the *0th and 1st (nonabelian) sheaf cohomology sets* as follows:

- The zeroth sheaf cohomology group $H^0(\mathcal{C}, J, \mathcal{G}) := \Gamma(\mathcal{G})$ is the group of global sections (Definition 5.3.11) of $\mathcal{G}$, assuming that it exists (which is always the case when $\mathcal{C}$ is essentially small (Definition 1.1.9) for example since the category of groups is closed under small projective limits (♠ TODO:)).
- The first sheaf cohomology set $H^1(\mathcal{C}, J, \mathcal{G})$ is the pointed set of isomorphism classes of $\mathcal{G}$ -torsors (Definition 5.4.15) on the site $(\mathcal{C}, J)$.

Proposition 5.4.17. See [Sta25, Tag 03AJ] for a statement⁴ Let $(\mathcal{C}, J)$ be a site (Definition 5.1.13). Assume that global sections objects $\Gamma(\mathcal{G})$ (Definition 5.3.11) exist for all objects $\mathcal{G}$ of $\mathbf{Sh}(\mathcal{C}, \mathbf{Ab})$ ⁵ so that Γ is a functor

$$\mathbf{Sh}(\mathcal{C}, \mathbf{Ab}) \rightarrow \mathbf{Ab}.$$

Let $\mathcal{G}$ be a sheaf (Definition 5.1.32) of abelian groups on $(\mathcal{C}, J)$. For $i = 0, 1$, there is a canonical bijection between the set of isomorphism classes of $\mathcal{G}$ -torsors (Definition 5.4.15) and the (abelian) sheaf cohomology group $H^i(\mathcal{C}, J; \mathcal{G})$ (Definition 5.4.13). In other words, for $i = 0, 1$, there is a canonical bijection between the i th nonabelian sheaf cohomology (Definition 5.4.16) and the i th abelian sheaf cohomology (Definition 5.4.13) of $\mathcal{G}$.

Proof. In the case of $i = 0$, both are identifiable with the groups of global sections. In the case of $i = 1$, see [Sta25, Tag 03AJ]. □

⁴While the Stacks Project defines a site as a category equipped with a Grothendieck *pretopology*, its proof of this statement should be applicable for more general sites

⁵for example, this occurs when $\mathcal{C}$ is essentially small (Definition 1.1.9)

6. HOMOLOGICAL ALGEBRA

Definition 6.0.1 (Chain complex in a preadditive category). Let $\mathcal{A}$ be a preadditive category (Definition 3.0.1) and let I be a totally ordered set (typically $\mathbb{Z}$, but $I \subseteq \mathbb{Z}$ is also allowed).

1. A **chain complex** (or **homological (chain) complex**) $(K_\bullet, d_\bullet)$ in $\mathcal{A}$ indexed by I is the homological convention for sequences with decreasing degrees. It consists of:
 - Objects $\{K_i\}_{i \in I}$ in $\mathcal{A}$, called the **terms in degree i** ,
 - Morphisms $d_i : K_i \rightarrow K_{i-1}$ in $\mathcal{A}$, called the **boundary maps** or **differentials in degree i** ,

such that for every $i \in I$, $d_{i-1} \circ d_i = 0$. That is,

$$K_\bullet : \cdots \xrightarrow{d_{i+1}} K_i \xrightarrow{d_i} K_{i-1} \xrightarrow{d_{i-1}} K_{i-2} \rightarrow \cdots$$

with $d_{i-1}d_i = 0$ for all i . We typically use the notation $K_\bullet = (K_i, d_i)_{i \in I}$.

2. Dually, a **cochain complex** (or **cohomological complex**) $(K^\bullet, d^\bullet)$ in $\mathcal{A}$ follows the cohomological convention with increasing degrees. It consists of objects $\{K^i\}_{i \in I}$ and **coboundary maps** $d^i : K^i \rightarrow K^{i+1}$ such that $d^{i+1} \circ d^i = 0$:

$$K^\bullet : \cdots \xrightarrow{d^{i-1}} K^i \xrightarrow{d^i} K^{i+1} \xrightarrow{d^{i+1}} K^{i+2} \rightarrow \cdots$$

We typically use the notation $K^\bullet = (K^i, d^i)_{i \in I}$.

3. Let $K_\bullet = (K_i, d_i^K)$ and $L_\bullet = (L_i, d_i^L)$ be chain complexes (Definition 6.0.1) in $\mathcal{A}$ indexed by the same set I . A **morphism of chain complexes** (or **chain map**)

$$f_\bullet : K_\bullet \rightarrow L_\bullet$$

consists of morphisms $f_i : K_i \rightarrow L_i$ for all $i \in I$, such that for every $i \in I$, the following diagram commutes:

$$\begin{array}{ccc} K_i & \xrightarrow{d_i^K} & K_{i-1} \\ \downarrow f_i & & \downarrow f_{i-1} \\ L_i & \xrightarrow{d_i^L} & L_{i-1} \end{array}$$

i.e., $d_i^L \circ f_i = f_{i-1} \circ d_i^K$.

A **morphism of cochain complexes** $f^\bullet : K^\bullet \rightarrow L^\bullet$ is defined similarly, satisfying the commutativity condition $d_L^i \circ f^i = f^{i+1} \circ d_K^i$.

The collection of these objects and morphisms forms a category. Notation for these categories is as follows:

- $\mathbf{Ch}(\mathcal{A})$ or $\mathbf{Ch}(\mathcal{A})$ is often used as a general term.
- To be explicit about the indexing convention, one uses $\mathbf{Ch}_\bullet(\mathcal{A})$ for chain complexes and $\mathbf{Ch}^\bullet(\mathcal{A})$ (or sometimes $\mathbf{CoCh}(\mathcal{A})$) for cochain complexes.
- The set of chain maps between two complexes is denoted by $\mathbf{Hom}_{\mathbf{Ch}(\mathcal{A})}(K_\bullet, L_\bullet)$; it is an abelian group under pointwise addition $(f + g)_i = f_i + g_i$.

Notations used to write chain complexes are *homological conventions* whereas notations used to write cochain complexes are *cohomological conventions*. In homological conventions, indices are written as subscripts and decrease when “following the differential maps”. In cohomological conventions, indices are written as superscripts and increase when “following the differential maps”. The theories of chain complexes and of cochain complexes are not genuinely different, however, as the conventions are essentially notational — starting from a chain complex $(K_\bullet, d_\bullet)$, one can obtain a cochain complex $(K^\bullet, d^\bullet)$ simply by defining $K^i = K_{-i}$ and d^i as d_{-i} and vice versa. As such, cochain complexes may be simply called **chain complexes** when careful distinctions between the conventions are unnecessary. Homological and cohomological notational conventions persist throughout homological algebra, see e.g. spectral sequences, double complexes, etc, but again the distinction between such conventions are merely notational.

If k is a commutative ring (Definition C.0.3) such that $\text{Hom}_{\mathcal{A}}(X, Y)$ is enriched in (Definition 4.0.1) the category of k -modules (Definition C.0.8), then $\text{Ch}(\mathcal{A})$ can be equipped with (Definition 11.4.3) the structure of a dg-category over k (Definition 11.4.2).

Definition 6.0.2. A **quiver** is a quadruple $Q = (Q_0, Q_1, s, t)$, where:

- Q_0 is a collection of **vertices**.
- Q_1 is a collection of **arrows**.
- $s, t : Q_1 \rightarrow Q_0$ are functions assigning to each arrow $\alpha \in Q_1$ its **source** $s(\alpha)$ and its **target** $t(\alpha)$.

Definition 6.0.3. Let Q be a quiver (Definition 6.0.2). The **path category generated by Q** , denoted $\mathcal{F}(Q)$, is the category (Definition 1.0.1) defined as follows:

- The objects of $\mathcal{F}(Q)$ are the vertices Q_0 .
- For any two objects $x, y \in Q_0$, the set of morphisms $\text{Hom}_{\mathcal{F}(Q)}(x, y)$ consists of all paths from x to y — A **path of length $n \geq 1$ from x to y** is a sequence of arrows $\alpha_n \dots \alpha_1$ such that $s(\alpha_1) = x$, $t(\alpha_n) = y$, and $s(\alpha_{i+1}) = t(\alpha_i)$ for all $1 \leq i < n$. Additionally, for each vertex x , there is a path e_x of length 0, which serves as the identity morphism.
- Composition of morphisms is defined by the concatenation of paths.

Definition 6.0.4. Let Q be a quiver (Definition 6.0.2) whose collection of arrows is small.

The **preadditive category generated by Q** , denoted $\mathbb{Z}Q$, is the preadditive category (Definition 3.0.1), i.e. the category enriched over (Definition 4.0.1) the category of abelian groups (Definition 1.0.4) defined as follows:

- The objects of $\mathbb{Z}Q$ are the vertices Q_0 .
- For any objects $x, y \in Q_0$, the morphism set $\text{Hom}_{\mathbb{Z}Q}(x, y)$ is the free abelian group generated by the set of all paths from x to y in Q .
- Composition is the unique bilinear extension of the path concatenation in $\mathcal{F}(Q)$. That is, for paths u, v, w where concatenation is defined, composition satisfies $(u + v) \circ w = u \circ w + v \circ w$ and $w \circ (u + v) = w \circ u + w \circ v$.

Definition 6.0.5. Let Q_{chain} be the quiver (Definition 6.0.2) with vertex set $Q_0 = \mathbb{Z}$ and arrow set $Q_1 = \{d_n : n \rightarrow n-1 \mid n \in \mathbb{Z}\}$. (♠ **TODO: quotient of a category**) The **walking chain**

complex category, denoted $\mathcal{I}$, is the quotient of the preadditive category (Definition 3.0.1) $\mathbb{Z}Q_{\text{chain}}$ by the ideal generated by the relations $d_{n-1} \circ d_n = 0$ for all $n \in \mathbb{Z}$. Explicitly:

- Objects are the integers $\mathbb{Z}$.
- Morphisms are $\mathbb{Z}$ -linear combinations of paths, subject to the relation that any path containing a subsegment $d_{n-1}d_n$ is identified with the zero morphism.

Definition 6.0.6. Let $\mathcal{A}$ and $\mathcal{B}$ be preadditive categories (Definition 3.0.1) (categories enriched over (Definition 4.0.1) the category of abelian groups (Definition 1.0.4)). The **additive functor category** $\text{Add}(\mathcal{A}, \mathcal{B})$ is the functor category where:

- Objects are additive functors (Definition 3.0.3) $F : \mathcal{A} \rightarrow \mathcal{B}$, i.e. is a functor (Definition 1.1.1) such that for any $x, y \in \text{Ob}(\mathcal{A})$, the map $F : \text{Hom}_{\mathcal{A}}(x, y) \rightarrow \text{Hom}_{\mathcal{B}}(F(x), F(y))$ is a homomorphism (Definition C.0.7) of abelian groups (Definition 1.0.4).
- Morphisms are natural transformations (Definition 1.1.3) between additive functors (Definition 3.0.3).

It is itself preadditive (Lemma 6.0.8).

Proposition 6.0.7. Let $\mathcal{B}$ be a preadditive category (Definition 3.0.1). The category $\text{Ch}(\mathcal{B})$ of chain complexes (Definition 6.0.1) in $\mathcal{B}$ is isomorphic to the category $\text{Add}(\mathcal{I}, \mathcal{B})$ (Definition 6.0.6) where $\mathcal{I}$ is the walking chain complex category (Definition 6.0.5).

An additive functor $F : \mathcal{I} \rightarrow \mathcal{B}$ corresponds to the chain complex defined by $C_n = F(n)$ and differentials $\partial_n = F(d_n)$.

Lemma 6.0.8. Let $\mathcal{A}, \mathcal{B}$ be preadditive categories (Definition 3.0.1) with $\mathcal{A}$ small (Definition 1.0.9).

1. The additive functor category (Definition 6.0.6) $\text{Add}(\mathcal{A}, \mathcal{B})$ is preadditive. If $\mathcal{B}$ is additionally additive (Definition 3.0.1)/abelian (Definition 3.0.4), then so is $\text{Add}(\mathcal{A}, \mathcal{B})$.
2. If $\mathcal{B}$ is an abelian category with property ABn for $n = 3, 4, 5, 6$ or ABn^* for $n = 3, 4, 5$ (Definition 3.0.5), then $\text{Add}(\mathcal{A}, \mathcal{B})$ possesses the same property.

Proof. 1. Let F, G be objects of $\text{Add}(\mathcal{A}, \mathcal{B})$. We show that $\text{Hom}_{\text{Add}(\mathcal{A}, \mathcal{B})}(F, G)$ is an abelian group. By definition, $\text{Hom}_{\text{Add}(\mathcal{A}, \mathcal{B})}(F, G)$ consists of the natural transformations (Definition 1.1.3) $F \Rightarrow G$ such that for all objects X, Y of $\mathcal{A}$. Note that the collection of natural transformations $F \Rightarrow G$ forms a set (Lemma 5.0.5), so $\text{Hom}_{\text{Add}(\mathcal{A}, \mathcal{B})}(F, G)$, which is a subcollection of this collection, is itself a set. Say that $\eta_1, \eta_2 : F \Rightarrow G$ are two such natural transformations. Define $\eta_1 + \eta_2 : F \Rightarrow G$ as follows — for an object X of $\mathcal{A}$, let

$$(\eta_1 + \eta_2)_X : F(X) \rightarrow G(X)$$

be the morphism $(\eta_1)_X + (\eta_2)_X$ from $F(X)$ to $G(X)$ in $\mathcal{B}$. Since $\mathcal{B}$ is itself a pre-additive category and hence $\text{Hom}_{\mathcal{B}}(F(X), G(X))$ is an abelian group, this is well defined. We show that this is indeed a natural transformation $F \rightarrow G$, i.e. the diagram

$$\begin{array}{ccc}
F(X) & \xrightarrow{F(f)} & F(Y) \\
(\eta_1 + \eta_2)_X \downarrow & & \downarrow (\eta_1 + \eta_2)_Y \\
G(X) & \xrightarrow{G(f)} & G(Y)
\end{array}$$

commutes. For all morphisms $f : X \rightarrow Y$ in $\mathcal{A}$, we have

$$(\eta_1 + \eta_2)_Y \circ F(f) = ((\eta_1)_Y + (\eta_2)_Y) \circ F(f),$$

which is in turn equal to

$$(J) \quad (\eta_1)_Y \circ F(f) + (\eta_2)_Y \circ F(f)$$

because the composition map

$$\mathrm{Hom}_{\mathcal{B}}(F(Y), G(Y)) \times \mathrm{Hom}_{\mathcal{B}}(F(X), F(Y)) \rightarrow \mathrm{Hom}_{\mathcal{B}}(F(X), G(Y))$$

is bilinear (Definition C.0.9). (J) thus equals

$$G(f) \circ (\eta_1)_X \circ F(f) + G(f) \circ (\eta_2)_X = G(f) \circ ((\eta_1)_X + (\eta_2)_X) = G(f) \circ (\eta_1 + \eta_2)_X$$

as desired.

Moreover, the addition that we defined on $\mathrm{Hom}_{\mathrm{Add}(\mathcal{A}, \mathcal{B})}(F, G)$ is associative and commutative because addition in $\mathrm{Hom}_{\mathcal{B}}(F(X), G(X))$ is. There is also a natural transformation $0 : F \Rightarrow G$ given on each object X of $\mathcal{A}$ by letting $0_X : F(X) \rightarrow G(X)$ be the element 0 in the abelian group $\mathrm{Hom}_{\mathcal{B}}(F(X), G(X))$. Lastly, for all $\eta \in \mathrm{Hom}_{\mathrm{Add}(\mathcal{A}, \mathcal{B})}(F, G)$

We show that composition of morphisms in $\mathrm{Add}(\mathcal{A}, \mathcal{B})$ is bilinear, i.e. that the composition maps

$$\mathrm{Hom}_{\mathrm{Add}(\mathcal{A}, \mathcal{B})}(G, H) \times \mathrm{Hom}_{\mathrm{Add}(\mathcal{A}, \mathcal{B})}(F, G) \rightarrow \mathrm{Hom}_{\mathrm{Add}(\mathcal{A}, \mathcal{B})}(F, H)$$

are bilinear for any objects $F, G, H : \mathcal{A} \rightarrow \mathcal{B}$ of $\mathrm{Add}(\mathcal{A}, \mathcal{B})$. Given $\eta_1, \eta_2 \in \mathrm{Hom}_{\mathrm{Add}(\mathcal{A}, \mathcal{B})}(F, G)$ and $\mu \in \mathrm{Hom}_{\mathrm{Add}(\mathcal{A}, \mathcal{B})}(G, H)$, we have that $(\mu \circ (\eta_1 + \eta_2))_X$ is given by assigning $X \in \mathcal{A}$ to the morphism

$$\mu_X \circ (\eta_1 + \eta_2)_X : F(X) \rightarrow G(X) \rightarrow H(X)$$

By the construction of the group structure on $\mathrm{Hom}_{\mathrm{Add}(\mathcal{A}, \mathcal{B})}(F, G)$,

$$\mu_X \circ (\eta_1 + \eta_2)_X = \mu_X \circ ((\eta_1)_X + (\eta_2)_X).$$

Since the composition map

$$\mathrm{Hom}_{\mathcal{B}}(G(X), H(X)) \times \mathrm{Hom}_{\mathcal{B}}(F(X), G(X)) \rightarrow \mathrm{Hom}_{\mathcal{B}}(F(X), H(X))$$

is bilinear, this equals

$$\mu_X \circ (\eta_1)_X + \mu_X \circ (\eta_2)_X,$$

which is the morphism that the natural transformation $(\mu \circ \eta_1 + \mu \circ \eta_2)$ assigns X to. Thus, $\mu \circ (\eta_1 + \eta_2) = \mu \circ \eta_1 + \mu \circ \eta_2$, so the composition maps are indeed bilinear as desired.

We show that there is a zero object of $\mathrm{Add}(\mathcal{A}, \mathcal{B})$ and it is the zero functor $0 : \mathcal{A} \rightarrow \mathcal{B}$ sending all objects of $\mathcal{A}$ to the zero object $0_{\mathcal{B}}$ of $\mathcal{B}$ and hence all morphisms of $\mathcal{B}$ to the unique (zero) morphism $0_{\mathcal{B}} \rightarrow 0_{\mathcal{B}}$. It is indeed an object of $\mathrm{Add}(\mathcal{A}, \mathcal{B})$ because for any objects X and Y of $\mathcal{A}$, the map $0 : \mathrm{Hom}_{\mathcal{A}}(X, Y) \rightarrow \mathrm{Hom}_{\mathcal{B}}(0(X), 0(Y)) = \mathrm{Hom}_{\mathcal{B}}(0_{\mathcal{B}}, 0_{\mathcal{B}}) = 0$ (Lemma 3.0.2) is the zero group homomorphism. We now show

that it is indeed the zero object, i.e. is initial and final (Definition 2.0.6). Given any object F of $\text{Add}(\mathcal{A}, \mathcal{B})$, there are natural transformations $0 \Rightarrow F$ and $F \Rightarrow 0$ respectively by assigning an object X of $\mathcal{A}$ to the unique morphisms $0_{\mathcal{B}} \rightarrow F(X)$ and $F(X) \rightarrow 0_{\mathcal{B}}$. In fact, such natural transformations are unique because morphisms to and from $0_{\mathcal{B}}$ in $\mathcal{B}$ are unique. Therefore, $0 : \mathcal{B} \rightarrow \mathcal{B}$ is indeed the zero object of $\text{Add}(\mathcal{A}, \mathcal{B})$.

We show that $\text{Add}(\mathcal{A}, \mathcal{B})$ is additive whenever $\mathcal{B}$ is additive. (♠ TODO:)

We show that $\text{Add}(\mathcal{A}, \mathcal{B})$ is abelian whenever $\mathcal{B}$ is abelian. (♠ TODO:)

2.

□

Proposition 6.0.9. Let $\mathcal{A}$ be an additive category (Definition 3.0.1).

1. The category $\text{Ch}(\mathcal{A})$ (Definition 6.0.1) of chain complexes is itself an additive category.
2. If $\mathcal{A}$ is an abelian category (Definition 3.0.4), then $\text{Ch}(\mathcal{A})$ is an abelian category.
3. If $\mathcal{A}$ is an abelian category (Definition 3.0.4) satisfying Grothendieck's axiom $\text{AB}n$ (resp. $\text{AB}n^*$) (Definition 3.0.5) for $n \in \{3, 4, 5, 6\}$, then $\text{Ch}(\mathcal{A})$ also satisfies $\text{AB}n$ (resp. $\text{AB}n^*$). If $\mathcal{A}$ is a Grothendieck abelian category (Definition 3.0.5), then so is $\text{Ch}(\mathcal{A})$.

Proof. Combine Proposition 6.0.7 and Lemma 6.0.8. □

7. DERIVED CATEGORIES

7.1. Homotopy category of chain complexes of an additive category. We will define homotopy categories and derived categories using the cohomological convention (Definition 6.0.1)

Definition 7.1.1 (Homotopy category $K(\mathcal{A})$). Let $\mathcal{A}$ be an additive category (Definition 3.0.1). (♠ TODO: describe the distinguished triangles of this category)

1. The *homotopy category* $K(\mathcal{A})$ has as objects the (cochain) complexes over $\mathcal{A}$, and as morphisms the cochain maps (Definition 6.0.1) modulo cochain homotopy equivalence. More explicitly,

$$\text{Hom}_{K(\mathcal{A})}(C^\bullet, D^\bullet) := \frac{\{\text{cochain maps } C^\bullet \rightarrow D^\bullet\}}{\{\text{cochain homotopies}\}},$$

where morphisms differing by a cochain homotopy are identified.

2. Let

$$F : \mathcal{A} \rightarrow \mathcal{B}$$

be an additive functor (Definition 3.0.3) of additive categories.

The functor F naturally induces a functor between the homotopy categories of complexes,

$$K(F) : K(\mathcal{A}) \rightarrow K(\mathcal{B}),$$

by applying F degreewise to complexes and cochain maps. That is, for a complex $(C^\bullet, d^\bullet)$ in $\mathcal{A}$, define

$$K(F)(C^\bullet) := (F(C^\bullet), F(d^\bullet)),$$

and for a cochain map $f^\bullet : C^\bullet \rightarrow D^\bullet$, define

$$K(F)(f^\bullet) := (F(f^\bullet)).$$

The functor $K(F)$ respects cochain homotopies, thus it is well-defined on the homotopy category.

3. (See [DBG⁺77, Catégories dérivées Chapitre 1 2-3] for notation) We write $K^+(\mathcal{A})$, $K^-(\mathcal{A})$, and $K^b(\mathcal{A})$ for the full subcategories of bounded below, bounded above, and bounded complexes respectively.
4. (See [DBG⁺77, Catégories dérivées Chapitre 1 2-3] for notation) If $\mathcal{A}$ is abelian, we may also write $K^{\infty,+}(\mathcal{A})$, $K^{\infty,-}(\mathcal{A})$, $K^{\infty,b}(\mathcal{A})$, and $K^{\infty,\emptyset}(\mathcal{A})$ for the full subcategories of cohomologically bounded below, cohomologically bounded above, cohomologically bounded, and acyclic complexes respectively. We may further write $K^{?,??}(\mathcal{A})$ for the full subcategory where $? \in \{+, -, b\}$ indicates the boundedness of the complex and $?? \in \{+, -, b, (\text{blank})\}$ indicates the cohomological boundedness of the complex.

The various categories here are (Proposition 7.1.6) triangulated categories.

Definition 7.1.2 (Triangulated category). A *triangulated category* is a triple $(\mathcal{T}, [1], \Delta)$ where:

- $\mathcal{T}$ is an additive category,
- $[1] : \mathcal{T} \rightarrow \mathcal{T}$ (also denoted by notations such as Σ or T) is an additive auto-equivalence called the *shift* or *suspension functor*,
- Δ is a class of *distinguished triangles* of the form

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

with objects $X, Y, Z \in \mathcal{T}$ and morphisms f, g, h in $\mathcal{T}$,

satisfying the following axioms:

- (TR1) The class Δ is closed under isomorphisms of triangles and contains the triangles isomorphic to

$$A \xrightarrow{\text{id}} A \rightarrow 0 \rightarrow A[1].$$

Moreover, for every morphism $f : X \rightarrow Y$ in $\mathcal{T}$, there exists a distinguished triangle

$$X \xrightarrow{f} Y \rightarrow Z \rightarrow X[1].$$

- (TR2) (Rotation) If

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

is a distinguished triangle, then so are

$$Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-f[1]} Y[1]$$

and

$$Z[-1] \xrightarrow{-h[-1]} X \xrightarrow{f} Y \xrightarrow{g} Z.$$

(TR3) (Octahedral axiom) Given morphisms $X \xrightarrow{f} Y$ and $Y \xrightarrow{g} Z$ in $\mathcal{T}$, there exist distinguished triangles

$$\begin{aligned} X &\xrightarrow{f} Y \xrightarrow{u} C(f) \xrightarrow{v} X[1], \\ Y &\xrightarrow{g} Z \xrightarrow{w} C(g) \xrightarrow{x} Y[1], \\ X &\xrightarrow{g \circ f} Z \xrightarrow{y} C(g \circ f) \xrightarrow{z} X[1], \end{aligned}$$

and morphisms

$$s : C(f) \rightarrow C(g \circ f), \quad t : C(g \circ f) \rightarrow C(g)$$

such that the following diagram commutes and its rows and columns are distinguished triangles:

$$\begin{array}{ccccccc} X & \xrightarrow{f} & Y & \xrightarrow{u} & C(f) & \xrightarrow{v} & X[1] \\ \parallel & & \downarrow g & & \downarrow s & & \parallel \\ X & \xrightarrow{g \circ f} & Z & \xrightarrow{y} & C(g \circ f) & \xrightarrow{z} & X[1] \\ & & \downarrow w & & \downarrow t & & \\ & & C(g) & \xlongequal{\quad} & C(g) & & \\ & & \downarrow x & & \downarrow & & \\ & & Y[1] & \xrightarrow{f[1]} & X[1][1] & & \end{array}$$

In particular,

- The compositions $s \circ u = y$, $t \circ s = w$, and $x \circ t = v[1]$ hold,
- All rows and columns form distinguished triangles.

(TR4) The class Δ is closed under the shift functor $[1]$: if

$$X \rightarrow Y \rightarrow Z \rightarrow X[1]$$

is distinguished, then so is

$$X[1] \rightarrow Y[1] \rightarrow Z[1] \rightarrow X[2].$$

Definition 7.1.3. A functor $T : D_1 \rightarrow D_2$ between triangulated categories (Definition 7.1.2) is called *exact* if it is additive, is translation preserving, and transforms distinguished triangles (Definition 7.1.2) to distinguished triangles. Oftentimes, a *morphism between triangulated categories* or a *triangulated functor* refers to an exact functor between triangulated categories.

Definition 7.1.4. In the context of a triangulated category (Definition 7.1.2) $\mathcal{D}$, a multiplicative system (Definition 7.2.1) S is *compatible with the triangulation* if:

1. For any $s \in S$, $T(s) \in S$ where T is the translation functor.
2. For any commutative square between the first two steps of two distinguished triangles where the vertical arrows are in S , there exists a morphism in S between the third objects that makes the whole diagram of triangles commute.

Definition 7.1.5. Let I be a finite set and $\varepsilon : I \rightarrow \{1, -1\}$ a variance function. Let $(\mathcal{A}_i)_{i \in I}$ and $\mathcal{A}$ be triangulated categories (Definition 7.1.2). A multifunctor (Definition 1.3.16) $F : \prod_{i \in I} \mathcal{A}_i \rightarrow \mathcal{A}$ is an **exact multifunctor of variance ε** if for every index $k \in I$, the partial functor

$$F_k : \mathcal{A}_k \rightarrow \mathcal{A}$$

(obtained by fixing all objects A_j for $j \neq k$) satisfies the following:

1. If $\varepsilon(k) = 1$, F_k is a covariant exact functor: it maps any distinguished triangle $X \rightarrow Y \rightarrow Z \rightarrow T_k X$ in $\mathcal{A}_k$ to a distinguished triangle

$$F_k(X) \rightarrow F_k(Y) \rightarrow F_k(Z) \rightarrow T F_k(X)$$

in $\mathcal{A}$.

2. If $\varepsilon(k) = -1$, F_k is a contravariant exact functor: it maps any distinguished triangle $X \rightarrow Y \rightarrow Z \rightarrow T_k X$ in $\mathcal{A}_k$ to a distinguished triangle

$$F_k(Z) \rightarrow F_k(Y) \rightarrow F_k(X) \rightarrow F_k(T_k^{-1} X)$$

in $\mathcal{A}$, using the isomorphism $F_k(T_k^{-1} X) \cong T F_k(X)$.

Furthermore, F must be a graded functor equipped with natural isomorphisms $\varphi_i : T \circ F \xrightarrow{\sim} F \circ T_i^{\varepsilon_i}$ for each $i \in I$. These must satisfy the anticommutativity condition: for any pair of distinct indices $i, j \in I$ ($i \neq j$), the following diagram of natural transformations commutes with a factor of -1 :

$$(K) \quad \begin{array}{ccc} T^2 F & \xrightarrow{T \varphi_i} & T F T_i^{\varepsilon_i} \\ \downarrow T \varphi_j & & \downarrow \varphi_j T_i^{\varepsilon_i} \\ T F T_j^{\varepsilon_j} & \xrightarrow{-\varphi_i T_j^{\varepsilon_j}} & F T_i^{\varepsilon_i} T_j^{\varepsilon_j} \end{array}$$

Equivalently, the following square is anticommutative:

$$(L) \quad (\varphi_j T_i^{\varepsilon_i}) \circ (T \varphi_i) = -(\varphi_i T_j^{\varepsilon_j}) \circ (T \varphi_j)$$

as morphisms of functors $T^2 F \rightarrow F T_i^{\varepsilon_i} T_j^{\varepsilon_j}$.

In particular in the case that $\varepsilon(i) = 1$ for all i , we are reduced to the following definition: Let $(\mathcal{A}_i)_{1 \leq i \leq n}$ and $\mathcal{A}$ be triangulated categories (Definition 7.1.2). A multifunctor (Definition 1.3.16) $F : \prod \mathcal{A}_i \rightarrow \mathcal{A}$ is called an **exact covariant multifunctor** if it is covariant in each argument and if for every index $i \in \{1, \dots, n\}$ and every fixed collection of objects $(A_j)_{j \neq i}$, the partial functor $F(\dots, A_{i-1}, -, A_{i+1}, \dots) : \mathcal{A}_i \rightarrow \mathcal{A}$ is an exact functor (Definition 7.1.3) of triangulated categories.

Proposition 7.1.6. Let $\mathcal{A}$ be an additive category (Definition 3.0.1).

1. The homotopy category $K(\mathcal{A})$ (Definition 7.1.1) is a triangulated category (Definition 7.1.2). (♠ TODO: shift of chain complexes, describe what the distinguished triangles are)
2. The subcategory $K^?(\mathcal{A})$ is a triangulated subcategory of $K(\mathcal{A})$ for $? \in \{+, -, b\}$.
3. Let $\mathcal{A}$ be an abelian category. The subcategory $K^{?,??}(\mathcal{A})$ is a triangulated subcategory of $K(\mathcal{A})$ for $? \in \{\infty, +, -, b\}$ and $?? \in \{+, -, b, (\text{blank})\}$.

(♠ TODO: How homotopy categories of chain complexes relate to topological homotopy categories)

Theorem 7.1.7 (Exact functor induces a functor on homotopy categories, cf. [DA73, Exposé XVII, Proposition 1.1.7.]). Let $F : \prod_{i=1}^n \mathcal{A}_i \rightarrow \mathcal{B}$ be an additive multifunctor of abelian categories (Definition 3.0.4). F induces exact multifunctors (Definition 7.1.5)⁶ between the corresponding homotopy categories (Definition 7.1.1):

$$\begin{aligned} K(\mathcal{A}) &\rightarrow K(\mathcal{B}) \\ K^?(\prod \mathcal{A}_i) &\rightarrow K^?(\mathcal{B}) \\ K^{?,??}(\prod \mathcal{A}_i) &\rightarrow K^{?,??}(\mathcal{B}) \end{aligned}$$

(Definition 7.1.1). We write these functors by $K(F)$.

7.2. Multiplicative systems in categories.

Definition 7.2.1 (Multiplicative system). (♠ TODO: Write a precise definition) Let $\mathcal{C}$ be a category. A *multiplicative system S in $\mathcal{C}$* consists of a collection of morphisms in $\mathcal{C}$ satisfying:

- S contains all identity morphisms,
- S is closed under composition,
- (Ore conditions) For any morphism f in $\mathcal{C}$ and any morphism $s \in S$ with suitable domain and codomain, there exist morphisms to form commutative squares allowing localization.

The precise Ore condition varies depending on context but guarantees the localization exists.

Definition 7.2.2 (Localization by a multiplicative system). Given a category $\mathcal{C}$ and a multiplicative system (Definition 7.2.1) $S \subseteq \text{Mor}(\mathcal{C})$, the *localization $\mathcal{C}[S^{-1}]$* is a category equipped with a functor

$$Q : \mathcal{C} \rightarrow \mathcal{C}[S^{-1}]$$

that sends every morphism in S to an isomorphism satisfying the universal property: any functor from $\mathcal{C}$ sending the morphisms in S to isomorphisms factors uniquely through Q .

Definition 7.2.3. A multiplicative system (Definition 7.2.1) S in a category (Definition 1.0.1) $\mathcal{C}$ is called *saturated* if it satisfies the following condition: any morphism f in $\mathcal{C}$ whose image $Q(f)$ in the localized category (Definition 7.2.2) $\mathcal{C}(S^{-1})$ is an isomorphism belongs to S .

Proposition 7.2.4. Let $\mathcal{C}$ be a category (Definition 1.0.1) and S a multiplicative system (Definition 7.2.1). There exists a unique saturated multiplicative system (Definition 7.2.3) $\bar{S}$ containing S such that the canonical functors $\mathcal{C}(S^{-1}) \rightarrow \mathcal{C}(\bar{S}^{-1})$ define an equivalence of categories (Definition 1.1.8). $\bar{S}$ consists of all morphisms f such that there exist morphisms g, h with $g \circ f \in S$ and $f \circ h \in S$.

⁶wlog covariant in each variable, but this can be made multi-variant by taking opposite categories (Definition 1.0.12)

If F is contravariant in the j -th variable, the definitions are adjusted by replacing $\mathcal{A}_j$ with its opposite category $\mathcal{A}_j^{\text{op}}$, thereby interchanging K^+ with K^- and RF with LF for that specific variable.

7.3. Derived categories of abelian categories. (♠ TODO: describe the objects and morphisms in a derived category)

Definition 7.3.1 (Derived category). Let $\mathcal{A}$ be an abelian category. Consider the class S of quasi-isomorphisms in the homotopy category $K(\mathcal{A})$ (Definition 7.1.1) (Definition 7.1.1).

The *derived category* $D(\mathcal{A})$ is the localization (Definition 7.2.2) of the homotopy category at the multiplicative system (Definition 7.2.1) S :

$$D(\mathcal{A}) := K(\mathcal{A})[S^{-1}],$$

We often write the localization functor (Definition 7.2.2) $K(\mathcal{A}) \rightarrow D(\mathcal{A})$ by q or Q .

- We often write $D^+(\mathcal{A})$, $D^-(\mathcal{A})$, and $D^b(\mathcal{A})$ for the full subcategories of cohomologically bounded below, cohomologically bounded above, and cohomologically bounded complexes respectively.

In general, $D(\mathcal{A})$ (and the aforementioned subcategories) may only exist as a large category (Definition 1.0.1), rather than a locally small category. When $\mathcal{A}$ is a Grothendieck abelian category (Definition 3.0.5), however, $D(\mathcal{A})$ is a locally small category (see [Sta25, Tag 09PA]).

Convention 7.3.2. Given an abelian category (Definition 7.3.1) $\mathcal{A}$, any object M of $\mathcal{A}$ may be identified with an object of the homotopy category $K(\mathcal{A})$ (Definition 7.1.1) and the derived category $D(\mathcal{A})$ (Definition 7.3.1) — simply take the chain complex concentrated at degree 0 with the object M . In fact, this derived object can often be regarded as an object of “nice” enough subcategories of $D(\mathcal{A})$, such as the bounded derived categories $D^+(\mathcal{A})$, $D^-(\mathcal{A})$, $D^b(\mathcal{A})$.

(♠ TODO: State how a derived category is a triangulated category)

Definition 7.3.3. Let $\mathcal{A}$ be an additive category (Definition 3.0.1). For $? \in \{+, -, b, (\text{blank})\}$, write $K^?(\mathcal{I})$ (resp. $K^?(\mathcal{P})$) for the full subcategory of $K^?(\mathcal{A})$ (Definition 7.1.1) whose objects are appropriately bounded complexes of injectives (resp. projectives).

Even though derived categories of abelian categories may not be guaranteed to be locally small, the following gives sufficient conditions for the appropriately (cohomologically) bounded derived categories to be locally small. The following also shows that appropriately bounded complexes under these circumstances can be represented in the derived category by a complex of injective or projective objects; we might think of such complexes of injectives or projectives as acting like an injective/projective resolution.

Theorem 7.3.4 (see e.g. [Wei94, Theorem 10.4.8]). Let $\mathcal{A}$ be an abelian category.

1. If $\mathcal{A}$ has enough injectives (Definition 5.4.4), then the category $D^+(\mathcal{A})$ (Definition 7.3.1) is equivalent to the category $K^+(\mathcal{I})$ (Definition 7.3.3), which is locally small (Definition 1.0.9). In fact, an equivalence is given by

$$K^+(\mathcal{I}) \xrightarrow{q} D^+(\mathcal{I}) \hookrightarrow D^+(\mathcal{A}).$$

2. If $\mathcal{A}$ has enough projectives (Definition 5.4.4), then the category $D^-(\mathcal{A})$ (Definition 7.3.1) is equivalent to the category $K^-(\mathcal{P})$ (Definition 7.3.3), which is locally small (Definition 1.0.9).

$$K^-(\mathcal{P}) \xrightarrow{q} D^-(\mathcal{P}) \hookrightarrow D^-(\mathcal{A}).$$

7.4. Total derived functors. The following subsection mostly takes the point of view of [Wei94, Chapter 10]

Definition 7.4.1 (cf. [Wei94, Definition 10.5.1]). Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $F : K(\mathcal{A}) \rightarrow K(\mathcal{B})$ (Definition 7.1.1) be an exact functor (Definition 7.1.3) of triangulated categories. Write $q_{\mathcal{A}} : K(\mathcal{A}) \rightarrow D(\mathcal{A})$ and $q_{\mathcal{B}} : K(\mathcal{B}) \rightarrow D(\mathcal{B})$ for the localization functors (Definition 7.3.1),

1. A *(total) right derived functor on F on $K(\mathcal{A})$* is an exact functor $RF : D(\mathcal{A}) \rightarrow D(\mathcal{B})$ together with a natural transformation

$$\xi : q_{\mathcal{B}}F \Rightarrow (RF)q_{\mathcal{A}},$$

which is universal in the sense that if $G : D(\mathcal{A}) \rightarrow D(\mathcal{B})$ is another morphism equipped with a natural transformation $\zeta : q_{\mathcal{B}}F \Rightarrow Gq_{\mathcal{A}}$, then there exists a unique natural transformation $\eta : RF \Rightarrow G$ so that $\zeta_A = \eta_{q_{\mathcal{A}}A} \circ \xi_A$ for all A in $K(\mathcal{A})$.

2. Dually, a *(total) left derived functor on F on $K(\mathcal{A})$* is an exact functor $LF : D(\mathcal{A}) \rightarrow D(\mathcal{B})$ together with a natural transformation

$$\xi : (LF)q_{\mathcal{A}} \Rightarrow q_{\mathcal{B}}F,$$

which is universal in the sense that if $G : D(\mathcal{A}) \rightarrow D(\mathcal{B})$ is another exact functor equipped with a natural transformation $\zeta : Gq_{\mathcal{A}} \Rightarrow q_{\mathcal{B}}F$, then there exists a unique natural transformation $\eta : G \Rightarrow LF$ such that

$$\zeta_A = \xi_A \circ \eta_{q_{\mathcal{A}}A}$$

for all A in $K(\mathcal{A})$.

The universal properties guarantee that if the total derived functors exist, then they are unique up to natural isomorphism, and that

These definitions/ideas may be extended to exact functors $K^?(\mathcal{A}) \rightarrow K(\mathcal{B})$ for $? \in \{+, -, b\}$ and to exact functors $K^{?,??}(\mathcal{A}) \rightarrow K(\mathcal{B})$ for $? \in \{\infty, +, -, b\}$ and $?? \in \{+, -, b, (\text{blank})\}$ (Definition 7.1.1)

See Definition 7.5.4 for a more general treatment of derived functors

In particular, the above definitions are specializations of those of Definition 7.5.8, which fit into the framework of Definition 7.5.4.

Proposition 7.4.2. See e.g. [Wei94, 10.5.2] Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an exact functor (Definition 3.0.7) of abelian categories (Definition 3.0.4). There is an induced functor

$$D(\mathcal{A}) \rightarrow D(\mathcal{B})$$

(Definition 7.3.1) also denoted by $\mathbf{F}$ by abuse of notation.

This functor is constructible in the “simple” way —

- For an object M of $D(\mathcal{A})$, take a chain complex K representing it. The object FM of $D(\mathcal{B})$ is the object represented by the chain complex FK .
- For a morphism $f : M \rightarrow N$ in $D(\mathcal{A})$, take a morphism $\tilde{f} : K \rightarrow L$ representing it. The morphism $Ff : FM \rightarrow FN$ is the morphism represented by $F\tilde{f} : FK \rightarrow FL$.

Due to the exactness of F , these do not depend on the choices of representatives.

The below shows that right and left total derived functors exist for nice enough abelian categories on sufficiently (cohomologically) bounded objects and can be computed using injective/projective resolutions.

Theorem 7.4.3 (See e.g. [Wei94, 10.5.6]). Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $F : K^+(\mathcal{A}) \rightarrow K(\mathcal{B})$ (Definition 7.1.1) be a morphism of triangulated categories (Definition 7.1.3).

1. If $\mathcal{A}$ has enough injectives (Theorem 7.1.7), then the right derived functor RF (Definition 7.4.1) exists (as a functor $D^+(\mathcal{A}) \rightarrow D(\mathcal{B})$ (Definition 7.3.1)), and if I is a bounded below complex of injectives, then

$$RF(I) \cong qF(I)$$

where $q : K(\mathcal{B}) \rightarrow D(\mathcal{B})$ is the localization functor (Definition 7.3.1).

2. Dually, if $\mathcal{A}$ has enough projectives, then the left derived functor LF exists (as a functor $D^-(\mathcal{A}) \rightarrow D(\mathcal{B})$), and if I is a bounded above complex of projectives, then

$$LF(P) \cong qF(P)$$

Corollary 7.4.4. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 3.0.3) between abelian categories (Definition 3.0.4). Write $q : K(\mathcal{B}) \rightarrow D(\mathcal{B})$ (Definition 7.1.1 Definition 7.3.1) for the localization functor (Definition 7.2.2).

1. Assuming that $\mathcal{A}$ has enough injectives (Definition 5.4.4), there exists a right derived functor (Definition 7.4.1) $RF : D^+(\mathcal{A}) \rightarrow D(\mathcal{B})$ and if $I \in K^+(\mathcal{I})$ (Definition 7.3.3) is a bounded below complex of injectives, then

$$RF(I) \cong q(K(F))(I).$$

where $K(F) : K^+(\mathcal{A}) \rightarrow K(\mathcal{B})$ is the exact functor (Definition 7.1.3) induced by F (Theorem 7.1.7).

If F is additionally left exact (Definition 3.0.7), then for all objects A of $\mathcal{A}$ we have

$$H^i(RF(A)) \cong R^iF(A)$$

(Definition 5.4.8) for all integers i where we regard A as the chain complex (Definition 6.0.1) object of $D^+(\mathcal{A})$ with A concentrated in degree 0 (Convention 7.3.2).

2. Assuming that $\mathcal{A}$ has enough projectives (Definition 5.4.4), there exists a left derived functor (Definition 7.4.1) $LF : D^-(\mathcal{A}) \rightarrow D(\mathcal{B})$ and if $P \in K^-(\mathcal{P})$ (Definition 7.3.3) is a bounded above complex of projectives, then

$$LF(P) \cong q(K(F))(P).$$

where $K(F) : K^-(\mathcal{A}) \rightarrow K(\mathcal{B})$ is the exact functor (Definition 7.1.3) induced by F (Theorem 7.1.7).

If F is additionally right exact (Definition 3.0.7), then for all objects A of $\mathcal{A}$ we have

$$H^{-i}(LF(A)) \cong L_i F(A)$$

(Definition 5.4.8) for all integers i where we regard A as the chain complex (Definition 6.0.1) object of $D^+(\mathcal{A})$ with A concentrated in degree 0 (Convention 7.3.2).

Proof. Apply Theorem 7.4.3 to the exact functor $K(F)$ to obtain the isomorphisms

$$RF(I) \cong q(K(F))(I)$$

$$LF(P) \cong q(K(F))(P).$$

If F is left exact and A is an object of $\mathcal{A}$, then the right derived functors $R^i F(A)$ (Definition 5.4.8) are defined. Recall that they are defined first by taking an injective resolution (Definition 5.4.3)

$$0 \rightarrow A \rightarrow I^0 \rightarrow I^1 \rightarrow \dots$$

Since the injective resolution is acyclic, The object A and the complex $0 \rightarrow I^0 \rightarrow I^1 \rightarrow \dots$ are equivalent as objects of $D^+(\mathcal{A})$ as an object of $D^+(\mathcal{A})$ are equivalent objects. Moreover, we have the isomorphism

$$RF(I) \cong q(K(F))(I).$$

In fact, the i th cohomology of the right hand side object precisely describes $R^i F(A)$ by definition. Therefore,

$$H^i(RF(A)) \cong R^i F(A)$$

as desired.

The analogous statement for $L_i F(A)$ holds similarly. □

(♠ TODO: define hyper-derived functors)

Corollary 7.4.5 (See e.g. [Wei94, Corollary 10.5.7, Remark 10.5.8]). Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 3.0.3) between abelian categories (Definition 3.0.4).

1. If $\mathcal{A}$ has enough injectives (Definition 5.4.4), the hyper-derived functors $\mathbb{R}^i F(X)$ are the cohomology of $\mathbf{R}F(X) : \mathbb{R}^i F(X) \cong H^i \mathbf{R}F(X)$ for all i .
2. If $\mathcal{A}$ has enough projectives (Definition 5.4.4), the hyper-derived functors $\mathbb{L}_i F(X)$ are the cohomology of $\mathbf{L}F(X) : \mathbb{L}_i F(X) \cong H^{-i} \mathbf{L}F(X)$ for all i .

7.5. General derived functors for exact multifunctors between triangulated categories. The following subsection mostly takes the point of view of [DA73, Exposé XVII, 1.2]

Definition 7.5.1 (Ind-category). Let $\mathcal{C}$ be a locally small category (Definition 1.0.9).

1. The *Ind-category of $\mathcal{C}$* , denoted $\mathbf{Ind}(\mathcal{C})$, is defined as follows:

- Objects of $\text{Ind}(\mathcal{C})$ are formal filtered colimits (Definition 1.2.12) of objects in $\mathcal{C}$. More precisely, an object is given by a filtered (Definition 1.2.8) small category I and a functor

$$X : I \rightarrow \mathcal{C}.$$

- Morphisms between objects $X : I \rightarrow \mathcal{C}$ and $Y : J \rightarrow \mathcal{C}$ are defined by

$$\text{Hom}_{\text{Ind}(\mathcal{C})}(X, Y) := \varinjlim_{i \in I} \varprojlim_{j \in J} \text{Hom}_{\mathcal{C}}(X_i, Y_j),$$

(Definition 1.2.12) where X_i and Y_j denote the images of $i \in I$ and $j \in J$ under X and Y , respectively.

The composition of morphisms is induced naturally from composition in $\mathcal{C}$. Hence, $\text{Ind}(\mathcal{C})$ is the completion of $\mathcal{C}$ under filtered colimits. Objects of $\text{Ind}(\mathcal{C})$ are called *Ind-objects of $\mathcal{C}$* .

2. The *Pro-category of $\mathcal{C}$* , denoted $\text{Pro}(\mathcal{C})$, is defined as follows:

- Objects of $\text{Pro}(\mathcal{C})$ are formal cofiltered limits (Definition 1.2.12) of objects in $\mathcal{C}$. More precisely, an object is given by a cofiltered small category I and a functor

$$X : I \rightarrow \mathcal{C}.$$

- Morphisms between objects $X : I \rightarrow \mathcal{C}$ and $Y : J \rightarrow \mathcal{C}$ are defined by

$$\text{Hom}_{\text{Pro}(\mathcal{C})}(X, Y) := \varprojlim_{j \in J} \varinjlim_{i \in I} \text{Hom}_{\mathcal{C}}(X_i, Y_j),$$

where X_i and Y_j denote the images of $i \in I$ and $j \in J$ under X and Y , respectively.

The composition of morphisms is induced naturally from composition in $\mathcal{C}$.

Hence, $\text{Pro}(\mathcal{C})$ is the completion of $\mathcal{C}$ under cofiltered limits. Objects of $\text{Pro}(\mathcal{C})$ are called *Pro-objects of $\mathcal{C}$* .

Since **Sets** has all limits and colimits (♠ TODO:) and hence has all projective and inductive limits and since $\mathcal{C}$ is locally small, $\text{Ind}(\mathcal{C})$ and $\text{Pro}(\mathcal{C})$ are locally small.

The *canonical embedding of a locally small category $\mathcal{C}$ into its Ind-category or Pro-category* is defined by treating each object as a "constant" diagram. Let $\mathbf{1}$ be the category with a single object $*$ and its identity morphism.

1. The Embedding $\iota_{\text{Ind}} : \mathcal{C} \rightarrow \text{Ind}(\mathcal{C})$:

- Objects: An object $C \in \mathcal{C}$ is mapped to the functor $X : \mathbf{1} \rightarrow \mathcal{C}$ sending $*$ to C . This represents C as a formal filtered colimit over the trivial index category.
- Morphisms: Given objects X (representing C) and Y (representing D), the Hom-set in the Ind-category simplifies:

$$\text{Hom}_{\text{Ind}(\mathcal{C})}(X, Y) \cong \varinjlim_{i \in \mathbf{1}} \varprojlim_{j \in \mathbf{1}} \text{Hom}_{\mathcal{C}}(X_i, Y_j) \cong \text{Hom}_{\mathcal{C}}(C, D).$$

2. The Embedding $\iota_{\text{Pro}} : \mathcal{C} \rightarrow \text{Pro}(\mathcal{C})$:

- Objects: An object $C \in \mathcal{C}$ is mapped to the functor $X : \mathbf{1} \rightarrow \mathcal{C}$ sending $*$ to C , representing C as a formal cofiltered limit.
- Morphisms: Similarly, the Pro-morphism definition reduces to:

$$\text{Hom}_{\text{Pro}(\mathcal{C})}(X, Y) \cong \varprojlim_{j \in \mathbf{1}} \varinjlim_{i \in \mathbf{1}} \text{Hom}_{\mathcal{C}}(X_i, Y_j) \cong \text{Hom}_{\mathcal{C}}(C, D).$$

In both cases, the functor ι is fully faithful, allowing us to identify $\mathcal{C}$ with a full subcategory of its respective completion. Given an object C of $\mathcal{C}$, the pro/ind-object $\iota(C)$ may be called the *constant pro/ind-object of C* .

Definition 7.5.2. Let $\mathcal{C}$ be a locally small category (Lemma 5.0.5).

1. An Ind-object (Definition 7.5.1) $X = \varinjlim X_i$ is said to be *essentially constant* if it is isomorphic in $\text{Ind}(\mathcal{C})$ to an object in the image of the canonical fully faithful functor $h : \mathcal{C} \rightarrow \text{Ind}(\mathcal{C})$, which maps an object $A \in \text{ob}(\mathcal{C})$ to the Ind-object represented by the constant diagram.
2. A Pro-object (Definition 7.5.1) $X = \varprojlim X_i$ is said to be *essentially constant* if it is isomorphic in $\text{Pro}(\mathcal{C})$ to an object in the image of the canonical fully faithful functor $h : \mathcal{C} \rightarrow \text{Pro}(\mathcal{C})$, which maps an object $A \in \text{ob}(\mathcal{C})$ to the Pro-object represented by the constant diagram.

In either definition, the object A may be referred to as the *value/limit/constant value of the Ind/Pro object* or the *object representing the Ind/Pro object*.

Proposition 7.5.3. Let $\mathcal{C}$ be a locally small category (Lemma 5.0.5).

1. An Ind-object (Definition 7.5.1) $X = \varinjlim_{i \in I} X_i$ (where I is a filtered (Definition 1.2.8) index category) is essentially constant (Definition 7.5.2) and isomorphic to $A \in \text{ob}(\mathcal{C})$ if and only if there exists a family of morphisms $(f_i : X_i \rightarrow A)_{i \in I}$ such that for every morphism $u : i \rightarrow j$ in I , the diagram

$$\begin{array}{ccc} X_i & \xrightarrow{X(u)} & X_j \\ & \searrow f_i & \downarrow f_j \\ & & A \end{array}$$

commutes, and the induced morphism $\varinjlim X_i \rightarrow A$ (Definition 1.2.12) is an isomorphism in $\text{Ind}(\mathcal{C})$.

2. A Pro-object (Definition 7.5.1) $X = \varprojlim_{i \in I} X_i$ (where I is a filtered (Definition 1.2.8) index category) is essentially constant (Definition 7.5.2) and isomorphic to $A \in \text{ob}(\mathcal{C})$ if and only if there exists a family of morphisms $(g_i : A \rightarrow X_i)_{i \in I}$ such that for every morphism $u : i \rightarrow j$ in I , the diagram

$$\begin{array}{ccc} A & \xrightarrow{g_i} & X_i \\ & \searrow g_j & \downarrow X(u) \\ & & X_j \end{array}$$

commutes, and the induced morphism $A \rightarrow \varprojlim X_i$ (Definition 1.2.12) is an isomorphism in $\text{Pro}(\mathcal{C})$.

Definition 7.5.4 (Mixed-Variance Derived Multifunctors, cf. [DA73, Exposé XVII, Définition 1.2.1., 1.2.5.]). Let $(\mathcal{A}_i)_{1 \leq i \leq n}$ and $(\mathcal{B}_j)_{1 \leq j \leq m}$ be families of triangulated categories (Definition 7.1.2), and let $\mathcal{C}$ be a triangulated category. Let

$$F : \prod_{i=1}^n \mathcal{A}_i \times \prod_{j=1}^m \mathcal{B}_j \rightarrow \mathcal{C}$$

be an exact multifunctor (Definition 7.1.5), covariant in $\mathcal{A}_i$ and contravariant in $\mathcal{B}_j$. Let S_i, T_j, U be saturated multiplicative systems (Definition 7.2.3) in $\mathcal{A}_i, \mathcal{B}_j, \mathcal{C}$ respectively assumed compatible with the triangulated structure (Definition 7.1.4) of its respective category. Write

$$\begin{aligned} Q_{A_i} &: \mathcal{A}_i \rightarrow \mathcal{A}_i(S_i^{-1}) \\ Q_{B_j} &: \mathcal{B}_j \rightarrow \mathcal{B}_j(T_j^{-1}) \\ Q_C &: \mathcal{C} \rightarrow \mathcal{C}(U^{-1}) \end{aligned}$$

for the localization functors (Definition 7.2.2).

1. The *right derived functor RF of F relative to (S_i, T_j) and U* is the functor

$$RF : \prod_{i=1}^n \mathcal{A}_i(S_i^{-1}) \times \prod_{j=1}^m \mathcal{B}_j(T_j^{-1}) \rightarrow \text{Ind}(\mathcal{C}(U^{-1}))$$

which makes the following diagram commutative

$$\begin{array}{ccc} \prod_{i=1}^n \mathcal{A}_i \times \prod_{j=1}^m \mathcal{B}_j & \xrightarrow{F} & \mathcal{C} \\ \downarrow \prod Q_{A_i} \times \prod Q_{B_j} & & \downarrow \overline{Q}_C \\ \prod_{i=1}^n \mathcal{A}_i(S_i^{-1}) \times \prod_{j=1}^m \mathcal{B}_j(T_j^{-1}) & \xrightarrow{RF} & \text{Ind}(\mathcal{C}(U^{-1})) \end{array}$$

where $\overline{Q}_C$ is the composition

$$\mathcal{C} \xrightarrow{Q_C} \mathcal{C}(U^{-1}) \xrightarrow{\iota_{\text{Ind}}} \text{Ind}(\mathcal{C}(U^{-1}))$$

of Q_C and the canonical embedding (Definition 7.5.1) ι_{Ind} .

2. The *left derived functor LF of F relative to (S_i, T_j) and U* is the functor

$$LF : \prod_{i=1}^n \mathcal{A}_i(S_i^{-1}) \times \prod_{j=1}^m \mathcal{B}_j(T_j^{-1}) \rightarrow \text{Pro}(\mathcal{C}(U^{-1}))$$

which makes the following diagram commutative:

$$\begin{array}{ccc} \prod_{i=1}^n \mathcal{A}_i \times \prod_{j=1}^m \mathcal{B}_j & \xrightarrow{F} & \mathcal{C} \\ \downarrow \prod Q_{A_i} \times \prod Q_{B_j} & & \downarrow \overline{Q}_C \\ \prod_{i=1}^n \mathcal{A}_i(S_i^{-1}) \times \prod_{j=1}^m \mathcal{B}_j(T_j^{-1}) & \xrightarrow{LF} & \text{Pro}(\mathcal{C}(U^{-1})) \end{array}$$

where $\overline{Q}_C$ is the composition

$$\mathcal{C} \xrightarrow{Q_C} \mathcal{C}(U^{-1}) \xrightarrow{\iota_{\text{Pro}}} \text{Pro}(\mathcal{C}(U^{-1}))$$

of Q_C and the canonical embedding (Definition 7.5.1) ι_{Pro} .

Definition 7.5.5 (Derivability and Adapted Families). Let $(\mathcal{A}_i)_{1 \leq i \leq n}$ and $(\mathcal{B}_j)_{1 \leq j \leq m}$ be families of triangulated categories (Definition 7.1.2), and let $\mathcal{C}$ be a triangulated category. Let

$$F : \prod_{i=1}^n \mathcal{A}_i \times \prod_{j=1}^m \mathcal{B}_j \rightarrow \mathcal{C}$$

be an exact multifunctor (Definition 7.1.5), covariant in $\mathcal{A}_i$ and contravariant in $\mathcal{B}_j$. Let S_i, T_j, U be saturated multiplicative systems (Definition 7.2.3) in $\mathcal{A}_i, \mathcal{B}_j, \mathcal{C}$ respectively assumed compatible with the triangulated structure (Definition 7.1.4) of its respective category. Write

$$\begin{aligned} Q_{A_i} : \mathcal{A}_i &\rightarrow \mathcal{A}_i(S_i^{-1}) \\ Q_{B_j} : \mathcal{B}_j &\rightarrow \mathcal{B}_j(T_j^{-1}) \\ Q_C : \mathcal{C} &\rightarrow \mathcal{C}(U^{-1}) \end{aligned}$$

for the localization functors (Definition 7.2.2). Write ι for the appropriate canonical embedding functors of the localized categories into their Ind or Pro categories (Definition 7.5.1).

1. Let RF be the right derived functor (Definition 7.5.4) of F relative to the multiplicative systems.
 - (a) RF is *defined at a family* (A_i, B_j) if the Ind-object (Definition 7.5.1) $RF(A_i, B_j)$ is essentially constant (Definition 7.5.2), i.e. isomorphic to an object of $\mathcal{C}(U^{-1})$. This object of $\mathcal{C}(U^{-1})$ is the *value of RF at (A_i, B_j)* .
 - (b) F is *right derivable* if RF is defined at all objects. In this case, we view RF as a functor into $\mathcal{C}(U^{-1})$.
 - (c) A family (A_i, B_j) is *adapted for RF* (or *deployed for RF* (*déployée pour RF* in French) or *unfolded for RF* , cf. [Spa88]) if the natural morphism $(\iota \circ Q_C \circ F)(A_i, B_j) \rightarrow RF(Q_{A_i}A_i, Q_{B_j}B_j)$ is an isomorphism in $\text{Ind}(\mathcal{C}(U^{-1}))$, i.e. the Ind-object $RF(A_i)$ of $A(S^{-1})$ is essentially constant (Definition 7.5.2) with value $F((A_i), (B_j))$.
2. Let LF be the left derived functor (Definition 7.5.4) of F relative to the multiplicative systems.
 - (a) LF is *defined at a family* (A_i, B_j) if the Pro-object (Definition 7.5.1) $LF(A_i, B_j)$ is essentially constant (Definition 7.5.2), i.e. isomorphic to an object of $\mathcal{C}(U^{-1})$. This object of $\mathcal{C}(U^{-1})$ is the *value of LF at (A_i, B_j)* .
 - (b) F is *left derivable* if LF is defined at all objects. In this case, we view LF as a functor into $\mathcal{C}(U^{-1})$.
 - (c) A family (A_i, B_j) is *adapted for LF* (or *deployed for LF* (*déployée pour LF* in French) or *unfolded for LF* , cf. [Spa88]) if the natural morphism $LF(Q_{A_i}A_i, Q_{B_j}B_j) \rightarrow (\iota \circ Q_C \circ F)(A_i, B_j)$ is an isomorphism in $\text{Pro}(\mathcal{C}(U^{-1}))$, i.e. the Pro-object $LF(A_i)$ of $A(S^{-1})$ is essentially constant (Definition 7.5.2) with value $F((A_i), (B_j))$.

In particular, note that if a family (A_i, B_j) is adapted for RF resp. LF , then RF resp. LF is defined at (A_i, B_j) .

Proposition 7.5.6 (cf. [Spa88]). Let $(\mathcal{A}_i)_{1 \leq i \leq n}$ and $(\mathcal{B}_j)_{1 \leq j \leq m}$ be families of triangulated categories (Definition 7.1.2), and let $\mathcal{C}$ be a triangulated category. Let

$$F : \prod_{i=1}^n \mathcal{A}_i \times \prod_{j=1}^m \mathcal{B}_j \rightarrow \mathcal{C}$$

be an exact multifunctor (Definition 7.1.5), covariant in $\mathcal{A}_i$ and contravariant in $\mathcal{B}_j$. Let S_i, T_j, U be saturated multiplicative systems (Definition 7.2.3) in $\mathcal{A}_i, \mathcal{B}_j, \mathcal{C}$ respectively assumed compatible with the triangulated structure (Definition 7.1.4) of its respective category.

A family (A_i, B_j) is adapted (or deployed/unfolded) (Definition 7.5.5) for RF (Definition 7.5.4) if and only if the following holds: For every family of morphisms $s_i : A_i \rightarrow X_i$ in S_i and $t_j : Y_j \rightarrow B_j$ in T_j , there exist morphisms $\sigma_i : X_i \rightarrow \mathcal{X}_i$ in S_i and $\tau_j : \mathcal{Y}_j \rightarrow Y_j$ in T_j such that the induced morphism in $\mathcal{C}$

$$F(\sigma_i \circ s_i, t_j \circ \tau_j) : F(A_i, B_j) \rightarrow F(\mathcal{X}_i, \mathcal{Y}_j)$$

is an isomorphism in the localized category $\mathcal{C}(U^{-1})$ (i.e., its image under Q_C is an isomorphism, or equivalently, it is an element of U).

See also Proposition 7.7.1

Proof. (♠ TODO: verify this proof) Let $\mathcal{D} = \prod_i \mathcal{A}_i \times \prod_j \mathcal{B}_j^{op}$ and let S be the product multiplicative system $\prod_i S_i \times \prod_j T_j^{op}$. The right derived functor RF evaluated at an object $\mathbf{A} = (A_i, B_j)$ is defined as the Ind-object in $\mathcal{C}(U^{-1})$ represented by the diagram:

$$RF(\mathbf{A}) = \varinjlim_{\mathbf{A} \rightarrow \mathbf{X} \in S} Q_C(F(\mathbf{X}))$$

By definition, $\mathbf{A}$ is *adapted* (deployed) for RF if the natural morphism from the constant Ind-object to the system is an isomorphism:

$$\eta : \iota(Q_C(F(\mathbf{A}))) \xrightarrow{\cong} RF(Q_{\mathcal{D}}(\mathbf{A})) \quad \text{in } \text{Ind}(\mathcal{C}(U^{-1}))$$

($\implies$) Suppose $\mathbf{A}$ is adapted. An isomorphism in an Ind-category $\text{Ind}(\mathcal{M})$ between a constant object M and a filtered system $\{N_k\}_{k \in K}$ implies that for every index $k \in K$ (representing a resolution $\mathbf{A} \rightarrow \mathbf{X}$), there exists a morphism $k \rightarrow \ell$ in the filtering category (representing a further resolution $\mathbf{X} \rightarrow \mathcal{X}$) such that the composite map $M \rightarrow N_\ell$ is an isomorphism in $\mathcal{M}$.

Translating this to our categories: for any family of morphisms $s_i : A_i \rightarrow X_i$ and $t_j : Y_j \rightarrow B_j$ (defining the resolution $\mathbf{A} \rightarrow \mathbf{X}$), there must exist a further resolution $\mathbf{X} \rightarrow \mathcal{X}$ (defined by $\sigma_i : X_i \rightarrow \mathcal{X}_i$ and $\tau_j : \mathcal{Y}_j \rightarrow Y_j$) such that the induced morphism $Q_C(F(\mathbf{A})) \rightarrow Q_C(F(\mathcal{X}))$ is an isomorphism in $\mathcal{C}(U^{-1})$. This is exactly the condition stated.

($\impliedby$) Conversely, assume the condition holds. Let J be the category of resolutions of $\mathbf{A}$. The condition implies that the constant object $Q_C(F(\mathbf{A}))$ is *cofinal* in the system defining $RF(\mathbf{A})$.

Specifically, the assumption guarantees that for every $\mathbf{X} \in J$, there is a $\mathcal{X} \in J$ further in the diagram such that $Q_C(F(\mathbf{A}))$ and $Q_C(F(\mathcal{X}))$ are isomorphic via the transition maps of the system. In the category of Ind-objects, if a filtered system contains a cofinal subsystem of objects isomorphic to a constant object M , then the Ind-limit of that system is isomorphic to M . Thus, $RF(\mathbf{A}) \cong \iota(Q_C(F(\mathbf{A})))$, which means $\mathbf{A}$ is adapted. $\square$

Proposition 7.5.7 (cf. [Spa88]). Let $(\mathcal{A}_i)_{1 \leq i \leq n}$ and $(\mathcal{B}_j)_{1 \leq j \leq m}$ be families of abelian categories, and let $\mathcal{C}$ be an abelian category. Let

$$F : \prod_{i=1}^n \mathcal{A}_i \times \prod_{j=1}^m \mathcal{B}_j \rightarrow \mathcal{C}$$

be an additive multifunctor, covariant in $K(\mathcal{A}_i)$ and contravariant in $K(\mathcal{B}_j)$.

A family of complexes $(A_i^\bullet, B_j^\bullet)$ is adapted (or deployed/unfolded) (Definition 7.5.5) for the right derived functor RF if and only if the following holds: For every family of quasi-isomorphisms $s_i : A_i^\bullet \rightarrow X_i^\bullet$ and $t_j : Y_j^\bullet \rightarrow B_j^\bullet$, there exist quasi-isomorphisms $\sigma_i : X_i^\bullet \rightarrow \mathcal{X}_i^\bullet$ and $\tau_j : \mathcal{Y}_j^\bullet \rightarrow Y_j^\bullet$ such that the induced morphism in $K(\mathcal{C})$

$$F(\sigma_i \circ s_i, t_j \circ \tau_j) : F(A_i^\bullet, B_j^\bullet) \rightarrow F(\mathcal{X}_i^\bullet, \mathcal{Y}_j^\bullet)$$

is a quasi-isomorphism (i.e., an isomorphism in the derived category $D(\mathcal{C})$).

Proof. This follows from Proposition 7.5.6. □

7.5.1. Derived multifunctors of additive multifunctors between abelian categories.

Definition 7.5.8. Let $F : \prod_{i=1}^n \mathcal{A}_i \rightarrow \mathcal{B}$ be an additive multifunctor of abelian categories (Definition 3.0.4).

The *right (resp. left) derived multifunctors of F* are the derived functors (Definition 7.5.4) RF (resp. LF) of the induced multifunctors (Theorem 7.1.7)

- $K(\prod \mathcal{A}_i) \rightarrow K(\mathcal{B})$
- $K^+(\prod \mathcal{A}_i) \rightarrow K^+(\mathcal{B})$
- $K^-(\prod \mathcal{A}_i) \rightarrow K^-(\mathcal{B})$

(Recall Proposition 7.1.6) relative to the systems (Definition 7.2.3) of quasi-isomorphisms in the respective homotopy categories (Definition 7.1.1). In particular, the functor RF a priori may refer to one of the functors

$$D^+(\prod \mathcal{A}_i) \rightarrow \text{Ind } D^+(\mathcal{B})$$

$$D^-(\prod \mathcal{A}_i) \rightarrow \text{Ind } D^-(\mathcal{B})$$

$$D(\prod \mathcal{A}_i) \rightarrow \text{Ind } D(\mathcal{B})$$

and the functor LF refers to one of the functors

$$D^+(\prod \mathcal{A}_i) \rightarrow \text{Pro } D^+(\mathcal{B})$$

$$D^-(\prod \mathcal{A}_i) \rightarrow \text{Pro } D^-(\mathcal{B})$$

$$D(\prod \mathcal{A}_i) \rightarrow \text{Pro } D(\mathcal{B})$$

(Definition 7.3.1) (Definition 7.5.1).

Definition 7.5.9 ([DA73, Definition 1.2.3]). Let $F : \prod_{i=1}^n \mathcal{A}_i \rightarrow \mathcal{B}$ be an additive multifunctor of abelian categories (Definition 3.0.4).

1. An object $(A_1, \dots, A_n)$ with $A_i \in \text{ob}(\mathcal{A}_i)$ is **right acyclic for F** or **acyclic for RF** if the complex concentrated in degree 0 associated to (A_i) is adapted for (Definition 7.5.5) RF .
2. Similarly, (A_i) is **left acyclic for F** or **acyclic for LF** if the complex concentrated in degree 0 associated to (A_i) is adapted for (Definition 7.5.5) LF .

In the case that $n = 1$ and $A_1 \in \mathcal{A}_1$ has a projective resolution (Definition 5.4.3) (resp. injective resolution), the notion of left acyclic for F (resp. right acyclic for F) is equivalent to the notion of F -acyclicity. (♠ TODO: justify)

Proposition 7.5.10 (cf. [DA73, Exposé XVII, Proposition 1.2.4.]). Let $F : \prod_{i=1}^n \mathcal{A}_i \rightarrow \mathcal{B}$ be an additive multifunctor of abelian categories (Definition 3.0.4).

1. The following diagram of categories is commutative for the right derived multifunctor:

$$\begin{array}{ccc} D^+(\prod \mathcal{A}_i) & \longrightarrow & D(\prod \mathcal{A}_i) \\ \downarrow RF & & \downarrow RF \\ \text{Ind } D^+(\mathcal{B}) & \longrightarrow & \text{Ind } D(\mathcal{B}) \end{array}$$

(Definition 7.3.1) (??) (Definition 7.5.8) where the horizontal arrows are the natural functors induced by inclusions. If RF is everywhere defined on $D(\prod \mathcal{A}_i)$, its restriction to $D^+(\prod \mathcal{A}_i)$ factors through $D^+(\mathcal{B})$ [DA73, Proposition 1.2.4].

2. Dually, the following diagram is commutative for the left derived multifunctor:

$$\begin{array}{ccc} D^-(\prod \mathcal{A}_i) & \longrightarrow & D(\prod \mathcal{A}_i) \\ \downarrow LF & & \downarrow LF \\ \text{Pro } D^-(\mathcal{B}) & \longrightarrow & \text{Pro } D(\mathcal{B}) \end{array}$$

If LF is everywhere defined on $D(\prod \mathcal{A}_i)$, its restriction to $D^-(\prod \mathcal{A}_i)$ factors through $D^-(\mathcal{B})$.

Theorem 7.5.11 ([DA73, Exposé XVII, Proposition 1.2.7.]). (♠ TODO:) Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor (Definition 3.0.3) between abelian categories (Definition 3.0.4).

If $\mathcal{A}$ has enough objects that are acyclic for (Definition 7.5.9) LF (resp. for RF), the derived functor $LF : D^-(\mathcal{A}) \rightarrow D^-(\mathcal{B})$ (resp. $RF : D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$) is left (resp. right) derivable (Definition 7.5.5) and every homologically upper bounded (resp. homologically lower bounded) complex of acyclics for LF (resp. for RF) is adapted for LF (resp. for RF) (Definition 7.5.5).

(♠ TODO:

- **Address the Biadditive/Multifunctor Gap**

- Add a Lemma or Remark following Definition 5.5.5 clarifying that for a biadditive functor $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$, the condition of a pair of complexes $(M^\bullet, N^\bullet)$ being adapted (deployed) is what allows the representation of LF via the Total Complex construction.
- Explicitly define the Total Complex functor $\text{Tot} : K(\mathcal{A}) \times K(\mathcal{B}) \rightarrow K(\mathcal{C})$ used to induce the multifunctor in Definition 5.5.6. Specify the sign convention for the total differential: $d(x \otimes y) = d_x(x) \otimes y + (-1)^p x \otimes d_y(y)$ for x of degree p .

- **Formulate the Balancing Results**

- Add a Proposition (The Balancing Lemma) stating that if $\mathcal{P} \subset \mathcal{A}$ is a class of objects adapted for $F(-, N)$ and $\mathcal{Q} \subset \mathcal{B}$ is a class adapted for $F(M, -)$, there exist canonical isomorphisms in $D(\mathcal{C})$:

$$\mathrm{Tot}(F(P^\bullet, N^\bullet)) \cong \mathrm{Tot}(F(M^\bullet, Q^\bullet)) \cong \mathrm{Tot}(F(P^\bullet, Q^\bullet))$$

- Cite [SGA 4, XVII, 1.2.7] to justify that these isomorphisms are well-defined because complexes of adapted objects are "deployed".

- **Incorporate the “Everywhere Defined” Logic**

- Include the “Finite Dimension” case from [SGA 4, XVII, Prop 1.2.10] to explain the conditions under which LF and RF can be extended to the unbounded derived category $D(\mathcal{A})$.

- **Technical Refinements**

- Provide a precise proof or reference for the property: “If F is bi-right-exact, then projective objects are left-acyclic (adapted) for F .”
- Define the term “Deployed in the first variable”: A complex $P^\bullet$ is deployed in the first variable if the family $(P^\bullet, N^\bullet)$ is deployed for the multifunctor F for every $N^\bullet \in K(\mathcal{A}_2)$.
- Resolve the ♠ TODO regarding the existence of limits/colimits in **Sets** to ensure Ind/Pro categories are locally small.

- **Categorical Equivalence**

- Add a Remark confirming the equivalence of categories $D(\mathcal{A}_1) \times D(\mathcal{A}_2) \simeq D(\mathcal{A}_1 \times \mathcal{A}_2)$, noting that quasi-isomorphisms are defined component-wise.

- **derived tensor product**

- Do proposition 4.1.7. expose XVII

)

7.6. K -flat resolutions of chain complexes with respect to bi-additive functors.

Definition 7.6.1. Let $\mathcal{A}$ be an abelian category (Definition 3.0.4).

1. ([Spa88, After Theorem B]) Let $f : A^\bullet \rightarrow B^\bullet$ be a quasi-isomorphism of chain complexes (Definition 6.0.1) of objects in $\mathcal{A}$. We say that $A^\bullet$ is a *left resolution of $B^\bullet$* and that $B^\bullet$ is a *right resolution of $A^\bullet$* .
2. Given a chain complex (Definition 6.0.1) $K^\bullet$ of objects in $\mathcal{A}$ and a class $\mathcal{Y}$ of chain complexes of objects in $\mathcal{A}$, an *$\mathcal{Y}$ -left resolution* may refer to a left resolution $Y^\bullet \rightarrow K^\bullet$ where $Y^\bullet \in \mathcal{Y}$ and an *$\mathcal{Y}$ -right resolution* may refer to a right resolution $K^\bullet \rightarrow Y^\bullet$ where $Y^\bullet \in \mathcal{Y}$.

Definition 7.6.2. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 3.0.4), and let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a bi-additive functor.

Let us say that a complex (Definition 6.0.1) $K^\bullet$ of $\mathcal{A}$ (resp. of $\mathcal{B}$) is *K -flat with respect to F on the left (resp. right) for direct sum total complexes* if for every acyclic complex $M^\bullet$ of $\mathcal{B}$ (resp. of $\mathcal{A}$), the direct sum total complex $\mathrm{Tot}^\oplus(F(K^\bullet, M^\bullet))$ (resp. $\mathrm{Tot}^\oplus(F(M^\bullet, K^\bullet))$) is acyclic.

Similarly, let us say that $K^\bullet$ is K -flat with respect to F on the left (resp. right) for product total complexes if for every acyclic complex $M^\bullet$ of $\mathcal{B}$ (resp. of $\mathcal{A}$), the product total complex $\text{Tot}\Pi(F(K^\bullet, M^\bullet))$ (resp. $\text{Tot}\Pi(F(M^\bullet, K^\bullet))$) is acyclic.

Definition 7.6.3. Let $\mathcal{A}$ be an additive category (Definition 3.0.1).

Given (cochain) complexes (Definition 6.0.1) $K^\bullet, L^\bullet$ of objects in $\mathcal{A}$, the *internal Hom complex* $\text{Hom}^\bullet(K^\bullet, L^\bullet)$, also called the *(product) total Hom complex* or the *complex of morphisms from $K^\bullet$ to $L^\bullet$* and also denoted by notations such as $\underline{\text{Hom}}(X, Y)$ depending on the context, is the product total complex $\text{Tot}\Pi(\text{Hom}(K^\bullet, L^\bullet))$ where $\text{Hom}(K^\bullet, L^\bullet)$ is the double complex associated to (??) the biadditive functor $\text{Hom} : \mathcal{A}^{\text{op}} \times \mathcal{A} \rightarrow \text{Ab}$. More explicitly, $\text{Hom}^\bullet(K^\bullet, L^\bullet)$ is the complex of objects in Ab given by

- Objects: In each degree $n \in \mathbb{Z}$, the object is the product

$$\text{Hom}^n(K^\bullet, L^\bullet) = \prod_{p \in \mathbb{Z}} \text{Hom}_{\mathcal{A}}(K^p, L^{p+n}).$$

An element $f \in \text{Hom}^n(K^\bullet, L^\bullet)$ is thus a family of morphisms $(f_p)_{p \in \mathbb{Z}}$ where $f_p : K^p \rightarrow L^{p+n}$.

- Differential: The differential $d^n : \text{Hom}^n(K^\bullet, L^\bullet) \rightarrow \text{Hom}^{n+1}(K^\bullet, L^\bullet)$ is defined for each component $f = (f_p)$ by

$$(d^n f)_p = d_L^{p+n} \circ f_p - (-1)^n f_{p+1} \circ d_K^p,$$

where d_K and d_L are the differentials of the complexes $K^\bullet$ and $L^\bullet$ respectively.

Under this construction, the 0-cocycles of $\text{Hom}^\bullet(K^\bullet, L^\bullet)$ correspond precisely to the chain maps from $K^\bullet$ to $L^\bullet$, and the 0-coboundaries correspond to those chain maps which are null-homotopic.

If $K_\bullet$ and $L_\bullet$ are chain complexes written with homological conventions instead, then the internal Hom complex should be denoted by $\underline{\text{Hom}}_\bullet(K_\bullet, L_\bullet)$ instead; explicitly, $\text{Hom}_\bullet(K_\bullet, L_\bullet)$ is the complex of objects in Ab given by

- Objects: In each degree $n \in \mathbb{Z}$, the object is the product

$$\text{Hom}_n(K_\bullet, L_\bullet) = \prod_{p \in \mathbb{Z}} \text{Hom}_{\mathcal{A}}(K_p, L_{p+n}).$$

An element $f \in \text{Hom}_n(K_\bullet, L_\bullet)$ is a family of morphisms $(f_p)_{p \in \mathbb{Z}}$ such that $f_p : K_p \rightarrow L_{p+n}$.

- Differential: The differential $d_n : \text{Hom}_n(K_\bullet, L_\bullet) \rightarrow \text{Hom}_{n-1}(K_\bullet, L_\bullet)$ is defined for a family $f = (f_p)$ by

$$(d_n f)_p = d_L^{p+n} \circ f_p - (-1)^n f_{p-1} \circ d_K^p,$$

where $d_K^p : K_p \rightarrow K_{p-1}$ and $d_L^q : L_q \rightarrow L_{q-1}$ are the homological differentials.

As in the cohomological case, the 0-cycles of $\text{Hom}_\bullet(K_\bullet, L_\bullet)$ are precisely the chain maps between the complexes, and the 0-boundaries are the null-homotopic chain maps.

Definition 7.6.4. Let $\mathcal{A}$ be an abelian category. A chain complex $A^\bullet$ of objects in $\mathcal{A}$ is called *K-projective* (resp. *K-injective*) if for every acyclic complex $X^\bullet$ of objects in $\mathcal{A}$, the complex of abelian groups $\text{Hom}^\bullet(A^\bullet, X^\bullet)$ (Definition 7.6.3) (resp. $\text{Hom}^\bullet(X^\bullet, A^\bullet)$) is acyclic.

Equivalently, $A^\bullet$ is *K-injective* if and only if it is *K-flat* with respect to Hom on the left for product total complexes and $A^\bullet$ is *K-projective* if and only if it is *K-flat* with respect to Hom on the right for product total complexes.

Proposition 7.6.5 ([Spa88, 1.2. Proposition]). Let $\mathcal{A}$ be an abelian category (Definition 3.0.4) and let $A^\bullet \in \mathbf{Ch}(\mathcal{A})$ be a complex (Definition 6.0.1) such that $A^i = 0$ for all $i \neq 0$. The complex (Definition 6.0.1) $A^\bullet$ is *K-projective* (resp. *K-injective*) if and only if A^0 is a projective (resp. injective) object of $\mathcal{A}$.

Proposition 7.6.6 ([Spa88, 1.3. Proposition]). Let $\mathcal{A}$ be an abelian category (Definition 3.0.4). If two of the vertices of a distinguished triangle (Definition 7.1.2) of $K(\mathcal{A})$ (Definition 7.1.1) are *K-projective* (Definition 7.6.4) (resp. *K-injective* (Definition 7.6.4)), then so is the third.

Proposition 7.6.7 ([Spa88, 1.3. Proposition]). Let $\mathcal{A}$ be an abelian category (Definition 3.0.4). A complex $A^\bullet$ of objects in $\mathcal{A}$ is *K-projective* (resp. *K-injective*) if and only if $A^\bullet[n]$ is as well, where n is any integer.

Proposition 7.6.8 ([Spa88, 1.4. Proposition, 1.5. Proposition]). Let $\mathcal{A}$ be an abelian category (Definition 3.0.4), and let $K(\mathcal{A})$ (Definition 7.1.1) and $D(\mathcal{A})$ (Definition 7.3.1) be its homotopy and derived categories, respectively.

1. For every complex (Definition 6.0.1) $A^\bullet$ in $\mathcal{A}$, the following conditions are equivalent:
 - (a) $A^\bullet$ is *K-projective* (Definition 7.6.4).
 - (b) For every $S^\bullet \in K(\mathcal{A})$, the natural homomorphism

$$\text{Mor}_{K(\mathcal{A})}(A^\bullet, S^\bullet) \rightarrow \text{Mor}_{D(\mathcal{A})}(A^\bullet, S^\bullet)$$

is an isomorphism.

- (c) For every diagram in $K(\mathcal{A})$

$$\begin{array}{ccc} & & A^\bullet \\ & \swarrow g & \downarrow f \\ X^\bullet & \xrightarrow{s} & S^\bullet \end{array}$$

where s is a quasi-isomorphism, there exists a unique morphism $g : A^\bullet \rightarrow X^\bullet$ such that $s \circ g = f$ in $K(\mathcal{A})$.

- (d) For every quasi-isomorphism $u : S^\bullet \rightarrow A^\bullet$ in $K(\mathcal{A})$, there exists a morphism $v : A^\bullet \rightarrow S^\bullet$ such that $u \circ v = 1_{A^\bullet}$ in $K(\mathcal{A})$.
2. Similarly, for every $A^\bullet$ in $\mathcal{A}$, the following conditions are equivalent:
 - (a) $A^\bullet$ is *K-injective* (Definition 7.6.4).
 - (b) For every $S^\bullet \in K(\mathcal{A})$, the natural homomorphism

$$\text{Mor}_{K(\mathcal{A})}(S^\bullet, A^\bullet) \rightarrow \text{Mor}_{D(\mathcal{A})}(S^\bullet, A^\bullet)$$

is an isomorphism.

(c) For every diagram in $K(\mathcal{A})$

$$\begin{array}{ccc} S^\bullet & \xrightarrow{s} & X^\bullet \\ f \downarrow & \swarrow g & \\ A^\bullet & & \end{array}$$

where s is a quasi-isomorphism, there exists a unique morphism $g : X^\bullet \rightarrow A^\bullet$ such that $g \circ s = f$ in $K(\mathcal{A})$.

(d) For every quasi-isomorphism $u : A^\bullet \rightarrow S^\bullet$ in $K(\mathcal{A})$, there exists a morphism $v : S^\bullet \rightarrow A^\bullet$ such that $v \circ u = 1_{A^\bullet}$ in $K(\mathcal{A})$.

Proposition 7.6.9. Let $\mathcal{A}$ be an abelian category (Definition 3.0.4).

1. If $P_\bullet$ is a bounded above homological chain complex (or equivalently a bounded below cohomological chain complex) of projective objects, then $P_\bullet$ is K -projective (Definition 7.6.4).

$$\cdots \rightarrow P_{i+1} \rightarrow P_i \rightarrow 0 \rightarrow 0 \rightarrow \cdots$$

2. If $I^\bullet$ is a bounded below cohomological chain complex (or equivalently a bounded above homological chain complex) of injective objects, then $I^\bullet$ is K -injective (Definition 7.6.4).

$$\cdots \rightarrow 0 \rightarrow 0 \rightarrow I^i \rightarrow I^{i+1} \rightarrow \cdots$$

Proof. (♠ TODO: verify the following) We prove (1). By the characterization of K -projectivity, it suffices to show that for any acyclic complex $N^\bullet$, every chain map $f : P^\bullet \rightarrow N^\bullet$ is homotopic to zero ($f \sim 0$).

Assume without loss of generality that $P^n = 0$ for $n > 0$. We construct a homotopy $h^n : P^n \rightarrow N^{n-1}$ such that $f^n = d_N^{n-1}h^n + h^{n+1}d_P^n$ by descending induction on n .

Base Case: For $n = 0$, since $P^1 = 0$, the chain map condition $d_N^0 f^0 = f^1 d_P^0$ implies $d_N^0 f^0 = 0$. Thus $\text{im}(f^0) \subseteq \ker(d_N^0)$. By the acyclicity of $N^\bullet$, $\ker(d_N^0) = \text{im}(d_N^{-1})$. Because P^0 is projective, f^0 factors through the surjection $N^{-1} \rightarrow \text{im}(d_N^{-1})$, providing $h^0 : P^0 \rightarrow N^{-1}$ such that $d_N^{-1}h^0 = f^0$. Setting $h^1 = 0$, we have $f^0 = d_N^{-1}h^0 + h^1 d_P^0$.

Inductive Step: Suppose we have constructed $\{h^k\}_{k>n}$ satisfying the homotopy relation. Consider the map $g = f^n - h^{n+1}d_P^n : P^n \rightarrow N^n$. We check its image under d_N^n :

$$\begin{aligned} d_N^n g &= d_N^n f^n - d_N^n h^{n+1} d_P^n \\ &= f^{n+1} d_P^n - (f^{n+1} - h^{n+2} d_P^{n+1}) d_P^n \\ &= f^{n+1} d_P^n - f^{n+1} d_P^n + h^{n+2} d_P^{n+1} d_P^n = 0. \end{aligned}$$

Thus $\text{im}(g) \subseteq \ker(d_N^n)$. By acyclicity, $\ker(d_N^n) = \text{im}(d_N^{n-1})$. Since P^n is projective, g factors through the surjection d_N^{n-1} , yielding $h^n : P^n \rightarrow N^{n-1}$ such that $d_N^{n-1}h^n = g = f^n - h^{n+1}d_P^n$.

This construction provides a well-defined sequence of maps $(h^n)_{n \in \mathbb{Z}}$ defining a homotopy $h : f \sim 0$. Thus $\text{Mor}_{K(\mathcal{A})}(P^\bullet, N^\bullet) = 0$, proving $P^\bullet$ is K -projective. The proof for (2) is dual, using ascending induction starting from the lowest non-zero degree of $I^\bullet$. $\square$

Theorem 7.6.10 ([Spa88, 3.4. Theorem, 3.8. Proposition]). 1. Let $\mathcal{A}$ be an AB5 (Definition 3.0.5) abelian category (Definition 3.0.4). Suppose that $\mathfrak{P}$ is a class of objects in $\mathbf{Ch}(\mathcal{A})$ satisfying the following:

Every (cohomologically) bounded above complex $A^\bullet \in \mathbf{Ch}(\mathcal{A})$ has a left $\mathfrak{P}$ -resolution (Definition 7.6.1).

Then every complex in $\mathbf{Ch}(\mathcal{A})$ has a left $\mathfrak{P}$ -resolution.

2. Dually, let $\mathcal{A}$ be an AB5* (Definition 3.0.5) abelian category (Definition 3.0.4). Suppose that $\mathfrak{I}$ is a class of objects in $\mathbf{Ch}(\mathcal{A})$ satisfying the following:

Every (cohomologically) bounded below complex $A^\bullet \in \mathbf{Ch}(\mathcal{A})$ has a left $\mathfrak{I}$ -resolution (Definition 7.6.1).

Then every complex in $\mathbf{Ch}(\mathcal{A})$ has a right $\mathfrak{I}$ -resolution.

Corollary 7.6.11 ([Spa88, 3.5. Corollary, 3.9. Corollary]). 1. Let $\mathcal{A}$ be an AB5 (Definition 3.0.5) abelian category (Definition 3.0.4). If $\mathcal{A}$ has enough projectives (Definition 5.4.4), then every complex in $\mathcal{C}$ has a left K -projective (Definition 7.6.4) resolution (Definition 7.6.1). (♠ TODO: bounded complex of projectives is K -projective)

2. Dually, let $\mathcal{A}$ be an AB5* (Definition 3.0.5) abelian category (Definition 3.0.4). If $\mathcal{A}$ has enough injectives (Definition 5.4.4), then every complex in $\mathcal{C}$ has a right K -injective (Definition 7.6.4) resolution (Definition 7.6.1).

Proof. 1. Since $\mathcal{A}$ has enough projectives, any bounded above homological complex in $\mathcal{A}$ has a left-resolution ?? by a bounded above homological complex of projective objects, which is K -projective by Proposition 7.6.9. Therefore, any complex in $\mathcal{A}$ has a left-resolution by a K -projective object by Theorem 7.6.10

2. This is dual to the previous part.

□

Theorem 7.6.12 ([Ser03, Theorem 3.13.]). Let $\mathcal{A}$ be a Grothendieck category (Definition 3.0.5).

For every chain complex (Definition 6.0.1) $A^\bullet$ in $\mathcal{A}$, there is a right resolution (Definition 7.6.1) $A^\bullet \rightarrow I^\bullet$ such that $I^\bullet$ is K -injective (Definition 7.6.4).

Lemma 7.6.13. Let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a bi-additive functor where $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are additive categories (Definition 3.0.1). Let J be a category (Definition 1.0.1).

1. Assume that $F(X, -) : \mathcal{B} \rightarrow \mathcal{C}$ preserves colimits (Definition 1.2.2) of diagrams of shape J (Definition 1.2.1) for all objects $X \in \mathcal{A}$. Then the induced functor $F(X^\bullet, -) : \mathbf{Ch}(\mathcal{B}) \rightarrow \mathbf{DC}(\mathcal{C})$ (??) (Definition 6.0.1) also preserves colimits of diagrams of shape J for all objects $X^\bullet \in \mathbf{Ch}(\mathcal{A})$.
2. Dually, assume that $F(-, Y) : \mathcal{A} \rightarrow \mathcal{C}$ preserves colimits (Definition 1.2.2) of diagrams of shape J (Definition 1.2.1) for all objects $Y \in \mathcal{B}$. Then the induced functor $F(-, Y^\bullet) : \mathbf{Ch}(\mathcal{A}) \rightarrow \mathbf{DC}(\mathcal{C})$ (??) (Definition 6.0.1) also preserves colimits of diagrams of shape J for all objects $Y^\bullet \in \mathbf{Ch}(\mathcal{B})$.
3. Assume that $F(X, -) : \mathcal{B} \rightarrow \mathcal{C}$ preserves limits (Definition 1.2.2) of diagrams of shape J (Definition 1.2.1) for all objects $X \in \mathcal{A}$. Then the induced functor $F(X^\bullet, -) :$

$\mathbf{Ch}(\mathcal{B}) \rightarrow \mathbf{DC}(\mathcal{C})$ (??) (Definition 6.0.1) also preserves limits of diagrams of shape J for all objects $X^\bullet \in \mathbf{Ch}(\mathcal{A})$.

4. Dually, assume that $F(-, Y) : \mathcal{A} \rightarrow \mathcal{C}$ preserves limits (Definition 1.2.2) of diagrams of shape J (Definition 1.2.1) for all objects $Y \in \mathcal{B}$. Then the induced functor $F(-, Y^\bullet) : \mathbf{Ch}(\mathcal{A}) \rightarrow \mathbf{DC}(\mathcal{C})$ (??) (Definition 6.0.1) also preserves limits of diagrams of shape J for all objects $Y^\bullet \in \mathbf{Ch}(\mathcal{B})$.

Proof. We prove that if $F(X, -) : \mathcal{B} \rightarrow \mathcal{C}$ preserves colimits of diagrams of shape J for all $X \in \mathcal{A}$, then so does $F(X^\bullet, -) : \mathbf{Ch}(\mathcal{B}) \rightarrow \mathbf{DC}(\mathcal{C})$ for all $X^\bullet \in \mathbf{Ch}(\mathcal{A})$. Given $X^\bullet \in \mathbf{Ch}(\mathcal{A})$ and $Y^\bullet \in \mathbf{Ch}(\mathcal{B})$, the term of $F(X^\bullet, Y^\bullet)$ at position (p, q) is $F(X^p, Y^q)$. Let $J \rightarrow \mathbf{Ch}(\mathcal{B}) : j \mapsto Y_{(j)}^\bullet$ be some diagram (Definition 1.2.1) whose colimit is $Y^\bullet$. In fact, since colimits of chain complexes are computed termwise (??), there are canonical isomorphisms

$$Y^q \cong \operatorname{colim}_{j \in J} Y_{(j)}^q$$

in $\mathcal{B}$.

We wish to show that $F(X^\bullet, Y^\bullet)$ is the colimit of the diagram $j \mapsto F(X^\bullet, Y_{(j)}^\bullet)$ in $\mathbf{DC}(\mathcal{C})$. The colimit in $\mathbf{DC}(\mathcal{C})$ is also computed termwise (note that $\mathbf{DC}(\mathcal{C})$ is identifiable as $\mathbf{Ch}(\mathbf{Ch}(\mathcal{C}))$), it suffices to verify the isomorphism at each position (p, q) and ensure it respects the differentials.

Consider the term at position (p, q) . The functor $F(X^\bullet, -)$ maps the colimit diagram in $\mathbf{Ch}(\mathcal{B})$ to a cocone (Definition 1.2.2) in $\mathbf{DC}(\mathcal{C})$. This induces a canonical comparison morphism:

$$\phi : \operatorname{colim}_{j \in J} F(X^\bullet, Y_{(j)}^\bullet) \longrightarrow F(X^\bullet, \operatorname{colim}_{j \in J} Y_{(j)}^\bullet) = F(X^\bullet, Y^\bullet).$$

At the position (p, q) , this morphism is the map:

$$\phi^{p,q} : \operatorname{colim}_{j \in J} F(X^p, Y_{(j)}^q) \longrightarrow F(X^p, Y^q).$$

By hypothesis, $F(X^p, -) : \mathcal{B} \rightarrow \mathcal{C}$ preserves colimits of shape J . Therefore, $\phi^{p,q}$ is an isomorphism for all $p, q \in \mathbb{Z}$.

Since a morphism of double complexes is an isomorphism if and only if it is an isomorphism at every term (p, q) , we conclude that ϕ is an isomorphism. Thus, $F(X^\bullet, -)$ preserves colimits of shape J .

□

Lemma 7.6.14. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories (Definition 3.0.4), and let $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ be a bi-additive functor. Assume that $\mathcal{C}$ is Ab5 (Definition 3.0.5).

1. Suppose that $F(X, -) : \mathcal{B} \rightarrow \mathcal{C}$ preserves filtered colimits (Definition 1.2.12) for all objects $X \in \mathcal{A}$. Let $P^\bullet$ be a (cohomologically) bounded above complex of objects of $\mathcal{A}$ that are flat with respect to F on the left. The complex $P^\bullet$ is K -flat with respect to F on the left for direct sum total complexes (Definition 7.6.2).
2. Suppose that $F(-, Y) : \mathcal{A} \rightarrow \mathcal{C}$ preserves filtered colimits (Definition 1.2.12) for all objects $Y \in \mathcal{B}$. Let $P^\bullet$ be a bounded above complex of objects in $\mathcal{B}$ that are flat with respect to F on the right. The complex $P^\bullet$ is K -flat with respect to F on the right for direct sum total complexes.

Proof. We show the claimed statement in the case that $P^\bullet$ is a complex of flat objects of $\mathcal{A}$ with respect to F on the left; the other case can be argued symmetrically. Let $M^\bullet$ be an acyclic complex of objects in $\mathcal{A}$. Writing $\tau_{\leq k} M^\bullet$ for the brutal truncation of $M^\bullet$, note that $M^\bullet$ is the filtered colimit of these truncations (??):

$$M^\bullet = \varinjlim_k \tau_{\leq k} M^\bullet.$$

Since $F(X, -) : \mathcal{B} \rightarrow \mathcal{C}$ preserves filtered colimits, so does $F(X^\bullet, -) : \mathbf{Ch}(\mathcal{B}) \rightarrow \mathbf{DC}(\mathcal{C})$ (Definition 6.0.1) by Lemma 7.6.13. Therefore,

$$\mathrm{Tot}^\oplus F(P^\bullet, M^\bullet) \cong \mathrm{Tot}^\oplus F(P^\bullet, \varinjlim_k \tau_{\leq k} M^\bullet) \cong \varinjlim_k \mathrm{Tot}^\oplus F(P^\bullet, \tau_{\leq k} M^\bullet).$$

Since $\tau_{\leq k} M^\bullet$ is bounded above and is of flat objects, the double complex $F(P^\bullet, \tau_{\leq k} M^\bullet)$ is bounded above, and each row is acyclic. Since $\mathcal{C}$ satisfies Ab5, $\mathrm{Tot}^\oplus F(P^\bullet, \tau_{\leq k} M^\bullet)$ is acyclic by Lemma ???. That $\mathcal{C}$ satisfies AB5 further implies that

$$H^n(\mathrm{Tot}^\oplus F(P^\bullet, M^\bullet)) \cong \varinjlim_k H^n \mathrm{Tot}^\oplus F(P^\bullet, \tau_{\leq k} M^\bullet) = 0,$$

i.e. $\mathrm{Tot}^\oplus F(P^\bullet, M^\bullet)$ is acyclic. Therefore, $P^\bullet$ is K -flat as desired. $\square$

7.7. General conditions for defining derived functors.

Proposition 7.7.1 (see [Sta25, Tag 06XN, 05S8]). Suppose that $F : \mathcal{D} \rightarrow \mathcal{D}'$ is an exact functor (Definition 7.1.3) of triangulated categories (Definition 7.1.2) and that S is a saturated multiplicative system (Definition 7.2.3) in $\mathcal{D}$ compatible with (Definition 7.1.4) the structure of the triangulated category (Definition 7.1.2) on $\mathcal{D}$, i.e. $s : X \rightarrow Y$ is in S if and only if $s[1] : X[1] \rightarrow Y[1]$ is in S and for every commutative diagram in $\mathcal{D}$ where the rows are distinguished triangles:

$$\begin{array}{ccccccc} X & \xrightarrow{f} & Y & \rightarrow & Z & \rightarrow & X[1] \\ \downarrow^{s_1} & & \downarrow^{s_2} & & \exists \downarrow^{s_3} & & \downarrow^{s_1[1]} \\ X' & \xrightarrow{f'} & Y' & \rightarrow & Z' & \rightarrow & X'[1] \end{array}$$

if $s_1, s_2 \in S$, there exists a morphism $s_3 \in S$ making the entire diagram commute.

If there exists a subcollection $\mathcal{I} \subset \mathrm{Ob}(\mathcal{D})$ such that

1. for all $X \in \mathrm{Ob}(\mathcal{D})$ there exists $s : X \rightarrow X'$ in S with $X' \in \mathcal{I}$, and
2. for every arrow $s : X \rightarrow X'$ in S with $X, X' \in \mathcal{I}$ the map $F(s) : F(X) \rightarrow F(X')$ is an isomorphism,

then RF (Definition 7.5.4) is everywhere defined (Definition 7.5.5) and every $X \in \mathcal{I}$ computes RF .

Dually, if there exists a subcollection $\mathcal{P} \subset \mathrm{Ob}(\mathcal{D})$ such that

1. for all $X \in \mathrm{Ob}(\mathcal{D})$ there exists $s : X' \rightarrow X$ in S with $X' \in \mathcal{P}$, and
2. for every arrow $s : X \rightarrow X'$ in S with $X, X' \in \mathcal{P}$ the map $F(s) : F(X) \rightarrow F(X')$ is an isomorphism,

then LF is everywhere defined and every $X \in \mathcal{P}$ computes LF . (♠ TODO: Proposition 7.5.6 should be relevant)

Proposition 7.7.2 (see [Sta25, Tag 070K]). Let $\mathcal{A}$ be an abelian category (Definition 3.0.4). Assume that every complex (Definition 6.0.1) of objects in $\mathcal{A}$ has a right resolution (Definition 7.6.1) by a K -injective (Definition 7.6.4) complex. For any exact functor (Definition 7.1.3) $F : K(\mathcal{A}) \rightarrow \mathcal{D}'$ (Definition 7.1.1) of triangulated categories (Definition 7.1.2), the right derived functor (Definition 7.5.8)

$$RF : D(\mathcal{A}) \rightarrow \mathcal{D}'$$

is defined (Definition 7.5.5) and $RF(I^\bullet) = F(I^\bullet)$ for K -injective complexes $I^\bullet$.

Proof. By Proposition 7.7.1, it suffices to show that if $I^\bullet \rightarrow J^\bullet$ is a quasi-isomorphism of K -injective complexes, then $F(I^\bullet) \rightarrow F(J^\bullet)$ is an isomorphism in $\mathcal{D}'$ (♠ TODO: continue). $\square$

7.7.1. Adjunction of derived functors of adjoint functors.

Proposition 7.7.3 ([Sta25, Tag 0FND]). Let $F : \mathcal{D} \rightarrow \mathcal{D}'$ and $G : \mathcal{D}' \rightarrow \mathcal{D}$ be exact functors of triangulated categories (Definition 7.1.3). Assume F is right adjoint (Definition 1.1.13) to G . Let S and S' respectively be multiplicative systems (Definition 7.2.3) for $\mathcal{D}$ and $\mathcal{D}'$ compatible with the triangulated structures (Definition 7.1.4). Let $K \in \text{Ob}(\mathcal{D})$ and $M \in \text{Ob}(\mathcal{D}')$. If RF (Definition 7.5.4) is defined (Definition 7.5.5) at K and LG is defined at M , then there is a canonical isomorphism

$$\text{Hom}_{\mathcal{D}'[(S')^{-1}]}(M, RF(K)) = \text{Hom}_{\mathcal{D}[S^{-1}]}(LG(M), K)$$

which is functorial (Definition 1.1.1) in both variables on the triangulated subcategories (Definition 7.1.2) of $\mathcal{D}[S^{-1}]$ and $\mathcal{D}'[(S')^{-1}]$ where RF and LG are defined.

Proposition 7.7.4 ([Sta25, Tag 0DVC]). Let $F : \mathcal{A} \rightarrow \mathcal{B}$ and $G : \mathcal{B} \rightarrow \mathcal{A}$ be functors (Definition 1.1.1) of abelian categories (Definition 3.0.4) such that F is a right adjoint (Definition 1.1.13) to G ; in particular, F and G are additive (Definition 3.0.3) (??). Let $K^\bullet$ be a complex (Definition 6.0.1) of $\mathcal{A}$ and let $M^\bullet$ be a complex of $\mathcal{B}$. If RF (Definition 7.5.8) is defined (Definition 7.5.5) at $K^\bullet$ and LG is defined at $M^\bullet$, then there is a canonical isomorphism

$$\text{Hom}_{D(\mathcal{B})}(M^\bullet, RF(K^\bullet)) = \text{Hom}_{D(\mathcal{A})}(LG(M^\bullet), K^\bullet)$$

This isomorphism is functorial (Definition 1.1.1) in both variables on the triangulated subcategories (Definition 7.1.2) of $D(\mathcal{A})$ (Definition 7.3.1) and $D(\mathcal{B})$ where RF and LG are defined.

Proposition 7.7.5 ([Sta25, Tag 09T5]). Let $F : \mathcal{A} \rightarrow \mathcal{B}$ and $G : \mathcal{B} \rightarrow \mathcal{A}$ be functors (Definition 1.1.1) of abelian categories (Definition 3.0.4) such that F is a right adjoint (Definition 1.1.13) to G . If the derived functors (Definition 7.5.8) $RF : D(\mathcal{A}) \rightarrow D(\mathcal{B})$ and $LG : D(\mathcal{B}) \rightarrow D(\mathcal{A})$ exist, then RF is a right adjoint to LG .

7.8. Having enough K -injectives and flats for sheaves of modules.

Theorem 7.8.1. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site (Definition 5.4.9) where the underlying site $(\mathcal{C}, J)$ is essentially small. In particular, $\text{Mod}_{\mathcal{O}}$ is an abelian category (Definition 3.0.4) and in fact is a Grothendieck category (Definition 3.0.5) (??).

For every chain complex (Definition 6.0.1) $A^{\bullet}$ in $\text{Mod}_{\mathcal{O}}$ (Definition 5.3.9) (resp. ${}_{\mathcal{O}}\text{Mod}$), there is a right resolution (Definition 7.6.1) $A^{\bullet} \rightarrow I^{\bullet}$ such that $I^{\bullet}$ is K -injective (Definition 7.6.4).

Proof. This follows from Theorem 7.6.12. $\square$

Definition 7.8.2. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site (Definition 5.4.9) whose underlying site (Definition 5.1.13) is essentially small.

A left (resp. right) $\mathcal{O}$ -module (Definition 5.3.9) $\mathcal{F}$ is called *flat* if the functor $- \otimes_{\mathcal{O}} \mathcal{F} : \text{Mod}_{\mathcal{O}} \rightarrow \text{Sh}(\mathcal{C}, J; \text{Ab})$ (resp. $\mathcal{F} \otimes_{\mathcal{O}} - : {}_{\mathcal{O}}\text{Mod} \rightarrow \text{Sh}(\mathcal{C}, J; \text{Ab})$) (Definition 5.1.52) (Definition 5.1.32) is exact (Definition 3.0.7).

Equivalently, $\mathcal{F}$ is flat if and only if it is flat with respect to $\otimes$.

Proposition 7.8.3. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site (Definition 5.4.9) whose underlying site (Definition 5.1.13) is essentially small.

The categories $\text{Mod}_{\mathcal{O}}$ and ${}_{\mathcal{O}}\text{Mod}$ have enough flat $\mathcal{O}$ -modules (Definition 7.8.2) on the left.
(♠ TODO: prove)

Proposition 7.8.4. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site (Definition 5.4.9) whose underlying site (Definition 5.1.13) is essentially small.

There are enough K -flat resolutions to the left of chain complexes (Definition 6.0.1) in of objects in $\text{Mod}_{\mathcal{O}}$ and ${}_{\mathcal{O}}\text{Mod}$. More precisely, for any chain complex (Definition 6.0.1) $A_{\bullet} \in \mathbf{Ch}(\text{Mod}_{\mathcal{O}})$ (resp. $\mathbf{Ch}({}_{\mathcal{O}}\text{Mod})$), there is a K -flat (Definition 7.6.2) (with respect to $\otimes : \text{Mod}_{\mathcal{O}} \times {}_{\mathcal{O}}\text{Mod} \rightarrow \text{Sh}(\mathcal{C}, J; \text{Ab})$) chain complex $F_{\bullet} \in \text{Mod}_{\mathcal{O}}$ and a left resolution (Definition 7.6.1) $F_{\bullet} \rightarrow A_{\bullet}$.

Proof. Since $\text{Mod}_{\mathcal{O}}$ and ${}_{\mathcal{O}}\text{Mod}$ have enough flats on the left by Proposition 7.8.3, every (homologically) bounded below complex in $\text{Mod}_{\mathcal{O}}$ and ${}_{\mathcal{O}}\text{Mod}$ has a left resolution by a (homologically) bounded below complex of flat objects; a (homologically) bounded below complex of flats is K -flat with respect to $\otimes$ by Lemma 7.6.14. To see that Lemma 7.6.14 applies, note that $\otimes$ is a left adjoint by Theorem 5.1.54 and hence commutes with colimits by Theorem 1.2.29, and note that $\text{Sh}(\mathcal{C}, J; \text{Ab})$ is AB5. Theorem 7.6.10 then applies to conclude that chain complex of objects in $\text{Mod}_{\mathcal{O}}$ and ${}_{\mathcal{O}}\text{Mod}$ has a left K -flat resolution. $\square$

7.9. Definitions of derived functors for modules of ringed sites.

Notation 7.9.1. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site (Definition 5.4.9) where the underlying site $(\mathcal{C}, J)$ is essentially small or let $(\mathcal{C}, \mathcal{O})$ be a ringed topos (Definition 5.4.11). In particular, $\text{Mod}_{\mathcal{O}}$ is an abelian category (Definition 3.0.4).

Denote by $D(\mathcal{O})$, $D^+(\mathcal{O})$, $D^-(\mathcal{O})$, $D^b(\mathcal{O})$ the derived categories $D(\mathcal{A})$, $D^+(\mathcal{A})$, $D^-(\mathcal{A})$, $D^b(\mathcal{A})$ (Definition 7.3.1) where $\mathcal{A}$ is either the abelian category ${}_{\mathcal{O}}\text{Mod}$ or $\text{Mod}_{\mathcal{O}}$.

Definition 7.9.2. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site (Definition 5.4.9) whose underlying site (Definition 5.1.13) is essentially small.

Given (cochain) complexes (Definition 6.0.1) $K^\bullet, L^\bullet$ of (say right) $\mathcal{O}$ -modules (Definition 5.3.9), the *internal sheaf/sheafy Hom complex* $\mathcal{H}om^\bullet(K^\bullet, L^\bullet)$, also called the *(product) total sheaf Hom complex*, but not to be confused with the internal hom complex (Definition 7.6.3) $\text{Hom}^\bullet(K^\bullet, L^\bullet)$, is the product total complex $\text{Tot}^\Pi(\mathcal{H}om(K^\bullet, L^\bullet))$, where $\mathcal{H}om(K^\bullet, L^\bullet)$ is the double complex associated to (??) the sheaf hom functor (Definition 5.1.53)

$$\mathcal{H}om : \text{Mod}_{\mathcal{O}}^{\text{op}} \times \text{Mod}_{\mathcal{O}} \rightarrow \text{Sh}(\mathcal{C}, J; \mathbb{Z})$$

(Definition 5.3.9) (Definition 5.1.32).

More explicitly, $\mathcal{H}om^\bullet(K^\bullet, L^\bullet)$ is the complex of objects in $\text{Sh}(\mathcal{C}, J; \mathbb{Z})$ given by

- **Objects:** In each degree $n \in \mathbb{Z}$, the sheaf of $\mathbb{Z}$ -modules is the product

$$\mathcal{H}om^n(K^\bullet, L^\bullet) = \prod_{p \in \mathbb{Z}} \mathcal{H}om_{\mathcal{O}}(K^p, L^{p+n}).$$

A local section $f \in \mathcal{H}om^n(K^\bullet, L^\bullet)(U)$ for an object $U \in \mathcal{C}$ is a family of $\mathcal{O}|_U$ -linear morphisms $f_p : K^p|_U \rightarrow L^{p+n}|_U$.

- **Differential:** The differential $d^n : \mathcal{H}om^n(K^\bullet, L^\bullet) \rightarrow \mathcal{H}om^{n+1}(K^\bullet, L^\bullet)$ is defined for a family of local morphisms $f = (f_p)$ by

$$(d^n f)_p = d_L^{p+n} \circ f_p - (-1)^n f_{p+1} \circ d_K^p,$$

where $d_K^p : K^p \rightarrow K^{p+1}$ and $d_L^q : L^q \rightarrow L^{q+1}$ are the cohomological differentials of the complexes.

Definition 7.9.3 (cf. [Sta25, Tag 08J7] for a discussion when $\mathcal{O}$ is assumed commutative). Let $(\mathcal{C}, J, \mathcal{O})$ be an essentially small ringed site (Definition 5.4.9) with a pretopology (Definition 5.1.13). (♠ TODO: the “pretopology” part should be loosened)

The *derived (sheafy/sheaf) internal hom functor* is the functor $R\mathcal{H}om : D(\mathcal{O})^{\text{op}} \times D(\mathcal{O}) \rightarrow D(\text{Sh}(\mathcal{C}, J; \mathbb{Z}))$ (Notation 7.9.1), also denoted by notations such as $R\mathcal{H}om(\mathcal{M}, \mathcal{N})$, $R\mathcal{H}om(\mathcal{M}, \mathcal{N})$, or $R\mathcal{H}om(\mathcal{M}, \mathcal{N})$, defined as the right derived functor (Definition 7.5.8) of the sheaf hom functor (Definition 5.1.53)

$$\mathcal{H}om : \text{Mod}_{\mathcal{O}}^{\text{op}} \times \text{Mod}_{\mathcal{O}} \rightarrow \text{Sh}(\mathcal{C}, J; \mathbb{Z}).$$

(♠ TODO: 07A5 for justifying why such a thing exists)

(♠ TODO: This is defined because...) Explicitly, given objects $L, M \in D(\mathcal{O})$, $R\mathcal{H}om(L, M)$ is given as follows: let $\mathcal{L}^\bullet$ be any complex of $\mathcal{O}$ -modules representing L and let $\mathcal{I}^\bullet$ be any (right) K -injective (Definition 7.6.4) resolution (Definition 7.6.1) of M , which exists by Theorem 7.8.1. The object $R\mathcal{H}om(L, M)$ is then given as the object of $D(\text{Sh}(\mathcal{C}, J; \mathbb{Z}))$ represented by the internal sheaf hom object (Definition 7.9.2) $\mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet)$.

Definition 7.9.4 (cf. [Sta25, Tag 06YU] for a simplified discussion when $\mathcal{O}$ is assumed commutative). Let $(\mathcal{C}, J, \mathcal{O})$ be an essentially small ringed site (Definition 5.4.9) with a pretopology (Definition 5.1.13). (♠ TODO: the “pretopology” part should be loosened)

The **derived tensor product** is the functor

$$- \otimes_{\mathcal{O}}^L - : D(\text{Mod}_{\mathcal{O}}) \times D({}_{\mathcal{O}}\text{Mod}) \rightarrow D(\text{Sh}(C, J; \text{Ab}))$$

(Definition 5.3.9) (Notation 7.9.1) also denoted by notations such as $\overset{L}{\otimes}$ defined as the left derived functor (Definition 7.5.8) of the tensor product functor (Definition 5.1.52)

$$- \otimes_{\mathcal{O}} - : \text{Mod}_{\mathcal{O}} \times_{\mathcal{O}} \text{Mod} \rightarrow \text{Sh}(C, J; \text{Ab})$$

Explicitly, given objects $L, M \in D(\mathcal{O})$, $L \otimes_{\mathcal{O}} M$ is given as follows: let $\mathcal{L}^{\bullet}$ be any complex of right $\mathcal{O}$ -modules representing L and let $\mathcal{M}^{\bullet}$ be any complex of left $\mathcal{O}$ -modules representing M , such that $\mathcal{L}^{\bullet}$ is K -flat (Definition 7.6.2) with respect to $\otimes_{\mathcal{O}}$ on the left for direct sum total complexes or $\mathcal{M}^{\bullet}$ is K -flat with respect to $\otimes_{\mathcal{O}}$ on the left for direct sum total complexes; note that such complexes exist by Proposition 7.8.4. The object $L \otimes_{\mathcal{O}}^L M$ is then given as the object of $D(\text{Sh}(C, J; \text{Ab}))$ represented by the total complex $\text{Tot}^{\oplus}(\mathcal{L}^{\bullet} \otimes_{\mathcal{O}} \mathcal{M}^{\bullet})$. (♠ TODO: justify definedness, i.e. why does this actually define the derived functor)

Theorem 7.9.5 (cf. [Sta25, Tag 08J9]). Let $(\mathcal{C}, J, \mathcal{O})$ be an essentially small ringed site (Definition 5.4.9) with a pretopology (Definition 5.1.13). (♠ TODO: this pretopology condition should be loosened)

Let K, L, M be objects of $D(\mathcal{O})$ (Notation 7.9.1). There is a canonical isomorphism

$$R\mathcal{H}om(K, R\mathcal{H}om(L, M)) \cong R\mathcal{H}om(K \otimes_{\mathcal{O}}^L L, M)$$

(Definition 7.9.3) (Definition 7.9.4) in $D(\mathcal{O})$ functorial in K, L, M . Taking the zeroth sheaf cohomology (or equivalently global sections) of both sides, one obtains the adjunction

$$\text{Hom}_{D(\mathcal{O})}(K, R\mathcal{H}om(L, M)) \cong \text{Hom}_{D(\mathcal{O})}(K \otimes_{\mathcal{O}}^L L, M).$$

Definition 7.9.6. Let $((\mathcal{C}, J), \mathcal{O}_{\mathcal{C}})$ and $((\mathcal{D}, K), \mathcal{O}_{\mathcal{D}})$ be ringed sites (Definition 5.4.9).

A **morphism of ringed sites** $f : ((\mathcal{C}, J), \mathcal{O}_{\mathcal{C}}) \rightarrow ((\mathcal{D}, K), \mathcal{O}_{\mathcal{D}})$ is a pair $(f, f^{\#})$ where:

1. $f : (\mathcal{C}, J) \rightarrow (\mathcal{D}, K)$ is a morphism of sites (Definition 5.2.5).
2. $f^{\#} : \mathcal{O}_{\mathcal{D}} \rightarrow f_*\mathcal{O}_{\mathcal{C}}$ (Definition 5.3.2) is a morphism of sheaves of rings (Definition 5.1.32) on the site $(\mathcal{D}, K)$.

Equivalently, the morphism $f^{\#}$ a morphism can be described by the adjoint morphism (Theorem 5.3.4) $f^{\#} : f^{-1}\mathcal{O}_{\mathcal{D}} \rightarrow \mathcal{O}_{\mathcal{C}}$ of sheaves of rings on $(\mathcal{C}, J)$, where f^{-1} is the inverse image functor (Definition 5.3.3) of sheaves.

Definition 7.9.7 (cf. [Sta25, Tag 07A5]). Let $f : (\mathcal{C}, \mathcal{O}_{\mathcal{C}}) \rightarrow (\mathcal{D}, \mathcal{O}_{\mathcal{D}})$ be a morphism of ringed topoi (Definition 5.4.12).

The **(total) derived direct image** or **(total) derived pushforward functor** Rf_* is the functor

$$Rf_* : D(\mathcal{O}_{\mathcal{C}}) \rightarrow D(\mathcal{O}_{\mathcal{D}})$$

(Notation 7.9.1) defined as the right derived functor (Definition 7.5.8) of the direct image functor (Definition 5.1.56) f_* . Explicitly, given an object $K \in D(\mathcal{O}_{\mathcal{C}})$, $Rf_*(K)$ is given as follows: Let $\mathcal{I}^{\bullet}$ be any (right) K -injective (Definition 7.6.4) resolution (Definition 7.6.1) of M , which exists by Theorem 7.8.1. The object $Rf_*(K)$ is then given as the object of

$D(\mathcal{O}_{\mathcal{D}})$ represented by the chain complex $f_*(\mathcal{I}^\bullet)$ (Proposition 7.4.2). This is well defined by Proposition 7.7.2.

Definition 7.9.8 (cf. [Sta25, Tag 06YY]). Let $f : (\mathcal{C}, \mathcal{O}_{\mathcal{C}}) \rightarrow (\mathcal{D}, \mathcal{O}_{\mathcal{D}})$ be a morphism of ringed topoi (Definition 5.4.12).

The *(total) derived direct image* or *(total) derived pushforward functor* Lf^* is the functor

$$Lf^* : D(\mathcal{O}_{\mathcal{D}}) \rightarrow D(\mathcal{O}_{\mathcal{C}})$$

(Notation 7.9.1) defined as the left derived functor (Definition 7.5.8) of the inverse image functor (Definition 5.1.56) f^* . Explicitly, given an object $K \in D(\mathcal{O}_{\mathcal{D}})$, $Lf^*(K)$ is given as follows: Let $\mathcal{F}^\bullet$ be any (left) K -flat (Definition 7.6.2) (with respect to $\otimes_{\mathcal{O}_{\mathcal{D}}}$ either on the left or right, depending on whether we are considering right or left $\mathcal{O}_{\mathcal{D}}$ -modules) resolution (Definition 7.6.1) of M , which exists by Proposition 7.8.4 (♠ TODO: justify the well definedness)

Theorem 7.9.9. Let $f : (\mathcal{C}, \mathcal{O}_{\mathcal{C}}) \rightarrow (\mathcal{D}, \mathcal{O}_{\mathcal{D}})$ be a morphism of ringed topoi (Definition 5.4.12).

The functors Rf_* (Definition 7.9.7) and Lf^* (Definition 7.9.8) are adjoint, i.e. for all $L \in D(\mathcal{O}_{\mathcal{C}})$ and $M \in D(\mathcal{O}_{\mathcal{D}})$, we have natural isomorphisms

$$\mathrm{Hom}_{D(\mathcal{O}_{\mathcal{C}})}(Lf^*M, L) \cong \mathrm{Hom}_{D(\mathcal{O}_{\mathcal{D}})}(M, Rf_*L)$$

(Definition 7.9.7) (Definition 7.9.8)

♠ TODO: total derived sheaf cohomology

8. SIMPLICIAL CATEGORIES

♠ TODO:)

Notation 8.0.1. Let $\mathbf{Cat}$ denote the category of (small) categories. Let Δ denote the simplex category, whose objects are finite nonempty totally ordered sets $[n] := \{0, 1, \dots, n\}$ for $n \geq 0$, and whose morphisms are all order-preserving functions $\theta : [m] \rightarrow [n]$. Let Δ^{op} denote its opposite category.

Definition 8.0.2. Let V be a real vector space of finite dimension. A *k -simplex in topology* (or a *geometric k -simplex*) is the convex hull of $k + 1$ affinely independent points $v_0, v_1, \dots, v_k \in V$, and is denoted by

$$[v_0, v_1, \dots, v_k] := \left\{ \sum_{i=0}^k t_i v_i \mid t_i \geq 0, \sum_{i=0}^k t_i = 1 \right\}.$$

It is also standard to talk of the *standard topological n -simplex* — the topological space, often denoted by notations such as Δ^n or $|\Delta^n|$ defined as the subset of Euclidean space $\mathbb{R}^{n+1}$ given by

$$|\Delta^n| = \left\{ (t_0, t_1, \dots, t_n) \in \mathbb{R}^{n+1} : \sum_{i=0}^n t_i = 1, \text{ and } t_i \geq 0 \text{ for all } i \right\}$$

equipped with the induced topology from the usual Euclidean topology on $\mathbb{R}^{n+1}$.

(♠ TODO: comment on how $|\Delta^n|$ makes sense via a geometric realization)

Definition 8.0.3. The *simplex category*, or *nonempty finite ordinal category*, denoted by Δ , is the locally small (Definition 1.0.9) category (Definition 1.0.1) whose

- objects are the finite nonempty totally ordered sets $[n] := \{0, 1, 2, \dots, n\}$ for each integer $n \geq 0$;
- morphisms are all order-preserving (non-decreasing) functions (Definition A.0.2) $\theta : [m] \rightarrow [n]$.

Composition in Δ is given by composition of functions.

Definition 8.0.4. Let $\mathcal{C}$ be a category (Definition 1.0.1).

1. The *category of simplicial objects in $\mathcal{C}$* , commonly denoted by notations such as $\mathbf{s}\mathcal{C}$, $\mathbf{Simp}\mathcal{C}$, $\mathcal{C}_\Delta$, or $(\Delta^{\text{op}})^{\mathcal{C}}$ (cf. Definition 1.2.1), or $\Delta^{\text{op}}\mathcal{C}$, is the functor category (Definition 1.2.1)

$$\mathbf{s}\mathcal{C} := \mathbf{Fun}(\Delta^{\text{op}}, \mathcal{C}).$$

(Definition 1.0.12) (Definition 8.0.3) In particular, a morphism between objects $X, Y : \Delta^{\text{op}} \rightarrow \mathcal{C}$ in this category is a natural transform $X \Rightarrow Y$ from X to Y as functors.

An object X of $\mathbf{s}\mathcal{C}$ is called a *simplicial object of $\mathcal{C}$* , and, by definition (Definition 1.1.1), consists of a family of objects $\{X_n\}_{n \geq 0}$ in $\mathcal{C}$ together with morphisms

$$X(\theta) : X_n \rightarrow X_m, \quad \text{for each } \theta : [m] \rightarrow [n] \text{ in } \Delta,$$

satisfying the functoriality conditions

$$X(\text{id}_{[n]}) = \text{id}_{X_n}, \quad X(\theta \circ \psi) = X(\psi) \circ X(\theta).$$

2. Dually, the *category of cosimplicial objects in $\mathcal{C}$* , commonly denoted by notations such as $\mathbf{c}\mathcal{C}$, $\mathbf{Cosimp}\mathcal{C}$, or $\Delta^{\mathcal{C}}$ (cf. Definition 1.2.1) is the functor category

$$\mathbf{c}\mathcal{C} := \mathbf{Fun}(\Delta, \mathcal{C}).$$

In particular, a morphism between objects $X, Y : \Delta \rightarrow \mathcal{C}$ in this category is a natural transform $X \Rightarrow Y$ from X to Y as functors.

An object Y of $\mathbf{c}\mathcal{C}$ is called a *cosimplicial object of $\mathcal{C}$* , and consists of a family of objects $\{Y^n\}_{n \geq 0}$ in $\mathcal{C}$ together with morphisms

$$Y(\theta) : Y^m \rightarrow Y^n, \quad \text{for each } \theta : [m] \rightarrow [n] \text{ in } \Delta,$$

satisfying the functoriality conditions

$$Y(\text{id}_{[n]}) = \text{id}_{Y^n}, \quad Y(\theta \circ \psi) = Y(\theta) \circ Y(\psi).$$

For instance, a *(co)simplicial set, group, topological space, ring, etc.* refers to a (co)simplicial object in the category of sets, of groups, of topological spaces, of rings, etc. and such categories are denoted by notations such as $\mathbf{Sets}_\Delta$, $\mathbf{Grps}_\Delta$, $\mathbf{Top}_\Delta$, $\mathbf{Rings}_\Delta$, etc. Accordingly, a *(co)simplicial map* between such (co)simplicial objects refers to a morphism in the appropriate (co)simplicial category.

If $\mathcal{C}$ is locally small (Definition 1.0.9), then both $\mathbf{s}\mathcal{C}$ and $\mathbf{c}\mathcal{C}$ are locally small as well.

8.0.1. Geometric realization and singular complex adjunction.

Definition 8.0.5. Let S be a simplicial set (Definition 8.0.4). The *geometric realization of S* , denoted $|S|$, is the compactly generated (Definition 11.3.1) Hausdorff (Definition E.0.18) topological space obtained as the coend (🔥 **TODO: coend**)

$$|S| = \int^{[n] \in \Delta} S_n \times |\Delta^n|,$$

(Definition 8.0.11) where $|\Delta^n|$ is the standard topological n -simplex (Definition 8.0.2), and the equivalence relation is generated by the simplicial structure maps in S and the face and degeneracy maps of the simplices $|\Delta^n|$. Concretely, $|S|$ is given by the quotient space

$$\left(\coprod_{n \geq 0} S_n \times |\Delta^n| \right) / \sim,$$

where for every morphism $\varphi : [m] \rightarrow [n]$ in the simplex category Δ , the identification

$$(s, \varphi^*(x)) \sim (\varphi_*(s), x)$$

holds for all $s \in S_n$ and $x \in |\Delta^m|$, with $\varphi^* : |\Delta^m| \rightarrow |\Delta^n|$ induced by φ , and $\varphi_* : S_n \rightarrow S_m$ the simplicial map induced by φ . This construction endows $|S|$ with the quotient topology from the disjoint union.

Definition 8.0.6. Let $\mathcal{C}$ be a topological space (Definition B.0.1). The *singular complex of $\mathcal{C}$* , denoted $\text{Sing}(\mathcal{C})$, is the simplicial set (Definition 8.0.4) defined as follows:

- For each integer $n \geq 0$, the set of n -simplices (Definition 8.0.11) is the set of continuous maps (Definition B.0.3) from the standard topological n -simplex $|\Delta^n|$ (Definition 8.0.2) to $\mathcal{C}$:

$$\text{Sing}(\mathcal{C})_n = \text{Hom}_{\text{Top}}(|\Delta^n|, \mathcal{C}).$$

- The face and degeneracy maps (Definition 8.0.11) of $\text{Sing}(\mathcal{C})$ are induced by the standard coface and codegeneracy maps of the simplices $|\Delta^n|$ in the usual way, making $\text{Sing}(\mathcal{C})$ a simplicial set.

Proposition 8.0.7. 1. The geometric realization functor (Definition 8.0.5) and the singular complex functor (Definition 8.0.6) form an adjoint pair (Definition 1.1.13)

$$|\cdot| : \mathbf{sSets} \dashv \text{Sing}$$

$$|\cdot| : \mathbf{sSets} \rightleftarrows \text{Top} : \text{Sing}.$$

(Definition 8.0.4) which infact restrict to an adjoint pair

$$|\cdot| : \mathbf{sSets} \rightleftarrows \mathcal{CGHaus} : \text{Sing}.$$

where $\mathcal{CGHaus}$ is the category of compactly generated (Definition 11.3.1) Hausdorff (Definition E.0.18) topological spaces (Definition B.0.1).

2. Moreover, for every topological space X , the counit map (Definition 1.1.13)

$$|\text{Sing } X| \rightarrow X$$

is a weak homotopy equivalence (Definition B.0.7) of topological spaces.

Theorem 8.0.8 (Quillen). The unit and counit morphisms (Definition 1.1.13)

$$\begin{aligned} S &\rightarrow \text{Sing } |S| \\ |\text{Sing } X| &\rightarrow X \end{aligned}$$

for simplicial sets (Definition 8.0.4) S and topological spaces X are weak homotopy equivalences (of simplicial sets (Definition 8.1.5) and of topological spaces (Definition B.0.7) respectively).

8.0.2. Homotopy groups of a Kan Simplicial Set.

Definition 8.0.9 (Basepoint of a Simplicial Set). Let X be a simplicial set (Definition 8.0.4). A *(base)point* or *basepoint vertex* of X is a chosen element

$$x \in X_0.$$

A *pointed simplicial set* is a simplicial set equipped with a choice of base point.

Definition 8.0.10 (Homotopy Groups of a Kan Simplicial Set). Let X be a Kan simplicial set (Definition 8.2.9), and fix a basepoint (Definition 8.0.9) $x \in X_0$.

For each integer $n \geq 1$, define the *n -th homotopy group of X at x* , denoted $\pi_n(X, x)$, as the set of homotopy classes of maps (♠ TODO: homotopy, boundary of Δ^n)

$$f : \partial\Delta^{n+1} \rightarrow X$$

that restrict to the constant map at x on the basepoint simplex, modulo homotopies relative to the boundary.

Equivalently, $\pi_n(X, x)$ can be described as the set of equivalence classes of n -simplices (Definition 8.0.4) whose faces are degenerate at x , with composition induced by the combinatorial structure of simplices.

These sets carry natural group structures for $n \geq 1$, with $\pi_1(X, x)$ being the fundamental group and $\pi_n(X, x)$ for $n \geq 2$ being abelian groups.

8.0.3.

Definition 8.0.11. Let $\mathcal{C}$ be a category (Definition 1.0.1).

1. Let X be a simplicial object (Definition 8.0.4) in $\mathcal{C}$. An object $X_n := X([n])$ of $\mathcal{C}$ is called the *n -simplices of X* . In case that $\mathcal{C}$ is some kind of category of sets, an element of X_n is called an *n -simplex of X* , so X_n is the *set of n -simplices of X* . In this case, a *vertex of X* moreover refers to a 0-simplex of X and a *edge of X* refers to a 1-simplex of X .

For each morphism $\theta : [m] \rightarrow [n]$ in Δ , the induced morphism

$$X(\theta) : X_n \rightarrow X_m$$

in $\mathcal{C}$ is called a *simplicial morphism*.

For each $0 \leq j \leq n$, the *jth face map of the n-simplicies* refers to the map

$$d_j = X(p_j) : X_n \rightarrow X_{n-1}, \quad p_j : [n-1] \rightarrow [n], \quad p_j(i) = \begin{cases} i & \text{if } i < j \\ i+1 & \text{if } i \geq j \end{cases}.$$

d_j is also denoted by ∂_j .

For each $0 \leq j \leq n$, the *jth degeneracy map of the n-simplicies* refers to the map

$$s_j = X(q_j) : X_n \rightarrow X_{n+1}, \quad q_j : [n+1] \rightarrow [n], \quad q_j(i) = \begin{cases} i & \text{if } i \leq j \\ i-1 & \text{if } i > j \end{cases}.$$

- Let $Y : \Delta \rightarrow \mathcal{C}$ be a cosimplicial object (Definition 8.0.4) of $\mathcal{C}$. An object $Y^n := Y([n])$ of $\mathcal{C}$ is called the *n-cosimplicies of Y*. In case that $\mathcal{C}$ is some kind of category of sets, an element of Y_n is called an *n-cosimplex of Y*, so Y^n is the *set of n-cosimplices of Y*.

For each morphism $\theta : [m] \rightarrow [n]$ in Δ , the induced morphism

$$Y(\theta) : Y^m \rightarrow Y^n$$

is called a *cosimplicial morphism*.

For each $0 \leq j \leq n$, the *jth coface map of the n-cosimplicies* refers to the map

$$d^j = Y(p_j) : Y^n \rightarrow Y^{n+1}, \quad p_j : [n] \rightarrow [n+1], \quad p_j(i) = \begin{cases} i & \text{if } i < j \\ i+1 & \text{if } i \geq j \end{cases}$$

d^j is also denoted by ∂^j .

For each $0 \leq j \leq n$, the *jth codegeneracy map of the n-cosimplicies* refers to the map

$$s^j = Y(q_j) : Y^n \rightarrow Y^{n-1}, \quad q_j : [n] \rightarrow [n-1], \quad q_j(i) = \begin{cases} i & \text{if } i \leq j \\ i-1 & \text{if } i > j \end{cases}$$

A *(co)face/degeneracy of a the n-(co)simplicies of a (co)simplicial object* may also refer to the images of the (co)face/degeneracy maps.

Definition 8.0.12. Let $n \geq 0$ be an integer. The representable functor (Definition 4.0.6)

$$\text{Hom}_\Delta(-, [n]) : \Delta^{\text{op}} \rightarrow \mathbf{Sets}$$

(Definition 8.0.3) (Definition 1.0.12) (Definition 1.0.3) is a simplicial set (Definition 8.0.4) often denoted by Δ^n , $\Delta^{[n]}$, or $\Delta[n]$, and is called the *standard n-simplex*. More generally, if J is a finite nonempty linearly ordered set, then we may speak of the simplicial set Δ^J given by the representable functor $\text{Hom}_\Delta(-, J)$.

The *Standard simplicial n-simplex functor* refers to the functor

$$\Delta^\bullet : \Delta \rightarrow \mathbf{sSets}$$

given by $[n] \mapsto \Delta^n$. By construct, note that $\Delta^\bullet$ is a cosimplicial object (Definition 8.0.4) in the category of simplicial sets.

Dually, the functor

$$\mathrm{Hom}_\Delta([n], -) : \Delta \rightarrow \mathbf{Sets}$$

is a cosimplicial set (Definition 8.0.4) called the *standard n -cosimplex*.

0-simplices of a simplicial set may be regarded like objects in a category and 1-simplices between such objects may be regarded like morphisms between the objects — however, unless the simplicial set satisfies more conditions, composition of morphisms may not exist and may not be unique.

Definition 8.0.13. Let X be a simplicial set (Definition 8.0.4).

1. An *object of X* refers to a vertex of X (Definition 8.0.11), i.e. a 0-simplex of X (Definition 8.0.11). We may write $x \in X$ or $x \in \mathrm{Ob} X$ to denote that x is an object of X .
2. A *(1-)morphism* $x \rightarrow y$ where $x, y : \Delta^0 \rightarrow X$ (Definition 8.0.12) (Lemma 8.0.17) are objects of X_0 is an edge (Definition 8.0.11) $f : \Delta^1 \rightarrow X$ such that $d_1(f) = x$ and $d_0(f) = y$, or equivalently $f(0) = x$ and $f(1) = y$. We may write such a morphism by $f : x \rightarrow y$.
3. Given an object x of X , the *identity morphism of x* is the edge $\mathrm{id}_x = s_0(X) : \Delta^1 \rightarrow X$.
4. Let $f : x \rightarrow y$ and $g : y \rightarrow z$ be morphisms in X . A *composition of f and g* (which may not exist and may not be unique) is a 2-simplex (Definition 8.0.11) $F : \Delta^2 \rightarrow X$ such that

$$d_2(F) = f, \quad \text{and} \quad d_0(F) = g.$$

(Definition 8.0.11). The *composite of f and g with respect to the composition F* is then the morphism $d_1(F) : x \rightarrow z$.

In particular, compositions of morphisms exist whenever X is a quasi-category (Definition 8.2.2) and are unique whenever X satisfies the conditions of Proposition 8.2.6, i.e. when X is isomorphic to the nerve (Definition 8.0.20) of a small category. Furthermore, if X is a quasi-category, any two compositions $F, F' : \Delta^2 \rightarrow X$ of the same morphisms f and g yield composites $d_1(F)$ and $d_1(F')$ that are homotopic in the sense of homotopic edges, meaning they define the same unique morphism in the homotopy category (Definition 8.1.11) $\mathrm{h}X$.

5. More generally, an *n -morphism of X* is an element of X_n . Geometrically, an n -morphism $\sigma : \Delta^n \rightarrow X$ acts as a directed higher cell whose source and target faces are determined by the face maps $d_i(\sigma)$.
6. Let $n \geq 1$. There are two primary geometric configurations for composing n -morphisms, depending on whether they are composed within a fixed mapping space or along an intermediate boundary. (♠ **TODO: verify**)
 - (a) Two n -morphisms $f, g \in X_n$ are *vertically composable* if they share identical source and target boundaries up to a shift in dimension; precisely, if there exists an index $0 < i < n + 1$ such that $d_j(f) = d_j(g)$ for all $j \neq i - 1, i$, and $d_i(f) = d_{i-1}(g)$. A *vertical composition of f and g* is an $(n + 1)$ -simplex $F : \Delta^{n+1} \rightarrow X$ satisfying the boundary conditions:

$$d_{i+1}(F) = f, \quad d_{i-1}(F) = g, \quad \text{and} \quad d_j(F) = s_{i-1}d_j(f) \text{ for all } j \neq i - 1, i, i + 1.$$

The *vertical composite of f and g with respect to F* is the n -morphism $d_i(F) \in X_n$.

- (b) Two n -morphisms $f, g \in X_n$ are *horizontally composable* if the primary target face of the first matches the primary source face of the second; precisely, if $d_0(f) = d_n(g) \in X_{n-1}$.

A *horizontal composition of f and g* is an $(n+1)$ -simplex $H : \Delta^{n+1} \rightarrow X$ satisfying the boundary conditions:

$$d_{n+1}(H) = f \quad \text{and} \quad d_0(H) = g.$$

The *horizontal composite of f and g with respect to H* is the n -morphism $d_n(H) \in X_n$ (or dually $d_1(H)$).

The notions of vertical and horizontal composition are structurally compatible and equivalent in the following sense. Let $\text{Map}_X(x, y)$ denote the mapping Kan complex between two objects $x, y \in X_0$.

Vertically, an n -morphism defines an $(n-1)$ -simplex in $\text{Map}_X(x, y)$, and the vertical composition F corresponds exactly to the strict combinatorial concatenation of cells inside this mapping space. Horizontally, composition defines the structural map of simplicial sets:

$$\circ : \text{Map}_X(y, z) \times \text{Map}_X(x, y) \rightarrow \text{Map}_X(x, z)$$

When X is a quasi-category, both operations are governed by the same inner horn filling condition $\Lambda_i^{n+1} \rightarrow X$. The equivalence of these notions is expressed by the Eckmann–Hilton argument for quasi-categories: the vertical composition of higher cells within a fixed mapping space corresponds up to coherent $(n+1)$ -morphism homotopy to the horizontal composition of those cells with the identity morphisms of their respective boundaries. Consequently, both operations induce the exact same associative and commutative graded ring structure on the higher homotopy groups $\pi_n(\text{Map}_X(x, y))$.

Definition 8.0.14 (Functor between quasi-categories). Let C and D be ∞ -categories/quasi-categories (Definition 8.2.2). A *functor of quasi-categories from C to D* is a simplicial map (Definition 8.0.4) $F : C \rightarrow D$.

Equivalently, a functor $C \rightarrow D$ is a 0-simplex (Definition 8.0.11) of the simplicial mapping object D^C (Definition 8.0.22).

Definition 8.0.15 (Simplicial category of quasi-categories). (♠ TODO: read this and make it more precise) Let $\mathbf{QCat}_{\mathcal{U}}^{\Delta}$ be the simplicial category defined as follows. Its objects are the $\mathcal{U}$ -small quasi-categories. For objects C, D , the mapping space is the Kan complex

$$\text{Map}_{\mathbf{QCat}_{\mathcal{U}}^{\Delta}}(C, D) := (D^C)^{\simeq},$$

where D^C is the exponential simplicial set (the functor simplicial set) and $(-)^{\simeq}$ denotes the largest Kan subcomplex. Composition

$$\text{Map}(C, D) \times \text{Map}(D, E) \longrightarrow \text{Map}(C, E)$$

is induced by the exponential composition $E^D \times D^C \rightarrow E^C$, and identities by the units $C \rightarrow C$ under the exponential adjunction; these maps restrict to Kan subcomplexes and equip $\mathbf{QCat}_{\mathcal{U}}^{\Delta}$ with a simplicial category structure.

Definition 8.0.16 (Infinity category of infinity categories). (♠ TODO: reformulate this definition)

1. Let $\mathcal{U}$ be a Grothendieck universe (Definition E.0.1). The ∞ -category of ∞ -categories (of $\mathcal{U}$ -small quasi-categories) is the quasi-category (Definition 8.2.2)

$$\mathrm{Cat}_{\infty}^{\mathcal{U}} := N_{\Delta}(\mathbf{QCat}_{\mathcal{U}}^{\Delta}).$$

(Definition 8.0.15) Concretely, the objects of $\mathrm{Cat}_{\infty}^{\mathcal{U}}$ are $\mathcal{U}$ -small quasi-categories, and for C, D the mapping space in $\mathrm{Cat}_{\infty}^{\mathcal{U}}$ is equivalent to $(D^C)^{\simeq}$, the largest Kan subcomplex (Definition 8.2.9) of the exponential simplicial set D^C (Definition 8.0.22), so that 0-simplices correspond to functors $C \rightarrow D$ and higher simplices encode their coherent homotopies.

2. The ∞ -category of small ∞ -categories is the large quasi-category (Definition 8.2.2)

$$\mathrm{Cat}_{\infty} := N_{\Delta}(\mathbf{QCat}^{\Delta}),$$

where $\mathbf{QCat}^{\Delta}$ is the simplicial category of small quasi-categories (Definition 8.0.15).

Concretely, the objects of Cat_{∞} are all small quasi-categories. For any two small quasi-categories C and D , the mapping space in Cat_{∞} is the small Kan complex (Definition 8.2.9)

$$\mathrm{Map}_{\mathrm{Cat}_{\infty}}(C, D) := (D^C)^{\simeq},$$

which is the largest Kan subcomplex of the exponential simplicial set D^C (Definition 8.0.22). Thus, 0-simplices correspond to functors $C \rightarrow D$ and higher simplices encode their coherent homotopies.

- Lemma 8.0.17.**
1. Let X be a simplicial set (Definition 8.0.4). The n -simplices (Definition 8.0.11) X^n is naturally isomorphic to $\mathrm{Hom}_{\mathrm{Simp}\mathbf{Sets}}(\Delta^n, X)$ (Definition 8.0.12).
 2. Let Y be a cosimplicial set (Definition 8.0.4). The n -cosimplices (Definition 8.0.11) Y^n is naturally isomorphic to $\mathrm{Hom}_{\mathrm{Cosimp}\mathbf{Sets}}(\mathrm{Hom}_{\Delta}([n], -), Y)$ (Definition 8.0.12).

Proof. This is a consequence of the Yoneda lemma (Theorem 4.0.7). □

- Definition 8.0.18.**
1. Let X and Y be simplicial sets (Definition 8.0.4), that is, functors $X, Y : \Delta^{\mathrm{op}} \rightarrow \mathbf{Set}$. A *simplicial subset of X* is a simplicial set Y together with a family of injective (Definition A.0.3) maps

$$\iota_n : Y_n \hookrightarrow X_n \quad (n \geq 0)$$

that form a natural transformation (Definition 1.1.3) $\iota : Y \Rightarrow X$. Equivalently, Y is a simplicial subset of X if and only if for every morphism $\theta : [m] \rightarrow [n]$ in Δ , the following diagram commutes:

$$\begin{array}{ccc} Y_n & \xrightarrow{Y(\theta)} & Y_m \\ \downarrow \iota_n & & \downarrow \iota_m \\ X_n & \xrightarrow{X(\theta)} & X_m \end{array}$$

and each ι_n is injective.

2. Let X and Y be cosimplicial sets, that is, functors $X, Y : \Delta \rightarrow \mathbf{Set}$. A *cosimplicial subset of X* is a cosimplicial set Y together with a family of injective maps

$$\iota^n : Y^n \hookrightarrow X^n \quad (n \geq 0)$$

that form a natural transformation $\iota : Y \Rightarrow X$. Equivalently, Y is a cosimplicial subset of X if and only if for every morphism $\theta : [m] \rightarrow [n]$ in Δ , the following diagram commutes:

$$\begin{array}{ccc} Y^m & \xrightarrow{Y(\theta)} & Y^n \\ \downarrow \iota^m & & \downarrow \iota^n \\ X^m & \xrightarrow{X(\theta)} & X^n \end{array}$$

and each ι^n is injective.

Definition 8.0.19. Let $n \geq 1$ be an integer and $\Delta[n]$ be the standard n -simplex. For each i with $0 \leq i \leq n$, the *i th horn of the n th standard simplex*, denoted by $\Lambda^i[n]$, is the simplicial subset of $\Delta[n]$ (Definition 8.0.12) defined as follows: the set of k -simplices (Definition 8.0.11) of $\Lambda^i[n]$ is

$$\Lambda^i[n]_k := \{ \theta \in \text{Hom}_\Delta([k], [n]) \mid \text{Im}(\theta) \text{ does not contain all of } [n] \setminus \{i\} \}.$$

Equivalently,

$$\Lambda^i[n] = \bigcup_{j \neq i} \text{Im}(d^j : \Delta[n-1] \rightarrow \Delta[n]),$$

where each d^j is the coface map (Definition 8.0.11) in the simplex category corresponding to the injective order-preserving inclusion $[n-1] \hookrightarrow [n]$ that omits j .

Definition 8.0.20. Let $\mathcal{C}$ be a (large) category (Definition 1.0.1). The *nerve of $\mathcal{C}$* is the simplicial “class” (Definition 8.0.4) $N(\mathcal{C})$ given as follows: for $n \geq 0$, the n -simplices (Definition 8.0.11) $N(\mathcal{C})_n$ is the class of all functors $[n] \rightarrow \mathcal{C}$ (Definition 8.0.3). In other words, $N(\mathcal{C})_n$ is the class of all composable sequences

$$C_0 \xrightarrow{f_1} C_1 \dots \xrightarrow{f_n} C_n$$

of morphisms of length n . The face map (Definition 8.0.11) $d_i : N(\mathcal{C})_n \rightarrow N(\mathcal{C})_{n-1}$ carries the above sequence to

$$C_0 \xrightarrow{f_1} C_1 \dots \xrightarrow{f_{i-1}} C_{i-1} \xrightarrow{f_{i+1} \circ f_i} C_{i+1} \xrightarrow{f_{i+2}} \dots \xrightarrow{f_n} C_n$$

while the degeneracy s_i carries it to

$$C_0 \xrightarrow{f_1} C_1 \dots \xrightarrow{f_i} C_i \xrightarrow{id_{C_i}} C_i \xrightarrow{f_{i+1}} C_{i+1} \xrightarrow{f_{i+2}} \dots \xrightarrow{f_n} C_n.$$

If $\mathcal{C}$ is a small category (Definition 1.0.9), then $N(\mathcal{C})$ is a simplicial set (Definition 8.0.4).

Note that the category $\mathcal{C}$ can be recovered, roughly up to isomorphism, from its nerve $N(\mathcal{C})$. The objects of $\mathcal{C}$ are simply the objects of $N(\mathcal{C})_0$. The morphisms $C_0 \rightarrow C_1$ are given by objects $\phi \in N(\mathcal{C})_1$ such that $d_1(\phi) = C_0$ and $d_0(\phi) = C_1$. The identity morphism $C \rightarrow C$ is given by the degenerate simplex $s_0(C)$. Moreover, given a diagram $C_0 \xrightarrow{\phi} C_1 \xrightarrow{\psi} C_2$, the object $N(\mathcal{C})_1$ corresponding to $\psi \circ \phi : C_0 \rightarrow C_2$ is uniquely characterized by the 2-simplex $\sigma \in N(\mathcal{C})_2$ such that $d_2(\sigma) = \phi$, $d_0(\sigma) = \psi$, and $d_1(\sigma) = \psi \circ \phi$.

Simplicial categories provide one framework for ∞ -categories.

Definition 8.0.21. A *simplicial category* is a category enriched over (Definition 4.0.1) the category of simplicial sets (Definition 8.0.4). Common notation for the category of simplicial categories include $\mathbf{Cat}_\Delta$ or $\mathbf{sCat}$. However, note that these would overlap with notation for the category of simplicial objects of the category $\mathbf{Cat}$ (Definition 8.0.4). Further note that all simplicial categories are regardable as simplicial objects in $\mathbf{Cat}$, but not vice versa. Unless otherwise specified, the notations here will take precedence; in other words, $\mathbf{Cat}_\Delta$ or $\mathbf{sCat}$ will mean the category of simplicial categories unless otherwise stated.

Definition 8.0.22. The category $\mathbf{sSets}$ of simplicial sets (Definition 8.0.4) can be enriched with the structure of a simplicial category (Definition 8.0.21) (i.e. $\mathbf{sSets}$ can be regarded as a category enriched (Definition 4.0.1) in $\mathbf{sSets}$) as follows: given simplicial sets X and Y , we let $\mathbf{Mor}(X, Y)$, which is also often denoted by notations such as Y^X , $\mathbf{Map}(X, Y)$, or $\mathbf{Fun}(X, Y)$, and referred to as the *exponential simplicial set from X to Y* , the *simplicial mapping object from X to Y* , or the *mapping simplicial set from X to Y*

$$\mathbf{Mor}(X, Y)_n = \mathbf{Hom}_{\mathbf{sSets}}(X \times \Delta^n, Y)$$

(Definition 8.0.11) (Definition 8.0.12) (♠ **TODO: discuss what the face and degeneracy maps do**) and we let composition be given by the simplicial set morphisms (Definition 8.0.4)

$$\circ : \mathbf{Mor}(Y, Z) \times \mathbf{Mor}(X, Y) \rightarrow \mathbf{Mor}(X, Z)$$

where X, Y, Z are simplicial sets as follows:

1. The n -simplices of the product simplicial set $\mathbf{Mor}(Y, Z) \times \mathbf{Mor}(X, Y)$ is the product set $\mathbf{Mor}(Y, Z)_n \times \mathbf{Mor}(X, Y)_n$
2. Since the n -simplices of $\mathbf{Mor}(Y, Z)$ and the n -simplices of $\mathbf{Mor}(X, Y)$ are respectively given by simplicial set morphisms

$$f : Y \times \Delta^n \rightarrow Z$$

$$g : X \times \Delta^n \rightarrow Y$$

respectively, the n -simplices of the product simplicial set are given by pairs (f, g) of such morphisms.

3. The composition $\circ(f, g)$ is the simplicial set morphism

$$X \times \Delta^n \xrightarrow{\text{id}_X \times \Delta} X \times (\Delta^n \times \Delta^n) \cong (X \times \Delta^n) \times \Delta^n \xrightarrow{g, \text{id}_{\Delta^n}} Y \times \Delta^n \xrightarrow{f} Z$$

where $\Delta : \Delta^n \rightarrow \Delta^n \times \Delta^n$ is the diagonal morphism of the standard n -simplex Δ^n .

This simplicial category may be called the *simplicial category of simplicial sets* and the same notations, such as $\mathbf{sSets}$ or $\mathbf{Sets}_\Delta$, used to denote the (ordinary) category of simplicial sets may be used to denote this simplicial category.

When the simplicial set Y is ∞ -categories (Definition 8.2.2), it is also customary to denote $\mathbf{Mor}(X, Y)$ by $\mathbf{Fun}(X, Y)$, and to call this the *functor ∞ -category from X to Y* , especially when both X and Y are ∞ -categories; it is itself an ∞ -category. See also Definition 8.0.14.

Definition 8.0.23. Let X be a simplicial set (Definition 8.0.4). Let $x, y \in X$ be objects (Definition 8.0.13). The *mapping space between x and y* is the simplicial set $\mathbf{Mor}(x, y)$ whose n -simplices (Definition 8.0.11) are given by

$$\mathbf{Mor}(x, y)_n = \{F : \Delta^n \times \Delta^1 \rightarrow X : F|_{\Delta^n \times \{0\}} = x, F|_{\Delta^n \times \{1\}} = y\}.$$

(Definition 8.0.12) (♠ TODO: comment on how composition of morphisms becomes well defined up to homotopy; it seems reasonable that compositions to be well defined on the homotopy categories $\mathbf{hMor}$ of the mapping spaces)

Definition 8.0.24. Let J be a finite nonempty linearly ordered set. Define the *simplicial path category* $\mathfrak{C}[\Delta^J]$ of Δ^J (Definition 8.0.12) as the following simplicial category (Definition 8.0.21):

- The objects of $\mathfrak{C}[\Delta^J]$ are the elements of J .
- If $i, j \in J$, then

$$\mathrm{Hom}_{\mathfrak{C}[\Delta^J]}(i, j) = \begin{cases} \emptyset & \text{if } j < i \\ N(P_{i,j}) & \text{if } i \leq j \end{cases}$$

(Definition 8.0.20) where $P_{i,j}$ is the partially ordered set

$$P_{i,j} = \{I \subseteq J : (i, j \in I) \wedge (\forall k \in I)[i \leq k \leq j])\}.$$

- If $i_0 \leq i_1 \cdots i_n$, then the composition

$$\mathrm{Hom}_{\mathfrak{C}[\Delta^J]}(i_0, i_1) \times \cdots \times \mathrm{Hom}_{\mathfrak{C}[\Delta^J]}(i_{n-1}, i_n) \rightarrow \mathrm{Hom}_{\mathfrak{C}[\Delta^J]}(i_0, i_n)$$

is induced by the map of partially ordered sets

$$\begin{aligned} P_{i_0, i_1} \times \cdots \times P_{i_{n-1}, i_n} &\rightarrow P_{i_0, i_n} \\ (I_1, \dots, I_n) &\mapsto I_1 \cup \cdots \cup I_n. \end{aligned}$$

The functor $\mathfrak{C} : \Delta \rightarrow \mathbf{Cat}_\Delta$ (Definition 8.0.3, Definition 8.0.21) extends uniquely (up to unique isomorphism) to a (small) colimit-preserving functor $\mathfrak{C} : \mathbf{Sets}_\Delta \rightarrow \mathbf{Cat}_\Delta$ (Definition 8.0.4, Lemma 11.3.8)

Definition 8.0.25. Let $\mathcal{C}$ be a simplicial category (Definition 8.0.21). The *(simplicial) nerve* $N(\mathcal{C})$ of $\mathcal{C}$ is the simplicial set (Definition 8.0.4) described by the formula

$$N(\mathcal{C})_n \cong \mathrm{Hom}_{\mathbf{Sets}_\Delta}(\Delta^n, N(\mathcal{C})) = \mathrm{Hom}_{\mathbf{Cat}_\Delta}(\mathfrak{C}[\Delta^n], \mathcal{C}).$$

(Lemma 8.0.17, Definition 8.0.4, Definition 8.0.11, Definition 8.0.21, Definition 8.0.24) It is synonymously called the *homotopy coherent nerve of $\mathcal{C}$* .

Definition 8.0.26. Let *Kan* be the full subcategory (Definition 1.3.2) of $\mathbf{sSets}$ (Definition 8.0.4) spanned by the collection of Kan complexes (Definition 8.2.9). Regard *Kan* as a simplicial category (Definition 8.0.21). (♠ TODO: how exactly is *Kan* regarded as a simplicial category?) In the context of ∞ -category theory, the simplicial nerve (Definition 8.0.25) of *Kan* is called the *∞ -category of spaces*.

8.1. Model categories.

Definition 8.1.1 (Model Category). (♠ TODO: I don't like the axioms as stated here) A *model category*, or synonymously a *closed model category*, is a complete and cocomplete category (Definition 1.3.9) $\mathcal{M}$ equipped with three distinguished classes of morphisms:

- *Weak equivalences* $\mathcal{W}$,
- *Fibrations* $\mathcal{F}$,

- **Cofibrations** $\mathcal{C}$,

1. **(Two-out-of-three)**: For any composable morphisms $f : X \rightarrow Y$, $g : Y \rightarrow Z$, if any two of f , g , or $g \circ f$ lie in $\mathcal{W}$, then so does the third.
2. **(Retracts)**: Each of the classes $\mathcal{W}, \mathcal{F}, \mathcal{C}$ is closed under retracts in the arrow category $\mathcal{M}^2$. That is, if f is a retract of g and g belongs to one of these classes, then f also belongs to that class.
3. **(Lifting)**: Given any commutative square

$$\begin{array}{ccc} A & \longrightarrow & X \\ i \downarrow & & \downarrow p \\ B & \longrightarrow & Y \end{array}$$

where $i \in \mathcal{C}$ and $p \in \mathcal{F}$, a diagonal filler (lift) exists making both triangles commute provided either

- i is also a weak equivalence (called an acyclic cofibration), or
- p is also a weak equivalence (called an acyclic fibration).

Formally, acyclic cofibrations have the left lifting property with respect to all fibrations, and cofibrations have the left lifting property with respect to all acyclic fibrations.

4. **(Factorization)**: Every morphism $f : X \rightarrow Y$ in $\mathcal{M}$ admits two functorial factorizations:
 - $f = p \circ i$, where $i \in \mathcal{C}$ is a cofibration and $p \in \mathcal{F} \cap \mathcal{W}$ is an acyclic fibration.
 - $f = q \circ j$, where $j \in \mathcal{C} \cap \mathcal{W}$ is an acyclic cofibration and $q \in \mathcal{F}$ is a fibration.

Here, an **acyclic fibration** (or **trivial fibration**) is a morphism in $\mathcal{F} \cap \mathcal{W}$, and an **acyclic cofibration** (or **trivial cofibration**) is a morphism in $\mathcal{C} \cap \mathcal{W}$.

Definition 8.1.2 (Proper Model Category). Let $\mathcal{M}$ be a model category (Definition 8.1.1) with classes of weak equivalences $\mathcal{W}$, cofibrations $\mathcal{Cof}$, and fibrations $\mathcal{Fib}$. (♠ **TODO: pushout, pullback**)

1. Then $\mathcal{M}$ is called **left proper** if weak equivalences are preserved under pushouts along cofibrations, i.e., if for every pushout diagram

$$\begin{array}{ccc} A & \xrightarrow{i} & B \\ \downarrow f \in \mathcal{W} & & \downarrow \\ C & \rightarrow & D \end{array}$$

with $i \in \mathcal{Cof}$, the induced map $C \rightarrow D$ lies in $\mathcal{W}$.

2. Dually, $\mathcal{M}$ is called **right proper** if weak equivalences are preserved under pullbacks along fibrations, i.e., if for every pullback diagram

$$\begin{array}{ccc} P & \rightarrow & X \\ \downarrow g & & \downarrow p \in \mathcal{Fib} \\ Y & \xrightarrow{h \in \mathcal{W}} & Z \end{array}$$

the induced map $P \rightarrow Y$ is in $\mathcal{W}$.

3. A model category $\mathcal{M}$ is **proper** if it is both left proper and right proper.

Lemma 8.1.3. A model category (Definition 8.1.1) always has a initial object and a final object (Definition 2.0.6).

Proof. This is because a model category, by definition, is complete and cocomplete (Definition 1.3.9). $\square$

Definition 8.1.4 (Fibrant Object). Let $(\mathcal{C}, \mathcal{W}, \mathcal{Cof}, \mathcal{Fib})$ be a model category (Definition 8.1.1).

1. An object $X \in \mathcal{C}$ is called **fibrant** if the unique morphism

$$X \longrightarrow *$$

to the terminal object (Definition 2.0.6) (Lemma 8.1.3) $*$ of $\mathcal{C}$ is a fibration (i.e., belongs to $\mathcal{Fib}$).

2. An object $X \in \mathcal{C}$ is called **cofibrant** if the unique morphism

$$\emptyset \longrightarrow X$$

from the initial object (Definition 2.0.6) (Lemma 8.1.3) $\emptyset$ of $\mathcal{C}$ is a cofibration (i.e., belongs to $\mathcal{Cof}$).

Definition 8.1.5 (Standard model category structure on simplicial sets). Let **sSet** denote the category of simplicial sets (Definition 8.0.4). The **standard model category structure on sSet**, also known as the **Kan model structure** or the **Quillen model structure** or the **Kan-Quillen model structure**, is the model category (Definition 8.1.1) given by specifying three classes of morphisms:

- **Weak equivalences:** A map $f : X \rightarrow Y$ in **sSet** is a weak equivalence, also called a **weak homotopy equivalence between the simplicial sets X and Y** , if the induced map of geometric realizations (Definition 8.0.5) $|f| : |X| \rightarrow |Y|$ is a weak homotopy equivalence (Definition B.0.7) of topological spaces (i.e., induces isomorphisms on all homotopy groups for all choices of basepoint).
- **Fibrations:** A map $f : X \rightarrow Y$ is a **(Kan) fibration** if it has the right lifting property with respect to the horn inclusions $\Lambda_k^n \hookrightarrow \Delta^n$ for all $n \geq 1$ and $0 \leq k \leq n$.
- **Cofibrations:** A map $f : X \rightarrow Y$ is a cofibration if it is a monomorphism (Definition 1.3.7) (i.e., injective at each level).

For this model category structure, all objects are cofibrant (Definition 8.1.4), and the fibrant objects (Definition 8.1.4) are exactly the Kan complexes (Definition 8.2.9).

Definition 8.1.6 (Model Structure on $\mathcal{Top}$). The category **Top** of topological spaces has a model structure (Definition 8.1.1), referred to such names as the **Quillen model structure (on the category of topological spaces)**, **classical model structure on the category of topological spaces**, **Serre model structure**, or **Quillen-Serre model structure**, given by the following three classes of morphisms

$$\mathcal{W}, \quad \mathcal{Cof}, \quad \mathcal{Fib}$$

such that:

- $\mathcal{W}$ is the class of weak homotopy equivalences (Definition B.0.7). (♠ **TODO: cofibrations, Serre fibrations of topological spaces**)

- Cof is the class of cofibrations: maps that are retracts of relative cell complexes (e.g., inclusions satisfying the homotopy extension property).
- Fib is the class of Serre fibrations: maps having the right lifting property with respect to all inclusions of disks into disks times the interval.

8.1.1. Homotopy categories of model categories.

Definition 8.1.7 (Homotopy category of a model category). Let $\mathcal{M}$ be a model category (Definition 8.1.1). The *homotopy category of $\mathcal{M}$* , denoted by notations such as $Ho(\mathcal{M})$, $h(\mathcal{M})$, etc., is the category (Definition 1.0.1) $\mathcal{M}[W^{-1}]$ whose objects are those of $\mathcal{M}$, and whose morphisms are equivalence classes of morphisms in $\mathcal{M}$ under the relation of left and right homotopy, localized (Definition 7.2.2) at the weak equivalences. Explicitly,

- The objects of $Ho(\mathcal{M})$ are the same as those in $\mathcal{M}$.
- For objects X, Y in $\mathcal{M}$, the morphism set $Hom_{Ho(\mathcal{M})}(X, Y)$ consists of maps in $\mathcal{M}$ modulo homotopy, with weak equivalences formally inverted.

Definition 8.1.8 (Homotopy category of the category of simplicial sets). Let $sSets$ denote the category of simplicial sets (Definition 8.0.4), equipped with its standard model structure (Definition 8.1.5). The *homotopy category of (the category of) simplicial sets*, denoted by notations such as $Ho(sSets)$, $h(Sets_{\Delta})$, etc., is the homotopy category (Definition 8.1.7) of the standard model category of simplicial sets (Definition 8.1.5). In other words, It is the category whose objects are simplicial sets, and whose morphisms are homotopy classes of maps between simplicial sets, with weak equivalences inverted. That is,

- Objects: simplicial sets X .
- Morphisms: equivalence classes of maps $f : X \rightarrow Y$ under simplicial homotopy, where weak equivalences (maps inducing isomorphism on all homotopy groups) are made invertible.

The homotopy category of the category of simplicial sets is equivalent (Definition 1.1.8) to the category obtained from the category $\mathcal{CG}$ of compactly generated (Definition 11.3.1), weakly Hausdorff (Definition 11.3.2) topological spaces by inverting (Definition 7.2.2) weak homotopy equivalences (Definition B.0.7) of spaces (Corollary 8.1.9).

The homotopy category of simplicial sets should not be confused with the homotopy category of a simplicial set (Definition 8.1.11).

Corollary 8.1.9. The category obtained from the category $\mathcal{CG}$ of compactly generated (Definition 11.3.1), weakly Hausdorff (Definition 11.3.2) topological spaces by inverting (Definition 7.2.2) weak homotopy equivalences (Definition B.0.7) of spaces is equivalent (Definition 1.1.8) to the homotopy category of the category of simplicial sets (Definition 8.1.8).

Proof. This follows from Theorem 8.0.8. □

Definition 8.1.10 ([Lur09, Before Definition 1.1.4.4]). Let $\mathcal{C}$ be a simplicial category (Definition 8.0.21). Let $\mathcal{H}$ be the homotopy category of the category of simplicial sets (Definition 8.1.8). The *homotopy category of $\mathcal{C}$* is the $\mathcal{H}$ -enriched category (Definition 4.0.1) $h\mathcal{C}$

obtained by applying the localization functor $h : \mathbf{Sets}_\Delta \rightarrow \mathcal{H}$ (Definition 8.0.4) to the Hom's of $\mathcal{C}$. In other words, $h\mathcal{C}$ is the $\mathcal{H}$ -enriched category whose

1. Objects are the objects of $\mathcal{C}$
2. Hom's are given for $X, Y \in \text{Ob } h\mathcal{C}$ by

$$\text{Hom}_{h\mathcal{C}}(X, Y) = h \text{Hom}_{\mathcal{C}}(X, Y).$$

(♠ TODO: homotopy category of the category of simplicial sets, equivalently the homotopy category of the category of compactly generated spaces)

Definition 8.1.11. Let S be a simplicial set (Definition 8.0.4). Its *homotopy category* hS is defined to be the homotopy category (Definition 8.1.10) $h\mathfrak{C}[S]$ of the simplicial category (Definition 8.0.21) $\mathfrak{C}[S]$ (Definition 8.0.24).

The homotopy category of a simplicial set should not be confused with the homotopy category of the category of simplicial sets (Definition 8.1.8).

Proposition 8.1.12 ([Lur09, Proposition 1.2.3.1]). (♠ TODO:) There is an adjunction (Definition 1.1.13) $N \dashv h$ between the functor of nerves of a small categories (Definition 8.0.20) and the functor of homotopy categories (Definition 8.1.11) of simplicial sets (Definition 8.0.4)

$$N : \mathbf{Cat} \rightleftarrows \mathbf{Sets}_\Delta : h$$

(Definition 8.0.4) (where we ignore that enriched category structure (Definition 4.0.1) of hS for simplicial sets (Definition 8.0.4) S)

Definition 8.1.13 (Joyal Model Structure, [Lur09, Theorem 2.2.5.1]). (♠ TODO: This description seems a bit incomplete; it should be necessary to say that the model structure is the left proper combinatorial model structure with these properties.) The *Joyal model structure* on the category $s\mathbf{Set}$ of simplicial sets (Definition 8.0.4) is characterized by: (♠ TODO: fibrant/cofibrant objects)

- Fibrant objects (Definition 8.1.4) are exactly the quasi-categories (Definition 8.2.2),
- Cofibrations are the monomorphisms,
- Weak equivalences are categorical equivalences, i.e., maps inducing equivalences of ∞ -categories.

This model category satisfies the axioms of a model category (Definition 8.1.1) and presents the homotopy theory of $(\infty, 1)$ -categories.

Definition 8.1.14 (Model for the Homotopy Theory of $(\infty, 1)$ -Categories). A model category (Definition 8.1.1) $\mathcal{M}$ is called a *model for the homotopy theory of $(\infty, 1)$ -categories* if:

- The fibrant objects of $\mathcal{M}$ are equivalent (under suitable equivalences) to $(\infty, 1)$ -categories,
- The weak equivalences correspond to equivalences of $(\infty, 1)$ -categories,
- $\mathcal{M}$ is Quillen equivalent to other established models of ∞ -categories (e.g., simplicial categories (Definition 8.0.21), Segal categories, complete Segal spaces). (♠ TODO: Quillen equivalent)

8.2. **Quasi-categories.** Quasi-categories provide one framework for ∞ -categories.

Definition 8.2.1. Let K be a simplicial set (Definition 8.0.4). We may say that K *satisfies the Kan condition/extension condition for the horn $\Lambda_i^n \subset \Delta^n$* (Definition 8.0.19) (Definition 8.0.12) if any morphism $\Lambda_i^n \rightarrow K$ of simplicial sets admits an extension $f : \Delta^n \rightarrow K$.

Definition 8.2.2. A *quasi-category* is a simplicial set (Definition 8.0.4) K satisfying the Kan extension condition (Definition 8.2.1) for the horns $\Lambda_i^n \subset \Delta^n$ for all n and all $0 < i < n$. More explicitly, for any $0 < i < n$, any map $f_0 : \Lambda_i^n \rightarrow K$ (Definition 8.0.19) admits an extension $f : \Delta^n \rightarrow K$ (Definition 8.0.12).

Equivalently, an quasi-category is a simplicial set K such that

$$\text{Map}(\Delta^2, K) \rightarrow \text{Map}(\Lambda_1^2, K)$$

(Definition 8.0.22) is a trivial Kan fibration (Definition 8.2.4) (Proposition 8.2.5).

In ∞ -category theory, a quasi-category is synonymously referred to as an $(\infty, 1)$ -category. Moreover, an ∞ -category, without further qualifications, often refers to a quasi-category. See Definition 8.0.13, which describes the framework to describe quasi-categories as ∞ -categories.

Definition 8.2.3. (♠ TODO: AI generated, verify) Let $p : X \rightarrow Y$ be a morphism of simplicial sets (Definition 8.0.4). The map p is called a *Kan fibration between simplicial sets* if for every integer $n \geq 0$ and every $0 \leq i \leq n$, the map p has the right lifting property with respect to the horn inclusion $\Lambda_i^n \hookrightarrow \Delta^n$ (Definition 8.0.19). That is, for any commutative diagram of simplicial sets

$$\begin{array}{ccc} \Lambda_i^n & \longrightarrow & X \\ \downarrow & \nearrow \exists & \downarrow p \\ \Delta^n & \longrightarrow & Y \end{array}$$

(Definition 8.0.12) there exists a lift $\Delta^n \rightarrow X$ making both triangles commute.

Definition 8.2.4. (♠ TODO: AI generated, verify) Let $p : X \rightarrow Y$ be a morphism of simplicial sets (Definition 8.0.4). The map p is called a *trivial Kan fibration between simplicial sets* if for every integer $n \geq 0$, the map p has the right lifting property with respect to the boundary inclusion $\partial\Delta^n \hookrightarrow \Delta^n$. That is, for any commutative diagram of simplicial sets

$$\begin{array}{ccc} \partial\Delta^n & \longrightarrow & X \\ \downarrow & \nearrow \exists & \downarrow p \\ \Delta^n & \longrightarrow & Y \end{array}$$

(♠ TODO: ∂) (Definition 8.0.12) there exists a lift $\Delta^n \rightarrow X$ making both triangles commute.

Proposition 8.2.5. Let K be a simplicial set (Definition 8.0.4). The following are equivalent:

1. K is a quasi-category (Definition 8.2.2).
2. The map of simplicial sets induced by the inner horn (Definition 8.0.19) inclusion $\Lambda_1^2 \hookrightarrow \Delta^2$,

$$\text{Map}(\Delta^2, K) \rightarrow \text{Map}(\Lambda_1^2, K),$$

(Definition 8.0.22) is a trivial Kan fibration (Definition 8.2.4).

A basic class of quasi-categories (Definition 8.2.2) would be nerves of categories (Definition 8.0.20) (Proposition 8.2.6).

Proposition 8.2.6 (see [Lur09, Proposition 1.1.2.2]). Let K be a simplicial set (Definition 8.0.4). Then the following conditions are equivalent:

- There exists a small category (Definition 1.0.9) $\mathcal{C}$ and an isomorphism $K \simeq N(\mathcal{C})$ (Definition 8.0.20).
- For each $0 < i < n$ and each diagram

$$\begin{array}{ccc} \Lambda_i^n & \xrightarrow{\quad} & K \\ \downarrow & \nearrow \text{dotted} & \\ \Delta^n & & \end{array}$$

there exists a *unique* dotted arrow rendering the diagram commutative.

Corollary 8.2.7. The nerve $N(\mathcal{C})$ (Definition 8.0.20) of any small category (Definition 1.0.9) is a quasi-category (Definition 8.2.2).

Proof. This is immediate by the definition of a quasi-category and Proposition 8.2.6. □

Definition 8.2.8. An ∞ -groupoid is an ∞ -category (Definition 8.2.2) whose homotopy category (Definition 8.1.11) $h\mathcal{C}$ is a groupoid, i.e. every morphism in $\mathcal{C}$ is an equivalence. (♠ **TODO: objects and morphisms, equivalence in an ∞ -category, which is a simplicial set**) An ∞ -groupoid is also often referred to as an $(\infty, 0)$ -category.

∞ -groupoids are equivalent to Kan-complexes (Definition 8.2.9) (Proposition 8.2.15).

An ∞ -groupoid may be referred to as an *anima* (in the Clausen-Scholze terminology), especially if the ∞ -groupoid is of the form $\text{Mor}_{\mathcal{C}}(x, y)$ (Definition 8.0.23) for an ∞ -category (Definition 8.2.2) $\mathcal{C}$ and objects (Definition 8.0.13)/0-simplices (Definition 8.0.11) x, y of $\mathcal{C}$, cf. Proposition 8.2.16.

Definition 8.2.9. A *Kan complex* or a *Kan simplicial set* is a simplicial set (Definition 8.0.4) K satisfying the Kan extension condition (Definition 8.2.1) for the horns (Definition 8.0.19) $\Lambda_i^n \subset \Delta^n$ for all n and all $0 \leq i < n$. More explicitly, for any $0 \leq i \leq n$, any map $f_0 : \Lambda_i^n \rightarrow K$ (Definition 8.0.19) admits an extension $f : \Delta^n \rightarrow K$ (Definition 8.0.12).

Kan-complexes are equivalent to ∞ -groupoids (Definition 8.2.8) (Proposition 8.2.15).

Lemma 8.2.10. Every Kan complex (Definition 8.2.9) is a quasi-category (Definition 8.2.2).

Proof. This is immediate by definition. □

Lemma 8.2.11. In the category of simplicial sets (Definition 8.0.4), Δ^0 (Definition 8.0.12) is the terminal object (Definition 2.0.6); Δ^0 is also alternatively denoted by $\blacksquare$.

Definition 8.2.12. The *empty simplicial set*, denoted by $\emptyset$, is the simplicial set (Definition 8.0.4) whose set of n -simplices (Definition 8.0.4) is the empty set for every integer $n \geq 0$.

Lemma 8.2.13. In the category of simplicial sets (Definition 8.0.4), the empty simplicial set (Definition 8.2.12) $\emptyset$ is the initial object (Definition 2.0.6).

Lemma 8.2.14. A simplicial set (Definition 8.0.4) K is a Kan complex (Definition 8.2.9) if and only if the simplicial set map $K \rightarrow \Delta^0$ (Definition 8.0.12) (Lemma 8.2.11) is a Kan fibration (Definition 8.2.3).

Proof. Since Δ^0 is the final object in the category of simplicial sets, that $K \rightarrow \Delta^0$ has the right lifting property with respect to a horn (Definition 8.0.19) inclusion $\Lambda_i^n \hookrightarrow \Delta^n$ for any $n \geq 0$ and $0 \leq i \leq n$ is equivalent simply to any morphism $\Lambda_i^n \rightarrow K$ extending to $\Delta^n \rightarrow K$, i.e. that K is a Kan complex. $\square$

Proposition 8.2.15 ([Joy02, Corollary 1.4], cf. [Lur09, Proposition 1.2.5.1]). Let $\mathcal{C}$ be a simplicial set (Definition 8.0.4). The following are equivalent:

1. $\mathcal{C}$ is an ∞ -groupoid (Definition 8.2.8).
2. $\mathcal{C}$ satisfies by Kan extension condition (Definition 8.2.1) for all horns (Definition 8.0.19) $\Lambda_i^n \subseteq \Delta^n$ (Definition 8.0.12) for all $0 \leq i < n$.
3. $\mathcal{C}$ satisfies by Kan extension condition (Definition 8.2.1) for all horns (Definition 8.0.19) $\Lambda_i^n \subseteq \Delta^n$ (Definition 8.0.12) for all $0 < i \leq n$.
4. $\mathcal{C}$ is a Kan complex (Definition 8.2.9).

Proposition 8.2.16. Let X be a quasi-category (Definition 8.2.2) and let $x, y \in X$ be objects (Definition 8.0.13). The mapping space $\text{Mor}(x, y)$ (Definition 8.0.23) is an ∞ -groupoid (Definition 8.2.8).

9. LIMITS AND COLIMITS IN ∞ -CATEGORIES

Definition 9.0.1. (♠ TODO: AI generated, verify) Let X and Y be simplicial sets (Definition 8.0.4). Their *join*, denoted by $X \star Y$, is the simplicial set determined by the following universal property on representables: for each pair of nonnegative integers m, n , there is a canonical identification

$$\Delta[m] \star \Delta[n] = \Delta[m + n + 1],$$

and $X \star Y$ is obtained by extending this assignment bi-simply. Concretely, for each $k \geq 0$, the set of k -simplices is

$$(X \star Y)_k = X_k \sqcup Y_k \sqcup \bigsqcup_{\substack{0 \leq i, j \\ i+j=k-1}} X_i \times Y_j,$$

where the face and degeneracy maps are induced by those on X and Y together with the standard simplex identifications above.

Proposition 9.0.2 ([Lur09, Proposition 1.2.9.2]). (♠ TODO: AI generated, verify) Let S and K be simplicial sets (Definition 8.0.4), and let $p : K \rightarrow S$ be a morphism. There exists

a simplicial set $S_{/p}$ with the following universal property: for every simplicial set Y , there is a natural bijection

$$\mathrm{Hom}_{s\mathrm{Set}}(Y, S_{/p}) \cong \mathrm{Hom}_p(Y \star K, S),$$

(Definition 9.0.1) where the subscript p on the right-hand side indicates that we consider only those morphisms $f : Y \star K \rightarrow S$ such that $f|_K = p$.

The simplicial set $S_{/p}$ is explicitly given by

$$(S_{/p})_n = \mathrm{Hom}_p(\Delta^n \star K, S).$$

Definition 9.0.3. (♠ TODO: AI generated, verify) Let $p : K \rightarrow \mathcal{C}$ be a morphism of simplicial sets (Definition 8.0.4), where $\mathcal{C}$ is a quasi-category (Definition 8.2.2).

1. The simplicial set $\mathcal{C}_{/p}$ from Proposition 9.0.2 is again a quasi-category. It is called the *overcategory of $\mathcal{C}$ with respect to p* , or the *∞ -category of objects of $\mathcal{C}$ over p* .
2. Dually, replacing the join $Y \star K$ by $K \star Y$, one obtains a simplicial set $\mathcal{C}_{p/}$ characterized by the natural bijection

$$\mathrm{Hom}_{s\mathrm{Set}}(Y, \mathcal{C}_{p/}) \cong \mathrm{Hom}_p(K \star Y, \mathcal{C}),$$

where the subscript p again indicates that only morphisms $f : K \star Y \rightarrow \mathcal{C}$ with $f|_K = p$ are considered. The simplicial set $\mathcal{C}_{p/}$ is called the *undercategory of $\mathcal{C}$ with respect to p* , or the *∞ -category of objects of $\mathcal{C}$ under p* .

In the special case where $K = \Delta^n$ and p classifies a simplex $\sigma \in \mathcal{C}_n$, we also write $\mathcal{C}_{\sigma/}$ for $\mathcal{C}_{p/}$. In particular, when X is an object of $\mathcal{C}$ (a 0-simplex), we write $\mathcal{C}_{X/}$.

Definition 9.0.4 (Joyal, [Lur09, Definition 1.2.13.4]). (♠ TODO: AI generated, verify) Let $\mathcal{C}$ be a quasi-category (Definition 8.2.2), and let $p : K \rightarrow \mathcal{C}$ be a morphism of simplicial sets (Definition 8.0.4).

1. A *colimit for p in $\mathcal{C}$* is an initial object (Definition 2.0.6) of the undercategory (Definition 9.0.3) $\mathcal{C}_{p/}$.
2. A *limit for p in $\mathcal{C}$* is a final object (Definition 2.0.6) of the overcategory (Definition 9.0.3) $\mathcal{C}_{/p}$.

Definition 9.0.5. (♠ TODO: AI generated, verify) Let $\mathcal{C}$ be an ∞ -category (Definition 8.2.2).

An object $\ast \in \mathcal{C}$ is called a *terminal/final object of $\mathcal{C}$* if it is a limit (Definition 9.0.4) of the empty diagram $\emptyset \rightarrow \mathcal{C}$. Equivalently, $\ast$ is terminal if for every object $X \in \mathcal{C}$, the mapping space (Definition 8.0.23) $\mathrm{Map}_{\mathcal{C}}(X, \ast)$ is contractible.

Dually, an object $\emptyset \in \mathcal{C}$ is called an *initial object of $\mathcal{C}$* if it is a colimit (Definition 9.0.4) of the empty diagram $\emptyset \rightarrow \mathcal{C}$. Equivalently, $\emptyset$ is initial if for every object $X \in \mathcal{C}$, the mapping space $\mathrm{Map}_{\mathcal{C}}(\emptyset, X)$ is contractible.

Definition 9.0.6. (♠ TODO: AI generated, verify) Let $\mathcal{C}$ be an ∞ -category (Definition 8.2.2).

A *pullback in $\mathcal{C}$* is a limit (Definition 9.0.4) of a diagram $p : K \rightarrow \mathcal{C}$ where K is the simplicial set corresponding to the span $\{1 \rightarrow 0 \leftarrow 2\}$ (the nerve of the category $\bullet \rightarrow \bullet \leftarrow \bullet$). Concretely, given morphisms $f : X \rightarrow Z$ and $g : Y \rightarrow Z$, the *pullback of f along g* (or *fiber*

product of X and Y over Z), denoted by $X \times_Z^h Y$, is an object equipped with morphisms $p_X : X \times_Z^h Y \rightarrow X$ and $p_Y : X \times_Z^h Y \rightarrow Y$ and an equivalence $f \circ p_X \simeq g \circ p_Y$, universal among such data.

The superscript h on $\times_Z^h$ emphasizes that this is a homotopy pullback, though in the context of ∞ -categories all limits are inherently homotopy limits, so the notation is sometimes omitted.

Dually, a *pushout in $\mathcal{C}$* is a colimit (Definition 9.0.4) of a cospan diagram $X \rightarrow Z \leftarrow Y$, denoted by $X \amalg_Z^h Y$.

10. ALGEBRAIC OBJECTS IN ∞ -CATEGORIES

Definition 10.0.1. (♠ TODO: AI generated, verify) Let $\mathcal{C}$ be an ∞ -category (Definition 8.2.2).

A *simplicial object in $\mathcal{C}$* is a functor (Definition 8.0.14)

$$X : N(\Delta^{\text{op}}) \rightarrow \mathcal{C},$$

where $N(\Delta^{\text{op}})$ is the nerve (Definition 8.0.20) of the opposite (Definition 1.0.12) of the simplex category (Definition 8.0.3) Δ .

By abuse of notation, we often write $X : \Delta^{\text{op}} \rightarrow \mathcal{C}$ rather than $X : N(\Delta^{\text{op}}) \rightarrow \mathcal{C}$. For each $n \geq 0$, we write X_n for the value of X on the object $[n] \in \Delta$. For each order-preserving map $\theta : [m] \rightarrow [n]$ in Δ , we write $X(\theta)$ for the induced morphism $X_n \rightarrow X_m$ in $\mathcal{C}$. In particular, the images of the coface and codegeneracy maps give *face maps* $d_i : X_n \rightarrow X_{n-1}$ and *degeneracy maps* $s_j : X_n \rightarrow X_{n+1}$ satisfying the simplicial identities up to coherent homotopy.

The ∞ -category of simplicial objects in $\mathcal{C}$ is denoted by $\mathbf{s}\mathcal{C}$ (or $\mathcal{C}^{\Delta^{\text{op}}}$).

Definition 10.0.2. (♠ TODO: AI generated, verify) Let $\mathcal{C}$ be an ∞ -category (Definition 8.2.2) admitting finite pullbacks (Definition 9.0.6).

A *category object in $\mathcal{C}$* is a simplicial object (Definition 10.0.1) $X : \Delta^{\text{op}} \rightarrow \mathcal{C}$ such that for every integer $n \geq 2$, the *Segal map*

$$X_n \rightarrow X_1 \times_{X_0}^h X_1 \times_{X_0}^h \cdots \times_{X_0}^h X_1$$

(n factors on the right) is an equivalence in $\mathcal{C}$. Here the fiber products are taken in the ∞ -category $\mathcal{C}$, and the Segal map is induced by the n injective order-preserving maps $\delta^i : [1] \rightarrow [n]$ defined by $\delta^i(0) = i$ and $\delta^i(1) = i + 1$ for $i = 0, \dots, n - 1$.

For a category object X , we interpret

- X_0 as the *object of objects*;
- X_1 as the *object of morphisms*;
- the face maps $d_0, d_1 : X_1 \rightarrow X_0$ as the *target* and *source* maps;
- the Segal equivalence for $n = 2$, $X_2 \simeq X_1 \times_{X_0}^h X_1$, as encoding *composition of morphisms*;

- the degeneracy map $s_0 : X_0 \rightarrow X_1$ as encoding *identity morphisms*.

The Segal conditions for $n \geq 3$ encode associativity of composition up to coherent homotopy.

Definition 10.0.3. (♠ TODO: AI generated, verify) Let $\mathcal{C}$ be an ∞ -category (Definition 8.2.2) admitting finite products, viewed as a cartesian monoidal ∞ -category.

A *semigroup object in $\mathcal{C}$* can be defined in two equivalent ways:

1. **(Via Segal Conditions)** A functor $X : N(\Delta_{\text{semi}}^{\text{op}}) \rightarrow \mathcal{C}$ from the nerve (Definition 8.0.20) of the opposite of the semi-simplex category Δ_{semi} (the category of non-empty finite linear orders and injective order-preserving maps) such that for every integer $n \geq 2$, the Segal map (Definition 10.0.2)

$$X_n \rightarrow \prod_{i=0}^{n-1} X_1 = \underbrace{X_1 \times X_1 \times \cdots \times X_1}_{n \text{ factors}}$$

is an equivalence in $\mathcal{C}$. Here, the map is induced by the n inert morphisms $\rho_i : [1] \rightarrow [n]$ in Δ_{semi} given by $\rho_i(0) = i$ and $\rho_i(1) = i + 1$. The underlying object of the semigroup is identified as $A := X_1$.

2. **(Via ∞ -Operads)** An algebra object in $\mathcal{C}$ over the non-unital associative ∞ -operad $\mathcal{A}ss^{\text{nu}}$. That is, an object of the ∞ -category $\text{Alg}_{\mathcal{A}ss^{\text{nu}}}(\mathcal{C})$. (♠ TODO: algebra object, non-unital associative ∞ -operad)

These two characterizations are equivalent via the *monoidal Rezk completion*. Specifically, the nerve $N(\Delta_{\text{semi}}^{\text{op}})$ serves as a simplicial model for the non-unital associative operad, and the ∞ -category of functors satisfying the Segal condition in (1) is equivalent to $\text{Alg}_{\mathcal{A}ss^{\text{nu}}}(\mathcal{C})$ under the cartesian symmetric monoidal structure.

A semigroup object is called *commutative* (or *abelian*) if it satisfies either of the following equivalent conditions:

- Under definition (1), the functor X extends from $N(\Delta_{\text{semi}}^{\text{op}})$ to the nerve of the opposite of the category of finite non-empty sets and surjective maps.
- Under definition (2), the algebra object extends to an algebra over the non-unital commutative ∞ -operad $\mathcal{C}om^{\text{nu}}$.

Definition 10.0.4. (♠ TODO: AI generated, verify) Let $\mathcal{C}$ be an ∞ -category (Definition 8.2.2) admitting finite products and a terminal object (Definition 9.0.5) $*$.

Let (A, μ) be a semigroup object (Definition 10.0.3) in $\mathcal{C}$. A *unital structure* on (A, μ) consists of a morphism $\eta : * \rightarrow A$ from the terminal object, together with a choice of homotopies rendering the diagrams

$$\begin{aligned} A &\simeq * \times A \xrightarrow{\eta \times \text{id}_A} A \times A \xrightarrow{\mu} A \\ A &\simeq A \times * \xrightarrow{\text{id}_A \times \eta} A \times A \xrightarrow{\mu} A \end{aligned}$$

equivalent to the identity id_A , compatible with the infinite hierarchy of higher associativity coherences of the underlying semigroup.

Formally, a **monoid object in $\mathcal{C}$** is a semigroup object $X : N(\Delta_{\text{semi}}^{\text{op}}) \rightarrow \mathcal{C}$ that extends to a functor $\tilde{X} : N(\Delta^{\text{op}}) \rightarrow \mathcal{C}$ out of the full simplex category, satisfying the Segal conditions.

Equivalently, under the operadic description, the forgetful functor from the associative ∞ -operad to the non-unital associative ∞ -operad induces a conservative, fully faithful functor on the subcategory of unit-admitting algebras:

$$\text{Alg}_{\mathcal{A}ss}(\mathcal{C}) \rightarrow \text{Alg}_{\mathcal{A}ss^{\text{nu}}}(\mathcal{C})$$

A monoid object is precisely a semigroup object that lies in the essential image of this forgetful functor.

Definition 10.0.5. (♠ TODO: AI generated, verify) Let $\mathcal{C}$ be an ∞ -category (Definition 8.2.2) admitting a terminal object (Definition 9.0.5) $*$ and finite pullbacks (Definition 9.0.6).

A **group object in $\mathcal{C}$** is a monoid object (Definition 10.0.4) M in $\mathcal{C}$ (presented by a simplicial object $X : \Delta^{\text{op}} \rightarrow \mathcal{C}$ with $X_0 \simeq *$) satisfying either of the following equivalent conditions:

1. For every object $Y \in \mathcal{C}$, the monoid of connected components $\pi_0(\text{Map}_{\mathcal{C}}(Y, M))$ is a group (Definition C.0.5).
2. The shear map $X_2 \rightarrow X_1 \times X_1$ induced by the face maps (d_0, d_2) is an equivalence in $\mathcal{C}$.

A group object G is called **abelian** if its underlying monoid structure extends to a commutative monoid object (an algebra over the commutative ∞ -operad $\mathcal{C}om$). This implies that for every $Y \in \mathcal{C}$, the group $\pi_0(\text{Map}_{\mathcal{C}}(Y, G))$ is abelian.

(♠ TODO: bar construction) The condition that M is a group object can also be expressed in terms of the simplicial object X presenting M : letting $BM := X$ denote the **bar construction** on M , M is a group object if and only if the canonical map to the loop space

$$M \rightarrow \Omega BM := * \times_{BM} *$$

(♠ TODO: need loop space definition for pointed simplicial set) is an equivalence, where the fiber product is the pullback along the basepoint $* \rightarrow BM$ in the ∞ -category of simplicial objects in $\mathcal{C}$.

(♠ TODO:)

Definition 10.0.6 (Ring objects in an ∞ -category). Let $\mathcal{C}$ be an ∞ -category with finite products.

1. A **ring object** in $\mathcal{C}$ is an object $R \in \mathcal{C}$ equipped with:
 - a group object structure $(R, +, 0, -)$ in $\mathcal{C}$ (additive structure),
 - a monoid object structure $(R, \cdot, 1_R)$ in $\mathcal{C}$ (multiplicative structure),
such that the distributivity, absorption and unital axioms for a ring hold up to coherent homotopy, expressed by the homotopy-commutativity of the diagrams corresponding to

$$a \cdot (b + c) \simeq a \cdot b + a \cdot c, \quad (a + b) \cdot c \simeq a \cdot c + b \cdot c,$$

and $0 \cdot a \simeq 0 \simeq a \cdot 0$, for all $a, b, c \in R$.

2. A **commutative ring object** in $\mathcal{C}$ is a ring object whose multiplicative monoid structure is commutative up to coherent homotopy (i.e. a commutative monoid object in the sense above).

Definition 10.0.7 (Module objects over a ring object in an ∞ -category). Let $\mathcal{C}$ be an ∞ -category with finite products and let R be a ring object in $\mathcal{C}$. An **R -module object in $\mathcal{C}$** is an object $M \in \mathcal{C}$ together with

- a group object structure $(M, +_M, 0_M, -_M)$ in $\mathcal{C}$,
- an action morphism in $\mathcal{C}$

$$\alpha : R \times M \rightarrow M,$$

such that the usual module axioms hold up to coherent homotopy, i.e. the diagrams expressing

$$\begin{aligned} r \cdot (m_1 +_M m_2) &\simeq r \cdot m_1 +_M r \cdot m_2, & (r_1 + r_2) \cdot m &\simeq r_1 \cdot m +_M r_2 \cdot m, \\ (r_1 \cdot r_2) \cdot m &\simeq r_1 \cdot (r_2 \cdot m), & 1_R \cdot m &\simeq m, & 0_R \cdot m &\simeq 0_M \simeq r \cdot 0_M \end{aligned}$$

commute in the homotopy-coherent sense in $\mathcal{C}$.

11. HIGHER CATEGORY THEORY

11.1. 2-categories.

Definition 11.1.1. A **strict 2-category** consists of:

1. A collection $\text{Ob}(\mathcal{C})$ of objects, referred to as the **0-cells**.
2. For each pair of objects $x, y \in \text{Ob}(\mathcal{C})$, a category (Definition 1.0.1) $\text{Hom}(x, y) = \text{Hom}_{\mathcal{C}}(x, y)$.
 - (a) The objects in this category are called the **1-morphisms** or **1-cells of $\mathcal{C}$** .
 - (b) The morphisms in this category are called the **2-morphisms** or **2-cells of $\mathcal{C}$** .
3. For each object $x \in \text{Ob}(\mathcal{C})$, an object

$$\text{id}_x \in \text{Hom}_{\mathcal{C}}(x, x)$$

called the **identity object**,

4. For each triple of objects $x, y, z \in \text{Ob}(\mathcal{C})$, A functor

$$\circ_{x,y,z} : \text{Hom}_{\mathcal{C}}(x, y) \times \text{Hom}_{\mathcal{C}}(y, z) \rightarrow \text{Hom}_{\mathcal{C}}(x, z)$$

(Definition 1.3.11) called **composition**

that satisfy the following:

1. **(Strict associativity)** For any quadruple of objects $w, x, y, z \in \text{Ob}(\mathcal{C})$, the two compositions

$$\circ_{w,y,z} \circ (1 \times \circ_{w,x,y}), \quad \circ_{w,x,z} \circ (\circ_{x,y,z} \times 1)$$

are *equal* as functors (♠ TODO: what is 1?)

$$\text{Hom}_{\mathcal{C}}(y, z) \times \text{Hom}_{\mathcal{C}}(x, y) \times \text{Hom}_{\mathcal{C}}(w, x) \rightarrow \text{Hom}_{\mathcal{C}}(w, z).$$

2. **(Strict unit laws)** for all $x, y \in \text{Ob}(\mathcal{C})$,

$$\circ_{x,x,y}(\text{id}_x, -) = 1_{\text{Hom}_{\mathcal{C}}(x,y)}, \quad \circ_{x,y,y}(-, \text{id}_y) = 1_{\text{Hom}_{\mathcal{C}}(x,y)}.$$

That is, composing with an identity morphism acts as the identity functor.

Roughly speaking, a strict 2-category is a strict n -category (Definition 11.2.2) for $n = 2$, at least when a strict 0-category is understood to be a set.

Definition 11.1.2. A **bicategory** or a **weak 2-category** $\mathcal{B}$ consists of the following data:

1. A collection of **0-cells** (or **objects**), denoted by $A, B, C, \dots$;
2. For each pair of 0-cells A, B , a category (Definition 1.0.1) $\mathcal{B}(A, B)$, whose objects are called **1-cells** (denoted $f : A \rightarrow B$) and whose morphisms are called **2-cells** (denoted $\alpha : f \Rightarrow g$);
3. For each triple of 0-cells A, B, C , a composition functor

$$\odot_{ABC} : \mathcal{B}(B, C) \times \mathcal{B}(A, B) \rightarrow \mathcal{B}(A, C),$$

denoted by $(g, f) \mapsto g \odot f$;

4. For each 0-cell A , an identity 1-cell $I_A \in \mathcal{B}(A, A)$;
5. For each quadruple of 0-cells A, B, C, D and 1-cells $f : A \rightarrow B, g : B \rightarrow C, h : C \rightarrow D$, a natural isomorphism called the **associator**,

$$a_{h,g,f} : (h \odot g) \odot f \xrightarrow{\cong} h \odot (g \odot f);$$

6. For each pair of 0-cells A, B and 1-cell $f : A \rightarrow B$, natural isomorphisms called the **left and right unitors**,

$$\lambda_f : I_B \odot f \xrightarrow{\cong} f \quad \text{and} \quad \rho_f : f \odot I_A \xrightarrow{\cong} f.$$

These must satisfy the following two coherence axioms:

1. **The Pentagon Identity:** For 1-cells f, g, h, k , the following diagram of 2-cells commutes:

$$\begin{array}{ccc}
 & (k \circ h) \circ (g \circ f) & \\
 \alpha \nearrow & & \searrow \alpha \\
 ((k \circ h) \circ g) \circ f & & k \circ (h \circ (g \circ f)) \\
 \alpha \circ 1 \downarrow & & \uparrow 1 \circ \alpha \\
 (k \circ (h \circ g)) \circ f & \xrightarrow{\alpha} & k \circ ((h \circ g) \circ f)
 \end{array}$$

2. **The Triangle Identity:** For 1-cells $f : A \rightarrow B$ and $g : B \rightarrow C$, the following diagram of 2-cells commutes:

$$\begin{array}{ccc}
 (g \circ I_B) \circ f & \xrightarrow{\alpha} & g \circ (I_B \circ f) \\
 \rho \circ 1 \searrow & & \swarrow 1 \circ \lambda \\
 & g \circ f &
 \end{array}$$

Lemma 11.1.3. Any strict 2-category (Definition 11.1.1) is a weak 2-category (Definition 11.1.2).

Proof. In a strict 2-category, the strict associativity and the strict unit laws in particular yield associators and left/right unitors. $\square$

Definition 11.1.4. The *category of small categories* is the category defined by the following data:

- The objects are all small categories (Definition 1.0.9).
- The morphisms between two small categories $\mathcal{C}$ and $\mathcal{D}$ are the functors (Definition 1.1.1) $F : \mathcal{C} \rightarrow \mathcal{D}$.
- The composition of morphisms is the standard composition of functors.
- The identity morphism for each object $\mathcal{C}$ is the identity functor $1_{\mathcal{C}}$.

This category is a large category (Definition 1.0.1) and is denoted by **Cat**

Definition 11.1.5. Let $\mathcal{U}$ be a fixed Grothendieck universe (Definition E.0.1). The *category of categories (relative to $\mathcal{U}$)* or the *category of $\mathcal{U}$ -small categories* is the category defined by:

- The objects are all $\mathcal{U}$ -small (Definition 1.0.9) categories (Definition 1.0.1) $\mathcal{C}$.
- The morphisms are functors (Definition 1.1.1) between such categories.

This category is denoted by $\mathcal{U} - \mathbf{Cat}$ or by **Cat**, if $\mathcal{U}$ is understood.

In the case that Tarski-Grothendieck (Definition E.0.15) set theory is assumed (so in particular, the axiom of universes is assumed), one often adopts the convention of denoting $U_n - \mathbf{Cat}$ (Definition E.0.16) by **Cat** and $U_{n+1} - \mathbf{Cat}$ (Definition E.0.16) by **CAT**. Thus, for $n = 1$, **Cat** would be the category of small categories, whereas **CAT** would serve as the “category of large categories” which consists of many common categories that are too large to be U_1 -categories such as **Set**, **Grp**, **Top** or even **Cat**.

Definition 11.1.6. Let $\mathcal{U}$ be a fixed Grothendieck universe (Definition E.0.1). The category $\mathcal{U} - \mathbf{Cat}$ (Definition 11.1.5) can be enriched with the structure of a strict 2-category (Definition 11.1.1) as follows:

- The objects/0-cells are all $\mathcal{U}$ -small categories (Definition 1.0.9).
- Given $\mathcal{U}$ -small categories $\mathcal{C}, \mathcal{D}$, the category $\mathrm{Hom}(\mathcal{C}, \mathcal{D}) = \mathrm{Hom}_{\mathcal{U} - \mathbf{Cat}}(\mathcal{C}, \mathcal{D})$ is the functor category (Definition 1.2.1) $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$; Recall that
 - the objects of $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ are functors (Definition 1.1.1) $\mathcal{C} \rightarrow \mathcal{D}$
 - the morphisms between functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$ are natural transformations (Definition 1.1.3) $F \Rightarrow G$.

note that $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ is a $\mathcal{U}$ -small category.

- The identity objects $\mathrm{id}_{\mathcal{C}} \in \mathrm{Ob}(\mathrm{Fun}(\mathcal{C}, \mathcal{C}))$ are given by the standard identity functors
- For every triple $\mathcal{C}, \mathcal{D}, \mathcal{E}$ of $\mathcal{U}$ -small categories, the composition functors

$$\mathrm{Fun}(\mathcal{C}, \mathcal{D}) \times \mathrm{Fun}(\mathcal{D}, \mathcal{E}) \rightarrow \mathrm{Fun}(\mathcal{C}, \mathcal{E})$$

are given by the standard composition of functors

Definition 11.1.7. A *pseudofunctor* $F : \mathcal{C} \rightarrow \mathcal{D}$ between weak 2-categories (Definition 11.1.2) $\mathcal{C}$ and $\mathcal{D}$ consists of:

- An assignment for each object $x \in \mathcal{C}$ to an object $Fx \in \mathcal{D}$.
- For each pair of objects $x, y \in \mathcal{C}$, a functor $F_{x,y} : \mathcal{C}(x, y) \rightarrow \mathcal{D}(Fx, Fy)$ between the mapping categories.
- For each object $x \in \mathcal{C}$, a 2-isomorphism $F_{id_x} : id_{Fx} \Rightarrow F(id_x)$ called the *identity constraint*.
- For each pair of composable 1-morphisms $f : x \rightarrow y$ and $g : y \rightarrow z$ in $\mathcal{C}$, a 2-isomorphism $F_{g,f} : Fg \circ Ff \Rightarrow F(g \circ f)$ called the *composition constraint*, which is natural in f and g .

The identity 2-isomorphism is often denoted by $\phi_x : id_{Fx} \cong F(id_x)$ and the composition 2-isomorphism by $\phi_{g,f} : Fg \circ Ff \cong F(g \circ f)$.

Theorem 11.1.8. Let $\mathcal{C}$ and $\mathcal{D}$ be weak 2-categories (Definition 11.1.2). For a pseudofunctor (Definition 11.1.7) $F : \mathcal{C} \rightarrow \mathcal{D}$ to be well-defined, the constraints must satisfy the following coherence conditions:

1. Associativity Coherence: For any three composable 1-morphisms f, g, h , the following diagram of 2-isomorphisms commutes:

$$\begin{array}{ccc}
 (Fh \circ Fg) \circ Ff & \xrightarrow{\alpha_{Fx, Fy, Fz}} & Fh \circ (Fg \circ Ff) \\
 \downarrow \phi_{h,g} \circ 1 & & \downarrow 1 \circ \phi_{g,f} \\
 F(h \circ g) \circ Ff & & Fh \circ F(g \circ f) \\
 \downarrow \phi_{h \circ g, f} & & \downarrow \phi_{h, g \circ f} \\
 F((h \circ g) \circ f) & \xrightarrow{F(\alpha_{x,y,z})} & F(h \circ (g \circ f))
 \end{array}$$

where α denotes the associator in the respective bicategories.

2. Unit Coherence: For any 1-morphism $f : x \rightarrow y$, the following diagrams commute:

$$\begin{array}{ccc}
 id_{Fy} \circ Ff & \xrightarrow{\ell_{Ff}} & Ff \\
 \downarrow \phi_y \circ 1 & & \uparrow F(\ell_f) \\
 F(id_y) \circ Ff & \xrightarrow{\phi_{id_y, f}} & F(id_y \circ f)
 \end{array}
 \quad \text{and} \quad
 \begin{array}{ccc}
 Ff \circ id_{Fx} & \xrightarrow{r_{Ff}} & Ff \\
 \downarrow 1 \circ \phi_x & & \uparrow F(r_f) \\
 Ff \circ F(id_x) & \xrightarrow{\phi_{f, id_x}} & F(f \circ id_x)
 \end{array}$$

where ℓ and r denote the left and right unitors.

Definition 11.1.9. Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be pseudofunctors (Definition 11.1.7) between weak 2-categories (bicategories) (Definition 11.1.2). A *pseudonatural transformation* $\alpha : F \Rightarrow G$ consists of:

- For each object $x \in \mathcal{C}$, a 1-morphism $\alpha_x : Fx \rightarrow Gx$ in $\mathcal{D}$.
- For each 1-morphism $f : x \rightarrow y$ in $\mathcal{C}$, an invertible 2-morphism (a 2-isomorphism) $\alpha_f : Gf \circ \alpha_x \Rightarrow \alpha_y \circ Ff$ in $\mathcal{D}$.

These data must satisfy two coherence conditions: (♠ TODO: make diagrams of these conditions)

1. Naturality with respect to 2-morphisms $\sigma : f \Rightarrow g$: The 2-morphisms α_f and α_g must be compatible with $F\sigma$ and $G\sigma$.
2. Compatibility with composition and identities: The 2-isomorphisms α_f must respect the composition constraints $\phi_{g,f}$ and identity constraints ϕ_x of the pseudofunctors.

The 2-isomorphism α_f in a pseudonatural transformation is often called the *filler of the naturality square*.

Definition 11.1.10. Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be pseudofunctors (Definition 11.1.7) between weak 2-categories (bicategories) (Definition 11.1.2). A *modification* $\Gamma : \alpha \Rightarrow \beta$ *between pseudonatural transformations* $\alpha, \beta : F \Rightarrow G$ is a collection of 2-morphisms $\Gamma_x : \alpha_x \Rightarrow \beta_x$ in $\mathcal{D}$ for each $x \in \mathcal{C}$, such that for every 1-morphism $f : x \rightarrow y$ in $\mathcal{C}$, the following square of 2-morphisms commutes:

$$\begin{array}{ccc} Gf \circ \alpha_x & \xrightarrow{\alpha_f} & \alpha_y \circ Ff \\ \downarrow 1 \circ \Gamma_x & & \downarrow \Gamma_y \circ 1 \\ Gf \circ \beta_x & \xrightarrow{\beta_f} & \beta_y \circ Ff \end{array}$$

Definition 11.1.11. Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be pseudofunctors (Definition 11.1.7) between weak 2-categories (bicategories) (Definition 11.1.2). A pseudofunctor (Definition 11.1.7) $F : \mathcal{C} \rightarrow \mathcal{D}$ is a *biequivalence* if there exists a pseudofunctor $G : \mathcal{D} \rightarrow \mathcal{C}$ and pseudonatural transformations (Definition 11.1.9) $\eta : id_{\mathcal{C}} \Rightarrow G \circ F$ and $\varepsilon : F \circ G \Rightarrow id_{\mathcal{D}}$ that are internal equivalences in the respective bicategories of functors. (♠ TODO: what does internal equivalences mean here?)

Theorem 11.1.12. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be pseudofunctors (Definition 11.1.7) between weak 2-categories (bicategories) (Definition 11.1.2). The pseudofunctor (Definition 11.1.7) $F : \mathcal{C} \rightarrow \mathcal{D}$ is a biequivalence (Definition 11.1.11) if and only if the following two conditions hold:

1. F is locally an equivalence: For every pair of objects $x, y \in \mathcal{C}$, the functor $F_{x,y} : \mathcal{C}(x, y) \rightarrow \mathcal{D}(Fx, Fy)$ is an equivalence of categories. (♠ TODO: biessentially surjective)
2. F is *biessentially* surjective on objects: For every object $d \in \mathcal{D}$, there exists an object $x \in \mathcal{C}$ and an equivalence $Fx \simeq d$ in the bicategory $\mathcal{D}$.

Theorem 11.1.13 (Coherence theorem/strictness theorem for bicategories). Every weak 2-category (Definition 11.1.2) is biequivalent (Definition 11.1.11) to a strict 2-category (Definition 11.1.1). (♠ TODO: a citation for this)

11.2. n -categories.

Notation 11.2.1. Let **Cat** denote the category of (small) categories and functors between them. For integers $n \geq 0$, we define inductively the category of n -categories.

In a strict n -category, the associativity and unit axioms are specified “strictly” — i.e. they are expressed in terms of literal equalities as opposed to looser isomorphisms.

Definition 11.2.2. A *strict 0-category* is a set. A *strict $(n+1)$ -category* is a category enriched in (Definition 4.0.1) the category of strict n -categories. Explicitly, a strict $(n+1)$ -category $\mathcal{C}$ consists of:

- a collection of objects $\text{Ob}(\mathcal{C})$; such an object is called a **0-cell** or synonymously a **0-morphism**,
- for each pair of objects $x, y \in \text{Ob}(\mathcal{C})$, a strict n -category $\text{Hom}_{\mathcal{C}}(x, y)$; inductively, this strict n -category itself contains **k -cells**, or synonymously **k -morphisms**, for $0 \leq k \leq n$; the k -cells of $\text{Hom}_{\mathcal{C}}(x, y)$ are exactly the $k + 1$ -cells of $\mathcal{C}$.
 - For each object/0-cell x of $\mathcal{C}$, there is an **identity** $\text{id}_x \in \text{Hom}_{\mathcal{C}}(x, x)$, which is an $n + 1$ cell of $\mathcal{C}$ and or equivalently an n cell of $\text{Hom}_{\mathcal{C}}(x, x)$.
- composition functors

$$\circ_{x,y,z} : \text{Hom}_{\mathcal{C}}(y, z) \times \text{Hom}_{\mathcal{C}}(x, y) \rightarrow \text{Hom}_{\mathcal{C}}(x, z)$$

that satisfy the following:

1. **(Strict associativity)** For any quadruple of objects w, x, y, z , the two possible compositions

$$\circ_{w,y,z} \circ (1 \times \circ_{w,x,y}), \quad \circ_{w,x,z} \circ (\circ_{x,y,z} \times 1)$$

are *equal* as functors (♠ TODO: what is 1?)

$$\text{Hom}_{\mathcal{C}}(y, z) \times \text{Hom}_{\mathcal{C}}(x, y) \times \text{Hom}_{\mathcal{C}}(w, x) \rightarrow \text{Hom}_{\mathcal{C}}(w, z).$$

2. **(Strict unit laws)** For each object $x \in \text{Ob}(\mathcal{C})$, there exists a distinguished identity element

$$\text{id}_x \in \text{Hom}_{\mathcal{C}}(x, x),$$

such that for all $x, y \in \text{Ob}(\mathcal{C})$,

$$\circ_{x,x,y}(\text{id}_x, -) = 1_{\text{Hom}_{\mathcal{C}}(x,y)}, \quad \circ_{x,y,y}(-, \text{id}_y) = 1_{\text{Hom}_{\mathcal{C}}(x,y)}.$$

That is, composing with an identity morphism acts as the identity functor.

3. **(Functoriality)** (♠ TODO: strict n -functor) Each composition map $\circ_{x,y,z}$ is a strict n -functor; it preserves all k -cell compositions and identities in the strict n -category structure of each $\text{Hom}_{\mathcal{C}}(x, y)$ strictly (i.e., on the nose).

In particular, collecting all the strict 0-categories yields the category of sets (Definition 1.0.3), i.e. the 1-category whose objects are sets and morphisms are functions between these sets. Moreover, a strict 1-category is equivalent to a locally small category (Definition 1.0.9).

Morphisms between strict $(n + 1)$ -categories are functors that preserve all levels of composition strictly.

Definition 11.2.3. A **weak n -category** (in the sense of Tamsamani and Simpson) is defined inductively on $n \geq 0$.

Base Case ($n = 0$): A weak 0-category is a set.

Inductive Step: For $n \geq 1$, a **weak n -category** is a simplicial object $X_{\bullet}$ in the category of weak $(n - 1)$ -categories, denoted $X : \Delta^{op} \rightarrow \text{Weak-}(n - 1)\text{-Cat}$, satisfying the Segal condition (composition) and the discreteness condition (identities).

Explicitly, X consists of:

- A sequence of weak $(n - 1)$ -categories X_k for each integer $k \geq 0$ (where X_0 represents the objects, X_1 the morphisms, X_2 composable pairs, etc.);

- Face maps $d_i : X_k \rightarrow X_{k-1}$ and degeneracy maps $s_i : X_k \rightarrow X_{k+1}$ in the category of weak $(n-1)$ -categories, satisfying the simplicial identities.

These data must satisfy the following properties:

1. **Discreteness (Units):** The object of 0-simplices X_0 is "discrete" in the sense that the underlying simplicial set of X_0 is constant (homotopically equivalent to a set).
2. **Segal Condition (Composition):** For every $k \geq 2$, the map induced by the simplicial segment inclusions

$$\mu_k : X_k \xrightarrow{\sim} X_1 \times_{X_0} X_1 \times_{X_0} \cdots \times_{X_0} X_1 \quad (k \text{ times})$$

is an *equivalence* of weak $(n-1)$ -categories.

Here, the fiber product is formed using the source and target maps $(d_1, d_0 : X_1 \rightarrow X_0)$. The "equivalence" is the notion of equivalence appropriate for $(n-1)$ -categories (defined inductively as maps inducing weak homotopy equivalences on all hom-spaces).

Definition 11.2.4. A *weak 3-category* (Definition 11.2.3) (or *tricategory* in the sense of Gordon-Power-Street) consists of:

- A collection of 0-cells (objects);
- For every pair of 0-cells A, B , a bicategory $\mathcal{C}(A, B)$;
- **Composition:** Pseudofunctors $\otimes : \mathcal{C}(B, C) \times \mathcal{C}(A, B) \rightarrow \mathcal{C}(A, C)$;
- **Units:** Identity 1-cells I_A (objects in $\mathcal{C}(A, A)$);

Equipped with the following coherent equivalence data (constraints):

- **Associator Adjoint Equivalence:** A pseudonatural equivalence in the bicategory of bicategories:

$$\mathbf{a} : \otimes \circ (\otimes \times 1) \xrightarrow{\sim} \otimes \circ (1 \times \otimes);$$

- **Unitor Adjoint Equivalences:** Pseudonatural equivalences $\mathbf{l}, \mathbf{r}$ for left and right units.
- **Modifications (The 3-dimensional Axioms):** Invertible 3-cells satisfying specific axioms:
 - The *pentagon modification* π (relating the various 3-cell associators);
 - The *middle triangle modification* μ ;
 - The *left/right triangle modifications* λ, ρ .

These modifications must satisfy the non-abelian 4-cocycle conditions, specifically the *non-abelian pentagon equation* (often called the associahedron equation K_5).

Definition 11.2.5. Let $\mathcal{C}$ and $\mathcal{D}$ be strict n -categories (Definition 11.2.2). For each pair of objects $x, y \in \text{Ob}(\mathcal{C})$, write

$$\text{Hom}_{\mathcal{C}}(x, y), \quad \text{Hom}_{\mathcal{D}}(Fx, Fy)$$

for the corresponding hom $((n-1))$ -categories.

A *functor between n -categories* (or *n -functor*)

$$F : \mathcal{C} \longrightarrow \mathcal{D}$$

consists of the following data:

- a function on objects

$$F_0 : \text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{D}),$$

- for each pair of objects $x, y \in \text{Ob}(\mathcal{C})$, a strict $(n - 1)$ -functor

$$F_{x,y} : \text{Hom}_{\mathcal{C}}(x, y) \rightarrow \text{Hom}_{\mathcal{D}}(Fx, Fy),$$

- these assignments must satisfy the following axioms.

1. **(Functoriality of composition)** For all $x, y, z \in \text{Ob}(\mathcal{C})$, the following square of strict $(n - 1)$ -functors commutes strictly:

$$\begin{array}{ccc} \text{Hom}_{\mathcal{C}}(y, z) \times \text{Hom}_{\mathcal{C}}(x, y) & \xrightarrow{\circ_{\mathcal{C}}} & \text{Hom}_{\mathcal{C}}(x, z) \\ \downarrow F_{y,z} \times F_{x,y} & & \downarrow F_{x,z} \\ \text{Hom}_{\mathcal{D}}(Fy, Fz) \times \text{Hom}_{\mathcal{D}}(Fx, Fy) & \xrightarrow{\circ_{\mathcal{D}}} & \text{Hom}_{\mathcal{D}}(Fx, Fz) \end{array}$$

That is,

$$F_{x,z} \circ \circ_{\mathcal{C}} = \circ_{\mathcal{D}} \circ (F_{y,z} \times F_{x,y}).$$

2. **(Preservation of identities)** For every object $x \in \text{Ob}(\mathcal{C})$, the identity n -cell is preserved strictly:

$$F_{x,x}(\text{id}_x) = \text{id}_{Fx}.$$

Morphisms between functors of n -categories are $(n + 1)$ -natural transformations, defined inductively in the same manner.

Definition 11.2.6. (♠ TODO: this definition is not very precise) A *weak n -category* is a structure that generalizes the notion of a strict n -category by relaxing the associativity and unit laws to hold only up to specified coherent equivalences. It can be defined inductively or through several equivalent formal models.

Formally, a weak n -category $\mathcal{C}$ consists of:

- for each $0 \leq k \leq n$, a collection of k -morphisms;
- source and target operations assigning to each k -morphism its $(k - 1)$ -boundary morphisms;
- composition operations of k -morphisms along $(k - 1)$ -boundaries;
- identity k -morphisms for each $(k - 1)$ -morphism;
- and coherent associativity, unit, and interchange equivalences expressing that composition is associative and unital up to higher morphisms, with these higher coherence conditions themselves satisfying higher coherence laws.

Several equivalent frameworks make this notion precise:

- In the **algebraic approach** (Batanin, Leinster), a weak n -category is encoded as a system of operations on n -cells governed by a globular operad, whose algebras satisfy the necessary coherence conditions.

- In the **simplicial or topological approach**, weak n -categories are modeled by higher Segal-type objects, such as Tamsamani–Simpson n -categories or Rezk’s $(n + k, n)$ – Θ –spaces, which are presheaves of simplicial sets on the category Θ_n satisfying Segal and completeness conditions.
- In the **truncated ∞ -categorical approach** (Baez–Dolan), a weak n -category is a weak ∞ -category with trivial cells above dimension n .

Thus, a weak n -category can be viewed as a category-like structure containing morphisms of all orders up to n , where composition is associative and unital only up to a coherent family of higher isomorphisms organized by the same structure.

Definition 11.2.7. (♠ TODO: this definition is not very precise) A *weak n -category* (or *weak higher category*) is a structure similar to a strict n -category, except that the associativity and unit axioms hold only up to coherent higher equivalences. That is, an n -category $\mathcal{C}$ consists of

- objects, 1-morphisms, 2-morphisms, $\dots$, n -morphisms,
- composition operations defined at each level,
- associativity, unit, and interchange data holding up to coherent $(k + 1)$ -morphisms for $k < n$.

Different models exist for weak n -categories (e.g. bicategories for $n = 2$, tricategories, quasicategories, Segal spaces, etc.), which satisfy equivalent homotopy-coherent conditions.

Definition 11.2.8. (♠ TODO: this definition is not very precise) An *∞ -category* (or *weak ∞ -category*) is a category-like structure that encodes morphisms of all dimensions, where composition is associative and unital only up to coherent higher homotopies. More precisely, one usually works with the notion of an *$(\infty, 1)$ -category*, in which all k -morphisms for $k > 1$ are invertible up to higher homotopy.

Formally, an ∞ -category can be defined in several equivalent ways, each specifying a precise model of this higher structure. Each of the standard models forms an *∞ -cosmos* — a simplicially enriched category whose hom-objects are themselves ∞ -categories — and all such models are equivalent up to categorical equivalence. Common frameworks include:

- **Quasicategories:** Simplicial sets C satisfying the *inner horn-filling condition*: for every $n \geq 2$ and $0 < i < n$, every map $\Lambda_i[n] \rightarrow C$ (where $\Lambda_i[n]$ is the i th inner horn of the standard simplex $\Delta[n]$) extends to a map $\Delta[n] \rightarrow C$. Such objects are the fibrant objects in the Joyal model structure on simplicial sets.
- **Complete Segal spaces:** Simplicial spaces $W_\bullet : \Delta^{\text{op}} \rightarrow \mathbf{sSet}$ satisfying the Segal condition

$$W_n \simeq W_1 \times_{W_0} \cdots \times_{W_0} W_1$$

(encoding associative composition up to homotopy) together with the completeness condition ensuring that equivalences are detected correctly. These are the fibrant objects in the Rezk model structure on simplicial spaces.

- **Simplicial categories:** Categories enriched in simplicial sets whose hom-objects are Kan complexes and satisfy homotopy-coherent associativity laws. These are related to quasicategories via the homotopy coherent nerve construction.

- **Other equivalent models:** Segal categories, naturally marked simplicial sets, and Θ -spaces, each giving a distinct but equivalent encoding of higher categorical coherence.

Morphisms between ∞ -categories, called **∞ -functors**, are maps in the chosen model that preserve the higher compositional structure up to coherent homotopy. The totality of ∞ -categories and ∞ -functors forms an $(\infty, 2)$ -category commonly denoted by **∞ -Cat**.

In summary, an ∞ -category formalizes the idea of a category enriched in homotopy types (or spaces) — a “category up to higher coherent homotopies.” All standard models of ∞ -categories (quasicategories, complete Segal spaces, simplicial categories, etc.) are equivalent in the sense of ∞ -cosmoi theory.

11.3. Topological categories. Topological categories (Definition 11.3.3) provide one framework for ∞ -categories.

Definition 11.3.1. (**♠ TODO: final topology**) A topological space (Definition B.0.1) X is said to be **compactly generated** (or a **k -space**) if a subset $U \subseteq X$ is open whenever for every compact (Definition E.0.17) subset $K \subseteq X$, the intersection $U \cap K$ is open in the subspace K . Equivalently, X is compactly generated if and only if the topology of X is the final topology with respect to the collection of inclusions $K \hookrightarrow X$ for compact $K \subseteq X$.

Definition 11.3.2. A topological space (Definition B.0.1) X is said to be **weakly Hausdorff** if for every continuous map (Definition B.0.3) $f : K \rightarrow X$ from a compact (Definition E.0.17) Hausdorff space K , the image $\text{Im}(f)$ is closed in X . Equivalently, X is weakly Hausdorff if and only if for every compact Hausdorff K , the induced map $f : K \rightarrow X$ is a closed map.

Definition 11.3.3. A **topological category** is a category (Definition 1.0.1) which is enriched over (Definition 4.0.1) $\mathcal{CG}$, the category of compactly generated (Definition 11.3.1) (and weakly Hausdorff (Definition 11.3.2)) topological spaces. The category of topological categories will often be denoted by **$\mathcal{Cat}_{\text{top}}$** .

More explicitly, a topological category $\mathcal{C}$ consists of a collection of objects, together with a (compactly generated) topological space $\text{Map}_{\mathcal{C}}(X, Y)$ for any pair of objects $X, Y \in \mathcal{C}$. These mapping spaces must be equipped with an associative composition law, given by continuous maps

$$\text{Map}_{\mathcal{C}}(X_0, X_1) \times \text{Map}_{\mathcal{C}}(X_1, X_2) \times \dots \times \text{Map}_{\mathcal{C}}(X_{n-1}, X_n) \rightarrow \text{Map}_{\mathcal{C}}(X_0, X_n)$$

(defined for all $n \geq 0$). Here the product is taken in the category of compactly generated topological spaces.

There are analogues to the geometric realization functor (Definition 8.0.5) of simplicial sets and the singular complex functor (Definition 8.0.6) of topological spaces:

Definition 11.3.4. Let $\mathcal{C}$ be a simplicial category (Definition 8.0.21). The **geometric realization of $\mathcal{C}$** , denoted **$|\mathcal{C}|$** , is the topological category (Definition 11.3.3) defined as follows:

- The objects of $|\mathcal{C}|$ are the same as the objects of $\mathcal{C}$:

$$\text{Ob}(|\mathcal{C}|) = \text{Ob}(\mathcal{C}).$$

- For any objects $X, Y \in \mathcal{C}$, the morphism space is the geometric realization (Definition 8.0.5) of the simplicial mapping space:

$$\mathrm{Hom}_{|\mathcal{C}|}(X, Y) = |\mathrm{Hom}_{\mathcal{C}}(X, Y)|.$$

(Definition 4.0.1)

- The composition maps

$$\mathrm{Hom}_{|\mathcal{C}|}(Y, Z) \times \mathrm{Hom}_{|\mathcal{C}|}(X, Y) \rightarrow \mathrm{Hom}_{|\mathcal{C}|}(X, Z)$$

are induced by applying the geometric realization functor to the composition maps of $\mathcal{C}$, using that geometric realization commutes with finite products.

Definition 11.3.5. Let $\mathcal{C}$ be a topological category (Definition 11.3.3). Its *singular simplicial category* $\mathrm{Sing} \mathcal{C}$ is the simplicial category (Definition 8.0.21) defined as follows:

- The objects of $\mathrm{Sing} \mathcal{C}$ are the same as those of $\mathcal{C}$:

$$\mathrm{Ob}(\mathrm{Sing} \mathcal{C}) = \mathrm{Ob}(\mathcal{C}).$$

- For any objects $X, Y \in \mathcal{C}$, the morphism simplicial set is given by the singular complex (Definition 8.0.6) of the topological morphism space:

$$\mathrm{Hom}_{\mathrm{Sing} \mathcal{C}}(X, Y) = \mathrm{Sing}(\mathrm{Hom}_{\mathcal{C}}(X, Y)).$$

Explicitly, this singular complex is given by

$$\mathrm{Sing}_n(\mathrm{Hom}_{\mathcal{C}}(X, Y)) = \mathrm{Top}(|\Delta^n|, \mathrm{Hom}_{\mathcal{C}}(X, Y)),$$

where $|\Delta^n|$ is the standard topological n -simplex (Definition 8.0.2).

- The composition maps in $\mathrm{Sing} \mathcal{C}$ are induced by the composition maps in $\mathcal{C}$ via functoriality of the singular complex:

$$\mathrm{Sing}(\mathrm{Hom}_{\mathcal{C}}(Y, Z)) \times \mathrm{Sing}(\mathrm{Hom}_{\mathcal{C}}(X, Y)) \rightarrow \mathrm{Sing}(\mathrm{Hom}_{\mathcal{C}}(X, Z)).$$

There is an adjunction analogous to that of Proposition 8.0.7:

Proposition 11.3.6. The geometric realization functor (Definition 11.3.4) and the singular simplicial category functor (Definition 11.3.5) form an adjoint pair (Definition 1.1.13)

$$|\cdot| : \mathbf{Cat}_{\Delta} \rightleftarrows \mathrm{Sing}$$

$$|\cdot| : \mathbf{Cat}_{\Delta} \rightleftarrows \mathrm{Cat}_{\mathrm{top}} : \mathrm{Sing}$$

(Definition 8.0.21, Definition 11.3.3).

Definition 11.3.7. If $\mathcal{C}$ is a topological category (Definition 11.3.3), we define the *(topological) nerve* $N(\mathcal{C})$ of $\mathcal{C}$ to be the simplicial nerve (Definition 8.0.25) of $\mathrm{Sing} \mathcal{C}$ (Definition 11.3.5).

Lemma 11.3.8 ([Lur09, between Examples 1.1.5.8 and 1.1.5.9]). The functor $\mathfrak{C} : \Delta \rightarrow \mathbf{Cat}_{\Delta}$ (Definition 8.0.24, Definition 8.0.3, Definition 8.0.21) extends uniquely (up to unique isomorphism) to a (small) colimit-preserving functor $\mathfrak{C} : \mathbf{Sets}_{\Delta} \rightarrow \mathbf{Cat}_{\Delta}$ (Definition 8.0.4).

Theorem 11.3.9 ([Lur09, Theorem 1.1.5.13]). Let $\mathcal{C}$ be a topological category (Definition 11.3.3), and let $X, Y \in \mathcal{C}$ be objects. The counit map (Definition 1.1.13) (obtained via the adjunction (Definition 1.1.13) of (♠ TODO: the adjunction should come from $\mathfrak{C}$ and N))

$$|\mathrm{Hom}_{\mathfrak{C}[N(\mathcal{C})]}(X, Y)| \rightarrow \mathrm{Hom}_{\mathcal{C}}(X, Y)$$

(Definition 8.0.5, Lemma 11.3.8, Definition 11.3.7) is a weak homotopy equivalence (Definition B.0.7) of topological spaces.

11.4. Derived categories via ∞ -categories.

Definition 11.4.1 (∞ -category localization). (♠ TODO: ∞ -category in what sense) Let $\mathcal{C}$ be an ∞ -category and let $W \subseteq \text{Mor}(\mathcal{C})$ be a collection of morphisms in $\mathcal{C}$, called *weak equivalences* or *localizing morphisms*.

A *localization of the ∞ -category $\mathcal{C}$ at W* is an ∞ -category $\mathcal{C}[W^{-1}]$ together with an ∞ -functor

$$L : \mathcal{C} \rightarrow \mathcal{C}[W^{-1}]$$

satisfying the following universal property:

- For every morphism $w \in W$, the image $L(w)$ is an equivalence in $\mathcal{C}[W^{-1}]$.
- For any ∞ -category $\mathcal{D}$, composition with L induces a fully faithful embedding of ∞ -categories

$$\text{Fun}(\mathcal{C}[W^{-1}], \mathcal{D}) \hookrightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$$

whose essential image consists precisely of the ∞ -functors that send every morphism in W to an equivalence in $\mathcal{D}$.

Definition 11.4.2 (dg-category). Let k be a commutative ring. A *dg-category* (differential graded category) $\mathcal{C}$ over k is a category enriched in (Definition 4.0.1) the category of chain complexes (Definition 6.0.1) of k -modules (Definition C.0.8). Explicitly, a dg-category $\mathcal{C}$ consists of the following data:

- A class of objects $\text{Ob}(\mathcal{C})$.
- For each pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, a complex (Definition 6.0.1) of k -modules $\text{Hom}_{\mathcal{C}}(X, Y) = (\text{Hom}_{\mathcal{C}}^n(X, Y), d)$; that is, a $\mathbb{Z}$ -graded k -module

$$\text{Hom}_{\mathcal{C}}(X, Y) = \bigoplus_{n \in \mathbb{Z}} \text{Hom}_{\mathcal{C}}^n(X, Y)$$

equipped with a differential $d : \text{Hom}_{\mathcal{C}}^n(X, Y) \rightarrow \text{Hom}_{\mathcal{C}}^{n+1}(X, Y)$ satisfying $d^2 = 0$.

- For each triple of objects $X, Y, Z \in \text{Ob}(\mathcal{C})$, a morphism of complexes (composition)

$$\circ : \text{Hom}_{\mathcal{C}}(Y, Z) \otimes_k \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{C}}(X, Z)$$

which is associative and satisfies the graded Leibniz rule with respect to the differential d :

$$d(f \circ g) = d(f) \circ g + (-1)^{|f|} f \circ d(g),$$

for homogeneous f, g with $|f|$ denoting the degree of f .

- For each object X , an identity morphism $1_X \in \text{Hom}_{\mathcal{C}}^0(X, X)$ such that for any $f \in \text{Hom}_{\mathcal{C}}(X, Y)$,

$$1_Y \circ f = f, \quad f \circ 1_X = f.$$

Definition 11.4.3 (category of chain complexes as a dg-category). Let $\mathcal{A}$ be an additive category (Definition 3.0.1) and suppose that $\mathcal{A}$ is enriched over (Definition 4.0.1) a commutative ring k . The category $\text{Ch}(\mathcal{A})$ (Definition 6.0.1) can be given the structure of a dg-category over k (Definition 11.4.2) as follows:

- An object in $\text{Ch}(\mathcal{A})$ is, as usual, a chain complex (Definition 6.0.1) of objects in $\mathcal{A}$.
- For a pair of chain complexes $X^\bullet, Y^\bullet \in \text{Ch}(\mathcal{A})$, the hom-complex $\text{Hom}_{\text{Ch}(\mathcal{A})}(X^\bullet, Y^\bullet) \in \text{Ob Ch}(\mathcal{A})$ is the complex of k -modules with

$$\text{Hom}_{\text{Ch}(\mathcal{A})}^n(X^\bullet, Y^\bullet) := \prod_{m \in \mathbb{Z}} \text{Hom}_{\mathcal{A}}(X^m, Y^{m+n}).$$

The differential on this complex, $\delta : \text{Hom}_{\text{Ch}(\mathcal{A})}^n(X^\bullet, Y^\bullet) \rightarrow \text{Hom}_{\text{Ch}(\mathcal{A})}^{n+1}(X^\bullet, Y^\bullet)$, is given by

$$\delta(f)^m := d_Y^{m+n} \circ f^m - (-1)^n f^{m+1} \circ d_X^m,$$

for $f = (f^m)_{m \in \mathbb{Z}}$.

- Composition of morphisms

$$\circ : \text{Hom}_{\text{Ch}(\mathcal{A})}^p(Y^\bullet, Z^\bullet) \otimes \text{Hom}_{\text{Ch}(\mathcal{A})}^q(X^\bullet, Y^\bullet) \rightarrow \text{Hom}_{\text{Ch}(\mathcal{A})}^{p+q}(X^\bullet, Z^\bullet)$$

is given by component-wise composition in $\mathcal{A}$ with the Koszul sign rule:

$$(g \circ f)^m := g^{m+q} \circ f^m.$$

- For each $X^\bullet$, the identity morphism is the family $(1_{X^m})_{m \in \mathbb{Z}}$ in degree zero.

Definition 11.4.4 (Moore complex of a simplicial object). Let $\mathcal{A}$ be an abelian category (Definition 3.0.4) and let $X_\bullet$ be a simplicial object (Definition 8.0.4) in $\mathcal{A}$, i.e., a functor

$$X_\bullet : \Delta^{op} \rightarrow \mathcal{A}$$

where Δ is the simplex category.

The *Moore complex*, synonymously called the *normalized chain complex* or *complex of normalized chains*, $M(X_\bullet) \in \text{Ch}_{\geq 0}(\mathcal{A})$ associated to $X_\bullet$ is the chain complex (Definition 6.0.1) given by:

- For each integer $n \geq 0$, the degree n component is

$$M_n(X_\bullet) := \bigcap_{i=1}^n \ker(d_i : X_n \rightarrow X_{n-1}),$$

where d_i are the face maps (Definition 8.0.11) of the simplicial object.

- The differential $d_n : M_n(X_\bullet) \rightarrow M_{n-1}(X_\bullet)$ is given by the restriction of the face map d_0 :

$$d_n := d_0|_{M_n(X_\bullet)}.$$

Theorem 11.4.5 (Dold-Kan correspondence). Let $\mathcal{A}$ be an abelian category with enough projectives. There exists an equivalence of categories, called the *Dold-Kan equivalence*, (♠ TODO: normalized chain complex functor, moore complex functor)

$$N : \text{s Ab}(\mathcal{A}) \rightleftarrows \text{Ch}_{\geq 0}(\mathcal{A}) : \Gamma,$$

(Definition 11.4.4) between the category of simplicial objects (Definition 8.0.4) in Ab , and the category of connective (non-negatively graded) chain complexes in $\mathcal{A}$.

This equivalence is exact, functorial, and compatible with the homotopy structures on both categories:

- The *normalized chain complex functor* N sends a simplicial object to its normalized chain complex.
- The inverse functor Γ takes a chain complex supported in degrees ≥ 0 to a simplicial object.
- The adjunction (N, Γ) restricts to an equivalence of categories.

Theorem 11.4.6 (Stable enhancement of the Dold-Kan correspondence). (♠ TODO: verify that the following is true) Let $\mathcal{A}$ be an abelian category with enough projectives. There exists an equivalence of stable ∞ -categories

$$\mathcal{D}(\mathcal{A}) \simeq \mathrm{Sp}(S_{\geq 0}(\mathcal{A})),$$

where

- $\mathcal{D}(\mathcal{A})$ denotes the derived ∞ -category of unbounded chain complexes in $\mathcal{A}$, obtained by inverting quasi-isomorphisms in $\mathrm{Ch}(\mathcal{A})$ and passing to the homotopy-coherent enhancement.
- $S_{\geq 0}(\mathcal{A})$ is the ∞ -category of connective (nonnegatively graded) simplicial objects in $\mathcal{A}$.
- $\mathrm{Sp}(S_{\geq 0}(\mathcal{A}))$ denotes the stable ∞ -category of spectrum objects in $S_{\geq 0}(\mathcal{A})$, i.e., the stabilization of the connective simplicial objects.

This equivalence *categorifies and stabilizes* the classical Dold-Kan correspondence and extends it from connective chain complexes to all unbounded chain complexes:

- The classical Dold-Kan equivalence identifies nonnegatively graded chain complexes with connective simplicial objects.
- The stable enhancement takes spectra in the simplicial world to model all chain complexes, thus producing a stable ∞ -category representing $\mathrm{Ch}(\mathcal{A})$ internally as an ∞ -category.

12. 6-FUNCTOR FORMALISM

(♠ TODO: read through these)

Notation 12.0.1 (Ambient data, sizes, pullbacks, spans). Fix a symmetric monoidal ∞ -category $(\mathcal{E}, \otimes, \mathbb{1}_{\mathcal{E}})$ and a class of morphisms $\mathcal{P}$ in an ∞ -category $\mathcal{C}$ such that:

- $\mathcal{C}$ admits pullbacks along morphisms in $\mathcal{P}$, and $\mathcal{P}$ is stable under pullback and composition.
- There is a functor $\omega: \mathcal{C} \rightarrow \mathcal{E}$ (often a coefficient or “sheaf/transfer” functor) that sends $\mathcal{P}$ -pullback squares in $\mathcal{C}$ to $\otimes$ -Beck–Chevalley squares in $\mathcal{E}$ (so that base-change along $\mathcal{P}$ exists and is coherent).

Write objects of $\mathcal{C}$ by X, Y, Z , and spans by diagrams $X \xleftarrow{p} U \xrightarrow{q} Y$. For a morphism $f: X \rightarrow Y$ in $\mathcal{C}$, its graph span is $X \xleftarrow{\mathrm{id}_X} X \xrightarrow{f} Y$. We use the following notation:

- $(\mathcal{E}, \otimes, \mathbb{1}_{\mathcal{E}})$

- $\mathcal{C}$
- $\mathcal{P} \subseteq \text{Mor}(\mathcal{C})$
- $\omega: \mathcal{C} \rightarrow \mathcal{E}$
- $X \xleftarrow{p} U \xrightarrow{q} Y$

Definition 12.0.2 (Category of $\mathcal{P}$ -correspondences). Given $\mathcal{C}$ and a class $\mathcal{P}$ as in the notation above, the ∞ -category of $\mathcal{P}$ -correspondences $\text{Corr}_{\mathcal{P}}(\mathcal{C})$ is defined as follows. Its objects are those of $\mathcal{C}$. For $X, Y \in \mathcal{C}$, the mapping anima $\text{Map}_{\text{Corr}_{\mathcal{P}}(\mathcal{C})}(X, Y)$ is the space of spans $X \xleftarrow{p} U \xrightarrow{q} Y$ with $p \in \mathcal{P}$, and morphisms given by equivalences of such spans. Composition is given by pullback: the composite of $X \xleftarrow{p} U \xrightarrow{q} Y$ and $Y \xleftarrow{p'} V \xrightarrow{q'} Z$ is

$$X \xleftarrow{\text{pr}_1} U \times_Y V \xrightarrow{q' \circ \text{pr}_2} Z,$$

where $\text{pr}_1 \in \mathcal{P}$ by stability of $\mathcal{P}$ under pullback, and associativity/unitality hold up to coherent homotopy. The unit at X is the identity span $X \xleftarrow{\text{id}_X} X \xrightarrow{\text{id}_X} X$.

Definition 12.0.3 (Coefficient systems on correspondences). A *coefficient system* (with values in $\mathcal{E}$ along $\mathcal{P}$) is the data of $(\omega, \mathcal{P})$ as in the notation, together with:

- for every span $X \xleftarrow{p \in \mathcal{P}} U \xrightarrow{q} Y$, a morphism in $\mathcal{E}$ written $\omega_!(p) \circ \omega^*(q): \omega(X) \rightarrow \omega(Y)$, where $\omega^*(q)$ is pullback along q and $\omega_!(p)$ is pushforward along p ;
- for every pullback square with left arrow in $\mathcal{P}$, a canonical base-change equivalence exhibiting Beck–Chevalley for $\omega_!$ and ω^* ;
- coherences making these assignments functorial for composition of spans.

We denote such a system by $(\omega, \mathcal{P})$.

Definition 12.0.4 (Symmetric monoidal structure on correspondences). Assume $\mathcal{C}$ admits finite products $\times$ preserved by pullbacks along $\mathcal{P}$, and that $(\mathcal{E}, \otimes, \mathbb{1}_{\mathcal{E}})$ is symmetric monoidal with ω lax symmetric monoidal. The *symmetric monoidal structure* on $\text{Corr}_{\mathcal{P}}(\mathcal{C})$ is given on objects by $X \otimes Y := X \times Y$, unit $\mathbb{1} := *$ (a terminal object of $\mathcal{C}$), and on morphisms by external product of spans:

$$(X \xleftarrow{p} U \xrightarrow{q} Y) \otimes (X' \xleftarrow{p'} U' \xrightarrow{q'} Y') := X \times X' \xleftarrow{p \times p'} U \times U' \xrightarrow{q \times q'} Y \times Y',$$

which is well-defined since $p \times p' \in \mathcal{P}$ by closure of $\mathcal{P}$ under products and pullbacks. Symmetry, associativity, and unit constraints are induced from those of $(\mathcal{C}, \times, *)$ and satisfy the ∞ -categorical coherences.

Definition 12.0.5 (Symmetric monoidal ∞ -category of correspondences with coefficients). Given $(\omega, \mathcal{P})$ and assumptions as above, the *symmetric monoidal ∞ -category of correspondences* is the symmetric monoidal ∞ -category

$$\text{Corr}(\mathcal{C}, \mathcal{E}) := (\text{Corr}_{\mathcal{P}}(\mathcal{C}), \otimes = \times, \mathbb{1} = *),$$

equipped with the symmetric monoidal functor $\Omega: \text{Corr}_{\mathcal{P}}(\mathcal{C}) \rightarrow \mathcal{E}$ determined on objects by $\Omega(X) = \omega(X)$ and on a span $X \xleftarrow{p \in \mathcal{P}} U \xrightarrow{q} Y$ by

$$\Omega(X \xleftarrow{p} U \xrightarrow{q} Y) := \omega_!(p) \circ \omega^*(q): \omega(X) \rightarrow \omega(Y),$$

with monoidal structure maps induced by the external product and the lax symmetric monoidal structure on ω together with Beck–Chevalley for base-change. This realizes $\text{Corr}(\mathcal{C}, \mathcal{E})$ as a symmetric monoidal ∞ -category encoding spans in $\mathcal{C}$ and their functorial “push–pull” action on coefficients in $\mathcal{E}$.

APPENDIX A. SET THEORY

Definition A.0.1. (♠ TODO: first order language) The *axioms of Zermelo–Fraenkel set theory (ZF)* are the following statements, formulated in the first-order language with symbol $\in$:

1. **Axiom of Extensionality:** Two sets are equal if and only if they have the same elements.

$$\forall A \forall B (\forall x (x \in A \leftrightarrow x \in B) \rightarrow A = B).$$

2. **Axiom of Empty Set:** There exists a set with no elements.

$$\exists S \forall x (x \notin S).$$

By extensionality, this set is unique; it is denoted by $\emptyset$.

3. **Axiom of Pairing:** For any sets x, y , there exists a set containing exactly x and y as elements.

$$\forall x \forall y \exists A \forall z [z \in A \leftrightarrow (z = x \vee z = y)].$$

4. **Axiom of Union:** For any set A , there exists a set that is the union of the elements of A .

$$\forall A \exists U \forall x [x \in U \leftrightarrow \exists B (x \in B \wedge B \in A)].$$

5. **Axiom of Power Set:** For any set A , there exists a set $\mathcal{P}(A)$ containing all subsets of A .

$$\forall A \exists P \forall B [B \in P \leftrightarrow B \subseteq A].$$

(♠ TODO: inductive set)

6. **Axiom of Infinity:** There exists an inductive set, that is, a set containing the empty set and closed under the successor operation.

$$\exists I [\emptyset \in I \wedge \forall x (x \in I \rightarrow x \cup \{x\} \in I)].$$

(♠ TODO: language of set theory)

7. **Axiom Schema of Separation:** For any property $\varphi(x)$ expressible in the language of set theory and any set A , there is a subset of A consisting of the elements of A satisfying $\varphi(x)$:

$$\forall A \exists B \forall x [x \in B \leftrightarrow (x \in A \wedge \varphi(x))].$$

One commonly writes B in the following manners:

$$\{x \in A : \varphi(x)\}$$

or

$$\{x \in A \mid \varphi(x)\}.$$

(♠ TODO: definable function)

8. **Axiom Schema of Replacement:** For any definable function given by a formula $\varphi(x, y)$ ensuring that for each x there exists a unique y with $\varphi(x, y)$, the image (Definition A.0.2) of any set under this function is also a set.

$$\forall A [\forall x \in A \exists! y \varphi(x, y)] \rightarrow \exists B \forall y (y \in B \leftrightarrow \exists x \in A \varphi(x, y)).$$

9. **Axiom of Foundation (Regularity):** Every nonempty set A contains an element x that is disjoint from A .

$$\forall A [A \neq \emptyset \rightarrow \exists x (x \in A \wedge x \cap A = \emptyset)].$$

The *Axiom of Choice (AC)* asserts that for every set X of nonempty sets, there exists a function $f : X \rightarrow \bigcup X$ such that $f(A) \in A$ for all $A \in X$. Formally:

$$\forall X [(\forall A \in X A \neq \emptyset) \rightarrow \exists f (\text{dom}(f) = X \wedge \forall A \in X f(A) \in A)].$$

When ZF is augmented by the Axiom of Choice, the resulting system is called *Zermelo–Fraenkel set theory with Choice (ZFC)*.

Definition A.0.2. Let X and Y be sets. A *map* (or *function*) from X to Y is a rule f assigning to each element $x \in X$ exactly one element $f(x) \in Y$. We write $f : X \rightarrow Y$. The element $f(x)$ of Y is called the *image of x under f* .

Equivalently, a function $f : X \rightarrow Y$ is identified with a binary relation $f \subseteq X \times Y$ such that for every element $x \in X$, there is exactly one element $y_x \in Y$ such that $(x, y_x) \in f$.

We say that X is the *domain* and that Y is the *codomain of f* . We say that f *maps into Y* .

Definition A.0.3. Let X and Y be sets and let $f : X \rightarrow Y$ be a function.

- The function f is said to be *injective* (or *one-to-one*) if for all $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
- The function f is said to be *surjective* (or *onto*) if for every $y \in Y$ there exists $x \in X$ such that $f(x) = y$.
- The map f is *bijective* if it is both injective and surjective. In this case, there exists a unique *inverse map* $f^{-1} : Y \rightarrow X$ such that for all $x \in X$ and $y \in Y$,

$$f^{-1}(f(x)) = x \text{ and } f(f^{-1}(y)) = y.$$

APPENDIX B. TOPOLOGY

Definition B.0.1 (Topology). Let X be a set. A *topology on X* is a collection $\mathcal{T}$ of subsets of X such that:

1. $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$,
2. For any collection $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ (with I arbitrary), the union $\bigcup_{i \in I} U_i \in \mathcal{T}$,
3. For any finite collection $\{U_1, \dots, U_n\} \subseteq \mathcal{T}$, the intersection $U_1 \cap \dots \cap U_n \in \mathcal{T}$.

If $\mathcal{T}$ is a topology on X , the pair $(X, \mathcal{T})$ is called a *topological space*. Members of $\mathcal{T}$ are called *open sets*.

A subset $C \subseteq X$ is **closed** (also called a **closed subset**) if its complement $X \setminus C$ is an open set in $\mathcal{T}$.

One very often refers to X as a topological space, omitting the notation of the topology $\mathcal{T}$.

The collection of all topologies on a set X may be denoted by notations such as **Top**(X), **Top**(X), or **Top**(X).

Definition B.0.2. Let (X, τ) be a topological space (Definition B.0.1). An **open covering** of X is a family of open sets

$$\mathcal{U} = \{U_i\}_{i \in I}$$

such that

$$\bigcup_{i \in I} U_i = X.$$

Here, each $U_i \in \tau$ is an open subset of X indexed by a set I , which can be finite or infinite.

Definition B.0.3. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces (Definition B.0.1). A map $f : X \rightarrow Y$ is called **continuous** if for every open set $V \in \mathcal{T}_Y$, the preimage $f^{-1}(V)$ is an open set in X , that is,

$$\forall V \in \mathcal{T}_Y, f^{-1}(V) \in \mathcal{T}_X.$$

Equivalently, f is continuous if and only if for every closed set $C \subseteq Y$, the preimage $f^{-1}(C)$ is closed in X .

A **map of topological spaces** usually refers to a continuous map between the topological spaces.

The set of continuous maps from X to Y is sometimes denoted by **C**(X, Y). Other standard notation include **Hom**_{Top}(X, Y) or **Top**(X, Y) coming from more general notation for morphisms between objects in a category (Definition 1.0.1).

Definition B.0.4 (Separation axioms). Let $(X, \mathcal{T})$ be a topological space (Definition B.0.1).

- $(X, \mathcal{T})$ is **T₀** (**Kolmogorov**) if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that, without loss of generality, $x \in U$ and $y \notin U$.
- $(X, \mathcal{T})$ is **T₁** (**Fréchet**) if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \in \mathcal{T}$ such that $x \in U$, $y \notin U$, and $y \in V$, $x \notin V$.
- $(X, \mathcal{T})$ is **T₂** or **Hausdorff** if for every pair of distinct points $x, y \in X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $y \in V$.
- $(X, \mathcal{T})$ is **regular** if it is T₁ and for each point $x \in X$ and closed set $F \subseteq X$ with $x \notin F$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $F \subseteq V$.
- $(X, \mathcal{T})$ is **T₃** (regular Hausdorff) if it is T₁ and regular.
- $(X, \mathcal{T})$ is **completely regular** if for each closed set $F \subseteq X$ and $x \notin F$, there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(x) = 0$ and $f|_F = 1$.
- $(X, \mathcal{T})$ is **T_{3½}** (completely regular Hausdorff) if it is T₁ and completely regular.
- $(X, \mathcal{T})$ is **normal** if it is T₁ and for each pair of disjoint closed sets $A, B \subseteq X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $A \subseteq U$ and $B \subseteq V$.
- $(X, \mathcal{T})$ is **T₄** (normal Hausdorff) if it is T₁ and normal.
- $(X, \mathcal{T})$ is **T₅** (completely normal Hausdorff) if it is T₁ and completely normal.

- $(X, \mathcal{T})$ is **perfectly normal** if every closed set is a $G\delta$ (countable intersection of open sets) and the space is normal.
- $(X, \mathcal{T})$ is **T_6** (perfectly normal Hausdorff) if it is T_1 and perfectly normal.

Definition B.0.5 (Homotopy of maps of topological spaces). Let X and Y be topological spaces and let $K \subseteq X$ be a subset. Let $C(X, Y)$ denote the set of all continuous maps $f : X \rightarrow Y$.

1. A **homotopy between two maps $f, g \in C(X, Y)$ relative to K** is a continuous map

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $x \in K$ and $t \in [0, 1]$,

$$H(x, t) = f(x) = g(x).$$

If such an H exists, we say f and g are **homotopic relative to K** , and we write **$f \simeq g \text{ rel } K$** or **$f \simeq_K g$** ; this is an equivalence relation.

In the case that $K = \emptyset$, a homotopy relative to K is simply called a **homotopy**. We write **$f \simeq g$** if a homotopy between f and g exists.

2. Let (X, x_0) and (Y, y_0) be pointed topological spaces and let $K \subseteq X$ be a subset with $x_0 \in K$. Let $C_*(X, Y)$ denote the set of all continuous based maps $f : X \rightarrow Y$ satisfying $f(x_0) = y_0$.

A **homotopy of based maps $f, g \in C_*(X, Y)$ relative to K** is a continuous map

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $k \in K$ and $t \in [0, 1]$,

$$H(k, t) = f(k) = g(k),$$

in particular fixing the basepoint throughout,

$$H(x_0, t) = y_0 \quad \text{for all } t \in [0, 1].$$

If such an H exists, we say f and g are **based homotopic relative to K** , and we write **$f \simeq g \text{ rel } K$** or **$f \simeq_K g$** . This is an equivalence relation.

When $K = \{x_0\}$, one calls a homotopy of based maps relative to K simply as a **homotopy of based maps** or **based homotopy**. We write **$f \simeq g$** if such a homotopy exists.

Definition B.0.6 (Homotopy groups of a pointed topological space). For any pointed topological space (X, x_0) and integer $n \geq 0$, the **n -th homotopy set of X at x_0** , denoted **$\pi_n(X, x_0)$** or simply **$\pi_n(X)$** when the base point x_0 is understood or a choice of base point does not really matter, is defined as the set of all homotopy classes (rel. ∂I^n) of paired maps

$$f : (I^n, \partial I^n) \rightarrow (X, \{x_0\}),$$

where $I^n = [0, 1]^n$ and ∂I^0 is taken to be the singleton I^0 instead of the literal boundary thereof. Equivalently, one can also define it as the set of all homotopy classes of pointed maps

$$f : (S^n, s_0) \rightarrow (X, x_0),$$

where S^n is the n -sphere (♠ TODO:) and $s_0 \in S^n$ is a choice of base point. In other words, $\pi_n(X, x_0)$ is by definition $[(S^n, s_0), (X, x_0)]_*$.

For $n \geq 1$, $\pi_n(X, x_0)$ is a group under concatenation of pointed maps, and for $n \geq 2$, it is abelian. As such, for $n \geq 1$, it is appropriate to refer to $\pi_n(X, x_0)$ as the *n th homotopy group of (X, x_0)* .

The *fundamental group of (X, x_0)* refers to $\pi_1(X, x_0)$. Equivalently, it is the group of homotopy classes (rel. endpoints) of loops $\gamma : [0, 1] \rightarrow X$ satisfying $\gamma(0) = \gamma(1) = x_0$.

For every $n \geq 0$, the assignment $(X, x_0) \mapsto \pi_n(X, x_0)$ is functorial — in fact

1. For $n = 0$, it is a functor $\mathbf{Top}_* \rightarrow \mathbf{Sets}$ (Definition 1.0.3),
2. For $n = 1$, it is a functor $\mathbf{Top}_* \rightarrow \mathbf{Grp}$ (Definition 1.0.4),
3. For $n \geq 2$, it is a functor $\mathbf{Top}_* \rightarrow \mathbf{Ab}$ (Definition 1.0.4).

Given a map $f : (X, x_0) \rightarrow (Y, y_0)$ of pointed spaces, the induced map $\pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$ is often denoted by f_* .

We further note that the set $\pi_n(X, x_0) = [(S^n, s_0), (X, x_0)]_*$ is identifiable with $\pi_0(\mathrm{Map}_*((S^n, s_0), (X, x_0)))$.

Definition B.0.7 (Weak homotopy equivalence). Let $f : X \rightarrow Y$ be a continuous map (Definition B.0.3) between topological spaces. The map f is a *weak homotopy equivalence* if, for every choice of basepoint $x_0 \in X$ with $y_0 = f(x_0)$, the induced maps on homotopy groups (Definition B.0.6)

$$f_* : \pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$$

are isomorphisms (Definition C.0.7) for all integers $n \geq 0$.

APPENDIX C. ALGEBRA

Definition C.0.1. A *ring* is a triple $(R, +, \cdot)$ where

1. $(R, +)$ is a commutative group (Definition C.0.5), and
2. $(R, \cdot)$ is a monoid.
3. $\cdot$ is distributive over $+$, i.e. for all $a, b, c \in R$, we have

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c.$$

Equivalently, a ring is a triple $(R, +, \cdot)$ where $+, \cdot : R \times R \rightarrow R$ are binary operations satisfying

1. $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$ for all $a, b, c \in R$
2. There exists an element $0 \in R$ such that $a + 0 = a = 0 + a$ for all $a \in R$.

3. For every $a \in R$, there exists an element $-a \in R$ such that $a + (-a) = 0 = (-a) + a$ for all $a \in R$.
4. There exists an element $1 \in R$ such that $a \cdot 1 = a = 1 \cdot a$ for all $a \in R$.
5. For all $a, b, c \in R$, we have

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c.$$

The operation $+$ is often called *addition* and the operation $\cdot$ is often called *multiplication*. Accordingly, the identity element 0 of $+$ is often called the *additive identity* and the identity element 1 of $\cdot$ is often called the *multiplicative identity*.

While other notations may be used to denote addition and multiplication, if no notation is specified explicitly, then $+$ and $\cdot$ are universally used as default notations for addition and multiplication in a ring.

Remark C.0.2. Some writers might not require a ring to have a multiplicative identity element, i.e. would define a ring so that $(R, +)$ is a commutative group, $(R, \cdot)$ is a semigroup, and $\cdot$ is distributive over $+$. Such writers would call the notion of ring in Definition C.0.1 a *unitary ring* to emphasize the existence of the multiplicative identity 1 .

Definition C.0.3. A *commutative (unital) ring* is a ring (Definition C.0.1) $(R, +, \cdot)$ such that $\cdot$ is a commutative operation, i.e. $a \cdot b = b \cdot a$.

For many writers (e.g. “commutative” algebraists or number theorists), a *ring* refers to a commutative ring as above.

Definition C.0.4. Let $(R, +, \cdot)$ and $(S, +, \cdot)$ be rings (Definition C.0.1), not assumed to be commutative. A function $f : R \rightarrow S$ is called a *ring homomorphism* if for all $r_1, r_2 \in R$ the following properties hold:

1. $f(r_1 + r_2) = f(r_1) + f(r_2)$,
2. $f(r_1 r_2) = f(r_1) f(r_2)$,
3. $f(1_R) = 1_S$ where 1_R and 1_S denote the multiplicative identities in R and S , respectively.

A ring homomorphism is said to be a *ring isomorphism* if it is invertible as a map of sets.

An *R -ring* refers to a ring S equipped with a ring homomorphism $f : R \rightarrow S$.

We note that a ring homomorphism $f : R \rightarrow S$ yields a natural left R -module (Definition C.0.8) structure on S and a natural right R -module structure on S respectively as follows for $r \in R$ and $s \in S$:

$$\begin{aligned} r \cdot s &= f(r) \cdot s \\ s \cdot r &= s \cdot f(r). \end{aligned}$$

However, these left and right module structures need not yield a two-sided R -module structure.

Definition C.0.5 (Groups). A *group* is a monoid such that every element has an inverse.

In other words, a group is a pair $(G, \cdot)$ where G is a set (Definition A.0.1) and $\cdot : G \times G \rightarrow G$ is a binary operation, subject to the following conditions:

1. (Associativity) For all $g, h, k \in G$ one has

$$(g \cdot h) \cdot k = g \cdot (h \cdot k).$$

2. (Identity element) There exists an identity element, i.e. an element $e \in G$ such that for all $g \in G$,

$$e \cdot g = g \cdot e = g.$$

3. (Inverse elements) For all $g \in G$, there exists an inverse element of g , i.e. an element $g^{-1} \in G$ such that

$$g \cdot g^{-1} = g^{-1} \cdot g = e.$$

Equivalently, a group is a group object (Definition 1.3.19) in the category of sets (Definition 1.0.3).

A group $(G, \cdot)$ is often simply written as G , when the notation for the binary operation $\cdot$ is clear.

Assuming that the notation for the identity element of a group is not explicitly specified,

1. if the group operation is written multiplicatively, then one usually implicitly writes e or 1 as the identity of G , and
2. If the group operation is instead written additively, then one usually implicitly write 0 as the identity of G .

An *abelian group* or synonymously, a *commutative group*, is a group $(G, \cdot)$ whose binary operation $\cdot$ is *abelian* or *commutative*, i.e. satisfies

$$g \cdot h = h \cdot g$$

for all $g, h \in G$. A group operation written additively, i.e. with $+$, is almost usually abelian. An abelian group's group operation may still often be written multiplicatively.

An abelian group is equivalent to a $\mathbb{Z}$ -module.

Definition C.0.6. (♠ TODO: AI generated, verify) A *trivial group* is a group (Definition C.0.5) consisting of exactly one element. That is, a trivial group is a group $(G, \cdot)$ such that $|G| = 1$. The unique element of a trivial group is necessarily the identity element (Definition C.0.5) of the group.

A trivial group is always an abelian group (Definition C.0.5) (since the unique element commutes with itself). When the fact that it is an abelian group is emphasized and its group structure is written additively, a trivial group is also called a *zero group* (or *zero object* in the category of abelian groups).

Up to isomorphism of groups, there is exactly one trivial group. In other words, the trivial group is the unique group of cardinality one up to isomorphism.

Definition C.0.7 (Group homomorphism). Let $(G, \cdot)$ and $(H, *)$ be groups (Definition C.0.5). A map $f : G \rightarrow H$ is called a *group homomorphism* if for all $g_1, g_2 \in G$ one has

$$f(g_1 \cdot g_2) = f(g_1) * f(g_2).$$

The collection of all groups with the group homomorphisms forms a locally small (Definition 1.0.9) category (Definition 1.0.1), called the *category of groups*.

If f is bijective (Definition A.0.3), then f is called a *group isomorphism*. Equivalently, a group isomorphism is an isomorphism (Definition 1.0.8) in the category of groups.

Definition C.0.8. Let R be a not-necessarily commutative ring (Definition C.0.1).

1. A *left R -module* is an abelian group $(M, +)$ together with an operation $R \times M \rightarrow M$, denoted $(r, m) \mapsto rm$, such that for all $r, s \in R$ and $m, n \in M$:
 - $r(m + n) = rm + rn$,
 - $(r + s)m = rm + sm$,
 - $(rs)m = r(sm)$,
 - $1_R m = m$ where 1_R is the multiplicative identity of R .
2. A *right R -module* is defined similarly as an abelian group $(M, +)$ with an operation $M \times R \rightarrow M$, denoted $(m, r) \mapsto mr$, such that for all $r, s \in R$ and $m, n \in M$:
 - $(m + n)r = mr + nr$,
 - $m(r + s) = mr + ms$,
 - $m(rs) = (mr)s$,
 - $m1_R = m$.
3. Let R and S be (not necessarily commutative) rings (Definition C.0.1).
 An *R - S -bimodule* (or an *R - S -module* or an (R, S) -module, etc.) is an abelian group (Definition C.0.5) $(M, +)$ equipped with
 - (a) a left action of R :

$$R \times M \rightarrow M, \quad (r, m) \mapsto r \cdot m,$$

making M a left R -module (Definition C.0.8),

- (b) a right action of S :

$$M \times S \rightarrow M, \quad (m, s) \mapsto m \cdot s,$$

making M a right S -module,

such that the left and right actions commute; that is, for all $r \in R$, $s \in S$, and $m \in M$,

$$r \cdot (m \cdot s) = (r \cdot m) \cdot s.$$

4. A *two-sided R -module* (or *R -bimodule*) is an R - R -bimodule.

If R is a commutative ring (Definition C.0.3), then a left/right R -module can automatically be regarded as a two-sided R -module. As such, we simply talk about *R -modules* in this case.

Any abelian group is equivalent to a two-sided $\mathbb{Z}$ -module. Moreover, any left R -module is equivalent to an $R - \mathbb{Z}$ -bimodule (Definition C.0.8) and any right R -module is equivalent to an $\mathbb{Z} - R$ -bimodule (Definition C.0.8). Given a left/right/two-sided R -module, its *natural bimodule structure* will refer to its structure as a $R - \mathbb{Z} / \mathbb{Z} - R / R - R$ bimodule. In this way, many definitions associated with the notions of left/right/two-sided R -modules can be defined as special cases for definitions for R - S -bimodules.

Definition C.0.9. (♠ TODO: AI generated, verify) Let $A_1, \dots, A_k, B$ be abelian groups (Definition C.0.5) (written additively). A function

$$\Phi : A_1 \times \dots \times A_k \rightarrow B$$

is called a *multilinear map of abelian groups* (or simply a *multilinear map* when the context is clear) if for each $j = 1, \dots, k$ and for all fixed $a_i \in A_i$ with $i \neq j$, the map

$$A_j \rightarrow B, \quad a_j \mapsto \Phi(a_1, \dots, a_j, \dots, a_k)$$

is a group homomorphism (Definition C.0.7) of abelian groups.

When $k = 2$, a multilinear map is called a *bilinear map of abelian groups* (or simply a *bilinear map*). Explicitly, a map $\Phi : A_1 \times A_2 \rightarrow B$ is bilinear if for all $a_1, a'_1 \in A_1$, $a_2, a'_2 \in A_2$:

$$\begin{aligned} \Phi(a_1 + a'_1, a_2) &= \Phi(a_1, a_2) + \Phi(a'_1, a_2), \\ \Phi(a_1, a_2 + a'_2) &= \Phi(a_1, a_2) + \Phi(a_1, a'_2). \end{aligned}$$

Similarly, one may speak of *trilinear maps* ($k = 3$), *quadrilinear maps* ($k = 4$), etc.

Definition C.0.10. Let R, S be (not-necessarily commutative) rings (Definition C.0.1).

1. Let M and N be R - S -bimodules (Definition C.0.8). A function $\varphi : M \rightarrow N$ is called an *R - S -bimodule homomorphism* or *R - S -linear* if it is a group homomorphism (Definition C.0.7) of the underlying abelian groups of M and N and respects the scalar actions as follows: for all $m_1, m_2 \in M$, $r \in R$, and $s \in S$,

$$\begin{aligned} \varphi(r \cdot m_1) &= r \cdot \varphi(m_1), \\ \varphi(m_1 \cdot s) &= \varphi(m_1) \cdot s. \end{aligned}$$

2. Let M and N be left/right/two-sided R -modules (Definition C.0.8). A function $\varphi : M \rightarrow N$ is called a *left/right/two-sided R -module homomorphism* if it is an bimodule homomorphism on the natural bimodule structures (Definition C.0.8) of M and N . Such a function is also called *R -linear*.

Modules and homomorphisms of a fixed type (i.e. R - S -bimodules or left/right/two-sided R -modules) form a locally small (Definition 1.0.9) category (Definition 1.0.1).

A module homomorphism is an *isomorphism* if it is a bijective (Definition A.0.3) map. Equivalently, a module isomorphism is an isomorphism (Definition 1.0.8) between the modules as objects in the appropriate module category (Definition C.0.11)

Definition C.0.11. Let R and S be (not necessarily commutative) rings (Definition C.0.1).

1. The *category of (R, S) -bimodules* (or R - S -bimodules), denoted by notations such as ${}_R\mathbf{Mod}_S$, is the category whose objects are (R, S) -bimodules (Definition C.0.8) and whose R - S -bimodule homomorphisms (Definition C.0.10).
2. The *category of left R -modules*, denoted by notations such as ${}_R\mathbf{Mod}$ or $R - \mathbf{Mod}$, is the category ${}_R\mathbf{Mod}_{\mathbb{Z}}$, i.e. the category whose objects are left R -modules (Definition C.0.8) and whose morphisms are left R -linear maps (Definition C.0.10).

3. The *category of right R -modules*, denoted by notations such as $\mathbf{Mod}_R$ or $\mathbf{Mod} - R$, is the category ${}_Z\mathbf{Mod}_R$, i.e. the category whose objects are right R -modules (Definition C.0.8) and whose morphisms are right R -linear maps (Definition C.0.10).

The category of bimodules can be canonically identified with module categories over tensor product rings:

- ${}_R\mathbf{Mod}_S$ is isomorphic to the category of left modules over the ring $R \otimes_{\mathbb{Z}} S^{\text{op}}$.
- ${}_R\mathbf{Mod}_S$ is isomorphic to the category of right modules over the ring $R^{\text{op}} \otimes_{\mathbb{Z}} S$.

Consequently, standard module-theoretic concepts (such as projective objects, injective objects, and flat objects) in ${}_R\mathbf{Mod}_S$ correspond exactly to the respective concepts in ${}_{R \otimes_{\mathbb{Z}} S^{\text{op}}}\mathbf{Mod}$.

Note that there are canonical isomorphisms of categories:

$${}_R\mathbf{Mod} \cong {}_R\mathbf{Mod}_{\mathbb{Z}} \quad \text{and} \quad \mathbf{Mod}_S \cong {}_{\mathbb{Z}}\mathbf{Mod}_S.$$

That is, left R -modules are exactly $(R, \mathbb{Z})$ -bimodules, and right S -modules are exactly $(\mathbb{Z}, S)$ -bimodules.

APPENDIX D. SCHEMES

Definition D.0.1. A morphism of schemes $f : X \rightarrow Y$ is called *étale* if it satisfies the following conditions:

- f is locally of finite presentation,
- f is flat,
- f is unramified, i.e., the sheaf of relative differentials $\Omega_{X/Y}$ equals 0.

(♠ **TODO: relative dimension**) Equivalently, a morphism of schemes is étale if and only if it is smooth of relative dimension 0.

APPENDIX E. GROTHENDIECK UNIVERSES

Recall that the collection of all sets within Zermelo-Fraenkel set theory is a proper class and not a set. We often use Grothendieck universes in categorical discussions as a way to restrict the sets considered so as to guarantee that only sets result in the categorical constructions that matter (e.g. to guarantee that the hom's in a newly constructed category form a set).

Definition E.0.1 (Grothendieck Universe). Let U be a set. We say U is a *Grothendieck universe* (or just a *universe*) if the following conditions hold:

1. If $x \in U$ and $y \in x$, then $y \in U$ (transitivity).
2. If $x, y \in U$, then $\{x, y\} \in U$ (closed under pair formation).
3. If $x \in U$, then the power set $\mathcal{P}(x) \in U$.
4. If $I \in U$ and $(x_\alpha)_{\alpha \in I}$ is a family with each $x_\alpha \in U$, then $\bigcup_{\alpha \in I} x_\alpha \in U$.

A set X is called *U -small* or a *U -set* if $X \in U$.

Remark E.0.2. Given a Grothendieck universe U , the collection of all U -small sets is U itself, which contrasts against the fact that the collection of all sets (for the ZF (and ZFC) axioms) is a proper class.

Remark E.0.3. Within ZF/ZFC alone, there is no guarantee that every given set S satisfies $S \in U$ for some universe S .

Remark E.0.4. While universes are a powerful tool for studying category theory, they are not universally used or necessary for doing so. Usage depends on the context and foundational preferences. For instance, Grothendieck universes can be used to simplify foundational technicalities when one needs to work with such things as “categories of all categories” or “functor categories” of large size.

Definition E.0.5. A set x is a **transitive set** if every element of x is also a subset of x . Formally, x is transitive if for all $y \in x$, the condition $y \subseteq x$ holds, which is equivalent to the statement that for all $z \in y$, it follows that $z \in x$.

Proposition E.0.6. For any set x , the following conditions are equivalent:

1. x is a transitive set (Definition E.0.5).
2. $\cup x \subseteq x$.
3. $x \subseteq \mathcal{P}(x)$, where $\mathcal{P}(x)$ denotes the power set of x .

Definition E.0.7. The **transitive closure of a set x** denoted $\text{tc}(x)$, is the unique minimal transitive set such that $x \subseteq \text{tc}(x)$. It can be constructed as the union $\bigcup_{n \in \mathbb{N}} \cup^n x$, where $\cup^0 x = x$ and $\cup^{n+1} x = \bigcup(\cup^n x)$.

Definition E.0.8. A set α is an **ordinal number** (or simply an **ordinal**) if it satisfies the following two conditions:

1. α is **transitive**, meaning that every element of α is also a subset of α . That is, if $x \in \alpha$, then $x \subseteq \alpha$.
2. α is **strictly well-ordered** by the membership relation $\in$. That is, the relation $<$ on α defined by $x < y \iff x \in y$ is a strict total ordering, and every non-empty subset of α has a least element under this ordering.

The class of all ordinal numbers is denoted by **Ord** or **ON**. For ordinals α and β , we write $\alpha < \beta$ when $\alpha \in \beta$, and we write and $\alpha \leq \beta$ when $\alpha \in \beta$ or $\alpha = \beta$.

Theorem E.0.9. An ordinal (Definition E.0.8) is a transitive set (Definition E.0.5) that is strictly well-ordered by the membership relation $\in$. (♠ TODO: strictly well-ordered)

Definition E.0.10. A set x is a **hereditarily finite set** if its transitive closure (Definition E.0.7) $\text{tc}(x)$ is a finite set.

The collection of all hereditarily finite sets is denoted by V_ω . It is a set (Proposition E.0.11).

Proposition E.0.11. The collection V_ω of all hereditarily finite sets (Definition E.0.10) is a set and it can be constructed via the cumulative hierarchy by defining $V_0 = \emptyset$, $V_{n+1} = \mathcal{P}(V_n)$ for $n \in \mathbb{N}$, and taking the union $V_\omega = \bigcup_{n < \omega} V_n$.

Theorem E.0.12. A set x is hereditarily finite (Definition E.0.10) if and only if x is finite and every element $y \in x$ is also a hereditarily finite set.

Proposition E.0.13. The structure $(V_\omega, \in)$ satisfies all the axioms of Zermelo-Fraenkel set theory (Definition A.0.1) excluding the Axiom of Infinity.

Theorem E.0.14. The set V_ω of all hereditarily finite sets (Definition E.0.10) is a Grothendieck universe (Definition E.0.1). Furthermore, V_ω is initial in the sense that it is contained in every Grothendieck universe (that may exist with respect to the chosen system of axioms containing ZF (Definition A.0.1)); more precisely, it is the unique minimal Grothendieck universe under the inclusion relation $\subseteq$.

Definition E.0.15. *Tarski-Grothendieck set theory*, denoted by **TG**, is the theory consisting of the axioms of ZFC (Definition A.0.1) together with *Tarski's Axiom of Universes*, which asserts that for every set x there exists a universe (Definition E.0.1) U such that $x \in U$:

$$\forall x \exists U (U \text{ is a Grothendieck universe} \wedge x \in U)$$

Definition E.0.16. Given an axiom system containing the Tarski-Grothendieck set theory (Definition E.0.15) axioms, the *hierarchy of Grothendieck universes* consists of the Grothendieck universes (Definition E.0.1) U_n for $n \in \mathbb{N} \cup \{0\}$ constructed as follows:

1. Base case: $U_0 := V_\omega$, the set of all hereditarily finite sets (Definition E.0.10).
2. Successor step: For each $n \geq 0$, U_{n+1} is the unique minimal Grothendieck universe such that $U_n \in U_{n+1}$.

In particular, $U_n \in U_{n+1}$ and $U_n \subset U_{n+1}$ for all n .

Definition E.0.17 (Compact topological space). A topological space $(X, \mathcal{T})$ is *compact* if every open cover of X admits a finite subcover; that is, for every collection $\{U_i\}_{i \in I}$ of open sets in $\mathcal{T}$ such that $X = \bigcup_{i \in I} U_i$, there exists a finite subcollection $\{U_{i_j}\}_{j=1}^n$ such that $X = \bigcup_{j=1}^n U_{i_j}$.

Some mathematicians, e.g. algebraic geometers, would refer to this property as *quasi-compactness*.

Definition E.0.18 (Separation axioms). Let $(X, \mathcal{T})$ be a topological space (Definition B.0.1).

- $(X, \mathcal{T})$ is **T_0** (*Kolmogorov*) if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that, without loss of generality, $x \in U$ and $y \notin U$.
- $(X, \mathcal{T})$ is **T_1** (*Fréchet*) if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \in \mathcal{T}$ such that $x \in U$, $y \notin U$, and $y \in V$, $x \notin V$.
- $(X, \mathcal{T})$ is **T_2** or *Hausdorff* if for every pair of distinct points $x, y \in X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $y \in V$.
- $(X, \mathcal{T})$ is *regular* if it is T_1 and for each point $x \in X$ and closed set $F \subseteq X$ with $x \notin F$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $F \subseteq V$.
- $(X, \mathcal{T})$ is **T_3** (regular Hausdorff) if it is T_1 and regular.
- $(X, \mathcal{T})$ is *completely regular* if for each closed set $F \subseteq X$ and $x \notin F$, there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(x) = 0$ and $f|_F = 1$.
- $(X, \mathcal{T})$ is **$T_{3\frac{1}{2}}$** (completely regular Hausdorff) if it is T_1 and completely regular.

- $(X, \mathcal{T})$ is **normal** if it is T_1 and for each pair of disjoint closed sets $A, B \subseteq X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $A \subseteq U$ and $B \subseteq V$.
- $(X, \mathcal{T})$ is **T_4** (normal Hausdorff) if it is T_1 and normal.
- $(X, \mathcal{T})$ is **T_5** (completely normal Hausdorff) if it is T_1 and completely normal.
- $(X, \mathcal{T})$ is **perfectly normal** if every closed set is a $G\delta$ (countable intersection of open sets) and the space is normal.
- $(X, \mathcal{T})$ is **T_6** (perfectly normal Hausdorff) if it is T_1 and perfectly normal.

Definition E.0.19. Let $\mathcal{C}$ be a category (Definition 1.0.1). A **category fibred in groupoids over $\mathcal{C}$** or a **CFG over $\mathcal{C}$** for short, is a fibered category $p : \mathcal{X} \rightarrow \mathcal{C}$ such that for every object U of $\mathcal{C}$, the fiber category $\mathcal{X}_U$ is a groupoid.

A **morphism of categories fibred in groupoids over $\mathcal{C}$** is a morphism as fibred categories over $\mathcal{C}$.

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